National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Prostate Cancer

Version 5.2026 — January 23, 2026

NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

NCCN Guidelines for Patients® available at www.nccn.org/patients

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NCCN Guidelines Version 5.2026 Prostate Cancer

NCCN Prostate Cancer Panel Members
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NCCN Categories of Evidence and Consensus: All recommendations are category 2A unless otherwise indicated.

See NCCN Categories of Evidence and Consensus.

NCCN Categories of Preference: All recommendations are considered appropriate.

See NCCN Categories of Preference.

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Updates in Version 5.2026 of the NCCN Guidelines for Prostate Cancer include:

PROS-14
- Treatment for Low-Volume M1 CSPC disease, regimen added to synchronous and metachronous settings
  - ADT with: Useful in certain circumstances, Niraparib/abiraterone (BRCA2m only) (category 2B)

PROS-15
- Treatment for High-Volume M1 CSPC disease, regimen added
  - ADT with: Useful in certain circumstances, Niraparib/abiraterone (BRCA2m only)

PROS-15A
- Footnote bbb added: Niraparib/abiraterone can be used following the completion of docetaxel in patients receiving triplet therapy or following EBRT to the primary tumor. The use of this combination in patients with low-volume disease is controversial because 78% of the patients in the trial had high-volume disease and there were very few patients with low-volume metachronous disease in particular. This regimen adds considerable toxicity, and patients with low-volume disease should be informed that the benefit in this setting is not clear.

PROS-G (4 of 6)
- ADT for patients with mCSPC, sub-header added: Additional option for patients with BRCA2m mCSPC
  - Bullet added: Orchiectomy, LHRH agonist, or LHRH antagonist plus niraparib and abiraterone
- Footnote g added: The fine-particle (category 2B; other recommended option) or standard formulation of abiraterone can be given with single-agent niraparib as a substitute for the combination niraparib/abiraterone tablet.

PROS-N (1 of 5)
- Non-hormonal systemic therapy for M1 castration-sensitive prostate cancer
  - Bullet 2 added: An additional option for patients with BRCA2-mutated mCSPC is the use of ADT with niraparib and abiraterone.
    - ◊ Sub-bullet 1 added: ADT with niraparib and abiraterone was compared to ADT plus abiraterone doublet therapy in a randomimzed phase 3 trial. Most patients had highvolume and/or synchronous disease. rPFS was improved in the BRCAm population; OS data are immature. Grade 3/4 adverse events were more common in the niraparib arm (75% vs. 59%) as were treatment-emergent adverse events leading to death (14 participants vs. 7 participants). The most common grade 3/4 adverse events were anemia (29.1% in the niraparib group versus 4.6% in the control group) and hypertension (26.5% versus 18.4%).
    - ◊ Sub-bullet 2 added: Niraparib/abiraterone can be used following the completion of docetaxel in patients receiving triplet therapy or following EBRT to the primary tumor.
    - ◊ Sub-bullet 3 added: The use of this combination in patients with low-volume disease is controversial because 78% of the patients in the trial had high-volume disease and there were very few patients with low-volume metachronous disease in particular. This regimen adds considerable toxicity, and patients with low-volume disease should be informed that the benefit in this setting is not clear.
    - ◊ Sub-bullet 4 added: The fine-particle (category 2B; other recommended option) or standard formulation of abiraterone can be given with single-agent niraparib as a substitute for the combination niraparib/abiraterone tablet.

MS-1
- The Discussion section was updated to reflect the changes in the algorithm.

Updates in Version 4.2026 of the NCCN Guidelines for Prostate Cancer include:

PROS-J (9 of 11)
- Table 1, sub-header modified: EBRT + Brachytherapy Boost *(combined with 45-50.4 Gy in 25-28 fx or 37.5 Gy in 15 fx).*

MS-1
- The remainder of the Discussion section was updated to reflect the changes in the algorithm.

Continued

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Updates in Version 3.2026 of the NCCN Guidelines for Prostate Cancer include:

PROS-G (3 of 6)
- ADT for Patients with M0 CSPC BCR2
  - Sub-heading modified: *For High-Risk BCR2* ~~Useful in Certain Circumstances~~

PROS-G (3 of 6)(continued)
  - Bullet 1 modified: Enzalutamide with or without Leuprolide ~~(for high-risk BCR2)~~
  - Bullet 2 modified: Apalutamide with LHRH agonist or LHRH antagonist ~~(for high-risk BCR2)~~

PROS-H
- Section extensively revised.

PROS-N (3 of 4)
- Immunotherapy, bullet 2, sub-bullet 3 added: Pembrolizumab and berahyaluronidase alfa-pmph subcutaneous injection may be substituted for IV pembrolizumab. Pembrolizumab and berahyaluronidase alfa-pmph has different dosing and administration instructions compared to IV pembrolizumab.

MS-1
- The section of the Discussion regarding "Advanced Risk Stratification Tools" was updated to reflect the changes in the algorithm.

Updates in Version 2.2026 of the NCCN Guidelines for Prostate Cancer from Version 1.2026 include:

MS-1
- The sections of the Discussion regarding mCSPC and mCRPC were updated to reflect the changes in the algorithm.

Updates in Version 1.2026 of the NCCN Guidelines for Prostate Cancer from Version 2.2025 include:

Global
- Very-low-risk group has been removed from the Guidelines.
- Androgen receptor pathway inhibitors ("ARPIs") has replaced "other hormonal agents" throughout the Guidelines.
- References have been updated throughout the Guidelines.

PROS-1
- Workup
  - Top pathway
    - ◊ Bullet 3 modified: Perform and/or collect prostate-specific antigen (PSA) ~~and calculate PSA density.~~
    - ◊ Bullet 4 modified: ~~Obtain and~~ Review diagnostic prostate biopsies.
  - Bottom pathway
    - ◊ Bullet 3 modified: *Consider* ~~Perform~~ DRE to confirm clinical stage.
    - ◊ Last column, pathway added: See Workup and Treatment of High-Volume M1 CSPC (PROS-15).
- Footnote d added: Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).
- Footnote h added: See "Number of Metastatic Sites" in the Principles of MDT (PROS-M). (Also for PROS-2A, 7A, -8A, -9A, -10A, -11, -12, -13, and -15A, and 18A)

PROS-2
- Additional Evaluation for Intermediate, High, and Very high, sub-bullets for soft-tissue imaging
  - Sub-bullet 1 modified: If regional ~~or distant~~ metastases are found, see PROS-7 ~~or~~
  - Sub-bullet added: If distant metastases are found, see PROS-14 for Low-Volume M1 (Metachronous or Synchronous) or Synchronous Oligometastatic CSPC or PROS-15 for High-Volume M1 CSPC.

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Updates in Version 1.2026 of the NCCN Guidelines for Prostate Cancer from Version 2.2025 include:

PROS-2A
- Footnote removed: An ultrasound-, MRI-, or DRE-targeted lesion that is biopsied more than once and demonstrates cancer (regardless of percentage core involvement or number of cores involved) can be considered as a single positive core.

PROS-3
- Footnote n added: Principles of Focal/Subtotal Therapy or Whole Gland Ablative Therapy (PROS-I). (Also for PROS-4, -5, -6, and -7A)

PROS-6
- ≤5 y and asymptomatic
  - Initial therapy modified: Observation or RT ± *ADT* ~~or ADT ± RT~~.

PROS-7A
- Footnote x modified: Adverse ~~laboratory/~~pathologic features include: positive margin(s), seminal vesicle invasion, *or* extracapsular extension~~, or detectable PSA~~.
- Footnote removed: The Panel remains concerned about the problems of overtreatment related to the increased diagnosis of early prostate cancer from PSA testing. See NCCN Guidelines for Prostate Cancer Early Detection. Active surveillance is recommended for this subset of patients.
- Footnote removed: The fine-particle formulation of abiraterone can be used instead of the standard form (category 2B; other recommended option). (Also for PROS-9A, -10A, -15A)

PROS-9
- PSA persistence/recurrence
  - Life expectancy >5 y: treatment recommendations extensively revised based on No evidence of M1 and M1 disease.

PROS-9A
- Footnote q, sentence 1 modified: Observation involves monitoring the course of disease with the expectation to deliver *definitive or* palliative therapy for the development of symptoms or a change in examination or PSA that suggests symptoms are imminent.
- Footnote jj added: Treatment for a first biochemical recurrence has historically been referred to as 'salvage' therapy. In efforts to use language that is more sensitive to patients, the NCCN Guidelines for Prostate Cancer refer to treatment in this setting as 'secondary' therapy. (Also for PROS-10A)
- Footnote ll added: For patients with PSA progression who have not received treatment for a first BCR, continue monitoring or consider treatment for first PSA persistence/recurrence. See PROS-9 for RP PSA persistence/recurrent or PROS-10 for RT recurrence. (Also for PROS-10A)
- Footnote removed: If considering treatment, reinitiate the PROS-10 algorithm.

PROS-10
- PSA recurrence or positive DRE
  - Life expectancy >5 y: treatment recommendations extensively revised based on No evidence of M1 and M1 disease.

PROS-11
- New page added: Second Biochemical Recurrence (BCR2).

PROS-12
- Page changed
  - New algorithm added: Treatment and Monitoring for Second Biochemical Recurrence (N0M0).
  - Algorithm removed: Treatment and Monitoring for Progressive M0 Castration-Sensitive Prostate Cancer (CSPC) After Maximal Pelvic Therapy.
- Footnote pp added: PSA level and PSADT should be considered when deciding whether to begin ADT for patients considered to have low risk disease. High-risk BCR has different definitions in different clinical trials that generally includes a PSADT ≤9 months and other adverse prognostic features.
- Footnote qq modified: ~~For patients with non-metastatic castration-sensitive disease, (by CT, MRI, or bone scan) who are not candidates for pelvic therapy, monitoring until diagnosis of metastatic disease is preferred. PSADT and Grade Group should be considered when deciding whether to begin ADT for patients with M0 disease. For ADT alone~~ *If favorable PSA response, an* intermittent ADT *approach should be considered*. ~~can be considered to reduce toxicity.~~

PROS-13
- New page added: Workup and Treatment of Metachronous Oligometastatic CSPC.

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Updates in Version 1.2026 of the NCCN Guidelines for Prostate Cancer from Version 2.2025 include:

PROS-14
- Page header modified: Workup and Treatment of *Low-Volume* M1 *(Metachronous or Synchronous)* or *Synchronous Oligometastatic* CSPC.
- Synchronous low-volume metastases or Synchronous oligometastatic disease, second option modified: ADT with docetaxel and one of the following *(low-volume disease only).*
- Progression modified: Progression *on ADT ± ARPI*.

PROS-15
- Page header modified: Workup and Treatment of *High-Volume* M1 CSPC.

PROS-15A
- Footnote zz added: Concurrent MDT can be considered in select patients with oligometastatic disease. See Principles of MDT (PROS-M).
- Footnote removed: Stereotactic body RT (SBRT) to metastases can be considered in appropriate clinical situations. See Principles of Radiation Therapy (PROS-I).

PROS-18
- Systemic Therapy for M1 CRPC: Adenocarcinoma
  - Page extensively revised and reorganized by Pre-ARPI, Post-ARPI/Pre-Docetaxel, Post-ARPI/Post-Docetaxel, Additional Options Irrespective of Prior ARPI or Prior Docetaxel.

PROS-18A
- Footnote kkk modified: ~~Novel hormone~~ *ARPI* therapies include abiraterone, enzalutamide, darolutamide, or apalutamide. Abiraterone given as part of neoadjuvant/concomitant/adjuvant ADT with EBRT is not considered *post-ARPI* ~~prior novel hormone therapy~~.
- Footnote mmm added: Principles of MDT (PROS-M).
- Footnote ooo, sentence 3 added: There are limited data to support the efficacy of Sipuleucel-T in the post-chemotherapy setting.
- Footnote removed: Principles of Radiation Therapy (PROS-I).
- Footnote removed: Visceral metastases refers to liver, lung, adrenal, peritoneal, and brain metastases. Soft tissue/lymph node sites are not considered visceral metastases.
- Footnote removed: Pan-cancer, tumor-agnostic treatments can be considered for patients with actionable mutations.
- Footnote removed: The fine-particle formulation of abiraterone can be used instead of the standard form (other recommended option).
- Footnote removed: The fine-particle (category 2B; other recommended option) or standard formulation of abiraterone can be given with single-agent niraparib as a substitute for the combination of niraparib/abiraterone tablet.

PROS-B
- Section renamed: Principles of Survivorship.

PROS-B (5 of 6)
- New page added: Cardiovascular Disease in Prostate Cancer.

PROS-B (6 of 6)
- New page added: Cardiovascular Disease in Prostate Cancer: Risk Assessment.

PROS-D
- Bullet 5 added: Consider geriatric assessment. See NCCN Guidelines for Older Adult Oncology for tools to aid optimal assessment and management of disease in older adults.
- Bullet 6 added: Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).

PROS-E (2 of 5)
- Bone Imaging
  - Bullet 2 modified: Plain films, CT, MRI, or PET/CT or PET/MRI with F-18 piflufolastat prostate-specific membrane antigen (PSMA), Ga-68 PSMA-11, F-18 flotufolastat PSMA, *F-18 fluciclovine*, F-18 sodium fluoride, or C-11 choline can be considered for equivocal results on initial bone scan.

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Updates in Version 1.2026 of the NCCN Guidelines for Prostate Cancer from Version 2.2025 include:

PROS-E (3 of 5)
- Full Body Imaging
  - Positron Emission Tomography
    - Bullet 1
      - Sub-bullet 1, last sentence modified: See Table *1 on PROS-E (5 of 5)* ~~2 in the discussion section~~ for more details.
      - Sub-bullet 3 modified: PSMA-PET/CT or PET/MRI can be considered as an alternative to CT, MRI, and bone scans for initial staging *in unfavorable intemediate-, high-, and very-high-risk disease,* the detection...
      - Sub-bullet 4 added: PSMA-PET imaging should only be used in the setting of M1 CRPC to determine if a patient is a candidate for Lu-177-PSMA-617.
      - Sub-bullet 5 added: Changes in systemic therapy should not be made solely based on a positive PSMA-PET in patients with M0 CRPC.
      - Sub-bullet 6 added: Bone scans can be considered to confirm osseous uptake on PET scans.
      - Sub-bullet 8 modified: C-11 choline *or F-18 fluciclovine* PET/CT or PET/MRI...
      - Sub-bullet 9 modified: Studies suggest that PSMA-PET imaging has a higher sensitivity than C-11 choline *or F-18 fluciclovine* PET imaging, especially at very low PSA levels *(eg, <0.5 ng/mL).*

PROS-E (4 of 5)
- Sub-bullet modified (continued from PROS-E [3 of 5]) ...interpretation of scans. *The use of a standardized reporting system is encouraged.* ~~Several reporting systems have been proposed but will not have been validated or widely used.~~
- Sub-bullet 3 modified: Table *1 on PROS-E (5 of 5)* ~~in the discussion section~~ provides a summary of the main imaging tracers utilized for study in prostate cancer ~~both before definitive therapy and at recurrence~~.
- Imaging as Workup for Progression
  - Bullet 2 modified: Imaging for patients with progressive ~~m~~*CRPC* should included chest CT, bone imaging, and abdomen/pelvis CT with contrast or abdomen/pelvis MRI with and without contrast.
  - Bullet 3 modified: There is a lack of evidenced to support the use of PET imaging in ~~this~~ *the CRPC* setting *and it should only be used in the setting of M1 CRPC to determine if a patient is a candidate for Lu-177-PSMA-617. Changes in treatment should not be made solely based on a positive PSMA-PET in patients with M0 CRPC.*

PROS-E (5 of 5)
- New table added: FDA-Cleared PET Imaging Tracers Studied in Prostate Cancer.
- Footnote a added: Interpret with caution. Wherever possible, studies were included that used histopathologic confirmation, but not all studies used confirmatory histology as the gold standard. Values may vary depending upon the site of the lesion and phase of the disease process.
- Footnote b added: CLR: Correct localization rate. Patient-level positive predictive value + anatomic lesion co-localization. Preferred over sensitivity and specificity in analyses of patients with BCR.

PROS-F
- Principles of Active Surveillance: Section extensively revised.

PROS-G (1 of 6)
- Neoadjuvant, Concurrent, and/or Adjuvant ADT with RT
  - Bullet 4 modified: *First* M0 RP PSA persistence/recurrence ~~(without maximal pelvic therapy)~~.
  - Notes
    - Bullet 1 added: ADT is not recommended with RT for most patients with favorable intermediate-risk prostate cancer. If it is given (see Principles of Risk Stratification), the duration should be short term (4–6 months).
    - Bullet 4 modified: For high-risk ~~and very-high-risk~~ prostate cancer...
    - Bullet 6 modified: *First* M0 PSA persistence/recurrence:
      - Sub-bullet 3 added: For patients treated with secondary RT in the setting of first RP recurrence, if ADT is given, it should be for a duration of 6 to 24 months.

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Updates in Version 1.2026 of the NCCN Guidelines for Prostate Cancer from Version 2.2025 include:

PROS-G (2 of 6)
- ADT for Patients with M0 CSPC, bullet 2 modified: *First* M0 RT recurrence ~~(without maximal pelvic therapy)~~.

PROS-G (3 of 6)
- ADT for Patients with M0 CSPC *BCR2* ~~After Maximal Pelvic Therapy~~.
  - Useful in Certain Circumstances
    - ◊ Bullet 1 modified: Enzalutamide with or without leuprolide *(for high-risk BCR2).*
    - ◊ Bullet 2 modified: Apalutamide with LHRH agonist or LHRH antagonist ~~(category 2B)~~ *(for high-risk BCR2)*.
  - Notes, bullet 1 modified: Monitoring until diagnosis of metastatic disease is preferred for patients with *M0 CSPC in the low-risk BCR2 setting* ~~non-metastatic castration-sensitive disease who are not candidates for pelvic therapy~~.
  - Footnote c added: For patients receiving MDT with ADT, the ADT options are LHRH agonist or LHRH antagonist.

PROS-G (5 of 6)
- ADT Monotherapy
  - Notes
    - ◊ Bullet 1 modified: ADT monotherapy is appropriate for patients with life expectancy ≤5 years whose cancer progressed on observation of localized disease, who are symptomatic or who have N1M0 disease ~~and in select patients with high- or very-high-risk disease, where complications, such as hydronephrosis or metastasis, can be expected within 5 years.~~
    - ◊ Bullet 2 modified: ADT monotherapy is also used for asymptomatic patients with ~~high-risk, very-high-risk, and~~ regional disease and life expectancy ≤5 years whether or not RT is given.
- Footnote h added: In the mCRPC setting, this option should only be used for select patients who are not candidates for other recommended mCRPC therapies.

PROS-G (6 of 6)
- Optimal ADT, bullet removed: Data are limited on long-term adherence to oral relugolix and the potential effects on optimal ADT. Ongoing monitoring for sustained suppression of testosterone (<50 ng/dL) can be considered, and relugolix may not be a preferred agent if adherence to the prescribed regimen is uncertain.

PROS-I
- New section added: Principles of Focal/Subtotal Therapy or Whole Gland Ablative Therapy.

PROS-J
- Principles of Radiation Therapy: Section extensively revised.

PROS-K
- Pelvic Lymph Node Dissection
  - Bullet 1, sub-bullet 5 modified: PLND can be performed using an open, laparascopic, or robotic technique *and can provide staging and prognostic information.*
  - Bullet removed: While PLND at the time of RP has not been shown to improve oncologic outcomes, it can provide staging and prognostic information.

PROS-L (1 of 2)
- Bullet 1 modified: Patients with biopsy-proven recurrence in the prostate after prior RT and without distant metastatic disease can be considered for local therapy. *Monitoring is also an option (see PROS-10). ADT may also be included, but it should not be reflexively ordered.*
- Bullet 3 modified: Local therapy options for patients with recurrence in the prostate ~~only~~ include:...
  - Sub-bullet 2 modified: High-intensity focused ultrasound (HIFU) ~~(category 2B)~~.
  - Sub-bullet 3 added: Irreversible electroporation (IRE) (category 2B).
  - Sub-bullet removed: Non-surgical strategies.
- Bullet and sub-bullets removed: Local therapy options for patients with recurrence in the regional nodes with or without prostate recurrence include: ADT + pelvic lymph node radiation (if not previously done), ADT + pelvic lymph node reirradiation (category 2B), ADT + PLND (category 2B), Pelvic lymph node radiation, PLND.

Continued

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Updates in Version 1.2026 of the NCCN Guidelines for Prostate Cancer from Version 2.2025 include:

PROS-L (1 of 2)(continued)
- Footnote a added: Reirradiation with LDR brachytherapy, HDR brachytherapy, and SBRT is supported by phase II trials with >36 months of median follow-up and should be strongly considered as an option in patients with >2 years interval from prior radiation who do not have ongoing moderate or higher grade radiation-associated toxicities. Phase II data with >36 months of median follow-up are available, albeit to a more limited scope, for secondary RP, cryotherapy, and HIFU.
- Footnote b added: Currently, all phase II and/or prospectively accrued registries reporting oncologic outcomes with >36 months of follow-up have pursued whole gland treatments. Focal therapy for prostate-only recurrences is the subject of active investigation. See Principles of Focal/Subtotal Therapy or Whole Gland Ablative Therapy (PROS-I).
- Footnote c added: There are no prospective data to guide the use and duration of ADT with reirradiation for prostate-only recurrences. The Panel recommends extrapolation from the definitive treatment setting to suggest that adding ADT or prolonging it in this setting likely offers an improvement in biochemical control. See Principles of Radiation Therapy (PROS-J).

PROS-M
- New section added: Principles of Metastasis-Directed Therapy.

PROS-N (1 of 4)
- Non-Hormonal Systemic Therapy for M1 Castration-Sensitive Prostate Cancer
  - Bullet 1 modified: ~~Patients with low-volume synchronous or high-volume castration-sensitive metastatic prostate cancer who are fit for chemotherapy should be considered for~~ Triple therapy with ADT, certain ARPIs, and docetaxel *is an option for certain patients with mCSPC who are fit for chemotherapy* based on phase 3 studies:

PROS-N (2 of 4)
- Non-Hormonal Systemic Therapy for M1 CRPC
  - Chemotherapy
    - Bullet 1
      - Sub-bullet 4 modified: Docetaxel retreatment can be attempted after progression on *an ARPI and docetaxel* ~~novel hormone therapy~~ in patients with mCRPC whose cancer has not yet demonstrated definitive evidence of progression on prior docetaxel therapy in *either* the castration-sensitive setting *or the mCRPC setting*.
      - Sub-bullet 5 added: Rare patients may have exposure to docetaxel without prior ARPI (ie, in the setting of EBRT to the primary tumor for low-volume synchronous mCSPC). Docetaxel rechallenge is not recommended for such patients in the pre-ARPI mCRPC setting.
  - Bullet 2, sub-bullet 5 added: Biweekly cabazitaxel at 16 mg/m^2 dose with G-CSF in patients ≥65 years in a Phase 3 trial. Clinical outcomes were comparable between the 2 groups.

PROS-N (3 of 4)
- Non-Hormonal Systemic Therapy for M1 CRPC
  - PARP Inhibitors With or Without ARPIs: Subsection extensively revised.
  - Immunotherapy
    - Bullet 2 modified: Pembrolizumab is an option for certain patients with *MSI-H/dMMR* mCRPC ~~and MSI-H, dMMR, or TMB ≥10 mut/mB~~ (PROS-18).
      - Sub-bullet 2 added: Limited data suggest that pembrolizumab may be associated with some benefit in patients with mCRPC and TMB ≥10 mut/mB. Lenis AT, et al. Clin Canc Res 2024;30:3894-3903.
    - Bullet removed: Patients with asymptomatic or minimally symptomatic mCRPC may consider immunotherapy.

PROS-N (4 of 4)
- Table added: Table 1. PARP Inhibitors with or without ARPIs.

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INITIAL PROSTATE CANCER DIAGNOSIS[a,b,c,d]

WORKUP

Clinically localized prostate cancer (Any T, N0, M0 or Any T, NX, MX) →

- Perform physical exam
- Perform digital rectal examination (DRE) to confirm clinical stage
- Perform and/or collect prostate-specific antigen (PSA)
- Review diagnostic prostate biopsies
- Estimate life expectancy (Principles of Life Expectancy Estimation [PROS-A])
- Inquire about known high-risk germline mutations and family history[e]
  - Perform somatic and/or germline testing as appropriate[e]
- Assess quality-of-life measures[f]

→ **Initial Risk Stratification and Staging Workup for Clinically Localized Disease (PROS-2)**

Regional prostate cancer (Any T, N1, M0) →

Metastatic prostate cancer (Any T, Any N, M1) →

- Perform physical examination
- Perform bone and soft tissue imaging for staging[g]
- Consider DRE to confirm clinical stage
- Perform and/or collect PSA and calculate PSA doubling time (PSADT)
- Estimate life expectancy (Principles of Life Expectancy Estimation [PROS-A])
- Inquire about known high-risk germline mutations and family history[e]
  - Perform somatic and/or germline testing as appropriate[e]
- Assess quality-of-life measures[f]

→ **Regional Prostate Cancer (PROS-7)**

→ **Workup and Treatment of Low-Volume M1 (Metachronous or Synchronous) or Synchronous Oligometastatic[h] Castration-Sensitive Prostate Cancer (CSPC) (PROS-14)**

→ **Workup and Treatment of High-Volume M1 CSPC (PROS-15)**

[a] See NCCN Guidelines for Older Adult Oncology for tools to aid optimal assessment and management of disease in older adults.
[b] NCCN Guidelines for Prostate Cancer Early Detection.
[c] Principles of Survivorship in Prostate Cancer (PROS-B).
[d] Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).
[e] Principles of Genetics and Molecular/Biomarker Analysis (PROS-C).
[f] Principles of Quality of Life and Shared Decision-Making (PROS-D).
[g] Principles of Imaging (PROS-E).
[h] See Number of Metastatic Sites in the Principles of MDT (PROS-M).

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INITIAL RISK STRATIFICATION AND STAGING WORKUP FOR CLINICALLY LOCALIZED DISEASE[i]

Risk Group	Clinical/Pathologic Features (Staging, ST-1)			Additional Evaluation[g,l]	Initial Therapy
Low[j]	Has all of the following: • cT1–cT2a • Grade Group 1 • PSA <10 ng/mL			• Confirmatory testing can be used to assess the appropriateness of active surveillance (PROS-F 2 of 5)	PROS-3
Intermediate[j]	Has all of the following: • No high-risk group features • No very-high-risk group features • Has one or more intermediate risk factors (IRFs): ▸ cT2b–cT2c ▸ Grade Group 2 or 3 ▸ PSA 10–20 ng/mL	Favorable intermediate	Has all of the following: • 1 IRF • Grade Group 1 or 2 • <50% biopsy cores positive (eg, <6 of 12 cores)[k]	• Confirmatory testing can be used to assess the appropriateness of active surveillance (PROS-F 2 of 5)	PROS-4
		Unfavorable intermediate	Has one or more of the following: • 2 or 3 IRFs • Grade Group 3 • ≥50% biopsy cores positive (eg, ≥6 of 12 cores)[k]	• Soft tissue imaging and consider bone imaging[g] ▸ If regional metastases are found, see PROS-7 ▸ If distant metastases are found, see PROS-14 for Low-Volume M1(Metachronous or Synchronous) or Synchronous Oligometastatic[h] CSPC or PROS-15 for High-Volume M1 CSPC	PROS-5
High	Has one or more high-risk features, but does not meet criteria for very high risk: • cT3–cT4 • Grade Group 4 or Grade Group 5 • PSA >20 ng/mL			Bone and soft tissue imaging[g] ▸ If regional metastases are found, see PROS-7 ▸ If distant metastases are found, see PROS-14 for Low-Volume M1 (Metachronous or Synchronous) or Synchronous Oligometastatic[h] CSPC or PROS-15 for High-Volume M1 CSPC	PROS-6
Very high	Has at least two of the following: • cT3–cT4 • Grade Group 4 or 5 • PSA >40 ng/mL			Bone and soft tissue imaging[g] ▸ If regional metastases are found, see PROS-7 ▸ If distant metastases are found, see PROS-14 for Low-Volume M1 (Metachronous or Synchronous) or Synchronous Oligometastatic[h] CSPC or PROS-15 for High-Volume M1 CSPC	PROS-6

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Footnotes on PROS-2A

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INITIAL RISK STRATIFICATION AND STAGING WORKUP FOR CLINICALLY LOCALIZED DISEASE

[g] Principles of Imaging (PROS-E).
[h] See Number of Metastatic Sites in the Principles of MDT (PROS-M).
[i] Tumor-based molecular assays and germline genetic testing are other tools that can assist with risk stratification. See CRIT-6 in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate and HRS-3 in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric to determine if a patient is an appropriate candidate for germline genetic testing, and see Principles of Risk Stratification and Biomarkers (PROS-H) to determine if a patient is an appropriate candidate for tumor-based molecular assays.
[j] For patients who are asymptomatic in low- and intermediate-risk groups with life expectancy ≤5 years, no imaging or treatment is indicated until the patient becomes symptomatic, at which time imaging can be performed [Principles of Imaging (PROS-E)] and androgen deprivation therapy (ADT) should be given [Principles of Androgen Deprivation Therapy (PROS-G)].
[k] Percentage of positive cores in the intermediate-risk group is based on biopsies that include systematic biopsies with or without targeted MRI-guided biopsies. The Panel considers biopsies from a single region of interest (ROI) to count as a single sample.
[l] Bone imaging should be performed for any patient with symptoms consistent with bone metastases.

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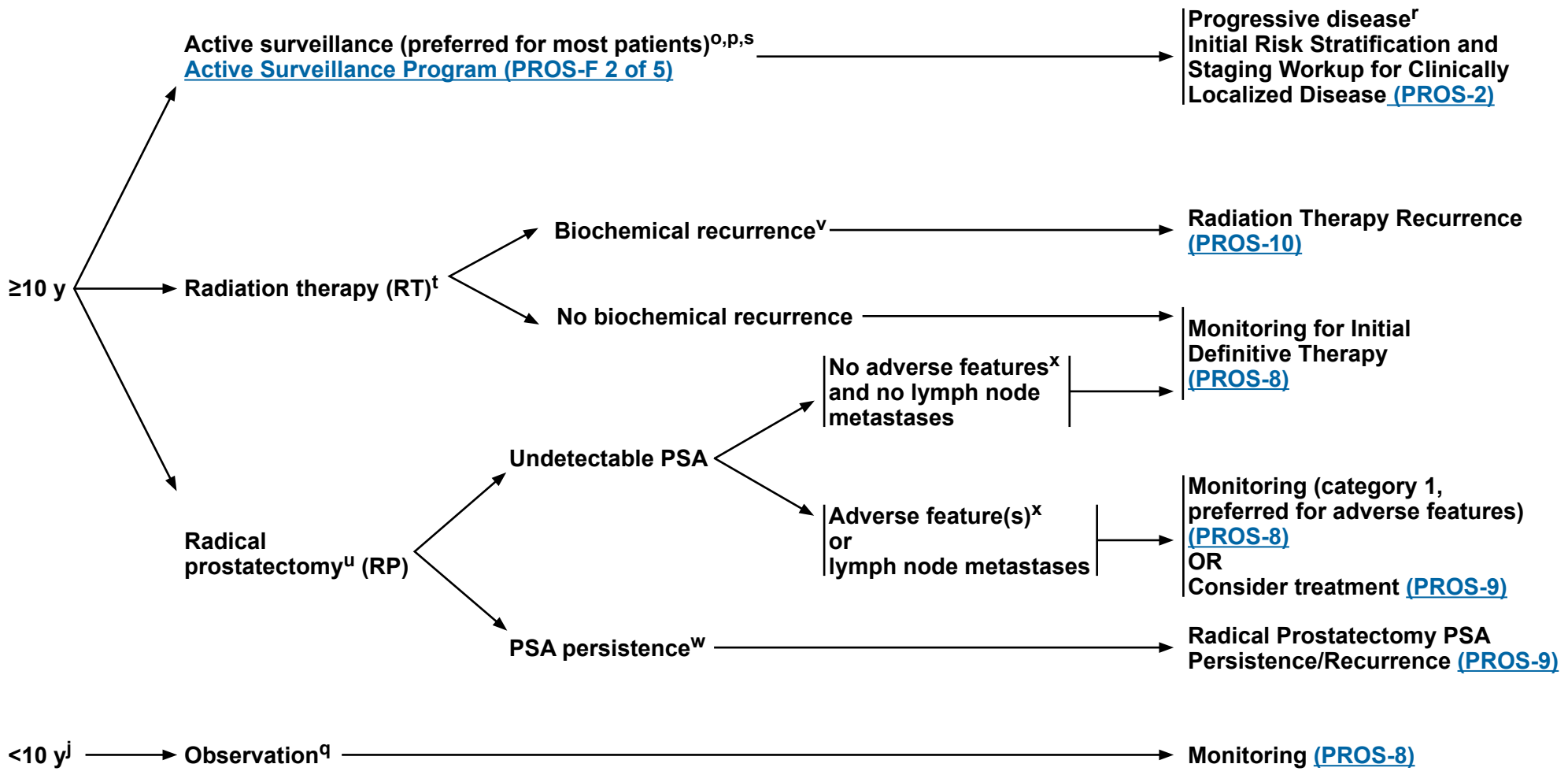
LOW-RISK GROUP



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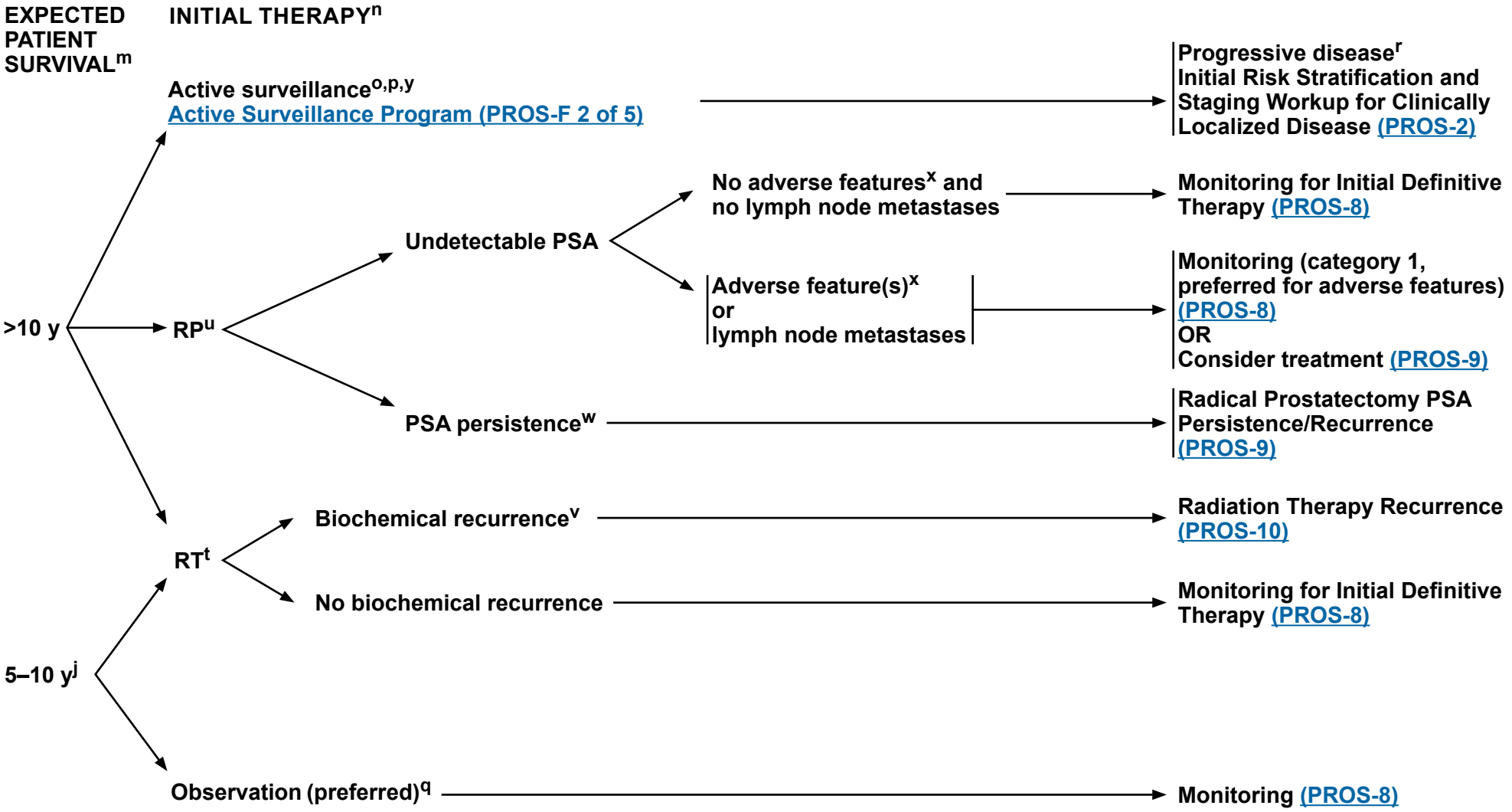
FAVORABLE INTERMEDIATE-RISK GROUP



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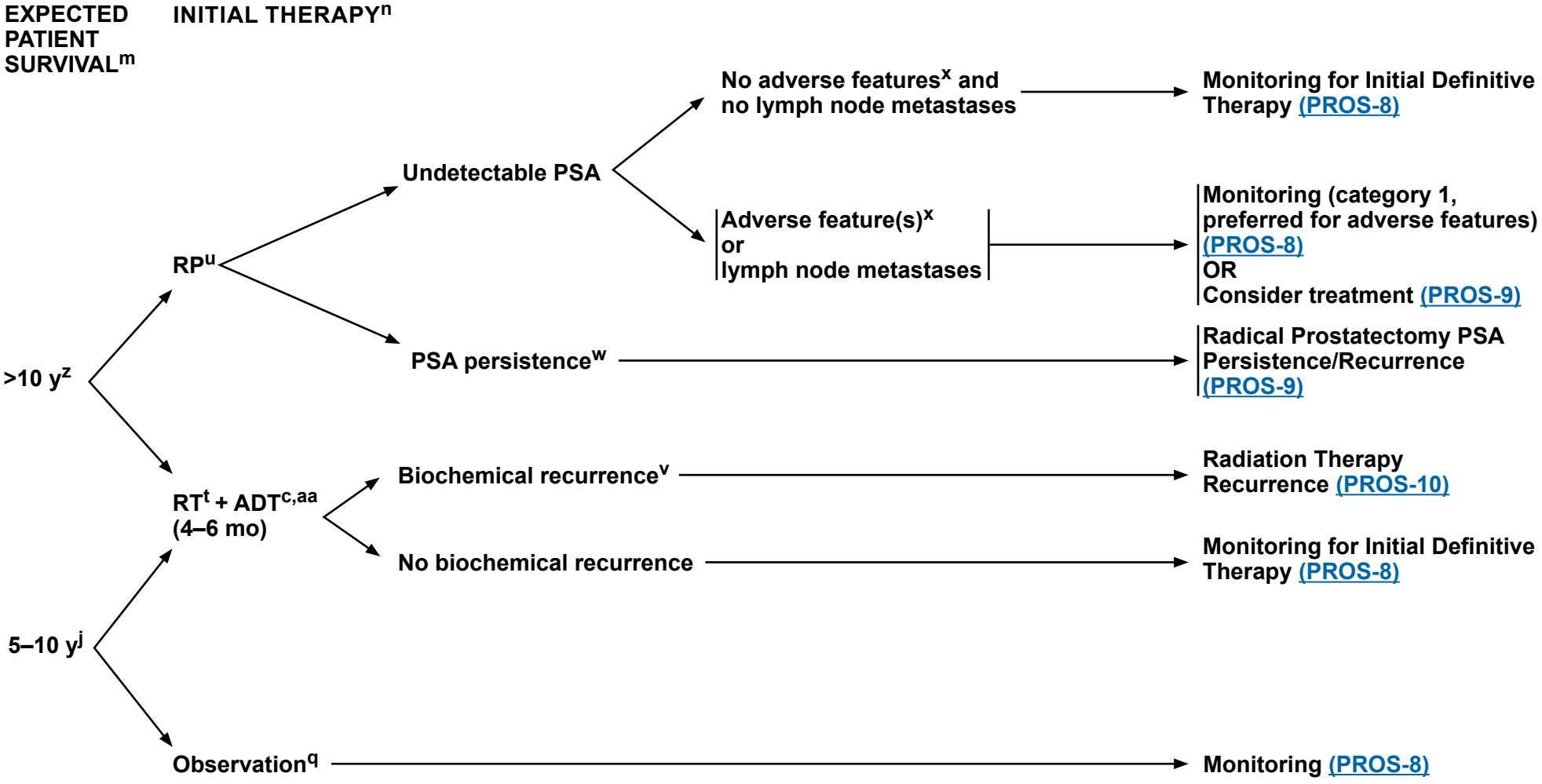
UNFAVORABLE INTERMEDIATE-RISK GROUP



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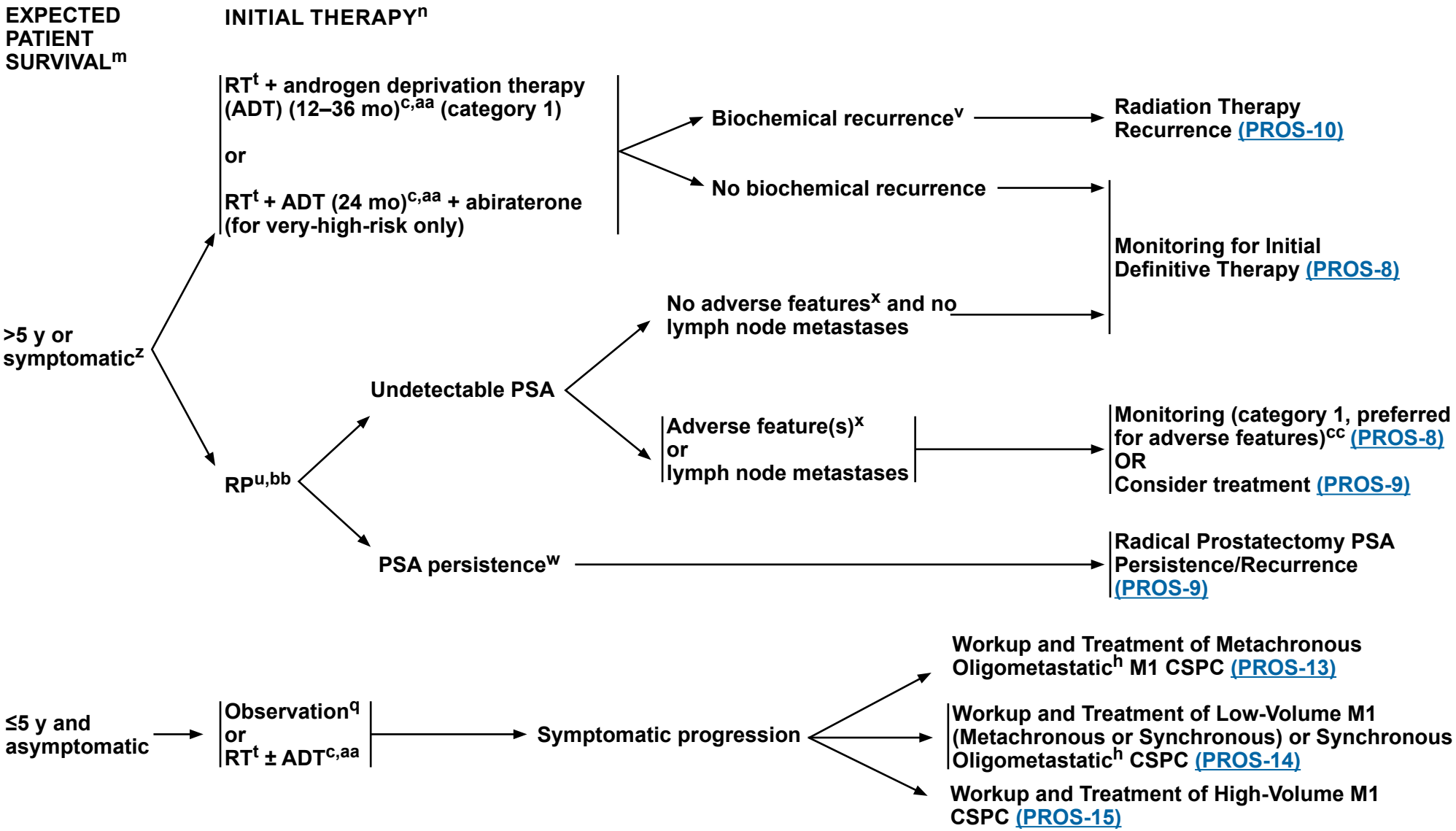
HIGH- OR VERY-HIGH-RISK GROUP



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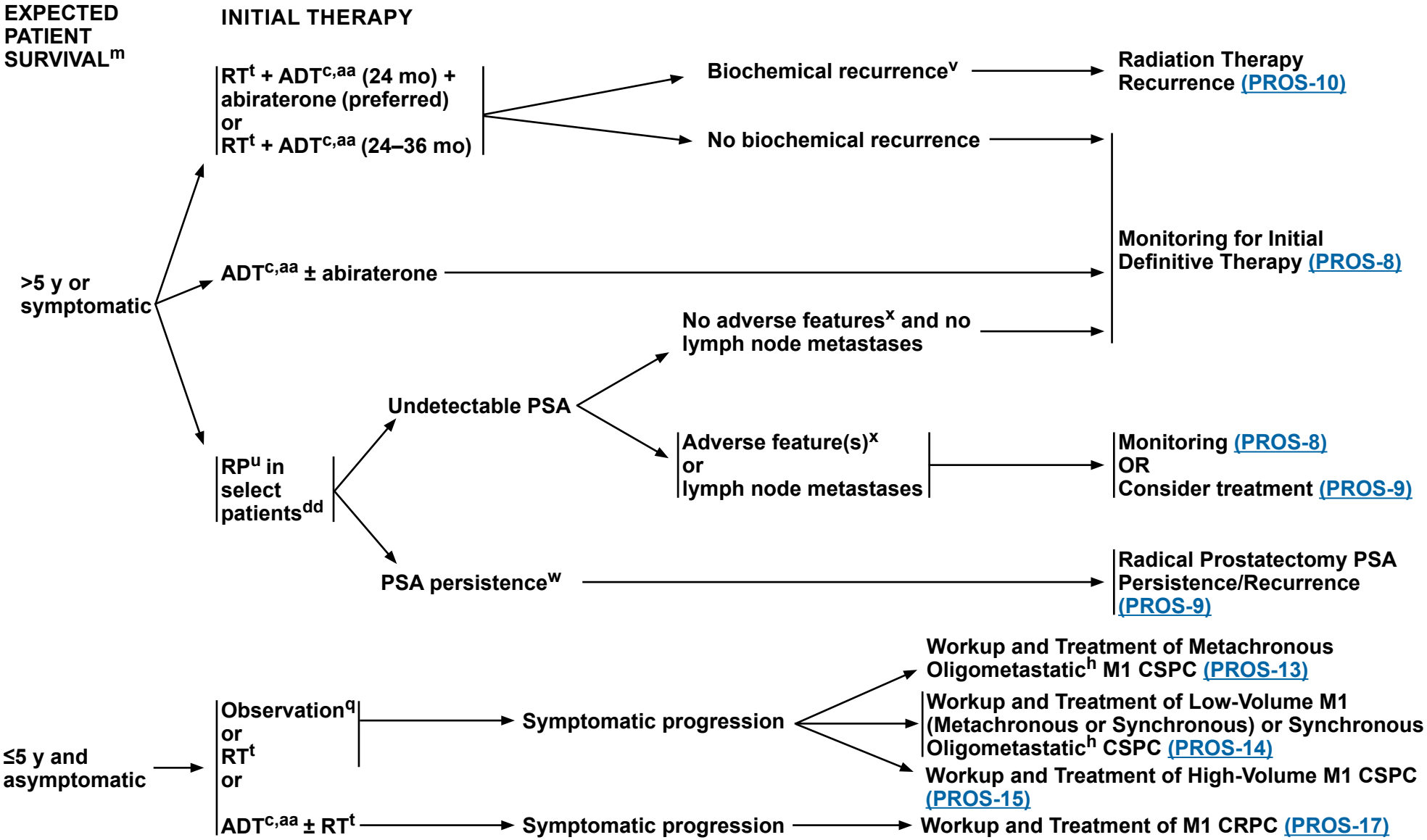
REGIONAL PROSTATE CANCER (ANY T, N1, M0)



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FOOTNOTES

[c] Principles of Survivorship in Prostate Cancer (PROS-B).
[h] See Number of Metastatic Sites in the Principles of MDT (PROS-M).
[j] For patients who are asymptomatic in low- and intermediate-risk groups with life expectancy ≤5 years, no imaging or treatment is indicated until the patient becomes symptomatic, at which time imaging can be performed [Principles of Imaging (PROS-E)] and ADT should be given [Principles of Androgen Deprivation Therapy (PROS-G)].
[m] Principles of Life Expectancy Estimation (PROS-A).
[n] Principles of Focal/Subtotal Therapy or Whole Gland Ablative Therapy (PROS-I).
[o] Active surveillance involves actively monitoring the course of disease with the expectation to intervene with potentially curative therapy if the cancer progresses. See Principles of Active Surveillance and Observation (PROS-F).
[p] Confirmatory testing can be used to assess the appropriateness of active surveillance (PROS-F 2 of 5). If higher grade and/or higher T stage is found during confirmatory testing, see PROS-2.
[q] Observation involves monitoring the course of disease with the expectation to deliver definitive or palliative therapy for the development of symptoms or a change in examination or PSA that suggests symptoms are imminent. See Principles of Active Surveillance and Observation (PROS-F).
[r] Criteria for progression are not well-defined and require physician judgment; however, a change in risk group strongly implies disease progression. See Discussion.
[s] The Panel recognizes that there is heterogeneity across the low-risk group, and that some factors may be associated with an increased probability of near-term grade reclassification, including high PSA density, a high number of positive cores (eg, ≥3), high genomic risk (from tissue-based molecular tumor analysis), and/or a known BRCA2 germline mutation. In some of these cases, upfront treatment with RP or prostate RT may be preferred based on shared decision-making with the patient. See Principles of Active Surveillance and Observation (PROS-F).
[t] Principles of Radiation Therapy (PROS-J).
[u] Principles of Surgery (PROS-K).
[v] RTOG-ASTRO (Radiation Therapy Oncology Group - American Society for Radiation Oncology) Phoenix Consensus: 1) PSA increase by ≥2 ng/mL above the nadir PSA is the standard definition for PSA recurrence after external beam RT (EBRT) with or without hormone therapy; and 2) a recurrence evaluation should be considered when PSA has been confirmed to be increasing after radiation even if the increase above nadir is <2 ng/mL, especially in candidates for secondary local therapy who are young and healthy. Retaining a strict version of the ASTRO definition allows comparison with a large existing body of literature. Rapid increase of PSA may warrant evaluation (prostate biopsy) prior to meeting the Phoenix definition, especially in patients who are younger or healthier.
[w] PSA persistence/recurrence after RP is defined as when PSA does not fall to undetectable levels (PSA persistence) or undetectable PSA after RP with a subsequent detectable PSA that increases on ≥2 determinations (PSA recurrence) or increases to PSA >0.1 ng/mL. Trials indicating noninferiority of early RT compared with adjuvant RT after RP have used a PSA threshold of 0.1 or 0.2 ng/mL to trigger treatment. Imaging and treatment at lower PSA levels may be appropriate in patients at high risk for progression based on pretreatment risk factors, pathologic parameters, timing of recurrence, and genomic classifier (GC) score, among other factors.
[x] Adverse pathologic features include: positive margin(s), seminal vesicle invasion, or extracapsular extension.
[y] Particular consideration to active surveillance may be appropriate for those patients in the favorable intermediate-risk group with a low percentage of Gleason pattern 4 cancer, low tumor volume, low PSA density, and/or low genomic risk (from tissue-based molecular tumor analysis). See Principles of Active Surveillance and Observation (PROS-F).
[z] Active surveillance of unfavorable intermediate and high-risk clinically localized cancers is not recommended in patients with a life expectancy >10 years (category 1).
[aa] For details on the use of ADT and androgen receptor pathway inhibitors (ARPIs) , see Principles of Androgen Deprivation Therapy (PROS-G) and Discussion.
[bb] RP + pelvic lymph node dissection (PLND) can be considered in patients who are younger and healthier without tumor fixation to the pelvic sidewall.
[cc] Monitoring is not preferred for patients with multiple high-risk features.
[dd] There is limited evidence that RP + PLND is beneficial in the setting of node-positive disease. Use of this approach should be limited to patients with >10-year life expectancy and resectable disease and should be used in the context of a clinical trial or planned multimodality approach.

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MONITORING
See NCCN Guidelines for Survivorship

RECURRENCE

Initial definitive therapy → • PSA every 6–12 mo for 5 y,[ee] then every year • Consider DRE if suspicion of recurrence

- Post-RP → PSA persistence/recurrence[w] → Radical Prostatectomy PSA Persistence/ Recurrence (PROS-9)
- Post-RT → PSA recurrence[v] or Positive DRE → Radiation Therapy Recurrence (PROS-10)
- Radiographic evidence of metastatic disease without PSA persistence/ recurrence → Biopsy of metastatic site
  - Workup and Treatment of Metachronous Oligometastatic[h] CSPC (PROS-13)
  - Workup and Treatment of Low-Volume M1 (Metachronous or Synchronous) or Synchronous Oligometastatic[h] CSPC (PROS-14)
  - Workup and Treatment of High-Volume M1 CSPC (PROS-15)

N1 on ADT
or
Localized on observation → • Physical examination + PSA every 3–6 mo • Imaging for symptoms or increasing PSA[g] → Progression[ff,gg]

- N1,M0 → Systemic Therapy for M0 CRPC (PROS-16)
- M1 → Workup and Treatment of M1 CRPC (PROS-17)

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Footnotes on PROS-8A

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MONITORING AND RECURRENCE FOOTNOTES

[g] Principles of Imaging (PROS-E).
[h] See Number of Metastatic Sites in the Principles of MDT (PROS-M).
[v] RTOG-ASTRO Phoenix Consensus: 1) PSA increase by ≥2 ng/mL above the nadir PSA is the standard definition for PSA recurrence after EBRT with or without hormone therapy; and 2) a recurrence evaluation should be considered when PSA has been confirmed to be increasing after radiation even if the increase above nadir is <2 ng/mL, especially in candidates for secondary local therapy who are young and healthy. Retaining a strict version of the ASTRO definition allows comparison with a large existing body of literature. Rapid increase of PSA may warrant evaluation (prostate biopsy) prior to meeting the Phoenix definition, especially in patients who are younger or healthier.
[w] PSA persistence/recurrence after RP is defined as when PSA does not fall to undetectable levels (PSA persistence) or undetectable PSA after RP with a subsequent detectable PSA that increases on ≥2 determinations (PSA recurrence) or increases to PSA >0.1 ng/mL. Trials indicating non-inferiority of early RT compared with adjuvant RT after RP have used a PSA threshold of 0.1 or 0.2 ng/mL to trigger treatment. Imaging and treatment at lower PSA levels may be appropriate in patients at high risk for progression based on pretreatment risk factors, pathologic parameters, timing of recurrence, and GC score, among other factors.
[ee] PSA as frequently as every 3 mo may be necessary to clarify disease status, especially in patients at high risk of recurrence.
[ff] Document castrate levels of testosterone if clinically indicated. Workup for progression should include bone and soft tissue evaluation. See Principles of Imaging (PROS-E).
[gg] Treatment for patients with life expectancy ≤5 y whose cancer progressed on observation of localized disease is ADT. See Principles of Androgen Deprivation Therapy (PROS-G).

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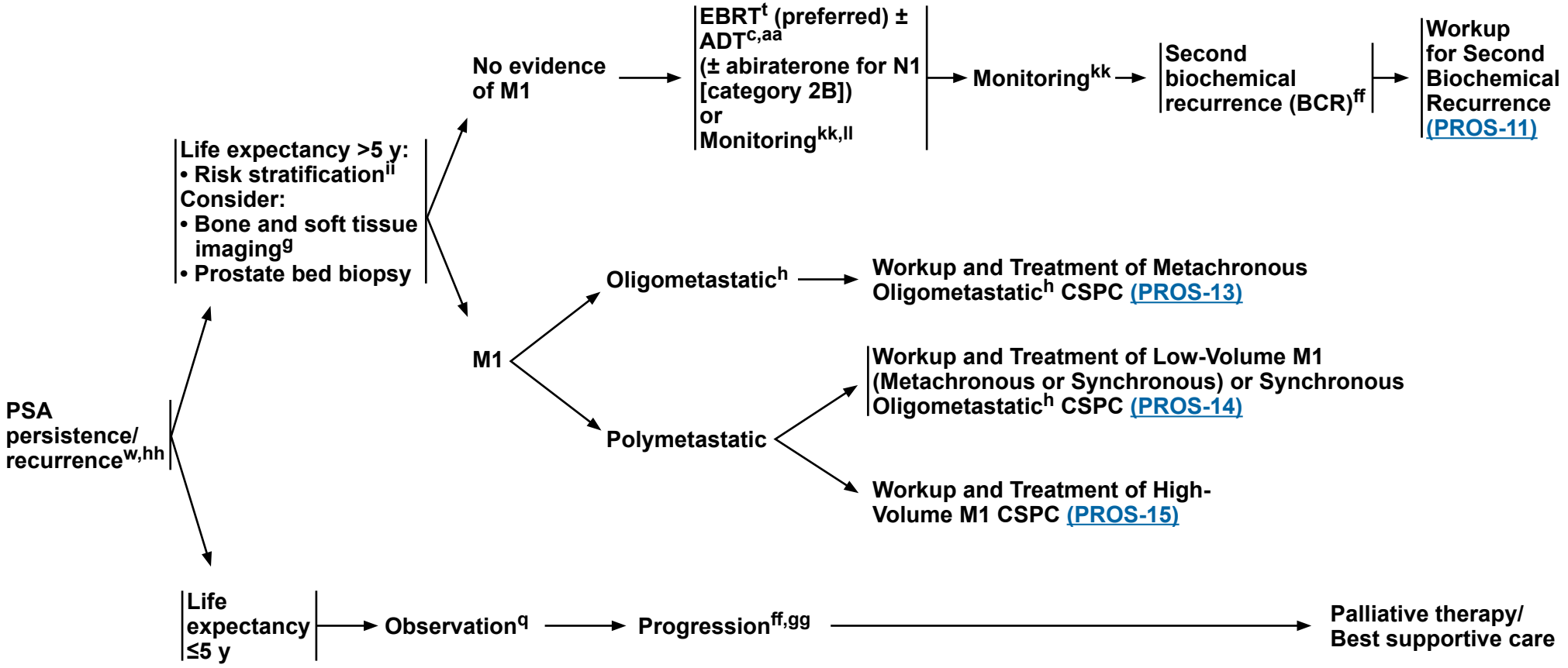
RADICAL PROSTATECTOMY PSA PERSISTENCE/RECURRENCE[hh]

TREATMENT FOR FIRST PSA PERSISTENCE/RECURRENCE[jj]



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RADICAL PROSTATECTOMY PSA PERSISTENCE/RECURRENCE FOOTNOTES

[c] Principles of Survivorship in Prostate Cancer (PROS-B).
[g] Principles of Imaging (PROS-E).
[h] See Number of Metastatic Sites in the Principles of MDT (PROS-M).
[q] Observation involves monitoring the course of disease with the expectation to deliver definitive or palliative therapy for the development of symptoms or a change in examination or PSA that suggests symptoms are imminent. See Principles of Active Surveillance and Observation (PROS-F).
[t] Principles of Radiation Therapy (PROS-J).
[w] PSA persistence/recurrence after RP is defined as when PSA does not fall to undetectable levels (PSA persistence) or undetectable PSA after RP with a subsequent detectable PSA that increases on ≥2 determinations (PSA recurrence) or increases to PSA >0.1 ng/mL. Trials indicating noninferiority of early RT compared with adjuvant RT after RP have used a PSA threshold of 0.1 or 0.2 ng/mL to trigger treatment. Imaging and treatment at lower PSA levels may be appropriate in patients at high risk for progression based on pretreatment risk factors, pathologic parameters, timing of recurrence, and GC score, among other factors.
[aa] For details on the use of ADT and ARPIs, see Principles of Androgen Deprivation Therapy (PROS-G) and Discussion.
[ff] Document castrate levels of testosterone if clinically indicated. Workup for progression should include bone and soft tissue evaluation. See Principles of Imaging (PROS-E).
[gg] Treatment for patients with life expectancy ≤5 y whose cancer progressed on observation of localized disease is ADT. See Principles of Androgen Deprivation Therapy (PROS-G).
[hh] Recommendations for RP PSA persistence/recurrence may also apply to patients with undetectable PSA with multiple adverse features or lymph node metastases if treatment is being considered.
[ii] Principles of Risk Stratification and Biomarkers (PROS-H).
[jj] Treatment for a first biochemical recurrence has historically been referred to as 'salvage' therapy. In efforts to use language that is more sensitive to patients, the NCCN Guidelines for Prostate Cancer refer to treatment in this setting as 'secondary' therapy.
[kk] Monitoring should include physical exam, PSA every 3–6 months, and imaging for symptoms or increasing PSA.
[ll] For patients with PSA progression who have not received treatment for a first biochemical recurrence (BCR), continue monitoring or consider treatment for first PSA persistence/recurrence. See PROS-9 for RP PSA persistence/recurrence or PROS-10 for RT recurrence.

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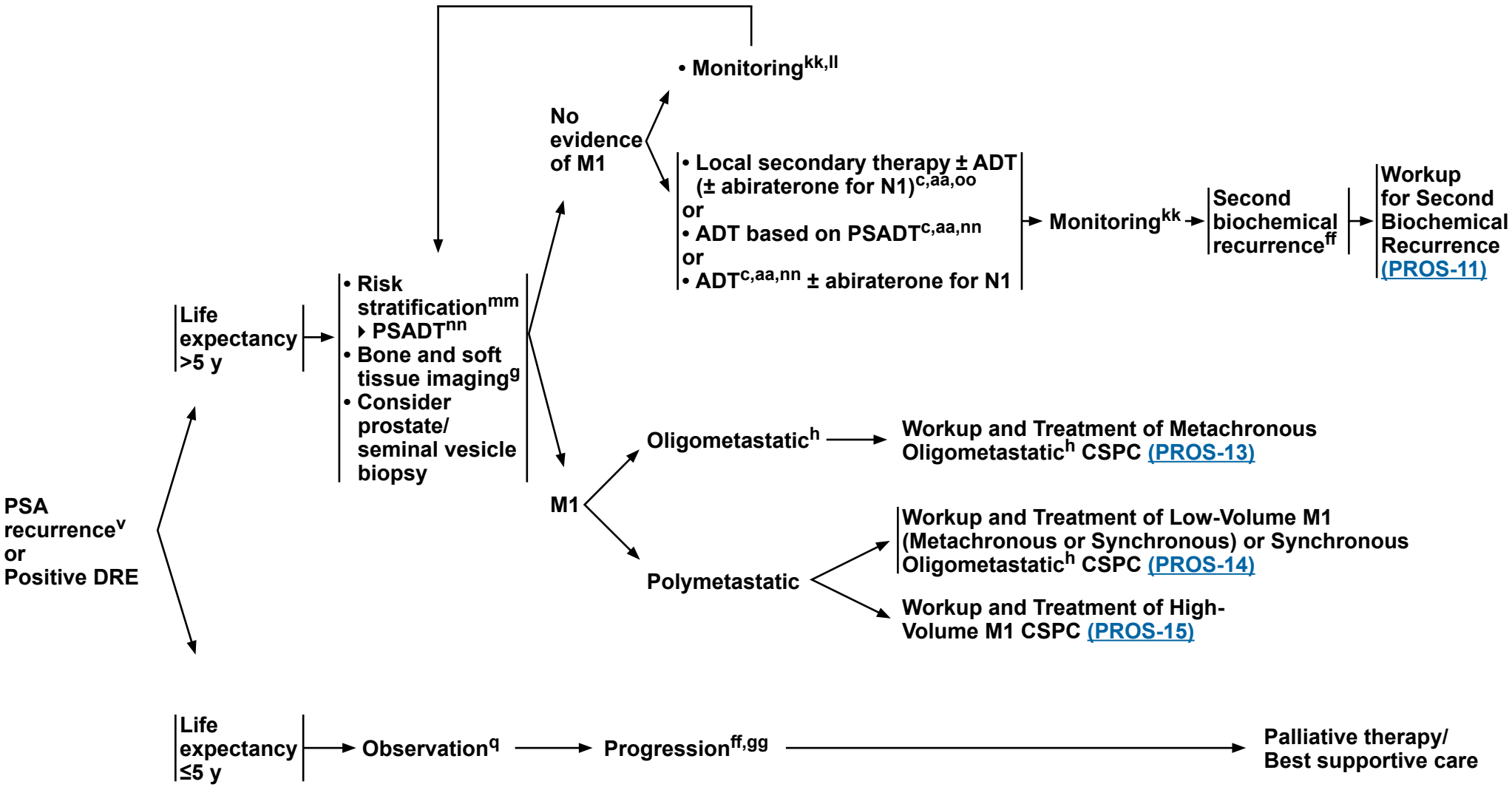
RADIATION THERAPY RECURRENCE

TREATMENT FOR FIRST RECURRENCE[jj]



Footnotes on PROS-10A

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RADIATION THERAPY RECURRENCE FOOTNOTES

[c] Principles of Survivorship in Prostate Cancer (PROS-B).
[g] Principles of Imaging (PROS-E).
[h] See Number of Metastatic Sites in the Principles of MDT (PROS-M).
[q] Observation involves monitoring the course of disease with the expectation to deliver definitive or palliative therapy for the development of symptoms or a change in examination or PSA that suggests symptoms are imminent. See Principles of Active Surveillance and Observation (PROS-F).
[v] RTOG-ASTRO Phoenix Consensus: 1) PSA increase by ≥2 ng/mL above the nadir PSA is the standard definition for PSA recurrence after EBRT with or without hormone therapy; and 2) A recurrence evaluation should be considered when PSA has been confirmed to be increasing after radiation even if the increase above nadir is <2 ng/mL, especially in candidates for secondary local therapy who are young and healthy. Retaining a strict version of the ASTRO definition allows comparison with a large existing body of literature. Rapid increase of PSA may warrant evaluation (prostate biopsy) prior to meeting the Phoenix definition, especially in patients who are younger or healthier.
[aa] For details on the use of ADT and ARPIs, see Principles of Androgen Deprivation Therapy (PROS-G) and Discussion.
[ff] Document castrate levels of testosterone if clinically indicated. Workup for progression should include bone and soft tissue evaluation. See Principles of Imaging (PROS-E).
[gg] Treatment for patients with life expectancy ≤5 y whose cancer progressed on observation of localized disease is ADT. See Principles of Androgen Deprivation Therapy (PROS-G).
[jj] Treatment for a first biochemical recurrence has historically been referred to as 'salvage' therapy. In efforts to use language that is more sensitive to patients, the NCCN Guidelines for Prostate Cancer refer to treatment in this setting as 'secondary' therapy.
[kk] Monitoring should include physical exam, PSA every 3–6 mo, and imaging for symptoms or increasing PSA.
[ll] For patients with PSA progression who have not received treatment for a first BCR, continue monitoring or consider treatment for first PSA persistence/recurrence. See PROS-9 for RP PSA persistence/recurrence or PROS-10 for RT recurrence.
[mm] PSADT can be calculated to inform nomogram use and counseling.
[nn] PSADT and Grade Group should be considered when deciding whether to begin ADT. See Principles of Androgen Deprivation Therapy (PROS-G).
[oo] Principles of Local Secondary Post-Recurrence Therapy (PROS-L).

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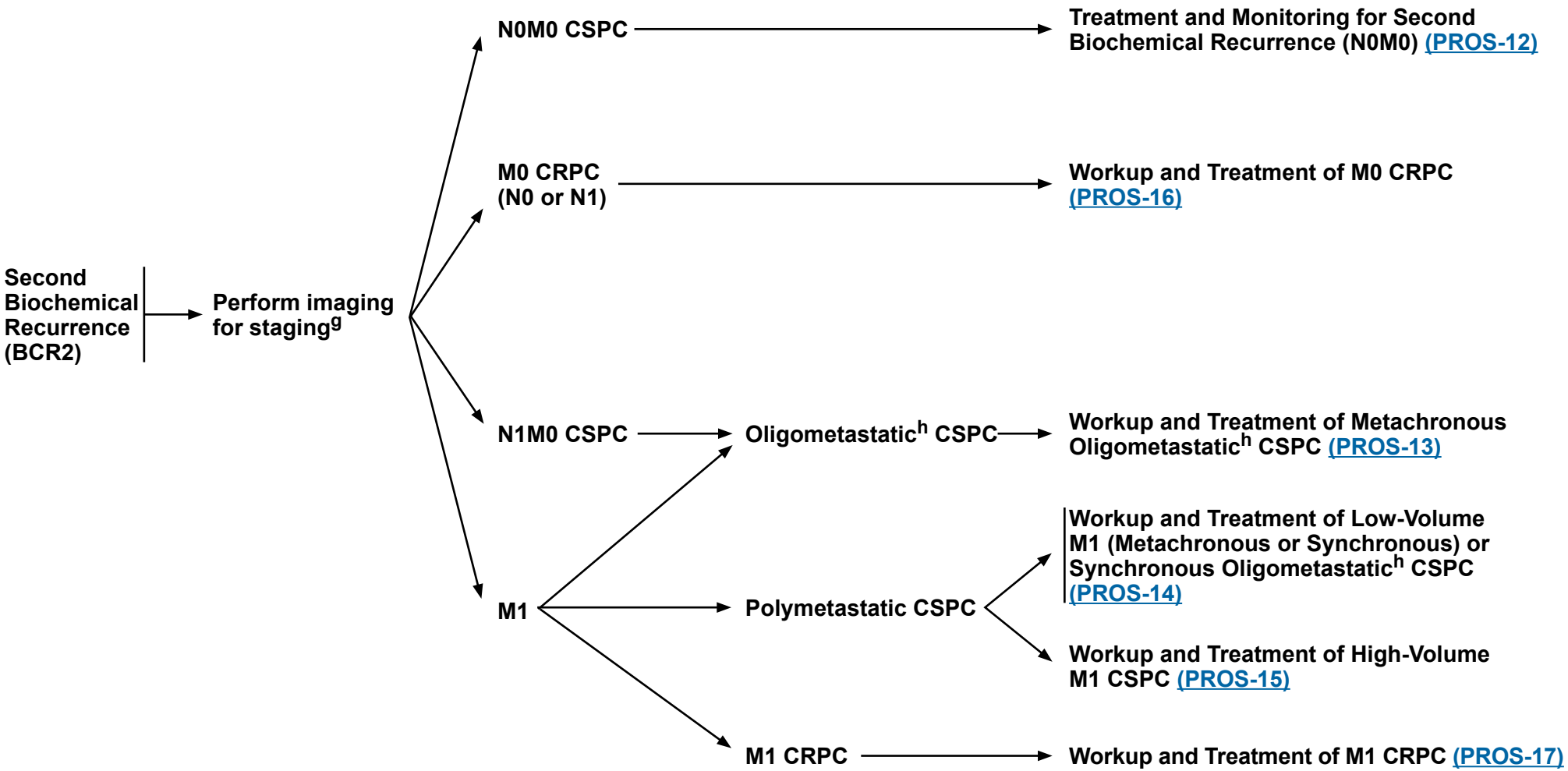
WORKUP FOR SECOND BIOCHEMICAL RECURRENCE (BCR2)



[g] Principles of Imaging (PROS-E).
[h] See Number of Metastatic Sites in the Principles of MDT (PROS-M).

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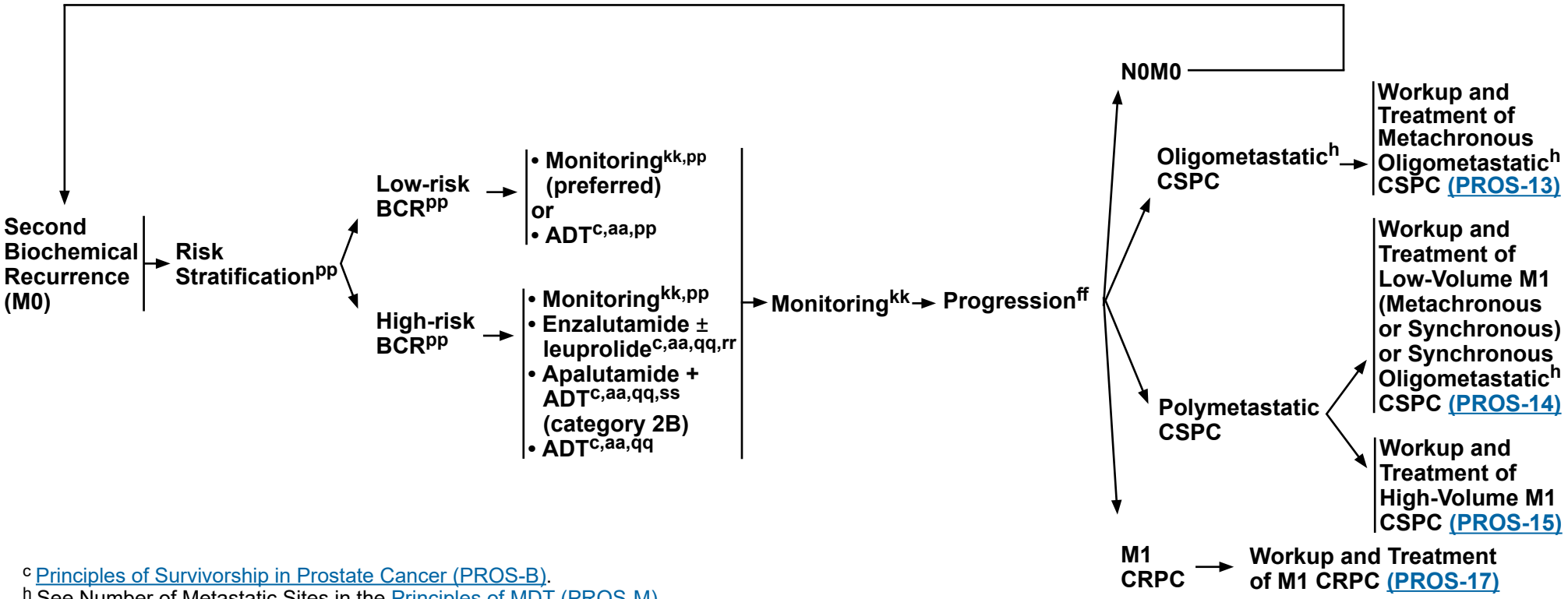
TREATMENT AND MONITORING FOR SECOND BIOCHEMICAL RECURRENCE (N0M0)



[c] Principles of Survivorship in Prostate Cancer (PROS-B).
[h] See Number of Metastatic Sites in the Principles of MDT (PROS-M).
[aa] For details on the use of ADT and ARPIs, see Principles of Androgen Deprivation Therapy (PROS-G) and Discussion.
[ff] Document castrate levels of testosterone if clinically indicated. Workup for progression should include bone and soft tissue evaluation. See Principles of Imaging (PROS-E).
[kk] Monitoring should include physical exam, PSA every 3–6 months, and imaging for symptoms or increasing PSA.
[pp] PSA level and PSADT should be considered when deciding whether to begin ADT for patients considered to have lower risk disease. High-risk BCR has different definitions in different clinical trials that generally include a PSADT ≤9 months and other adverse prognostic features.
[qq] If favorable PSA response, an intermittent ADT approach should be considered.
[rr] Enzalutamide with or without leuprolide is an option for patients who have the following high-risk criteria: M0 by CT, MRI, or bone scan; PSADT ≤9 months; PSA ≥2 ng/mL above nadir after RT or ≥1 ng/mL after RP with or without postoperative RT; and not considered a candidate for pelvic-directed therapy (Freedland SJ, et al. N Engl J Med 2023;389:1453-1465). See Principles of Androgen Deprivation Therapy (PROS-G).
[ss] Apalutamide plus ADT is an option for patients with biochemical recurrence after RP who meet the following high-risk criteria: PSADT ≤9 months; PSA ≥0.5 ng/mL; and prior adjuvant or secondary RT or not considered a candidate for RT (Aggarwal R, et al. J Clin Oncol 2024;42:1114-1123.) See Principles of Androgen Deprivation Therapy (PROS-G).

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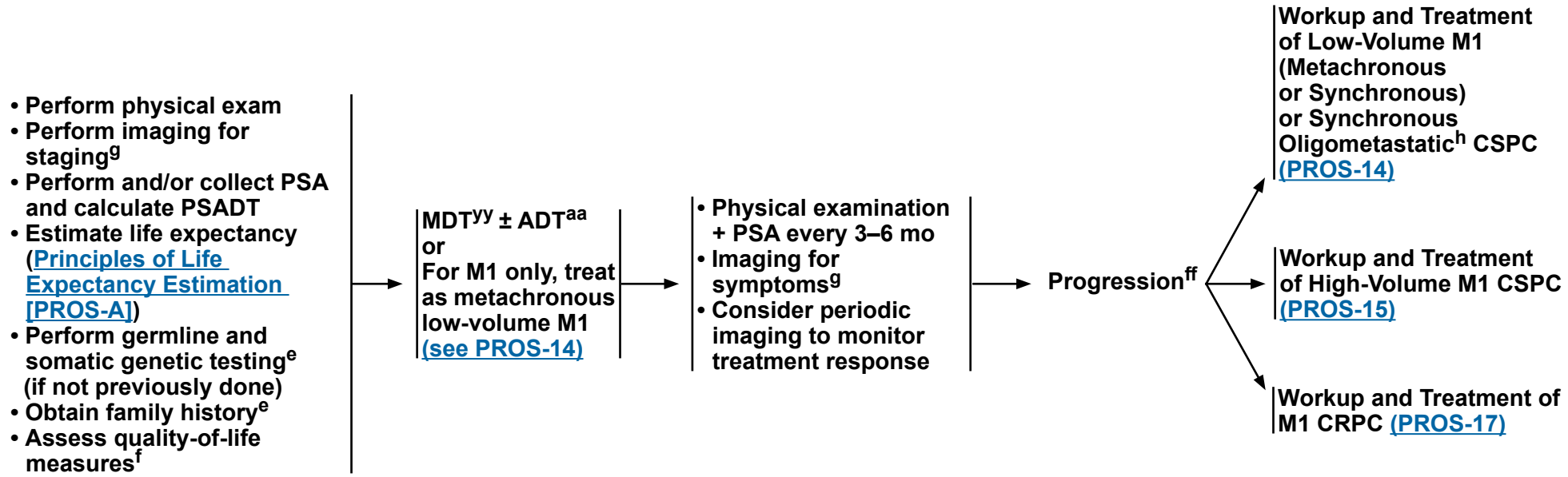
WORKUP AND TREATMENT OF METACHRONOUS OLIGOMETASTATIC CSPC[c,h,tt,uu,vv,ww]

WORKUP FOR METASTASES[xx]



[c] Principles of Survivorship in Prostate Cancer (PROS-B).
[e] Principles of Genetics and Molecular/Biomarker Analysis (PROS-C).
[f] Principles of Quality of Life and Shared Decision-Making (PROS-D).
[g] Principles of Imaging (PROS-E).
[h] See Number of Metastatic Sites in the Principles of MDT (PROS-M).
[aa] For details on the use of ADT and ARPIs, see Principles of Androgen Deprivation Therapy (PROS-G) and Discussion.
[ff] Document castrate levels of testosterone if clinically indicated. Workup for progression should include bone and soft tissue evaluation. See Principles of Imaging (PROS-E).
[tt] EBRT to sites of bone metastases can be considered if metastases are in weight-bearing bones or if the patient is symptomatic.
[uu] Bone antiresorptive therapy is indicated for elevated fracture risk based upon FRAX in the castration-sensitive setting. See PROS-B.
[vv] The term "castration-sensitive" is used to define disease in patients who have not been treated with ADT and those who are not on ADT at the time of progression. The NCCN Prostate Cancer Panel uses the term "castration-sensitive" even when patients have had neoadjuvant, concurrent, or adjuvant ADT as part of RT provided they have recovered testicular function.
[ww] ADT is strongly recommended in combination therapy for metastatic castration-sensitive disease. The use of ADT monotherapy in metastatic castration-sensitive disease is discouraged unless there are clear contraindications to combination therapy. If ADT monotherapy is given, intermittent ADT can be considered to reduce toxicity. See Principles of Androgen Deprivation Therapy (PROS-G).
[xx] ADT alone (PROS-G) or observation are recommended for asymptomatic patients with metastatic disease or M0 CRPC and life expectancy ≤5 years.
[yy] See Principles of MDT (PROS-M).

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WORKUP AND TREATMENT OF LOW-VOLUME M1 (METACHRONOUS OR SYNCHRONOUS) OR SYNCHRONOUS OLIGOMETASTIC CSPC[c,h,tt,uu,vv,ww]

WORKUP FOR METASTASES[xx]

- Perform physical exam
- Perform imaging for staging[g]
- Perform and/or collect PSA and calculate PSADT
- Estimate life expectancy (Principles of Life Expectancy Estimation [PROS-A])
- Perform germline and somatic genetic testing[e] (if not previously done)
- Obtain family history[e]
- Assess quality-of-life measures[f]

Synchronous low-volume metastases or Synchronous oligometastatic disease →

ADT[aa,zz] with one of the following:
- Preferred:
  - Abiraterone (category 1)[aa]
  - Apalutamide (category 1)[aa]
  - Enzalutamide (category 1)[aa]
- Other Recommended:
  - Darolutamide (category 2B)[aa]

or

ADT[aa,zz] with docetaxel and one of the following (low-volume disease only):
- Abiraterone (category 2B)[aa]
- Apalutamide (category 2B)[aa]
- Darolutamide (category 2B)[aa]
- Enzalutamide (category 2B)[aa]

or

ADT[aa,zz] with EBRT[t] to the primary tumor[aaa] alone or with one of the following:
- Abiraterone[aa]
- Apalutamide (category 2B)[aa]
- Docetaxel (category 2B)
- Enzalutamide (category 2B)[aa]

or

ADT[aa,zz] with:
- Useful in Certain Circumstances
  - Niraparib/abiraterone[bbb] (*BRCA2*m only) (category 2B)

Metachronous low-volume metastases →

ADT[aa,zz] with one of the following:
- Preferred:
  - Abiraterone (category 1)[aa]
  - Apalutamide (category 1)[aa]
  - Enzalutamide (category 1)[aa]
- Other Recommended:
  - Darolutamide (category 2B)[aa]

or

ADT[aa,zz] with:
- Useful in Certain Circumstances
  - Niraparib/abiraterone[bbb] (*BRCA2*m only) (category 2B)

→

- Physical examination + PSA every 3–6 mo
- Imaging for symptoms[g]
- Periodic imaging to monitor treatment response

→ **Progression[ff] on ADT ± androgen receptor pathway inhibitor (ARPI)** → **Workup and Treatment of M1 CRPC (PROS-17)**

Footnotes on PROS-15A

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WORKUP AND TREATMENT OF HIGH-VOLUME M1 CSPC[c,tt,uu,vv,ww,ccc]

WORKUP FOR METASTASES[xx]

- Perform physical exam
- Perform imaging for staging[g]
- Perform and/or collect PSA and calculate PSADT
- Estimate life expectancy (Principles of Life Expectancy Estimation [PROS-A])
- Perform germline and somatic genetic testing[e] (if not previously done)
- Obtain family history[e]
- Assess quality-of-life measures[f]

→ Synchronous or metachronous high-volume[ccc] metastases →

ADT[aa] with docetaxel and one of the following:
- Preferred:
  - Abiraterone (category 1)[aa]
  - Darolutamide (category 1)[aa]
- Other Recommended:
  - Apalutamide (category 2B)[aa]
  - Enzalutamide (category 2B)[aa]

or

ADT[aa] with one of the following:
- Preferred:
  - Abiraterone (category 1)[aa]
  - Apalutamide (category 1)[aa]
  - Enzalutamide (category 1)[aa]
- Other Recommended:
  - Darolutamide[aa]

or

ADT[aa] with:
- Useful in Certain Circumstances
  - Niraparib/abiraterone[bbb] (*BRCA2*m only)

→

- Physical examination + PSA every 3–6 mo
- Imaging for symptoms[g]
- Periodic imaging to monitor treatment response

→ Progression[ff] → Workup and Treatment of M1 CRPC (PROS-17)

Note: All recommendations are category 2A unless otherwise indicated.

Footnotes on PROS-15A

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FOOTNOTES

[c] Principles of Survivorship in Prostate Cancer (PROS-B).
[e] Principles of Genetics and Molecular/Biomarker Analysis (PROS-C).
[f] Principles of Quality of Life and Shared Decision-Making (PROS-D).
[g] Principles of Imaging (PROS-E).
[h] See Number of Metastatic Sites in the Principles of MDT (PROS-M).
[t] Principles of Radiation Therapy (PROS-J).
[aa] For details on the use of ADT and ARPIs, see Principles of Androgen Deprivation Therapy (PROS-G) and Discussion.
[ff] Document castrate levels of testosterone if clinically indicated. Workup for progression should include bone and soft tissue evaluation. See Principles of Imaging (PROS-E).
[tt] EBRT to sites of bone metastases can be considered if metastases are in weight-bearing bones or if the patient is symptomatic.
[uu] Bone antiresorptive therapy is indicated for elevated fracture risk based upon FRAX in the castration-sensitive setting. See PROS-B.
[vv] The term "castration-sensitive" is used to define disease in patients who have not been treated with ADT and those who are not on ADT at the time of progression. The NCCN Prostate Cancer Panel uses the term "castration-sensitive" even when patients have had neoadjuvant, concurrent, or adjuvant ADT as part of RT provided they have recovered testicular function.
[ww] ADT is strongly recommended in combination therapy for metastatic castration-sensitive disease. The use of ADT monotherapy in metastatic castration-sensitive disease is discouraged unless there are clear contraindications to combination therapy. If ADT monotherapy is given, intermittent ADT can be considered to reduce toxicity. See Principles of Androgen Deprivation Therapy (PROS-G).
[xx] ADT alone (PROS-G) or observation are recommended for asymptomatic patients with metastatic disease or M0 CRPC and life expectancy ≤5 years.
[zz] Concurrent MDT can be considered in select patients with oligometastatic disease. See Principles of MDT (PROS-M).
[aaa] EBRT to the primary tumor is associated with an overall survival (OS) benefit in patients with low metastatic burden at the time of diagnosis of metastatic disease, which is defined by bone scan and CT or MRI as either non-regional, lymph-node-only disease OR <4 bone metastases and without visceral/other metastasis (Ali A, et al. JAMA Oncol 2021;7:555-563). See Principles of Radiation Therapy (PROS-J).
[bbb] Niraparib/abiraterone can be used following the completion of docetaxel in patients receiving triplet therapy or following EBRT to the primary tumor. The use of this combination in patients with low-volume disease is controversial because 78% of the patients in the trial had high-volume disease and there were very few patients with low-volume metachronous disease in particular. This regimen adds considerable toxicity, and patients with low-volume disease should be informed that the benefit in this setting is not clear.
[ccc] High-volume disease in this setting is defined based on CHAARTED criteria (the presence of visceral metastasis or ≥4 bone lesions with ≥1 beyond the vertebral bodies and pelvis).

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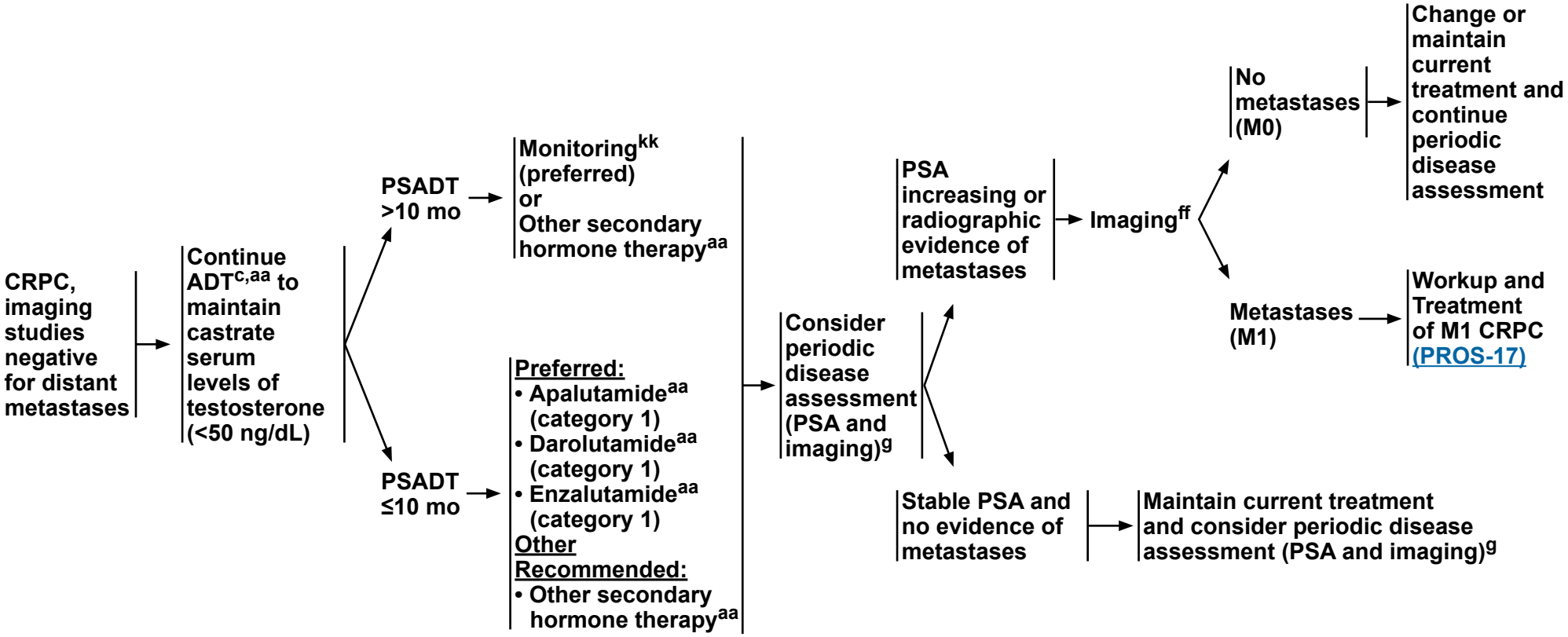
WORKUP AND TREATMENT OF M0 CASTRATION-RESISTANT PROSTATE CANCER (CRPC)[xx,ddd]



[c] Principles of Survivorship in Prostate Cancer (PROS-B).
[g] Principles of Imaging (PROS-E).
[aa] For details on the use of ADT and ARPIs, see Principles of Androgen Deprivation Therapy (PROS-G) and Discussion.
[ff] Document castrate levels of testosterone if clinically indicated. Workup for progression should include bone and soft tissue evaluation. See Principles of Imaging (PROS-E).
[kk] Monitoring should include physical exam, PSA every 3–6 mo, and imaging for symptoms or increasing PSA.
[xx] ADT alone (PROS-G) or observation are recommended for asymptomatic patients with metastatic disease or M0 CRPC and life expectancy ≤5 years.
[ddd] CRPC is prostate cancer that progresses clinically, radiographically, or biochemically despite castrate levels of serum testosterone (<50 ng/dL). Scher HI, et al. J Clin Oncol 2008;26:1148-1159.

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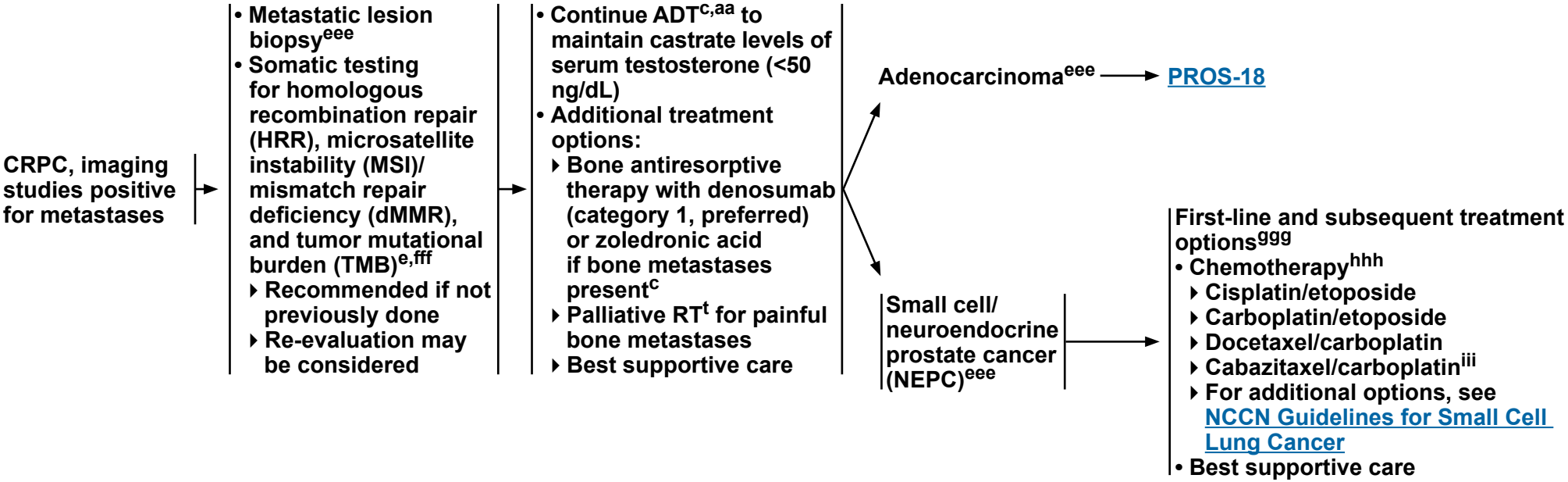
WORKUP AND TREATMENT OF M1 CRPC[xx,ddd]



[c] Principles of Survivorship in Prostate Cancer (PROS-B).
[e] Principles of Genetics and Molecular/Biomarker Analysis (PROS-C).
[t] Principles of Radiation Therapy (PROS-J).
[aa] For details on the use of ADT and ARPIs, see Principles of Androgen Deprivation Therapy (PROS-G) and Discussion.
[xx] ADT alone (PROS-G) or observation are recommended for asymptomatic patients with metastatic disease or M0 CRPC and life expectancy ≤5 years.
[ddd] CRPC is prostate cancer that progresses clinically, radiographically, or biochemically despite castrate levels of serum testosterone (<50 ng/dL). Scher HI, et al. J Clin Oncol 2008;26:1148-1159.
[eee] Histologic evidence of both adenocarcinoma and small cell carcinoma may be present, in which case treatment can follow either pathway. Treat as adenocarcinoma if biopsy is not feasible or not performed.
[fff] Germline testing for HRR mutations is recommended if not performed previously. See Principles of Genetics and Molecular/Biomarker Analysis (PROS-C).
[ggg] Document castrate levels of testosterone if progression occurs on ADT. See Principles of Imaging (PROS-E) and Discussion.
[hhh] For details on the efficacy and safety of these agents, see Principles of Non-Hormonal Systemic Therapy (PROS-N).
[iii] Cabazitaxel 20 or 25 mg/m² plus carboplatin area under the curve [AUC] 4 mg/mL per min with growth factor support can be considered for fit patients with aggressive variant metastatic CRPC (mCRPC) (ie, visceral metastases, low PSA and bulky disease, high lactate dehydrogenase [LDH], high carcinoembryonic antigen [CEA], lytic bone metastases, NEPC histology) or unfavorable genomics (defects in at least 2 of PTEN, TP53, and RB1). Corn PG, et al. Lancet Oncol 2019;20:1432-1443.

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SYSTEMIC THERAPY FOR M1 CRPC: ADENOCARCINOMA[g,aa,jjj,kkk]

Pre-ARPI[aa,lll]	Post-ARPI[lll]/Pre-Docetaxel[aa]	Post-ARPI[lll]/Post-Docetaxel[aa]
Preferred: • Abiraterone (category 1) • Enzalutamide (category 1) **Other Recommended:** • Docetaxel[hhh] (category 1) **Useful in Certain Circumstances:** • Molecular Biomarker-Directed Therapy ▸ *BRCA* mutation ◊ Niraparib/abiraterone[mmm] (category 1) ◊ Olaparib/abiraterone[mmm] (category 1) ◊ Talazoparib/enzalutamide[mmm] (category 1) ▸ *HRRm* (other than BRCA1/2) ◊ Talazoparib/enzalutamide[mmm] (category 1) • Disease State-Specific Therapy ▸ Bone metastases ◊ Radium-223[ooo]/enzalutamide	**Preferred:** • Docetaxel[hhh] (category 1) **Useful in Certain Circumstances:** • Molecular Biomarker-Directed Therapy ▸ *BRCA* mutation ◊ Olaparib[mmm] (category 1, preferred) ◊ Rucaparib[mmm] (category 1, preferred) ◊ Niraparib/abiraterone[mmm] (category 2B) ◊ Talazoparib/enzalutamide[mmm] (category 2B) ▸ *HRRm* (other than BRCA1/2) ◊ Olaparib[mmm] ◊ Talazoparib/enzalutamide[mmm] (category 2B) • Disease State-Specific Therapy ▸ PSMA-positive metastases ◊ Lutetium Lu 177 vipivotide tetraxetan (Lu-177-PSMA-617)[qqq] ▸ Aggressive variant[iii] ◊ Cabazitaxel/Carboplatin[hhh]	**Preferred:** • Cabazitaxel[hhh] (category 1) • Docetaxel rechallenge[hhh] **Useful in Certain Circumstances:** • Molecular Biomarker-Directed Therapy ▸ *BRCA* mutation ◊ Olaparib[mmm] (category 1) ◊ Rucaparib[mmm] ▸ *HRRm* (other than *BRCA1/2*) ◊ Olaparib[mmm] ▸ Other FDA-approved agents for tissue agnostic indications[hhh] • Disease State-Specific Therapy ▸ PSMA-positive metastases ◊ Lu-177-PSMA-617[qqq] (category 1) ▸ Aggressive variant[iii] ◊ Cabazitaxel/carboplatin[hhh] ▸ Palliation for symptomatic patients unable to tolerate other therapies ◊ Mitoxantrone[hhh]

Additional Options Irrespective of Prior ARPI or Prior Docetaxel (Useful in Certain Circumstances)	
• Disease State-Specific Therapy ▸ Asymptomatic without visceral metastases ◊ Sipuleucel-T[hhh,ppp] ▸ Oligometastatic[h]/Oligoprogressive disease ◊ Metastasis-directed therapy[nnn] with metastatic castration-resistant prostate cancer (mCRPC) systemic therapy ▸ Symptomatic bone-predominant metastases ◊ Radium-223[ooo] (category 1)	• Molecular Biomarker-Directed Therapy ▸ MSI-High (MSI-H)/dMMR ◊ Pembrolizumab[hhh] (category 2B)

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Footnotes on PROS-18A

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THERAPY FOR M1 CRPC: ADENOCARCINOMA
FOOTNOTES

[g] Principles of Imaging (PROS-E).
[h] See Number of Metastatic Sites in the Principles of MDT (PROS-M).
[aa] For details on the use of ADT and ARPIs, see Principles of Androgen Deprivation Therapy (PROS-G) and Discussion.
[hhh] For details on the efficacy and safety of these agents, see Principles of Non-Hormonal Systemic Therapy (PROS-N).
[iii] Cabazitaxel 20 or 25 mg/m² plus carboplatin AUC 4 mg/mL per min with growth factor support can be considered for fit patients with aggressive variant mCRPC (ie, visceral metastases, low PSA and bulky disease, high LDH, high CEA, lytic bone metastases, NEPC histology) or unfavorable genomics (defects in at least 2 of PTEN, TP53, and RB1). Corn PG, et al. Lancet Oncol 2019;20:1432-1443.
[jjj] Document castrate levels of testosterone if progression occurs on ADT. Consider metastatic lesion biopsy. If small cell neuroendocrine is found, see PROS-17.
[kkk] Patients can continue through all treatment options listed. Best supportive care, which can include androgen-directed therapy or steroid, is always an appropriate option.
[lll] ARPI therapies include abiraterone, enzalutamide, darolutamide, or apalutamide. Abiraterone given as part of neoadjuvant/concomitant/adjuvant ADT with EBRT is not considered post-ARPI.
[mmm] PARP inhibitors with or without ARPI have different biomarker and previous treatment requirements. See Principles of Non-Hormonal Systemic Therapy (PROS-N).
[nnn] Principles of MDT (PROS-M).
[ooo] Radium-223 should not be used in patients with visceral metastases. Concurrent use with systemic therapies other than enzalutamide should be pursued with caution. Concomitant use of denosumab or zoledronic acid is recommended. See Principles of Radiation Therapy (PROS-J).
[ppp] Sipuleucel-T is recommended only for asymptomatic or minimally symptomatic, no liver metastases, life expectancy >6 months, and ECOG performance status 0–1. Benefit with sipuleucel-T has not been reported in patients with visceral metastases and is not recommended if visceral metastases are present. There are limited data to support the efficacy of Sipuleucel-T in the post-chemotherapy setting. Sipuleucel-T is not recommended for patients with small cell prostate cancer/NEPC.
[qqq] Lu-177–PSMA-617 is a treatment option for patients with ≥1 PSMA-positive lesion and/or metastatic disease that is predominately PSMA-positive and with no dominant PSMA-negative metastatic lesions who have been treated previously with androgen receptor-directed therapy and a taxane-based chemotherapy or are considered appropriate to delay a taxane-based chemotherapy. Sartor O, et al. N Engl J Med 2021;385:1091-1103; Morris MJ, et al. Lancet 2024;404:1227-1239. See Principles of Radiation Therapy (PROS-J).

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PRINCIPLES OF LIFE EXPECTANCY ESTIMATION

- **Life expectancy estimation is critical to informed decision-making in prostate cancer early detection and treatment.**

- **Estimation of life expectancy is possible for groups of patients but challenging for individuals.**

- **Life expectancy can be estimated using:**
  - **The Social Security Administration tables (www.ssa.gov/OACT/STATS/table4c6.html)**
  - **The WHO's Life Tables by country (http://apps.who.int/gho/data/view.main.60000?lang=en)**
  - **The Memorial Sloan Kettering Male Life Expectancy tool (https://www.mskcc.org/nomograms/prostate)**
  - **University of California San Francisco (UCSF) Lee Schonberg Index (https://eprognosis.ucsf.edu/leeschonberg.php)**

- **If using a life expectancy table, life expectancy should be adjusted using the clinician's assessment of overall health as follows:**
  - **Best quartile of health - add 50%**
  - **Worst quartile of health - subtract 50%**
  - **Middle two quartiles of health - no adjustment**

- **Examples of upper, middle, and lower quartiles of life expectancy at selected ages are included in the NCCN Guidelines for Older Adult Oncology for life expectancy estimation.**

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PRINCIPLES OF SURVIVORSHIP
BONE HEALTH IN PROSTATE CANCER

Treatment-Related Bone Loss

- ADT increases the risk of bone loss, and this risk is exacerbated with more potent androgen suppression, longer duration of therapy or delayed testosterone recovery, and concurrent prednisone use.
- The goal of osteoporosis screening is to identify patients at increased risk of sustaining a low-trauma fracture who would benefit from intervention to minimize the fracture risk. Risk assessment for treatment-related bone loss should take place for all patients initiating ADT of any duration. Fracture risk can be assessed using the Fracture Risk Assessment Tool (FRAX), the algorithm released by The University of Sheffield (https://www.fraxplus.org/). FRAX was developed to estimate the 10-year probability of hip fracture or major osteoporotic fractures combined (hip, spine, shoulder, or wrist) for an untreated individual using easily obtainable clinical risk factors for fracture with or without information on bone mineral density. When utilizing the FRAX algorithm select YES for secondary osteoporosis for individuals with hypogonadism. ADT should be considered "secondary osteoporosis" when using the FRAX algorithm. A previous major osteoporotic fracture (hip fracture or spine fracture) is considered clinical osteoporosis and warrants bone antiresorptive drug therapy independent of bone mineral density.
- A baseline dual-energy x-ray absorptiometry (DEXA) scan should be obtained before starting ADT in patients at increased risk for fracture based on FRAX screening and being considered for antiresorptive therapy (see Table 1). For patients at low risk of fracture based on the FRAX risk assessment, baseline DEXA scan can be omitted. The exact FRAX fracture risk threshold has not been defined in this population. One approach is to set the threshold at 10-year risk of major osteoporotic fracture (calculated without DEXA) greater than that of a 65-year old white woman with no additional risk factors (defined as 8.4% in the United States).
- Treatment for osteoporosis is advised according to guidelines for the general population from the Bone Health and Osteoporosis Foundation.[1] These guidelines (see Table 1) include recommendations for: 1) calcium (1000–1200 mg daily from food, with supplements if intake is insufficient); 2) vitamin D3 (serum levels of 30–50 ng/mL with supplements prescribed if needed); and 3) pharmacologic treatment for patients aged ≥50 years with low bone mass (T-score between -1.0 and -2.5, osteopenia) at the femoral neck or total hip by DEXA scan with a 10-year probability of hip fracture ≥3% or a 10-year probability of a major osteoporosis-related fracture ≥20% based on FRAX screening (see Table 2).
- Antiresorptive medications that increase bone mineral density and reduce disease-related skeletal complications during ADT for prostate cancer include denosumab (60 mg subcutaneously [SQ] every 6 months), zoledronic acid (5 mg IV annually), and alendronate (70 mg PO weekly) (see Table 2). Treatment with either denosumab, zoledronic acid, or alendronate sodium is recommended when the absolute fracture risk warrants drug therapy.
  - Choice of agent may depend on underlying comorbidities, whether the patient has been treated with zoledronic acid previously, logistics, and/or cost considerations.
  - Bisphosphonates (zoledronic acid or alendronate) can cause side effects of acute phase reaction, joint pain, hypocalcemia, osteonecrosis of the jaw, nephrotoxicity with need for dose modification for renal insufficiency, ocular toxicities, and atypical femoral fractures with prolonged use (>3 to 5 years).
  - Denosumab can cause side effects of hypocalcemia, osteonecrosis of the jaw, and atypical femoral fractures with prolonged use. The risk factors for denosumab-associated hypocalcemia include blastic bone metastases, renal impairment, vitamin D deficiency, the lack of prophylactic supplementation of calcium and/or vitamin D, preexisting hypoparathyroidism, hypomagnesemia, and gastric bypass. Although renal monitoring is not required, denosumab is not recommended in patients with a creatinine clearance <30 mL/min given the risk of severe hypocalcemia. Calcium, creatinine, and vitamin D levels should be checked prior to initiating therapy. Periodic monitoring of serum calcium levels is recommended with denosumab use. Stopping denosumab therapy can result in rebound bone loss and fractures; therefore it is recommended to administer at least one dose of a potent bisphosphonate (zoledronic acid 4 or 5 mg) to prevent rebound bone loss and presumably rebound fracture.[2]

References on PROS-B 4 of 6

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PRINCIPLES OF SURVIVORSHIP
BONE HEALTH IN PROSTATE CANCER

Treatment-Related Bone Loss Continued

- **The risk of osteonecrosis of the jaw is increased in patients who have tooth extractions, poor dental hygiene, or a dental appliance. To prevent osteonecrosis of the jaw, it is recommended that all patients have a comprehensive dental evaluation prior to initiating an osteoclast inhibitor.[3] If invasive dental procedures are required, bone-targeted therapy should be withheld until the dentist indicates that the patient has healed completely from all dental procedure(s). Stopping denosumab represents a dilemma is this context, and the clinician must carefully weigh the risk of rebound spine fractures versus the risk of osteonecrosis of the jaw.**

- **Annual assessment of fracture risk using the FRAX risk assessment tool is recommended for all patients on ADT or those who remain hypogonadal after completion of ADT (see Table 1). Depending on the fracture risk and prior DEXA scan results, repeat DEXA scan in 1 to 2 years is recommended for those patients on ADT. For individuals initiated on antiresorptive therapy, a follow-up DEXA scan after 1 year of treatment is recommended by the International Society for Clinical Densitometry, although there is no consensus on the optimal approach to monitoring the effectiveness of bone treatment. Use of biochemical markers of bone turnover to monitor response to therapy is not recommended. There are currently no guidelines on how often to monitor vitamin D levels.**
- **For patients receiving antiresorptive therapy, there are currently no consensus guidelines on duration of treatment. Due to concerns of long-term risks of antiresorptive therapy, a "drug holiday" at 3 to 5 years can be considered based on agent utilized, stability of bone mineral density, prior fracture history, and future fracture risk. Bone mineral density should be monitored approximately every 1 to 2 years after suspending therapy, and therapy should generally be resumed if bone mineral density declines significantly or if the patient develops a new fragility fracture.**

Table 1: Risk Assessment and Monitoring

Clinical Scenario	Recommendation
Baseline at ADT initiation	DEXA recommended for most patients. In select individuals at low probably of fracture based on FRAX risk assessment tool, DEXA can be omitted
On ADT	DEXA every 1–2 years, dependent on FRAX risk assessment tool
On antiresorptive therapy	DEXA at 1 year

References on PROS-B 4 of 6

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PRINCIPLES OF SURVIVORSHIP
BONE HEALTH IN PROSTATE CANCER

Prevention of Symptomatic Skeletal-Related Events (SREs) in Patients with Bone-Metastatic CRPC

- **In patients with CRPC who have bone metastases, denosumab and zoledronic acid have been shown to prevent disease-related skeletal complications, which include fracture, spinal cord compression, or the need for surgery or RT to bone.**
- **When compared to zoledronic acid, denosumab was shown to be superior in prevention of SREs in patients with metastatic CRPC (mCRPC), albeit with numerically higher hypocalcemia and osteonecrosis of the jaw risks. Initial studies investigated zoledronic acid and denosumab administered every 4 weeks. Subsequent studies demonstrated that every–12-week dosing of zoledronic acid compared to every–4-week dosing did not increase the risk of skeletal events.[4,5] Every–12-week dosing of zoledronic acid is recommended for symptomatic SRE reduction when indicated. Every–12-week dosing of denosumab is under investigation and current data suggest noninferior symptomatic skeletal events compared to every–4-week dosing.[5] Utilization of zoledronic acid and denosumab for symptomatic SRE reduction requires consideration of degree of benefit and risk associated with therapy to optimize use, dose, and schedule. It is important to recognize that testing of zoledronic acid and denosumab in bone-metastatic CRPC was conducted during an era when treatment options for mCRPC were largely limited to docetaxel chemotherapy. Subsequent studies investigating abiraterone, enzalutamide, cabazitaxel, radium-223, and Lu-177–PSMA-617 have demonstrated improvement of SREs with treatment. While radium-223 did improve symptomatic SREs in patients with bone mCRPC, the combination of radium-223 with abiraterone was associated with increased frequency of bone fractures, particularly in individuals not receiving an antiresorptive agent.[6]**
- **A phase 3 clinical trial that assessed the role for zoledronic acid in patients with castration-sensitive disease beginning ADT for bone metastases was negative.[7] Therefore, use of osteoclast inhibitors for reduction of symptomatic SREs in metastatic castration-sensitive disease with bone metastases is not recommended. However, usage of these agents to prevent bone loss and fragility fractures at appropriate doses and dosing intervals should be utilized when clinically appropriate in this context (see Treatment-Related Bone Loss, PROS-B 1 of 4).**

Table 2: Optimization of Bone Health in Patients with Prostate Cancer

Patient Population	Category	Intervention
All patients receiving ADT	Lifestyle modifications	• Weight-bearing exercises (30 minutes per day), balance training, safe movement strategies • Limit alcohol consumption • Smoking cessation
	Calcium and vitamin D supplementation	• Calcium 1000–1200 mg daily from food with supplements if needed • Maintain serum vitamin D3 levels of 30–50 ng/mL with supplements if needed
For treatment-related bone loss in patients receiving ADT	Antiresorptive agents	• Alendronate 70 mg PO weekly • Denosumab 60 mg SQ every 6 months • Zoledronic acid 5 mg IV annually
For prevention of symptomatic SREs in patients with bone-metastatic CRPC	Antiresorptive agents	• Denosumab 120 mg SQ every 4 weeks • Zoledronic acid 4 mg IV every 12 weeks

References on PROS-B 4 of 6

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PRINCIPLES OF SURVIVORSHIP
BONE HEALTH IN PROSTATE CANCER: REFERENCES

[1] LeBoff MS, Greenspan SL, Insogna KL, et al. The clinician's guide to prevention and treatment of osteoporosis. Osteoporos Int 2022;33:2049-2102.
[2] Cummings SR, Ferrari S, Eastell R, et al. Vertebral fractures after discontinuation of denosumab: A post hoc analysis of the randomized placebo-controlled freedom trial and its extension. J Bone Miner Res 2018;33:190-198.
[3] Yarom N, Shapiro CL, Peterson DE, et al. Medication-related osteonecrosis of the jaw: MASCC/ISOO/ASCO Clinical practice guideline. J Clin Oncol 2019;37:2270-2290.
[4] Himelstein AL, Foster JC, Khatcheressian JL, et al. Effect of longer-interval vs standard dosing of zoledronic acid on skeletal events in patients with bone metastases: a randomized clinical trial. JAMA 2017;317:48-58.
[5] Clemons M, Ong M, Stober C, et al. A randomised trial of 4- versus 12-weekly administration of bone-targeted agents in patients with bone metastases from breast or castration-resistant prostate cancer. Eur J Cancer 2021;142:132-140.
[6] Smith M, Parker C, Saad F, et al. Addition of radium-223 to abiraterone acetate and prednisone or prednisolone in patients with castration-resistant prostate cancer and bone metastases (ERA 223): a randomised, double-blind, placebo-controlled, phase 3 trial. Lancet Oncol 2019;20:408-419.
[7] Smith MR, Halabi S, Ryan CJ, et al. Randomized controlled trial of early zoledronic acid in men with castration-sensitive prostate cancer and bone metastases: results of CALGB 90202 (alliance). J Clin Oncol 2014;32:1143-1150.

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PRINCIPLES OF SURVIVORSHIP
CARDIOVASCULAR DISEASE IN PROSTATE CANCER

- **Individuals with prostate cancer are at increased risk for cardiovascular disease (CVD) and cardiovascular events, and CVD is a leading cause of death in this population.[1]**
  - **The main cause for this increased risk is the cardiovascular effects of ADT and ARPIs.[2,3]**
  - **Shared risk factors for prostate cancer and CVD (ie, smoking, poor health behaviors) also likely contribute to the increased risk of CVD in this population.[4]**
- **CVD develops over time because of well-defined risk factors such as hypertension, hyperlipidemia, use of tobacco products, obesity, diabetes, and the cardiotoxic effects of cancer treatment.**
- **CVD-related morbidity and mortality can occur in both the short- and long-term in prostate cancer survivors.**
- **Control of modifiable risk factors can decrease the risk of subsequent cardiovascular events.**
- **Therefore, prostate cancer survivors should assessed for and counseled on:**
  - **Pre-existing and emerging CVD (eg, coronary artery disease, congestive heart failure, peripheral vascular disease, arrhythmias including atrial fibrillation)**
  - **Any increased risk of CVD based on prior treatment, comorbidity, or CVD risk factors (eg, hypertension, dyslipidemia, obesity, cigarette/tobacco use, diabetes mellitus)**
  - **Lifestyle modification and interventions for modifiable risk factors (see the ABCDEs of CVD Prevention, Table 1)**
- **Aggressive risk factor management is recommended for those deemed to be at higher risk, and referral to a cardiologist or cardio-oncologist in patients at elevated CVD risk should be considered at any stage of the prostate cancer journey.**
- **Cooperation and shared care with primary care providers, and cardiovascular specialists as needed, is key to optimizing cardiac and vascular outcomes in prostate cancer survivors.**
- **For further information on cardiovascular risk assessment in cancer survivors, see the NCCN Guidelines for Survivorship.**

[1] Sturgeon KM, Deng L, Bluethmann SM, et al. A population-based study of cardiovascular disease mortality risk in US cancer patients. Eur Heart J 2019;40:3889-3897.
[2] Okwuosa TM, Morgans A, Rhee JW, et al. Impact of hormonal therapies for treatment of hormone-dependent cancers (breast and prostate) on the cardiovascular system: effects and modifications: a scientific statement from the American Heart Association. Circ Genom Precis Med 2021;14:e000082.
[3] El-Taji O, Taktak S, Jones C, et al. Cardiovascular events and androgen receptor signaling inhibitors in advanced prostate cancer: A systematic review and meta-analysis. JAMA Oncol 2024;20:874-884.
[4] Bergengren O, Pekala KR, Matsoukas K, et al. 2022 update on prostate cancer epidemiology and risk factors - a systematic review. Eur Urol 2023;84:191-206.

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PRINCIPLES OF SURVIVORSHIP
CARDIOVASCULAR DISEASE IN PROSTATE CANCER: RISK ASSESSMENT[a]

Table 1: ABCDEs to Promote Cardiovascular Wellness in Cancer Survivors[b]	
A	• Awareness of risks and presentation of heart disease • Assessment of CVD and cardiovascular risk (consider use of a CVD risk assessment tool[c]) • Aspirin use as appropriate (indicated for secondary prevention; clinician-survivor discussion required for primary prevention with careful weighing of benefits and risks)[d]
B	• Blood pressure monitoring/management (with clinician-survivor discussion regarding the use of hypertension treatment and blood pressure goals)
C	• Cholesterol assessment/management (with clinician-survivor discussion regarding the use of statin therapy for primary prevention and lipid profile goals) • Cigarette/tobacco cessation (NCCN Guidelines for Smoking Cessation)
D	• Diet and weight management (see SNWM-1 in NCCN Guidelines for Survivorship) • Diabetes mellitus prevention/treatment
E	• Exercise (see SPA-1 in NCCN Guidelines for Survivorship)[e] • Echocardiogram (ECHO) and/or electrocardiogram (ECG) based on individual risk (consider use of a CVD risk assessment tool[c])

[a] Armenian SH, Lacchetti C, Barac A, et al. Prevention and monitoring of cardiac dysfunction in survivors of adult cancers: American Society of Clinical Oncology Clinical Practice Guideline. J Clin Oncol 2017;35:893-911.
[b] Adapted with permission from Montazeri K, Unitt C, Foody JM, et al. ABCDE steps to prevent heart disease in breast cancer survivors. Circulation 2014;130:e157-e159.
[c] The ASCVD Risk Estimator Plus from the American College of Cardiology is available at https://tools.acc.org/ascvd-risk-estimator-plus/#!/calculate/estimate and The American Heart Association's PREVENT Risk Calculator is available at https://professional.heart.org/en/guidelinesand-statements/prevent-calculator.
[d] U.S. Preventive Services Task Force; Davidson KW, Barry MJ, Mangione CM, et al. Aspirin use to prevent cardiovascular disease: U.S. Preventive Services Task Force Recommendation Statement. JAMA 2022;327:1577-1584.
[e] Gilchrist SC, et al. Circulation 2019;139:e997-e1012.

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PRINCIPLES OF GENETICS AND MOLECULAR/BIOMARKER ANALYSIS

GERMLINE TESTING

For details regarding the nuances of genetic counseling and testing, see Principles of Cancer Risk Assessment and Counseling (EVAL-A) in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic and Prostate

- **Pre-test Considerations**
  - **The Panel recommends inquiring about family and personal history of cancer, and known germline variants at time of initial diagnosis. Criteria for germline testing (see CRIT-6 in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic and Prostate and HRS-3 in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric) should be reviewed at time of initial diagnosis and, if relevant, at recurrence.**
  - **Germline testing is also recommended for patients with metastatic, regional (node positive), very-high-risk localized, or high-risk localized prostate cancer.**
  - **Germline testing should be considered in appropriate individuals where it is likely to impact the prostate cancer treatment and clinical trial options, management of risk of other cancers, and/or potential risk of cancer in family members.**
- **Testing**
  - **If criteria are met, multigene testing is recommended (see GENE-1 in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate).**
- **Post-test Considerations**
  - **Post-test genetic counseling is strongly recommended if a germline mutation (pathogenic/likely pathogenic variant) is identified. Cascade testing for relatives is critical to inform the risk for familial cancers in all relatives.**
  - **Post-test genetic counseling is recommended if positive family history but no pathogenic variant OR if only germline variants of uncertain significance (VUS) are identified. This is to ensure accurate understanding of family implications and review indications for additional testing and/or follow-up (including clinical trials of reclassification).**
  - **Resources are available to review the available data supporting pathogenic consequences of specific variants (eg, https://www.ncbi.nlm.nih.gov/clinvar/; https://brcaexchange.org/about/app).**
  - **Individuals should be counseled to inform providers of any updates to family cancer history.**

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PRINCIPLES OF GENETICS AND MOLECULAR/BIOMARKER ANALYSIS

SOMATIC TUMOR TESTING

- **Pre-test Considerations**
  - **At present, tumor molecular and biomarker analysis is recommended for patients with metastatic disease for treatment decision-making, including understanding eligibility for biomarker-directed treatments, genetic counseling, and eligibility for clinical trials. Clinical trials may include established and/or candidate molecular biomarkers for eligibility.**
  - **Tumor molecular profiles may change with subsequent treatments and re-evaluation may be considered at time of cancer progression for treatment decision-making.**
  - **Patients should be informed that tumor molecular analysis by DNA sequencing has the potential to uncover germline findings. Confirmatory germline testing may be recommended [see Post-test Considerations (below) and Tumor Testing:s Potential Implications for Germline Testing in the Principles of Cancer Risk Assessment and Counseling (EVAL-A) in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate].**
- **Testing**
  - **Somatic testing for alterations in DNA damage response:**
    - ◊ **Multigene tumor testing for alterations in HRR genes, including but not limited to *BRCA1, BRCA2, ATM, PALB2, FANCA, RAD51D, CHEK2*, and *CDK12*, is recommended in patients with metastatic prostate cancer. This testing can be considered in patients with regional prostate cancer.**
      - **– Loss of *BRCA1* and *BRCA2* may be especially associated with response to PARP inhibitor therapy compared to other HRR gene alterations.**
    - ◊ **Tumor testing for MSI-H or dMMR is recommended in patients with mCRPC and may be considered in patients with regional or castration-sensitive metastatic prostate cancer.**
    - ◊ **TMB testing is recommended in patients with mCRPC.**
- **Tumor Specimen and Assay Considerations**
  - **The Panel strongly recommends a metastatic biopsy for histologic and molecular evaluation. This could include lymph node biopsy for patients with N1 disease.**
    - ◊ **When metastatic biopsy is unsafe or unfeasible, plasma circulating tumor DNA (ctDNA) assay is an option, preferably collected during biochemical (PSA) and/or radiographic progression in order to maximize diagnostic yield. When diagnostic yield is low, the risk of false negatives is higher, so ctDNA collection is not recommended when PSA is undetectable.**
  - **Caution is needed when interpreting ctDNA-only evaluation due to potential interference from clonal hematopoiesis of indeterminate potential (CHIP), which can result in a false-positive biomarker signal.**
  - **DNA analysis for MSI and immunohistochemistry for mismatch repair (MMR) are different assays measuring different biological effects caused by dMMR function. If MSI is used, testing using a next-generation sequencing assay validated for prostate cancer is preferred.**
- **Post-test Considerations**
  - **Post-test genetic counseling is recommended if pathogenic/likely pathogenic variant (mutation) identified in any gene that has clinical implications if also identified in germline (eg, *BRCA1, BRCA2, ATM, PALB2, CHEK2, HOXB13, MLH1, MSH2, MSH6, PMS2*).**
  - **Post-test genetic counseling to assess for the possibility of Lynch syndrome is recommended if MSI-H or dMMR is found.**

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PRINCIPLES OF QUALITY OF LIFE AND SHARED DECISION-MAKING

- **Treatments for patients with localized prostate cancer have risks and side effects that must be considered in the context of the risk posed by the disease.[1-4]**
- **Baseline urinary, sexual, and bowel function are strongly associated with functional outcomes among patients undergoing treatment.[1-4]**
- **Thus, it is important to measure baseline disease-specific function (urinary, sexual, and bowel function), preferably using a standardized patient-reported outcomes instrument (eg, EPIC-26[5]).**
- **Shared decision-making in a multidisciplinary manner regarding initial management of localized prostate cancer should include an explanation of the potential benefits and potential harms of each option. The provider should explain the likelihood of cure, recurrence, disease progression, and disease-specific mortality with each management option, taking into account disease severity and competing risks. In addition to the primary intended effects of treatment, the clinician should discuss the side effects of each treatment and predicted impact on quality of life, including urinary, sexual, and bowel function. Patient preferences should be elicited and should be incorporated into the disease management decision.[6]**
- **Consider geriatric assessment. See NCCN Guidelines for Older Adult Oncology for tools to aid optimal assessment and management of disease in older adults.**
- **Refer to the NCCN Distress Thermometer and Problem List, which includes social determinants of health. See NCCN Guidelines for Distress Management (DIS-A).**

[1] Sanda MG, Dunn RL, Michalski J, et al. Quality of life and satisfaction with outcome among prostate-cancer survivors. N Engl J Med 2008;358:1250-1261.
[2] Chen RC, Basak R, Meyer AM, et al. Association between choice of radical prostatectomy, external beam radiotherapy, brachytherapy, or active surveillance and patient-reported quality of life among men with localized prostate cancer. JAMA 2017;317:1141-1150.
[3] Hoffman KE, Penson DF, Zhao Z, et al. Patient-reported outcomes through 5 years for active surveillance, surgery, brachytherapy, or external beam radiation with or without androgen deprivation therapy for localized prostate cancer. JAMA 2020;323:149-163.
[4] Donovan JL, Hamdy FC, Lane JA, et al; ProtecT Study Group. Patient-reported outcomes after monitoring, surgery, or radiotherapy for prostate cancer. N Engl J Med 2016;375:1425-1437.
[5] Szymanski KM, Wei JT, Dunn RL, Sanda MG. Development and validation of an abbreviated version of the expanded prostate cancer index composite instrument for measuring health-related quality of life among prostate cancer survivors. Urology 2010;76:1245-1250.
[6] Makarov D, Fagerlin A, Finkelstein J et al. AUA White Paper on Implementation of Shared Decision Making into Urological Practice. American Urological Association 2022. Available at: https://www.auanet.org//guidelines-and-quality/guidelines/best-practice-statements-and-whitepapers/shared-decision-making.

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PRINCIPLES OF IMAGING

Goals of Imaging

- Imaging is performed for the detection and characterization of disease to select treatment or guide change in disease management.
- Imaging techniques can evaluate anatomic or functional parameters.
  - Anatomic imaging techniques include ultrasound, CT, and MRI.
  - Functional imaging techniques include radionuclide bone scan, PET/CT, and advanced MRI techniques, such as spectroscopy and diffusion-weighted imaging (DWI).

Efficacy of Imaging

- The utility of imaging for patients with early PSA persistence/recurrence after RP depends on risk group prior to operation, pathologic Gleason grade and stage, PSA, and PSADT after recurrence. Low- and intermediate-risk groups with low serum PSAs postoperatively have a very low risk of positive bone scans or CT scans.
- Frequency of imaging should be based on individual risk, age, PSADT, Gleason score, and overall health.
- Bone scans are rarely positive in asymptomatic patients with PSA <10 ng/mL. The relative risk for bone metastasis or death increases as PSADT shortens. Bone imaging should be performed more frequently when PSADT is ≤8 months, where there appears to be an inflection point.

Ultrasound

- Ultrasound uses high-frequency sound waves to image small regions of the body.
  - Standard ultrasound imaging provides anatomic information.
  - Vascular flow can be assessed using Doppler ultrasound techniques.
- Endorectal ultrasound is used to guide transrectal biopsies of the prostate. Endorectal ultrasound can be considered for patients with suspected recurrence after RP to guide prostate bed biopsy.
- Advanced ultrasound techniques for imaging of the prostate and for differentiation between prostate cancer and prostatitis are under evaluation.

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PRINCIPLES OF IMAGING

Bone Imaging

- The use of the term "bone scan" refers to the technetium-99m-MDP bone scan in which technetium is taken up by bone that is turning over and imaged with a gamma camera using planar imaging or 3D imaging with single-photon emission CT (SPECT).
  - Sites of increased uptake imply accelerated bone turnover and may indicate metastatic disease.
  - Osseous metastatic disease may be diagnosed based on the overall pattern of activity, or in conjunction with anatomic imaging.
- Plain films, CT, MRI, or PET/CT or PET/MRI with F-18 piflufolastat prostate-specific membrane antigen (PSMA), Ga-68 PSMA-11, F-18 flotufolastat PSMA, F-18 fluciclovine, F-18 sodium fluoride, or C-11 choline can be considered for equivocal results on initial bone scan.
- Ga-68 PSMA-11, F-18 piflufolastat PSMA, or F-18 flotufolastat PSMA-PET/CT or PET/MRI (full body imaging) can be considered as an alternative to bone scan.
- Bone imaging is indicated in the initial evaluation of patients at high risk for skeletal metastases.
- Bone imaging can be considered for the evaluation of the patient post-prostatectomy when PSA does not fall to undetectable levels, or when there is undetectable PSA after RP with a subsequent detectable PSA that increases on ≥2 subsequent determinations.
- Bone imaging can be considered for the evaluation of patients with an increasing PSA or positive DRE after RT if the patient is a candidate for additional local therapy or systemic therapy.
- Bone scans are helpful to monitor metastatic prostate cancer to determine the clinical benefit of systemic therapy. However, new lesions seen on an initial post-treatment bone scan, compared to the pretreatment baseline scan, may not indicate disease progression.
- New lesions in the setting of a falling PSA or soft tissue response and in the absence of pain progression at that site may indicate bone scan flare or an osteoblastic healing reaction. For this reason, a confirmatory bone scan 8 to 12 weeks later is warranted to determine true progression from flare reaction. Additional new lesions favor progression. Stable scans make continuation of treatment reasonable. Bone scan flare is common, particularly on initiation of new hormonal therapy, and may be observed in nearly half of patients treated with the newer agents, enzalutamide and abiraterone. Similar flare phenomena may exist with other imaging modalities, such as CT or PET/CT imaging.
- Bone scans and soft tissue imaging (CT or MRI) in patients with metastatic or non-metastatic prostate cancer may be obtained regularly during systemic therapy to assess clinical benefit.
- Bone scans should be performed for symptoms and as often as every 6 to 12 months to monitor ADT. The need for soft tissue images remains unclear. In CRPC, 8- to 12-week imaging intervals appear reasonable.

- *Plain Radiography*
  - Plain radiography can be used to evaluate symptomatic regions in the skeleton. However, plain films will not detect a bone lesion until nearly 50% of the mineral content of the bone is lost or gained.
  - CT or MRI may be more useful to assess fracture risk as these modalities permit more accurate assessment of cortical involvement than plain films where osteoblastic lesions may obscure cortical involvement.

Soft Tissue Imaging

- Soft tissue imaging of the pelvis, abdomen, and chest can include:
  - Chest CT and abdomen/pelvis CT or abdomen/pelvis MRI or
  - PSMA-PET/CT or PSMA-PET/MRI for bone and soft tissue (full body) imaging.

- *Computed Tomography*
  - CT provides a high level of anatomic detail, and may detect gross extracapsular disease, nodal metastatic disease, and/or visceral metastatic disease.
  - CT is generally not sufficient to evaluate the prostate gland.
  - CT may be performed with IV contrast, and CT technique should be optimized to maximize diagnostic utility while minimizing radiation dose.
  - CT can be used for examination of the pelvis and/or abdomen for initial evaluation (PROS-2) and as part of workup for recurrence or progression (see PROS-9 through PROS-18).

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PRINCIPLES OF IMAGING

- *Magnetic Resonance Imaging*
  - The strengths of MRI include high soft tissue contrast and characterization, multiparametric image acquisition, multiplanar imaging capability, and the use of specific MRI sequences to assess function.
    - MRI can be performed with and without the administration of IV contrast material.
    - Resolution of MRI images in the pelvis can be augmented using a phased array/endorectal coil.
  - Standard MRI techniques can be used for examination of the pelvis and/or abdomen for initial evaluation (PROS-1) and as part of workup for recurrence or progression (see PROS-9 through PROS-18).
  - MRI may be considered in patients after RP when PSA does not fall to undetectable levels or when an undetectable PSA becomes detectable and increases on ≥2 subsequent determinations, or after RT for increasing PSA or positive DRE if the patient is a candidate for additional local therapy. MRI-ultrasound fusion biopsy may improve the detection of higher grade (Grade Group ≥2) cancers.
  - Multiparametric MRI (mpMRI) can be used in the staging and characterization of prostate cancer. mpMRI images are defined as images acquired with at least one more sequence in addition to the anatomical T2-weighted images, such as DWI or dynamic contrast-enhanced (DCE) images. mpMRI may be used to better risk stratify patients who are considering active surveillance. Additionally, mpMRI may detect large and poorly differentiated prostate cancer (Grade Group ≥2) and detect extracapsular extension (T staging) and is preferred over CT for abdomen/pelvis staging. mpMRI has been shown to be equivalent to CT scan for pelvic lymph node evaluation.

Full Body Imaging

- *Positron Emission Tomography*
  - PSMA-PET refers to a growing body of radiopharmaceuticals that target PSMA on the surface of prostate cells. There are multiple PSMA radiopharmaceuticals at various stages of investigation. At this time, the NCCN Guidelines only recommend the currently FDA-approved PSMA agents: F-18 piflufolastat PSMA (also known as F-18 DCFPyL), F-18 flotufolastat PSMA (also known as rh-PSMA-7.3), and Ga-68 PSMA-11 (also known as PSMA HBED-CC). Throughout these Guidelines, "PSMA-PET" refers to any of these FDA-approved PSMA ligands. See Table 1 on PROS-E (5 of 5) for more details.
  - F-18 flotufolastat PSMA is a PET imaging agent that is part of a class of tracers referred to as radiohybrid (rh) ligands. These tracers have two binding sites for radionuclides for both imaging and treatment, but the significance of this remains to be determined.
  - PSMA-PET/CT or PET/MRI can be considered as an alternative to CT, MRI, and bone scans for initial staging of unfavorable intermediate, high, and very-high-risk disease, the detection of biochemically recurrent disease, and as workup for progression.
  - In the setting of M1 CRPC, PSMA-PET imaging should only be used to determine if a patient is a candidate for Lu-177-PSMA-617.
  - Changes in systemic therapy should not be made solely based on a positive PSMA-PET in patients with M0 CRPC.
  - Bone scans can be considered to confirm osseous uptake on PET scans.
  - Synthesis of Ga-68 PSMA-11 requires that the PSMA-11 ligand is labeled with Ga-68 from a generator or cyclotron. Two commercial kits to perform this in nuclear pharmacies have been approved by the FDA.
  - C-11 choline or F-18 fluciclovine PET/CT or PET/MRI may be used to detect small-volume recurrent disease in soft tissues and in bone.
  - Studies suggest that PSMA-PET imaging has a higher sensitivity than C-11 choline or F-18 fluciclovine PET imaging, especially at very low PSA levels (eg, <0.5 ng/mL).
  - Because of the increased sensitivity and specificity of PSMA-PET tracers for detecting micrometastatic disease compared to CT, MRI, and bone scan at both initial staging and BCR, PSMA-PET/CT or PSMA-PET/MRI can serve as a more effective frontline imaging tool for these patients.
  - Histologic or radiographic confirmation of involvement detected by PET imaging is recommended whenever feasible due to the presence of false positives. Although false positives exist, literature suggests that these are outweighed by the increase in true positives detected by PET relative to CT, MRI, and bone scans. To reduce the false-positive rate, physicians should consider the

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PRINCIPLES OF IMAGING

intensity of PSMA-PET uptake and correlative CT findings in the interpretation of scans. The use of a standardized reporting system is encouraged.
- PSMA imaging should be done before initiation of ADT because ADT may affect detection sensitivity.
- High variability among PET/CT or PET/MRI equipment, protocols, interpretation, and institutions provides challenges for application and interpretation of the utility of PET/CT or PET/MRI.
- Table 1 on PROS-E (5 of 5) provides a summary of FDA-approved PET imaging tracers utilized for study in prostate cancer.
- PET/CT or PET/MRI results may change treatment but may not change oncologic outcome.
- When patients with the worst prognosis move from one risk group to the higher risk group, the average outcome of both risk groups will improve even if treatment has no impact on disease. This phenomenon is known as the Will Rogers effect, in which the improved outcomes of both groups could be falsely attributed to improvement in treatment, but would be due only to improved risk group assignment. As an example, F-18 sodium fluoride PET/CT may categorize some patients as M1b who would have been categorized previously as M0 using a bone scan (stage migration). Absent any change in the effectiveness of therapy, the overall survival (OS) of both M1b and M0 groups would improve. The definition of M0 and M1 disease for randomized clinical trials that added docetaxel or abiraterone to ADT was based on CT and radionuclide bone scans. Results suggest that OS of M1 disease is improved, whereas progression-free but not OS of M0 disease is improved. Therefore, a subset of patients now diagnosed with M1 disease using F-18 sodium fluoride PET/CT might not benefit from the more intensive therapy used in these trials and could achieve equivalent OS from less intensive therapy aimed at M0 disease. Carefully designed clinical trials using proper staging will be necessary to prove therapeutic benefit, rather than making assumptions compromised by stage migration.
- Fluorodeoxyglucose (FDG)-PET/CT should not be used routinely for staging prostate cancer since data are limited in this setting.
- F-18 FDG-PET has been shown to be prognostic in patients with progressive CRPC.[1,2]
- The increasing use of PSMA-PET has identified the potential for considerable biological diversity among disease foci within a given individual with prostate cancer, especially mCRPC, and that this heterogeneity can be detected with a combination of PSMA-PET and FDG-PET. Initial data suggest that metastases with PSMA-negative/FDG-positive mismatches may exist in patients with mCRPC undergoing Lu-PSMA radioligand therapy and that patients with these mismatches may have worse outcomes. Currently, no robust clinical trial data exist to support the incorporation of FDG-PET into routine clinical use alongside PSMA-PET. To overcome the limitations of PSMA-PET in PSMA-negative metastatic disease, the Panel currently recommends the use of contrast-enhanced CT or MRI in these patients, as the non-contrast CT component of PSMA-PET/CT is insufficient to detect visceral metastatic disease.

Imaging as Workup for Progression

- Workup for progression should include bone and soft tissue evaluation.
  - See Bone Imaging (PROS-E [2 of 5]).
  - See Soft Tissue Imaging (PROS-E [2 of 5]).
- Imaging for patients with progressive CRPC should include chest CT, bone imaging, and abdomen/pelvis CT with contrast or abdomen/pelvis MRI with and without contrast.
- There is a lack of evidence to support the use of PET imaging in the CRPC setting and it should only be used in the setting of M1 CRPC to determine if a patient is a candidate for Lu-177-PSMA-617. Changes in treatment should not be made solely based on a positive PSMA-PET in patients with M0 CRPC.

[1] Buteau JP, Martin AJ, Emmett L, et al. PSMA and FDG-PET as predictive and prognostic biomarkers in patients given [177Lu]Lu-PSMA-617 versus cabazitaxel for metastatic castration-resistant prostate cancer (TheraP): A biomarker analysis from a randomised, open-label, phase 2 trial. Lancet Oncol 2022;23:1389-1397.
[2] Pathmanandavel S, Crumbaker M, Nguyen A, et al. The prognostic value of posttreatment 68Ga-PSMA-11 PET/CT and 18F-FDG PET/CT in metastatic castration-resistant prostate cancer treated with 177Lu-PSMA-617 and NOX66 in a phase I/II trial (LuPIN). J Nucl Med 2023;64:69-74.

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PRINCIPLES OF IMAGING

Table 1. FDA-Cleared PET Imaging Tracers Studied in Prostate Cancer

Tracer	Half-Life (min)	Excretion	Detection Rates[a]	Panel Recommendation
Ga-68 PSMA-11 (PSMA-HBED-CC)	68	Renal	• 40% sensitivity and 95% specificity to detect nodal involvement in primary staging of patients with intermediate-, high-, and very-high-risk disease • 92% patient-level PPV in BCR	• May be considered as an alternative to CT, MRI, and bone scans for initial staging of unfavorable intermediate-, high-, and very-high-risk disease, the detection of biochemically recurrent disease, and as workup for progression • May be considered for equivocal results on initial bone scan
F-18 piflufolastat-PSMA (18F-DCFPyL)	110	Renal	• 31%–42% sensitivity and 96%–99% specificity to detect nodal involvement in primary staging of patients with unfavorable intermediate-risk, high-risk, and very-high-risk disease • 85%–87% patient-level CLR[b] in BCR	
F-18 flotufolastat PSMA (18F-rh-PSMA-7.3)[6,7]	110	Renal	• 23%–30% sensitivity and 93%–97% specificity to detect nodal involvement in primary staging of patients with unfavorable-intermediate-, high-, and very-high-risk disease • 82% patient-level PPV in BCR	
C-11 choline	20	Hepatic and Renal	• 53%–96% PPV in BCR	• May be used to detect small-volume recurrent disease in soft tissues and in bone • May be considered for equivocal results on initial bone scan
F-18 fluciclovine (FACBC)	110	Renal	• 87%–91% CLR[b] in BCR	
F-18 sodium fluoride	110	Renal	• 77%–94% sensitivity, 92%–99% specificity, and 82%–97% PPV for bone metastases	• May be considered for equivocal results on initial bone scan

[a] Interpret with caution. Wherever possible, studies were included that used histopathologic confirmation, but not all studies used confirmatory histology as the gold standard. Values may vary depending upon the site of the lesion and phase of the disease process.
[b] CLR: Correct localization rate. Patient-level positive predictive value (PPV) + anatomic lesion co-localization. Preferred over sensitivity and specificity in analyses of patients with BCR.

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PRINCIPLES OF ACTIVE SURVEILLANCE AND OBSERVATION

- **The NCCN Prostate Cancer Panel and the NCCN Prostate Cancer Early Detection Panel (NCCN Guidelines for Prostate Cancer Early Detection) remain concerned about overdiagnosis and overtreatment of prostate cancer. The Prostate Cancer Panel recommends that patients and their physicians carefully consider active surveillance based on the patient's prostate cancer risk profile and estimated life expectancy. In settings where the patient's age and comorbidities suggest a shorter life expectancy, observation may be more appropriate. Shared decision-making, after appropriate counseling on the risks and benefits of the various options, is critical.**
- **Life expectancy is a key determinant for the choice between observation and active surveillance. Multiple tools exist to assist in life expectancy estimation. See Principles of Life Expectancy Estimation (PROS-A).**
- **There remains no high-level evidence showing clinical benefit of advanced risk stratification tools (eg, gene expression biomarkers, germline or somatic testing, or artificial intelligence (AI)-based digital pathology models), to guide the candidacy for active surveillance.**

ACTIVE SURVEILLANCE[1]

- **Active surveillance involves actively monitoring the course of disease with the expectation to intervene with curative intent if the cancer progresses. It involves PSA monitoring and longitudinal serial pathologic assessments via a prostate biopsy, often guided by MRI.**
- **There are three phases of active surveillance: 1) candidacy, 2) confirmation of candidacy, and 3) active surveillance.**

- **Candidacy for Active Surveillance:**
  - **NCCN Low-Risk:**
    - ◊ **Active surveillance is the preferred management approach for patients with low-risk prostate cancer and a life expectancy ≥10 years.**
    - ◊ **The Panel recognizes that there is heterogeneity across this risk group.**
      - – **Some factors may be associated with an increased probability of near-term grade reclassification including:**
        - ▪ **High PSA density (≥0.15)[2]**
        - ▪ **Increased number of positive cores[3]**
        - ▪ **Higher body mass index (BMI)[3]**
        - ▪ **Faster PSA kinetics[3]**
      - – **Conversely a negative biopsy following diagnosis is associated with a decreased probability of grade reclassification.**
      - – **The Canary PASS risk calculator can be used for more precise estimates of upgrading probability on active surveillance (Canary Pass Risk Calculator)**
      - – **Higher risk features alone do not necessitate immediate treatment, but should be used as part of a shared-decision making discussion and in tailoring the intensity of active surveillance.**
    - ◊ **For patients who are at high-risk of suboptimal adherence to an active surveillance protocol, a careful discussion should be had about the potential risks of active surveillance. For some of these patients, upfront treatment with definitive RP or RT may be preferred.**

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References on PROS-F 5 of 5

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PRINCIPLES OF ACTIVE SURVEILLANCE AND OBSERVATION

- **Candidacy for Active Surveillance (continued):**
  - **NCCN Favorable Intermediate-Risk (FIR)**
    - ◊ **Patients with FIR prostate cancer and a life expectancy of >10 years may also consider active surveillance.**
    - ◊ **The FIR group was not created for optimal selection of active surveillance candidacy, and there is considerable heterogeneity in this risk group. Not all patients with FIR disease are suitable candidates for active surveillance. For example:**
      - **– A patient with Grade Group 1 in a single biopsy core, cT1c, PSA of 12 ng/mL, and PSA density of 0.15 ng/mL/g has FIR disease and a favorable prognostic outcome. Therefore, active surveillance is preferred.**
      - **– A patient with GG2 in 5 of 12 cores, cribriform architecture, cT1c, PSA of 8 ng/mL, and PSA density 0.30 ng/mL/g has FIR disease, but active surveillance would not be the preferred strategy.**
    - ◊ **In addition to prognostic factors listed for Low Risk, additional prognostic factors relevant for FIR include:**
      - **– Increased percentage pattern 4.**
      - **– Presence of unfavorable histology such as expansile/large cribriform histology and/or intraductal carcinoma.**
    - ◊ **Active surveillance is not recommended for patients with FIR and unfavorable histology (eg, expansile/large cribriform histology, intraductal carcinoma). See Discussion.**
    - ◊ **Intensity of active surveillance should generally be greater for patients with FIR disease.**
    - ◊ **Patients should be counseled that long-term data show that patients with GG2 disease managed with active monitoring or watchful waiting had an increase in the development of distant metastasis as compared to those receiving immediate radical treatment.[4] Continued refinements in active surveillance protocols and active surveillance candidacy are expected to improve the safety of active surveillance for GG2 FIR disease, and patients should be counseled on the importance of longitudinal follow-up.**

- **Confirmation of Candidacy:**
  - **Goals of confirmation testing are to help facilitate early identification of patients who may be at a higher risk of grade reclassification or cancer progression.**
  - **An initial prostate biopsy may underestimate tumor grade or volume, thus confirmatory testing is strongly recommended for patients who are considering active surveillance.**
  - **A repeat biopsy is a routine component of confirmatory testing.**
    - ◊ **The timing of repeat biopsy can be based on the patient's risk of progression. Patients with a higher risk should have a repeat biopsy earlier than those with a lower risk.**
    - ◊ **If the initial prostate biopsy was performed without mpMRI guidance, then the Panel recommends an earlier repeat biopsy (eg, within the first 6–12 months) and recommends mpMRI prior to the biopsy. A systematic biopsy should be included.**
    - ◊ **For most patients, the timing is generally within 6 to 24 months of diagnostic biopsy, but all patients on active surveillance should undergo a confirmatory prostate biopsy within 3 years of their diagnostic biopsy, irrespective of prior mpMRI findings.**
  - **Confirmatory testing should also include mpMRI with calculation of PSA density.**
  - **Advanced risk stratification tools lack high level evidence to obviate other testing at this time.**
  - **Other forms of imaging are discouraged.**

Note: All recommendations are category 2A unless otherwise indicated.

References on PROS-F 5 of 5

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PRINCIPLES OF ACTIVE SURVEILLANCE AND OBSERVATION

- **Active Surveillance Program**
  - Patients who choose active surveillance should have regular follow-up.
  - Key principles include:
    - PSA no more often than every 6 months, unless clinically indicated.
    - DRE no more often than every 12 months, unless clinically indicated.
    - Repeat prostate biopsy no more often than every 12 months, unless clinically indicated. While the intensity of surveillance may be tailored based on patient and tumor factors (eg, grade, tumor volume), most patients should have prostate biopsies every 1 to 3 years as part of their monitoring. Longer intervals between biopsies may be considered if maintained on active surveillance for >3 years without progression.
    - Consider repeat prostate MRI no more than every 12 months, unless clinically indicated.
    - In patients with a suspicious lesion on prostate MRI, MRI-targeted biopsy improves the detection of higher grade (GG ≥2) cancers.
    - Patients should be transitioned to observation when life expectancy is <10 years.
    - A metastatic staging evaluation (PSMA PET, bone scan, CT scan, or whole-body MRI) should not be performed unless patient develops unfavorable intermediate or higher-risk disease, or has concerning symptoms not explainable by other causes.

- **Considerations for Treatment of Patients on Active Surveillance**
  - Low risk: Grade reclassification on repeat biopsy to GG2 or higher is the most common factor to transition off active surveillance to treatment. PSA alone or growth in size of the tumor on MRI (assuming organ confined) should not be used alone to stop active surveillance without histologic confirmation of progression.
  - FIR: Grade reclassification, increase in percent pattern 4, increase in tumor volume on biopsy or imaging, or rise in PSA density are common factors to transition off active surveillance to treatment.
  - Patient anxiety is commonly cited as a reason to transition off active surveillance. However, studies indicate patient anxiety is closely linked with provider education on the safety of active surveillance. Thus, if a patient is an excellent candidate for active surveillance and has anxiety, the provider should provide appropriate education on the safety of active surveillance.

- **Advantages of Active Surveillance**
  - Approximately 50% of those eligible for active surveillance may safely avoid treatment for 10 years.[5-7] This varies by baseline risk.
  - Patients will avoid or delay possible side effects of treatment. Quality of life will be less affected while on active surveillance.
  - Risk of unnecessary treatment of small, indolent cancers will be reduced.

- **Limitations of Active Surveillance**
  - According to current data, the majority of patients on active surveillance receive treatment by 15 years post-diagnosis.
  - Although the risk is very low for patients diagnosed with GG1 disease (<1% at 10 years and 5% at 15 years), it is possible for cancer to progress to a regional or metastatic stage.[5-8]
  - The majority of high quality evidence with >10 years of follow-up primarily limited active surveillance to patients with low-risk disease, low-volume and low PSA density.
  - In early series, GG2 disease had higher rates of progression to metastatic disease.[8] However, there has been grade migration, stage migration with MRI, and refinement in active surveillance candidacy with assessment of percentage pattern 4 and unfavorable histologic variants. Long-term follow-up (10–15 years) of patients with GG2 FIR disease continues to develop.

Note: All recommendations are category 2A unless otherwise indicated.

References on PROS-F 5 of 5

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PRINCIPLES OF ACTIVE SURVEILLANCE AND OBSERVATION

OBSERVATION

- **Observation involves monitoring with a history and physical examination no more often than every 12 months (without surveillance biopsies, imaging, or PSA monitoring) until symptoms develop or are thought to be imminent.**
- **Low risk**
  - **Observation is recommended for asymptomatic patients with life expectancy <10 years.**
- **FIR**
  - **Observation is recommended for asymptomatic patients with a life expectancy ≤5 years.**
  - **Observation is preferred for asymptomatic patients with a life expectancy between 5 to 10 years.**
- **Observation may be considered for asymptomatic patients with high-risk, very-high-risk, regional, and metastatic prostate cancer and a life expectancy ≤5 years.**
- **If patients under observation become symptomatic, an assessment of disease burden can be performed, and treatment per the appropriate disease setting or palliation can be considered.**

- **Advantages of Observation**
  - **Patients will avoid possible side effects of unnecessary confirmatory testing and definitive therapy with a low probability that this would compromise their survival.**

- **Limitations of Observation**
  - **There may be risk of local or systemic symptoms (eg, urinary retention, pathologic fracture) that could have been avoided with upfront definitive treatment.**

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PRINCIPLES OF ACTIVE SURVEILLANCE AND OBSERVATION
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PRINCIPLES OF ANDROGEN DEPRIVATION THERAPY

Neoadjuvant, Concurrent, and/or Adjuvant ADT with RT[a]:
- **Clinically localized (N0, M0) disease**
- **Regional (N1, M0) disease**
- **Positive lymph nodes (pN1 disease) and/or adverse features post-RP**
- **First M0 RP PSA persistence/recurrence**

RT with one of the following ADT options:
- **Luteinizing hormone-releasing hormone (LHRH) agonist monotherapy**
  - Goserelin, leuprolide, or triptorelin
- **LHRH agonist (as above) plus first-generation antiandrogen**
  - Nilutamide, flutamide, or bicalutamide
- **LHRH antagonist**
  - Degarelix or relugolix

LHRH agonist or antagonist plus abiraterone[b] (2 years; only for very-high-risk localized disease or positive lymph nodes)

Notes:
- **ADT is not recommended with RT for most patients with favorable intermediate-risk prostate cancer. If it is given (see Principles of Risk Stratification), the duration should be short term (4–6 months).**
- **For unfavorable intermediate risk prostate cancer treated with RT, short-term ADT (ST-ADT) (4–6 months) is recommended. Concurrent/adjuvant ADT is preferred over neoadjuvant ADT in this setting.**
- **For high- and very-high-risk prostate cancer treated with EBRT alone, long-term ADT (LT-ADT) (18–36 months) is recommended.**
- **For high-risk prostate cancer treated with combination EBRT + brachytherapy, a shortened duration of ADT (12 months) can be considered.**
- **For additional details on the use of RT with ADT by risk group, see PROS-J.**
- **First M0 PSA persistence/recurrence:**
  - **The timing of secondary treatment for patients whose only evidence of cancer after definitive treatment is an increasing PSA is influenced by PSA velocity, patient anxiety, the short- and long-term side effects of ADT, and the underlying comorbidities of the patient.**
  - **Earlier treatment may be better than delayed treatment, although the definitions of early and late (what level of PSA) are controversial. Since the benefit of early treatment is not clear, treatment should be individualized until definitive studies are done. Patients with a shorter PSADT (or a rapid PSA velocity) and an otherwise long life expectancy should be encouraged to consider treatment earlier.**
  - **For patients treated with secondary RT in the setting of first RP recurrence, if ADT is given, it should be for a duration of 6 to 24 months.**
  - **Patients with prolonged PSADTs (>12 months) and who are older are candidates for observation or monitoring.**
  - **Patients who choose ADT monotherapy in the secondary treatment setting should consider intermittent ADT.**

[a] Specific recommendations, Categories of Evidence and Consensus, and Categories of Preference vary based on patient and disease characteristics (see PROS-5 through PROS-18). This chart delineates the forms of ADT that can be used and provides some additional details.

[b] The fine-particle formulation of abiraterone can be used instead of the standard form (category 2B; other recommended). Abiraterone should be given with concurrent steroid: prednisone 5 mg PO once daily (in the CSPC setting without docetaxel) or twice daily (in the CSPC setting with docetaxel and in the CRPC setting) with the standard formulation or methylprednisolone 4 mg PO twice daily with the fine-particle formulation. The standard formulation of abiraterone can be given at 250 mg/day following a low-fat breakfast in patients who will not take or cannot afford the standard dose of 1000 mg/day after an overnight fast.

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PRINCIPLES OF ANDROGEN DEPRIVATION THERAPY

ADT for Patients with M0 CSPC[a]:
- **Regional (N1, M0) disease**
- **First M0 RT recurrence**

ADT Options:
- **Orchiectomy**
- **LHRH agonist monotherapy**
  - Goserelin, leuprolide, or triptorelin
- **LHRH agonist (as above) plus first-generation antiandrogen**
  - Nilutamide, flutamide, or bicalutamide
- **LHRH antagonist**
  - Degarelix or relugolix
- **LHRH agonist, LHRH antagonist, or orchiectomy plus abiraterone[b] (only for positive lymph nodes)**

Notes:
- **M0 PSA persistence/recurrence:**
  - **The timing of secondary treatment for patients whose only evidence of cancer after definitive treatment is an increasing PSA is influenced by PSA velocity, patient anxiety, the short- and long-term side effects of ADT, and the underlying comorbidities of the patient.**
  - **Earlier treatment may be better than delayed treatment, although the definitions of early and late (what level of PSA) are controversial. Since the benefit of early treatment is not clear, treatment should be individualized until definitive studies are done. Patients with a shorter PSADT (or a rapid PSA velocity) and an otherwise long life expectancy should be encouraged to consider treatment earlier.**
  - **Patients with prolonged PSADTs (>12 months) and who are older are candidates for observation or monitoring.**
  - **Patients who choose ADT monotherapy in the secondary treatment setting should consider intermittent ADT.**

[a] Specific recommendations, Categories of Evidence and Consensus, and Categories of Preference vary based on patient and disease characteristics (see PROS-5 through PROS-18). This chart delineates the forms of ADT that can be used and provides some additional details.
[b] The fine-particle formulation of abiraterone can be used instead of the standard form (category 2B; other recommended). Abiraterone should be given with concurrent steroid: prednisone 5 mg PO once daily (in the CSPC setting without docetaxel) or twice daily (in the CSPC setting with docetaxel and in the CRPC setting) with the standard formulation or methylprednisolone 4 mg PO twice daily with the fine-particle formulation. The standard formulation of abiraterone can be given at 250 mg/day following a low-fat breakfast in patients who will not take or cannot afford the standard dose of 1000 mg/day after an overnight fast.

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PRINCIPLES OF ANDROGEN DEPRIVATION THERAPY

ADT for Patients with M0 CSPC BCR2[a]:

ADT Options[c]:
- **Orchiectomy**
- **LHRH agonist monotherapy**
  - Goserelin, leuprolide, or triptorelin
- **LHRH agonist (as above) plus first-generation antiandrogen**
  - Nilutamide, flutamide, or bicalutamide
- **LHRH antagonist**
  - Degarelix or relugolix

For High-Risk BCR2:
- Enzalutamide with or without Leuprolide[d]
- Apalutamide with LHRH agonist or LHRH antagonist[e]

Notes:
- **Monitoring until diagnosis of metastatic disease is preferred for patients with M0 CSPC in the low-risk BCR2 setting.**
- **PSADT and Grade Group should be considered when deciding whether to begin ADT for patients with M0 disease.**
- **ADT monotherapy is an option for these patients, and intermittent ADT can be considered to reduce toxicity.**

[a] Specific recommendations, Categories of Evidence and Consensus, and Categories of Preference vary based on patient and disease characteristics (see PROS-5 through PROS-18). This chart delineates the forms of ADT that can be used and provides some additional details.
[c] For patients receiving MDT with ADT, the ADT options are LHRH agonist or LHRH antagonist.
[d] Enzalutamide with or without leuprolide is an option for patients who have the following high-risk criteria: M0 by CT, MRI, or bone scan; PSADT ≤9 months; PSA ≥2 ng/mL above nadir after RT or ≥1 ng/mL after RP with or without postoperative RT; and not considered a candidate for pelvic-directed therapy.
[e] Apalutamide plus ADT is an option for patients with biochemical recurrence after RP who meet the following high-risk criteria: PSADT ≤ 9 months; PSA ≥0.5 ng/mL; and prior adjuvant or secondary RT or not considered a candidate for RT (Aggarwal R, et al. J Clin Oncol 2024;42:1114-1123).

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PRINCIPLES OF ANDROGEN DEPRIVATION THERAPY

ADT for Patients with mCSPC[a]:

ADT with treatment intensification (systemic therapy):
- Orchiectomy plus abiraterone,[b] apalutamide, darolutamide, or enzalutamide
- LHRH agonist plus abiraterone,[b] apalutamide, darolutamide, or enzalutamide
- LHRH antagonist plus abiraterone,[b] apalutamide, darolutamide, or enzalutamide
- Orchiectomy plus docetaxel and either abiraterone,[b] apalutamide, darolutamide, or enzalutamide
- LHRH agonist plus docetaxel and either abiraterone,[b] apalutamide, darolutamide, or enzalutamide
- LHRH antagonist plus docetaxel and either abiraterone,[b] apalutamide, darolutamide, or enzalutamide

ADT with treatment intensification (EBRT to the primary tumor):
EBRT with:
- Orchiectomy alone or with abiraterone,[b] apalutamide, docetaxel, or enzalutamide
- LHRH agonist alone or with abiraterone,[b] apalutamide, docetaxel,[f] or enzalutamide
- LHRH antagonist alone or with abiraterone,[b] apalutamide, docetaxel, or enzalutamide

Additional option for patients with *BRCA2*m mCSPC:
- Orchiectomy, LHRH agonist, or LHRH antagonist plus niraparib and abiraterone[g]

ADT alone for select patients[h]:
- **Orchiectomy**
- **LHRH agonist monotherapy[f]**
  - Goserelin, leuprolide, or triptorelin
- **LHRH agonist (as above) plus first-generation antiandrogen**
  - Nilutamide, flutamide, or bicalutamide
- **LHRH antagonist**
  - Degarelix or relugolix

[a] Specific recommendations, Categories of Evidence and Consensus, and Categories of Preference vary based on patient and disease characteristics (see PROS-5 through PROS-18). This chart delineates the forms of ADT that can be used and provides some additional details.

[b] The fine-particle formulation of abiraterone can be used instead of the standard form (category 2B; other recommended). Abiraterone should be given with concurrent steroid: prednisone 5 mg PO once daily (in the CSPC setting without docetaxel) or twice daily (in the CSPC setting with docetaxel and in the CRPC setting) with the standard formulation or methylprednisolone 4 mg PO twice daily with the fine-particle formulation. The standard formulation of abiraterone can be given at 250 mg/day following a low-fat breakfast in patients who will not take or cannot afford the standard dose of 1000 mg/day after an overnight fast.

[f] A first-generation antiandrogen must be given with LHRH agonist for ≥7 days to prevent testosterone flare if metastases are present in weight-bearing bone or if there is a large prostate mass.

[g] The fine-particle (category 2B; other recommended option) or standard formulation of abiraterone can be given with single-agent niraparib as a substitute for the combination niraparib/abiraterone tablet.

[h] ADT is strongly recommended in combination therapy for metastatic castration-sensitive disease. The use of ADT monotherapy in metastatic castration-sensitive disease is discouraged unless there are clear contraindications to combination therapy. If ADT monotherapy is given, intermittent ADT can be considered to reduce toxicity. Close monitoring of PSA and testosterone levels and possibly imaging is required when using intermittent ADT, especially during off-treatment periods, and patients may need to switch to continuous ADT upon signs of disease progression.

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PRINCIPLES OF ANDROGEN DEPRIVATION THERAPY

Secondary Hormone Therapy for Patients with M0 or M1 CRPC[a]:

Castrate levels of testosterone (<50 ng/dL) should be maintained while additional therapies are applied:
- Orchiectomy, LHRH agonist, or LHRH antagonist with specific therapies noted on PROS-16 [M0 CRPC], PROS-17 [NEPC], and PROS-18 [M1 CRPC]
- Orchiectomy, LHRH agonist, or LHRH antagonist with other secondary hormone options:
  - **Abiraterone[b] or enzalutamide following progression on ARPIs (M1 only)**
  - **Abiraterone[b] plus 0.5 mg/day dexamethasone following progression on either formulation of abiraterone (M1 only)**
  - **Antiandrogen withdrawal[i]**
  - **Corticosteroids** (hydrocortisone, prednisone, or dexamethasone)[i]
  - **First-generation anti-androgen** (nilutamide, flutamide, or bicalutamide)[i]
  - **Ketoconazole plus hydrocortisone[i]**

Notes:
- **Although the optimal sequence of therapies remains undefined, some data are emerging that can help with treatment selection in some cases. See Discussion.**

ADT Monotherapy[a]:

ADT Options:
- **Orchiectomy**
- **LHRH agonist monotherapy[f]**
  - Goserelin, leuprolide, or triptorelin
- **LHRH antagonist**
  - Degarelix or relugolix

Notes:
- **ADT monotherapy is appropriate for patients with life expectancy ≤5 years whose cancer progressed on observation of localized disease, who are symptomatic, or who have N1M0 disease.**
- **ADT monotherapy is also used for asymptomatic patients with regional disease and life expectancy ≤5 years whether or not RT is given.**

[a] Specific recommendations, Categories of Evidence and Consensus, and Categories of Preference vary based on patient and disease characteristics (see PROS-5 through PROS-18). This chart delineates the forms of ADT that can be used and provides some additional details.
[b] The fine-particle formulation of abiraterone can be used instead of the standard form (category 2B; other recommended). Abiraterone should be given with concurrent steroid: prednisone 5 mg PO once daily (in the CSPC setting without docetaxel) or twice daily (in the CSPC setting with docetaxel and in the CRPC setting) with the standard formulation or methylprednisolone 4 mg PO twice daily with the fine-particle formulation. The standard formulation of abiraterone can be given at 250 mg/day following a low-fat breakfast in patients who will not take or cannot afford the standard dose of 1000 mg/day after an overnight fast.
[f] A first-generation antiandrogen must be given with LHRH agonist for ≥7 days to prevent testosterone flare if metastases are present in weight-bearing bone or if there is a large prostate mass.
[i] In the mCRPC setting, this option should only be used for select patients who are not candidates for other recommended mCRPC therapies.

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PRINCIPLES OF ANDROGEN DEPRIVATION THERAPY

Optimal ADT:

- **Medical castration (ie, LHRH agonist or antagonist) and surgical castration (ie, bilateral orchiectomy) are equally effective.**
- **Patients who do not achieve adequate suppression of serum testosterone (<50 ng/dL) with medical or surgical castration can be considered for additional hormonal manipulations (with antiandrogens, LHRH antagonists, or steroids), although the clinical benefit remains uncertain. Consider monitoring testosterone levels 12 weeks after first dose of LHRH therapy, then upon increase in PSA. The optimal level of serum testosterone to affect "castration" has yet to be determined.**

Monitor/Surveillance:

- **ADT has a variety of adverse effects, including hot flashes, loss of libido, erectile dysfunction, shrinkage of penis and testicles, loss of muscle mass and strength, fatigue, anemia, breast enlargement and tenderness/soreness, depression and mood swings, hair loss, osteoporosis, greater incidence of clinical fractures, obesity, insulin resistance, alterations in lipids, and greater risk for diabetes and cardiovascular disease. The intensity and spectrum of these side effects vary greatly, and many are reversible or can be avoided or mitigated. See NCCN Guidelines for Survivorship. Patients and their medical providers should be advised about these risks prior to treatment.**

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PRINCIPLES OF RISK STRATIFICATION AND BIOMARKERS

General Principles:

- **Currently, the primary method for personalization of treatment from localized to advanced prostate cancer is based on prognostic risk stratification, rather than the use of predictive tools.**
- **NCCN uses multiple categories and subgroupings to capture prognostic risk to personalize treatment recommendations.**
- **The purpose of the NCCN categories and subgroupings are to provide a method for risk stratification to allow standardized treatment recommendations to be provided.**
  - **It is acknowledged that there are methods of risk stratification with superior prognostic performance to NCCN risk groups. However, they have not been routinely reported in clinical trials. This limits the ability to provide evidence-based guideline treatment recommendations using these methods. Thus, the NCCN Guidelines continue to use NCCN categories and subgroups of risk as a framework.**
  - **Clinical trials have established the benefit of various treatments in prostate cancer and have commonly enrolled patients across a spectrum of risk. Subgroup analyses, absolute benefit estimates, and expert opinion are used to provide treatment recommendations for each NCCN risk group or disease state.**
  - **There is intrinsic heterogeneity in prognosis within a given NCCN category and subgroup. Thus, treatment recommendations for adjacent subgroups or categories of risk may be appropriate when using additional risk stratification methods.**
  - **The Panel acknowledges the ability to personalize treatment decisions through additional tools and have created this section to assist.**
- **Tools that are prognostic or predictive in one disease state may not be in other disease states, or they may have other forms of clinical utility beyond prognostication and prediction of treatment benefit.**
  - **For example, germline homologous recombination deficiency (HRD) mutations do not have an established prognostic or predictive role in localized prostate cancer, but specific HRD mutations have been demonstrated to have a prognostic and predictive role in advanced disease. Additionally, the utility of germline testing extends to inform screening recommendations for other cancers and cascade germline testing for family members.**
- **Imaging is also a biomarker (ie, MRI, PSMA-PET/CT) and can aid in risk stratification. See Principles of Imaging (PROS-E).**

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PRINCIPLES OF RISK STRATIFICATION AND BIOMARKERS

Biomarker Categories:

- **Biomarkers and risk stratification methods are tools that may assist in personalization of treatment. For clarity these tools are separated by type and category:**

  - **Type:**
    - ◊ **Standard Tools: These include clinical and/or pathologic variables routinely collected to assign a patient to an NCCN category and/or subgroup. Examples include TNM stage, Grade Group, PSA, and metastatic volume of disease.**
    - ◊ **Clinical and Pathologic Tools: These include clinical and/or pathologic tools that are generally derived from standard tools. Examples include multivariable models or nomograms, histologic variants, and PSA kinetics.**
    - ◊ **Advanced Tools: These involve an additional test above what is collected to assign an NCCN category or subgroup. These may include, but are not limited to, germline or somatic tests, gene expression tests, digital histopathology-based tests, additional imaging, and circulating markers.**

  - **Category:**
    - ◊ **Prognostic: Discriminates the risk of developing an oncologic endpoint (eg, distant metastasis). The relative benefit of a treatment (ie, the treatment effect or hazard ratio) is generally similar across a prognostic spectrum, although the absolute benefit of an intervention may vary by risk (ie, number needed to treat [NNT]).**
      - **– Ideally, prognostic biomarkers independently discriminate and are associated with a clinically meaningful endpoint above and beyond standard tools relevant to that disease setting that ultimately helps guide a therapeutic decision.**
    - ◊ **Predictive: Discriminates a difference in the relative benefit of a specific treatment for an oncologic endpoint.**
      - **– Ideally, predictive biomarkers have been demonstrated to measure differential treatment effects that ultimately help guide a therapeutic decision in the context of a randomized trial, specifically randomizing the treatment of interest.**

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PRINCIPLES OF RISK STRATIFICATION AND BIOMARKERS

Clinical and Pathologic Tools:

- **An extensive number of prognostic clinical or pathologic tools have been reported based on highly variable evidence quality (retrospective or registry study vs. randomized trial), validation rigor, strength of endpoint (adverse pathology or BCR vs. distant metastasis), and univariable versus multivariable association with an outcome. Thus, while some of these tools may have value, these limitations hinder the ability to accurately provide guidance to specific treatment recommendations with confidence.**
- **A comprehensive list of these tools is outside the scope of this guideline.**
  - **Examples of such prognostic tools include multivariable models and nomograms (eg, CAPRA,[1] STAR-CAP,[2] MSKCC nomograms[3]), histopathology (ie, cribriform, intraductal carcinoma, absolute and percent Gleason pattern 4, total mm of cancer), and clinical variables (ie, PSA density, PSA velocity, PSA level, PSADT).**

Advanced Tools:

- **Advanced risk stratification tools generally offer either superior prognostic performance beyond clinical and pathologic tools AND/OR serve as predictive biomarkers for identifying patient groups that differentially benefit from a specific treatment. In general, these tools are only recommended when they have the potential ability to change management and should not be ordered reflexively.**
  - **Prognostic tools: Generally, the Panel recommends the use of prognostic tests that are validated in well-designed prospective studies with clinically meaningful endpoints based on disease settings that guide a specific treatment indication based on a specific score or result. These studies can be either prospective integral or integrated clinical trial(s) or post-hoc correlative analyses of prospective trials.**
  - **Predictive tools: Generally, the Panel recommends the use of predictive tests that are validated in a prospective, biomarker-directed, randomized clinical trial of the treatment of interest. Alternatively, if validation is post-hoc in a trial not designed to test the tool, it should be performed in more than one independent randomized trial of the treatment of interest.**
- **The Panel recognizes that there is an extensive number of advanced tools created with substantial variability in quality of reporting and model design, endpoint selection, and quality and caliber of validation. There are risks in using advanced tools to change management without robust validation, as they may drive patients or providers to inappropriate treatment options. If advanced tools are used, it is recommended to use tests that have robust validation, ideally with high-quality, long-term clinical trial data and across multiple clinical trials.**
- **Only advanced tools with high evidence quality are shown in Table 1. Other prognostic tools that are commonly used are described in the Discussion.**

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References

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PRINCIPLES OF RISK STRATIFICATION AND BIOMARKERS

Table 1. Advanced Prognostic Tools[a]

Intermediate-Risk Prostate Cancer		
Tool	**Category**	**Discussion**
22-gene genomic classifier (GC) (Decipher)[4-6]	Gene Expression	NRG/RTOG 0126 phase III randomized trial was profiled post-hoc with a prespecified analysis plan.[4] The study demonstrated the independent prognostic effect of GC on biochemical failure, secondary therapy, DM, PCSM, MFS, and OS. Patients receiving RT alone with low GC scores had 10-year DM rates of 4%, compared with 16% for GC high risk. These results suggest that the benefit of short-term ADT in NCCN intermediate-risk prostate cancer is likely to be smaller in patients with low GC scores (≤0.45) than in patients with high GC scores (≥0.60). A breakdown of outcomes between patients with favorable vs. unfavorable intermediate-risk disease was not provided.
Multimodal artificial intelligence (MMAI) (ArteraAI Prostate)[b,7-11]	AI-Pathology	A post-hoc meta-analysis of eight phase III randomized trials (NRG/RTOG 9202, 9408, 9413, 9902, 9910, 0126, 0415, and 0521) with MMAI scores have been reported.[7,8] There were a total of 630 patients with NCCN intermediate-risk disease who primarily received RT alone or RT+ST-ADT. Among patients with NCCN intermediate-risk disease, those with MMAI low-risk disease had 5-year and 10-year DM rates of 1% and 4%, respectively. In contrast, those with MMAI intermediate- and high-risk tumors had 5-and 10-year DM rates of 7% and 40%. MMAI was independently prognostic for DM, PCSM, and death after DM. These results suggest that the benefit of short-term ADT in NCCN intermediate-risk prostate cancer is likely to be smaller in patients with MMAI low-risk disease than in patients with MMAI intermediate- or high-risk disease. Neither a breakdown of favorable vs. unfavorable intermediate risk, nor a breakdown of outcomes after RT alone vs. RT+ST-ADT were provided.

DM = distant metastases; MFS = metastasis-free survival; PCSM = prostate cancer-specific mortality; ST-ADT = short-term ADT

[a] In the absence of prospective trials, caution is warranted if using these prognostic tools to influence treatment decisions. The Panel awaits future trials that confirm the initial results described here.

[b] There is also an MMAI predictive biomarker that was validated post-hoc in RTOG 9408 to predict benefit of ST-ADT added to RT in patients with intermediate-risk prostate cancer. While promising, due to differences in tissue sampling, grading, staging, and treatment, the Panel recommends further validation prior to using this predictive biomarker to guide treatment decisions in isolation.

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PRINCIPLES OF RISK STRATIFICATION AND BIOMARKERS

Table 1. Advanced Prognostic Tools[a]

Tool	Category	Discussion
High-Risk and Very-High-Risk Prostate Cancer		
Tool	**Category**	**Discussion**
22-gene GC[4-6]	Gene Expression	A meta-analysis of three phase III randomized trials (NRG/RTOG 9202, 9413, and 9902) that included patients with high-/very-high-risk prostate cancer were profiled post-hoc with a prespecified analysis plan.[5] The study demonstrated the independent prognostic effect of GC on biochemical failure, DM, MFS, PCSM, and OS. Patients with low GC scores had 10-year DM rates of 6%, compared with 26% for GC high risk. The absolute benefit of LT-ADT over ST-ADT was 11% for patients with high GC scores (NNT of 9), and 3% for patients with low GC scores (NNT of 33). These results suggest that the benefit of long-term ADT in NCCN high- and very-high-risk prostate cancer is likely to be smaller in patients with low GC scores (≤0.45) than in patients with high GC scores (≥0.60). A breakdown of outcomes between patients with high- vs. very high-risk disease was not provided.
MMAI[b,7-12]	AI-Pathology	Multiple post-hoc meta-analyses of patients with NCCN high- or very-high-risk disease from six randomized phase III randomized trials (NRG/RTOG 9202, 9408, 9413, 9902, 9910, and 0521) with MMAI scores have been reported.[8,10] A total of 1090 patients had NCCN high-/very-high-risk disease and were treated predominately with RT+LT-ADT (with or without chemotherapy; 68%) or RT+ST-ADT (30%), and a small number with RT alone (2%). Among patients with NCCN high-/very-high-risk disease, those with MMAI low-risk disease had 5-year rates of DM of 0% and 10-year rates of 3%. In contrast, those with MMAI intermediate- and high-risk tumors had 8% and 26% risk of metastasis at 10 years post-treatment, respectively. MMAI was independently prognostic for DM, PCSM, and death after DM. These results suggest that the benefit of ADT in NCCN high- and very-high-risk prostate cancer is likely to be smaller in patients with MMAI low-risk disease than in patients with MMAI intermediate- or high-risk disease. Neither a breakdown of high- vs. very high-risk, nor a breakdown of outcomes between cohorts receiving RT+ST-ADT vs. RT+LT-ADT, were provided. Published results from four phase III randomized trials from the STAMPEDE platform, which included 1575 patients with locally advanced, M0 disease (ie, NCCN very-high-risk disease), with post-hoc derivation of MMAI scores have been reported.[12] Overall, patients with locally advanced N0M0 disease had a 5% PCSM event rate, but when divided by MMAI quartile, there was a significant difference between the lower three quartiles (3%) vs. the highest quartile (11%). When restricting to patients receiving abiraterone, these rates were 2% vs. 5%. These results suggest that the benefit of abiraterone in NCCN very-high-risk prostate cancer is likely to be smaller in patients with MMAI low-risk disease than in patients with MMAI high-risk disease.
BCR Post-RP		
Tool	**Category**	**Discussion**
22-gene GC[13,14]	Gene Expression	Two phase III randomized trials post-RP were profiled post-hoc with prespecified analysis plans. NRG/RTOG 9601 demonstrated the independent prognostic effect of GC on DM, PCSM, and OS, and found that for patients with lower entry PSAs (<0.7 ng/mL), the 12-year DM rate benefit from hormone therapy for patients with GC lower risk vs. GC higher risk was 0.4% vs. 11.2%.[13] The SAKK 09/10 phase III trial tested post-RP lower vs. higher dose RT alone. The study demonstrated the independent prognostic effect of GC on biochemical progression, clinical progression, secondary hormone therapy, DM, and MFS.[c,14] These results suggest that the benefit of ADT added to RT for patients with RP recurrence planned for early secondary RT is likely to be smaller in those with low or intermediate GC scores (<0.6) than in those with high GC scores (≥0.60).

DM = distant metastases; LT-ADT = long-term ADT; MFS = metastasis-free survival; NNT = number needed to treat; PCSM = prostate cancer-specific mortality; ST-ADT = short-term ADT

[a] In the absence of prospective trials, caution is warranted if using these prognostic tools to influence treatment decisions. The Panel awaits future trials that confirm the initial results described here.

[b] There is also an MMAI predictive biomarker that was validated post-hoc in RTOG 9408 to predict benefit of ST-ADT added to RT in patients with intermediate-risk prostate cancer. While promising, due to differences in tissue sampling, grading, staging, and treatment, the Panel recommends further validation prior to using this predictive biomarker to guide treatment decisions in isolation.

[c] SAKK 09/10 combined GC low and intermediate risk due to relatively similar prognosis. NRG/RTOG 9601 dichotomized patients by GC low versus intermediate and high risk. However, due to the age of the tissue from NRG/RTOG 9601 (>20 years old), there is a known shifting of GC scores, and a more contemporary distribution of score distribution would approximate closer to combining GC low and intermediate risk together.

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PRINCIPLES OF FOCAL/SUBTOTAL THERAPY OR WHOLE GLAND ABLATIVE THERAPY

General

- **Ablative therapy refers to the use of a device to destroy tissue that can be directed at part of the prostate gland (termed partial gland, hemi-gland, or focal therapy [FT]) or the whole prostate gland.**
- **FT includes ablative therapies and non-ablative therapies, such as partial prostatectomy and focal radiation therapy.[a]**
- **Presently, FT is an experimental and emerging technology for the initial treatment of localized prostate cancer that lacks randomized controlled trial evidence with long-term follow-up showing its superiority or noninferiority to current recommended management strategies. As such, FT meets the criteria as an alternative therapy, or a non-standard treatment.**

Newly Diagnosed or Previously Untreated Prostate Cancer

- **Pathologic confirmation of prostate cancer and risk stratification is required prior to consideration of ablative or FT.**
- **Active surveillance is the preferred treatment for low-risk prostate cancer, and ablative and focal therapies are discouraged in this population.**
- **The Panel also believes that ablative FT should be discouraged in patients with high- or very-high-risk, regional, or metastatic prostate cancer outside of a clinical trial.**
- **There is currently insufficient comparative effectiveness evidence for FT to be recommended for patients with intermediate-risk prostate cancer, and the Panel believes there is uncertainty about the long-term efficacy and toxicity of these treatments. Therefore, ablative and focal therapies in this population should be utilized only in the context of a clinical trial.**
- **The following device categories are currently FDA approved or cleared for the initial treatment of prostate cancer, but randomized evidence to the superiority in long-term cancer control and/or quality of life are lacking when delivered as focal rather than whole gland therapy. These treatments may be considered as FT for patients with previously untreated, localized prostate cancer only in the context of a clinical trial:**
  - **External beam radiotherapy[a]**
  - **Brachytherapy**
  - **Cryotherapy ablation**
- **The following device categories are currently not FDA approved or cleared for the treatment of prostate cancer as focal or whole gland therapy, and should only be used for patients with previously untreated, localized prostate cancer in the context of a clinical trial:**
  - **High-intensity focused ultrasound (HIFU)**
  - **Transurethral ultrasound ablation (TULSA)**
  - **Focal laser ablation (FLA)**
  - **Radiofrequency ablation (RFA)**
  - **Irreversible electroporation (IRE)**
  - **Photodynamic therapy (PDT)**
  - **Histotripsy**
  - **Partial prostatectomy**

[a] Although ultra-hypofractionated EBRT, also known as stereotactic body radiotherapy (SBRT) or stereotactic ablative radiotherapy (SABR), is considered by some as an ablative technology, it does not fall within the general class of ablative therapies. EBRT is recommended in multiple contexts for the treatment of prostate cancer, and when delivered to the prostate it is recommended to be delivered to the whole prostate gland (discussed within Principles of Radiation Therapy (PROS-J). However, when delivered as FT it should be viewed within the context of other ablative therapies at this time given similar quality of evidence and risks of residual cancer and retreatment.

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PRINCIPLES OF FOCAL/SUBTOTAL THERAPY OR WHOLE GLAND ABLATIVE THERAPY

- **Patients should be counseled prior to receiving FT or whole gland ablative therapy as follows:**
  - **HIFU, TULSA, FLA, RFA, IRE, PDT, and histotripsy are not FDA approved or cleared for the treatment of prostate cancer.**
  - **Currently, no treatment delivered as FT, including radiotherapy or partial prostatectomy, have demonstrated superiority or non-inferiority to guideline-recommended therapies in randomized trials with long-term follow-up. The importance of long-term follow-up is of increased importance in this setting given the high rates of secondary therapy after FTs and potential for increased toxicity.**
  - **For patients with low- and intermediate-risk disease[1-7]:**
    - ◊ **Biopsy-detected (any) cancer after FT is common within 2 years post-treatment (20%–70%).**
    - ◊ **Erectile dysfunction (10%–40%) and urinary incontinence (0%–20%) have been reported to occur of variable severity within 2 years post-treatment.**
  - **FT or whole gland ablative therapy may impact the safety and/or efficacy of subsequent local therapy.**
  - **Moderate toxicity (grade 2) is common, and severe toxicity (grade 3 or higher) is possible with reported rates in trials from approximately 1%–14% in the first 2 years post-FT.[1-7]**
  - **Currently, there is no evidence-based, universally agreed upon follow-up protocol after FT. Patients often undergo serial imaging and prostate biopsies.**

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PRINCIPLES OF FOCAL/SUBTOTAL THERAPY OR WHOLE GLAND ABLATIVE THERAPY

Management of Recurrence After Focal Ablative or Whole Gland Ablative Therapy

- **Presently, the optimal management of residual or recurrent disease after receipt of these treatments is unknown and limited evidence exists. A multi-disciplinary discussion with experienced prostate cancer specialists is recommended.**
- **If there is suspicion of residual or recurrent prostate cancer, a biopsy should be performed to confirm the presence of residual or recurrent prostate cancer.**
  - **Histologic grading of the cancer is recommended if located in previously untreated areas of the prostate. Reliability of grading within a previously treated area of the prostate is unknown at this time, and patients should be informed of this uncertainty.**
- **Staging imaging should be performed based on the patient's NCCN risk group at time of recurrence (see PROS-2). However, PSA groupings as used for newly diagnosed prostate cancer may underestimate the burden of disease after prior FT or ablative therapy.**
  - **Interpretation of imaging (ie, assessment of cT3 disease by MRI near ablation zone) after FT or whole gland ablative therapy may be less reliable.[8]**
  - **Advanced risk stratification tools (eg, gene expression or digital pathology-based biomarkers) have not been adequately studied for use in the context of focal or whole gland ablative therapy and are not recommended to guide treatment decisions in this setting.**
- **Patients with a clinically significant recurrence after initial focal therapy should be offered radical prostatectomy or prostate RT if appropriate based on life expectancy.**
- **Patients should be counseled as follows:**
  - **There is limited evidence on how to optimally manage a FT or whole gland ablative therapy recurrence.**
  - **Patients are at an increased risk of toxicity from subsequent local therapy compared to the primary treatment setting.**
  - **It is not known if radical therapy has equivalent long-term success after FT or whole gland ablative therapy given lack of trials with long-term follow-up in this setting.**
  - **Secondary treatment should be reserved for patients with appropriate life expectancy and Grade Group 2 and higher recurrences detected on biopsy.**

Local Recurrence After Radiation Therapy

- **Patients with biopsy-proven recurrence in the prostate after prior RT and without distant metastatic disease can be considered for local secondary therapy.**
- **However, clinicians and patients should be aware that no local therapy currently has level 1 evidence for the management of a local recurrence after definitive RT.**
- **Treatments with higher quality of evidence are recommended. These are discussed in the Principles of Local Secondary Therapy Post-Radiation (PROS-L) and include select forms of FT or whole gland ablative therapy.**

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PRINCIPLES OF RADIATION THERAPY

All patients diagnosed with intermediate-, high-, very-high-, regional-risk, low-volume mCSPC, biochemically recurrent, low-volume metastatic castration-sensitive prostate cancer (mCSPC) and oligometastatic mCRPC are recommended to be evaluated by a radiation oncologist as part of a multidisciplinary discussion of treatment options.

General

- **EBRT**
  - **Treatment Planning: Intensity modulated RT (IMRT) is recommended over 3-dimensional conformal RT to improve dose conformality.**
  - **Image Guidance: Methods to improve accuracy of treatment delivery, thereby enabling reduction in planning target value (PTV) margins, which generally result in reduced toxicity are encouraged and may include one or more of the following:**
    - **Image guidance with daily 3D imaging is recommended with either a cone beam CT (CBCT) or MRI.**
    - **Devices that assist with optimal image guidance (fiducial markers) or reduce motion (endorectal balloons).**
    - **Real-time intrafraction volumetric tracking.**
    - **Online adaptive radiotherapy if target is near mobile organs (ie, performing a simultaneous integrated boost to a positive pelvic lymph node adjacent to bowel).**
  - **Beam Type: Photon and proton RT are acceptable forms of EBRT and appear to have similar outcomes in regard to toxicity, QOL, and tumor control. Potential financial toxicity should be discussed with patients.**
  - **Fractionation: An extensive list of fractionation schedules have been studied. These are grouped into three categories, conventional fractionation (1.8–2 Gy/fraction), moderate hypofractionation (>2.5 to 4 Gy/fraction), and ultra-hypofractionation (>6 Gy/fraction).[1] Common dose/fraction schedules are shown in Table 1. In general, iso-effective moderate hypofractionation dosing has demonstrated noninferior tumor control, toxicity, and QOL over conventional fractionation and is more convenient for patients and thus preferred. Similarly, ultra-hypofractionation has been demonstrated to be noninferior to both moderate hypofractionation and conventional fractionation.[2] Thus, use of conventionally fractionated radiotherapy is no longer preferred for localized prostate cancer.**
    - **Ultra-hypofractionation encompasses stereotactic body radiation therapy (SBRT), which requires precision treatment setup and required additional image guidance techniques.**
    - **SBRT is recommended and preferred specifically when:**
      - **Performing metastasis-directed radiotherapy (MDRT) (see Principles of MDT [PROS-M])**
      - **There is limited progression (eg, oligoprogression) or limited residual disease and the patient is on otherwise effective systemic therapy (eg, consolidation).**
      - **The lesion occurs in or immediately adjacent to a previously irradiated treatment field.**
      - **At physician discretion for more durable control of pain than achieved with typical palliative regimens.**

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PRINCIPLES OF RADIATION THERAPY

- **Brachytherapy**
  - **Patient Selection: Providers should consider aspects of gland size, baseline urinary symptoms, and prior procedures (ie, transurethral resection of prostate) that may increase risk of adverse effects. Neoadjuvant ADT to shrink a gland to allow treatment should be considered in context of the balance between its additional toxicity with this benefit.**
  - **Dose Rate: Both low-dose rate (LDR) and high-dose rate (HDR) are acceptable to be used as either monotherapy or as a boost combined with EBRT in higher risk disease. Data indicates that HDR may have lower toxicity than LDR, and while efficacy appears similar between HDR and LDR when used as a boost combined with EBRT, it remains unknown if HDR as monotherapy provides equivalent tumor control to LDR given lack of robust randomized comparisons. Post-implant dosimetry must be performed for LDR implants to verify dosimetry.**
  - **Monotherapy: Brachytherapy monotherapy has been shown to have fewer side effects than when used with EBRT without compromising tumor control for intermediate-risk disease.[3]**
  - **Combination Therapy: EBRT with an LDR brachytherapy boost has been shown to improve biochemical control over conventionally fractionated EBRT with ADT (12 months) without a focal/micro EBRT boost.[4] This was at the expense of an increase in high-grade toxicity.[5] Thus, careful patient selection and contemporary planning, use of recognized organ at risk (OAR) dose constraints, and use of high-quality ultrasound and other imaging are essential.**

- **Elective Nodal Irradiation (ENI)**
  - **ENI can be considered in select patients.**
  - **Use of IMRT with image guidance is recommended when performing ENI. Advanced techniques that include online adaptive radiotherapy may be considered to reduce ENI planning target volume margins.**
  - **If performing ENI, it is recommended to include the common, internal, and external iliac, pre-sacral, and obturator lymph node stations. Peri-rectal/mesorectal lymph nodes may be included if there is evidence of lymph node involvement in this nodal station. Para-aortic lymph nodes should not be included in patients who do not have evidence of para-aortic lymph node involvement.**

Definitive RT

- **Biocompatible and biodegradable perirectal spacers:**
  - **These devices may be implanted between the prostate and rectum in patients undergoing radiation therapy to the prostate in order to displace the rectum from high radiation dose regions for the purpose of toxicity reduction. Patients with grossly apparent true posterior extraprostatic extension should not undergo perirectal spacer implantation. NCCN high-risk disease without posterior extraprostatic extension is not a contraindication to spacer use.**
  - **These devices should be placed by practitioners who can maintain a moderate to high volume of use to ensure quality.**
  - **Additional caution should be used in the re-irradiation setting due to potential changes in tissue planes.**

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PRINCIPLES OF RADIATION THERAPY

Definitive RT (continued)

- **Low Risk**
  - **Patients with NCCN low-risk prostate cancer are encouraged to pursue active surveillance.[6]**
  - **Those electing treatment with RT may receive EBRT or brachytherapy as monotherapy.**
  - **The target should include the prostate and consideration of treatment of seminal vesicles (proximal).**
  - **Patients should not be treated with any of the following:**
    - **ENI**
    - **ADT**
    - **Combination brachytherapy boost with EBRT**

- **Favorable Intermediate Risk**
  - **The target should include the prostate and consideration of treatment of seminal vesicles (proximal or full).**
  - **Those electing treatment with RT may receive EBRT or brachytherapy as monotherapy.**
  - **Patients should not be treated with any of the following:**
    - **ENI**
    - **Combination brachytherapy boost with EBRT[3]**
  - **ADT is generally not recommended, but can be considered if additional risk assessments suggest aggressive tumor behavior or increased potential benefit from short-term ADT.[7,8]**

- **Unfavorable Intermediate Risk**
  - **The target should include the prostate and the seminal vesicles (proximal or full).**
  - **RT options include either EBRT, brachytherapy monotherapy, or EBRT with a brachytherapy boost.**
  - **Multiple randomized trials that included patients with unfavorable intermediate risk who did not demonstrate superiority of combination therapy over brachytherapy monotherapy.[3]**
  - **Focal boosting with isotoxic delivery can be considered if there is sufficient provider and practice expertise to delineate the dominant intraprostatic lesion (DIL) and OARs on MRI.[9]**
  - **ENI should not be used routinely.**
  - **ADT (level 1 data for ST-ADT) should be used unless additional risk assessments suggest less aggressive tumor behavior, or less predicted benefit, or if medically contraindicated.[8,10] Radiotherapy dose or modality has not clearly demonstrated the ability to obviate the benefit of ADT. ST-ADT is recommended to be used as a concurrent and adjuvant approach with RT rather than a neoadjuvant and concurrent approach.[11]**

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PRINCIPLES OF RADIATION THERAPY

Definitive RT (continued)

- **High Risk**
  - **The target should include the prostate and the seminal vesicles (full).**
  - **RT options include EBRT or brachytherapy boost combined with EBRT. Carefully selected patients may receive brachytherapy monotherapy.**
  - **Focal boosting with isotoxic delivery can be considered if there is sufficient provider and practice expertise to delineate the DIL and OARs on MRI.[9]**
  - **Clinical trials have demonstrated varied results regarding the benefit of ENI. Multiple randomized trials have not demonstrated improvement in outcomes from ENI. However, the benefit of ENI was shown in one trial of patients with high-risk prostate cancer staged primarily with PSMA-PET/CT and who had a risk of nodal involvement >20%. Currently, the use of ENI is at the discretion of the treating physician.**
  - **ADT (level 1 data for LT-ADT 12–36 months) is recommended for patients with life expectancy >5 years or who are symptomatic unless medically contraindicated. Radiotherapy dose or modality has not clearly demonstrated the ability to obviate the benefit of LT-ADT.[10]**

- **Very-High-Risk**
  - **The target should include the prostate and the seminal vesicles (full).**
  - **RT options include EBRT. Carefully selected patients may receive EBRT with a brachytherapy boost.**
  - **Focal boosting with isotoxic delivery can be considered if there is sufficient provider and practice expertise to delineate the DIL and OARs on MRI.**
  - **Combination brachytherapy boost with EBRT should not be used routinely. The primary randomized evidence to recommend combination brachytherapy boost with EBRT comes from the ASCENDE-RT trial, which excluded patients with PSA >40 ng/mL and T3b.[12]**
  - **ADT (level 1 data for LT-ADT 18–36 months) is recommended for patients with life expectancy >5 years or who are symptomatic unless medically contraindicated.**
  - **Currently, the use of ENI is at the discretion of the treating physician.**
  - **Addition of abiraterone should be used very selectively as the benefit in contemporary practice with modern staging is uncertain (see PROS-6 and PROS-G 1 of 5).[13] It is not recommended to be used routinely for patients with solely MRI defined T3a disease with low volume Gleason 8 disease. Studies validating the post-hoc subset analysis of STAMPEDE for the benefit of abiraterone in very high-risk prostate cancer using other ARPIs have not yet reported, and thus the benefit of using ARPIs with RT+LT-ADT remains uncertain in contemporary patients with very high risk N0M0 disease. Radiotherapy dose or modality has not clearly demonstrated the ability to obviate the benefit of LT-ADT.[10]**

- **Regional Disease**
  - **EBRT is recommended to include the prostate, seminal vesicles, and pelvic lymph nodes.**
  - **Simultaneous integrated boost to involved lymph nodes is recommended while respective adjacent OAR dose constraints.**
  - **Use of a brachytherapy boost is not recommended in these patients.**
  - **Addition of abiraterone is recommended (see PROS-7 and PROS-G [1 of 5]).[13]**

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PRINCIPLES OF RADIATION THERAPY

Postoperative RT:

- **Adjuvant RT: Adjuvant RT (aRT) involves the use of EBRT post-radical prostatectomy for patients with adverse pathologic features (ie, pT3 and/or positive surgical margins) in the setting of an undetectable postoperative PSA. This may result in some patients unnecessarily receiving aRT and experience potential additional side effects.**
  - **The early results and aggregate meta-analysis of three trials comparing aRT to early secondary RT (sRT) did not demonstrate a difference in biochemical recurrence for patients with generally a favorable prognosis. Thus, waiting for a detectable PSA and using an early sRT approach is recommended for most patients who undergo a RP rather than an immediate aRT approach to avoid over treatment and subsequent potential side effects.[14]**
  - **Use of aRT may be reasonable for patients with multiple adverse features and/or lymph node involvement.**
    - **Patients with multiple adverse features and/or lymph node involvement were underrepresented in these trials, and in the RADICALS-RT trial, the largest of the trials comparing aRT to early sRT, there was a significant reduction in prostate cancer-specific mortality with aRT and numerical trends in superior freedom-from-distant metastasis with aRT.[15] This raises uncertainty to the true noninferiority of early sRT in these patients with high risk of recurrence post-RP.**
  - **Multiple dose/fraction schedules have been used with aRT that include both conventionally fractionated (60–64 Gy in 30–36 fractions) and moderately hypofractionated regimens (52.5 Gy in 20 fractions).[16]**
- **Secondary RT is the use of EBRT post-RP when a patient has a detectable PSA; a lower limit is not defined and may be below 0.2 ng/mL.**
  - **Early sRT at a PSA of 0.1–0.2 ng/mL is recommended for most patients who experience BCR post-RP. The exception to this is in patients who have delayed recurrence many years post-RP with very slow PSA doubling times (ie, >24 months), especially when a patient has limited life expectancy.**
  - **Unlike in localized prostate cancer, the use of ADT in the post-RP setting has demonstrated survival improvements in only a small subset of patients, primarily in those with high pre-sRT PSAs receiving late sRT (PSA >0.7 ng/mL).[17] For patients receiving early sRT there is no clear OS benefit from the addition of ADT in any of the four published randomized trials, irrespective of ADT duration, or on meta-analysis.[18] ADT does significantly delay the incidence of biochemical recurrence.**
  - **Use of ADT with sRT should be personalized based on pre-RT PSA, clinicopathologic risk factors, patient age, life expectancy, comorbidities, and preferences, as well as advanced risk stratification tools (see PROS-H).[18,19]**
    - **Generally, ST-ADT is recommended to balance toxicity, QOL, and tumor control. However, LT-ADT may be preferred for patients receiving late sRT and/or those with multiple adverse pathologic features and/or lymph node involvement.**
    - **Treatment intensification with an ARPI has not demonstrated an improvement in metastasis-free survival or overall survival in this population in unselected patients.[20,21]**
  - **Multiple dose/fraction schedules have been used with sRT that include both conventionally fractionated (64–70.2 Gy in 32–39 fractions) and moderately hypofractionated regimens (52.5 Gy in 20 fractions or 62.5 Gy in 25 fractions).[16,22] The moderately hypofractionated regimens have demonstrated noninferior tumor control, toxicity, and QOL. Multiple single arm trials and early follow-up of phase II randomized trials have been reported using ultra-hypofractionated RT in the sRT setting, and generally are using 32.5–34 Gy in 5 fractions.[23,24] These studies often leverage either intrafraction motion monitoring, online adaptive radiotherapy, and strict image guidance with substantially modified contouring of target volumes to limit dose to OARs.**
  - **The addition of ENI to sRT plus ST-ADT has demonstrated improvements in biochemical control.[25] It appears the benefit is greater in patients with higher pre-sRT PSAs, as well as in patients who did not have their pelvic lymph nodes assessed, or minimally assessed, via a pelvic lymph node dissection. Use of PSMA PET/CT has not been shown to be sensitive enough to rule out the presence of pelvic lymph node micrometastases.**

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PRINCIPLES OF RADIATION THERAPY

RT in Advanced Disease:

- **Synchronous mCSPC**
  - **Treatment of the primary tumor:**
    - ◊ **Minimizing toxicity is paramount when delivering RT to the primary tumor in patients with metastatic disease. As such, it is unclear if dose escalation (ie, brachytherapy or focal boost) improves outcomes.**
    - ◊ **Low Volume: Treatment of the primary is recommended with EBRT.[26,27]**
      - **– Volume of disease should be defined according to CT and bone scan, if performed.**
      - **– PET imaging should not be used to exclude a patient from treatment of the primary tumor.**
      - **– The strongest data for a benefit of adding RT to the primary tumor are in patients receiving either ADT alone, ADT+ docetaxel, or ADT + abiraterone for those with ≤5 bony metastases.**
    - ◊ **High Volume: Treatment of the primary tumor with EBRT can be considered for select patients with high-volume mCSPC (category 2B).[27]**
      - **– RT to the primary tumor in high volume mCSPC has demonstrated improvements in time to mCRPC and a reduction in the development of severe genitourinary adverse events when given in the context of ADT with variable use of docetaxel and/or abiraterone.**
  - **Treatment of Metastases**
    - ◊ **MDT, which typically uses SBRT, may be used for patients who have a limited burden of lymph node or osseous disease. See Principles of MDT (PROS-M).**

- **Metachronous mCSPC**
  - **A greater level of evidence supports the role of MDT in the metachronous setting as compared to synchronous mCSPC. This commonly is delivered using SBRT, but moderately hypofractionated RT may be considered.**
  - **Use of MDT in this setting has been shown to delay the need for ADT compared to observation,[28] the addition of RT to ADT has been shown to prolong a treatment free eugonadal interval as compared to ADT alone,[29] and the addition of ADT to RT has improved progression-free survival (PFS).[30]**
  - **See Principles of MDT (PROS-M).**

  - **mCRPC**
    - ◊ **The use of MDT has been shown to prolong PFS and rPFS when used for patients with oligometastatic CRPC treated with ADT+ARPI as compared to systemic therapy alone. MDT may be considered when the goal is to prolong PFS and extend duration of systemic therapy prior to needing to switch therapies.[31] See Principles of MDT (PROS-M).**

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PRINCIPLES OF RADIATION THERAPY

Radiopharmaceutical Therapy

- **Radiopharmaceutical therapies for prostate cancer are suitable options for improving survival and/or rPFS in select patients with advanced castration-resistant disease. Due to prior therapy exposure, specific targets, and hematologic effects of these therapies, careful selection and sequencing strategy with other therapies is important. This section discusses the two currently FDA-approved agents in use (Ra-223, Lu-177–PSMA-617).**

- **Radium-223**
  - **Radium-223 has been shown to extend survival in patients who have CRPC with symptomatic bone metastases, but no visceral metastases.[32] Radium-223 alone has not been shown to extend survival in patients with visceral metastases or bulky nodal disease (>3 to 4 cm).**
  - **Radium-223 is administered IV once a month for 6 months by an appropriately licensed facility, usually in nuclear medicine or radiation oncology facilities.**
  - **Concurrent use with enzalutamide and bone health agents may be considered given the overall survival benefit over enzalutamide alone.[33] Concurrent use with other systemic therapies should be pursued with caution.**
  - **Bone fracture risk: Radium-223 may increase fracture risk when given concomitantly with an ARPI without use of a bone health agent. Concomitant use of denosumab or zoledronic acid is recommended.[34]**

- **Lu-177–PSMA-617[16]**
  - **In patients with PSMA-positive disease, Lu-177–PSMA-617 has been shown to improve OS in patients with progressive mCRPC previously treated with androgen receptor inhibitors and taxane chemotherapy.[35] It has also been shown to improve rPFS in patients who have not received taxanes with PSMA-positive mCRPC who were previously treated with an androgen receptor inhibitor compared with changing to a different androgen receptor inhibitor.**
  - **This agent is approved for patients with mCRPC who have been treated with ARPI therapy and have received prior taxane-based chemotherapy or are considered appropriate to delay receipt of chemotherapy.[36]**
  - **Lu-177–PSMA-617 is not recommended in patients with dominant PSMA-negative lesions. PSMA-negative lesions are defined as metastatic disease that lack PSMA uptake including bone with soft tissue components ≥1.0 cm, lymph nodes ≥2.5 cm in short axis, and solid organ metastases ≥1.0 cm in size.**
  - **Use of PSMA PET/CT SUVmean and use of an FDG PET/CT scan may aid in identifying patients with lower PSMA expression and non-PSMA avid disease who will derive less or potentially no benefit from 177-Lu-PSMA-617.**
  - **Lu-177–PSMA-617 is typically administered IV 200 mCi (7.4 GBq) every 6 weeks for up to 6 treatments by an appropriately licensed facility.**
  - **Use of post-treatment SPECT or PET/CT imaging can be considered to monitor PSMA expression for adaptive dosing.**
  - **Because Lu-177 also emits gamma radiation, appropriate precautions should be taken to minimize exposure to personnel administering the radiopharmaceutical. Treatment rooms should be monitored for potential contamination following treatments, and patients should be provided written instructions regarding radiation safety precautions following treatment.**
  - **For information regarding the use of PSMA-PET to define PSMA-positive disease, see Principles of Imaging (PROS-E).**

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PRINCIPLES OF RADIATION THERAPY

Table 1: Below are examples of regimens that have shown acceptable efficacy and toxicity. The optimal regimen for an individual patient warrants evaluation of comorbid conditions, voiding symptoms, and toxicity of therapy. Additional fractionation schemes may be used as long as sound oncologic principles and appropriate estimate of BED are considered.

See treatment pages and Principles of ADT (PROS-G) for other recommendations, including recommendations for neoadjuvant/concomitant/adjuvant ADT.

EBRT Regimen	Preferred Dose/Fraction	Definitive RT						Post-Treatment RT			Advanced Disease	
								Post-RP		Post-RT	Primary Tumor	Metastases
		Low	FIR	UIR	High	Very-High	Regional	aRT	sRT	sRT	mCSPC M0 CRPC mCRPC	MDT
Conventional	1.8–2 Gy x 37–45 fx			☼	☼	✓	✓				☼	
	1.8–2 Gy x 30–39 fx							✓	✓		☼	
Moderate Hypofractionation	3 Gy x 20 fx (preferred)[a] 2.7 Gy x 26 fx 2.5 Gy x 28 fx	☼	✓	✓	✓	✓	✓			☼	☼	☼
	2.63–2.75 Gy x 20 fx 2.5 Gy x 25 fx							✓	✓	☼	✓	☼
Ultra Hypofractionation (SBRT)	9.5 Gy x 4 fx 7.25–8 Gy x 5 fx 6 Gy x 6 fx 6.1 Gy x 7 fx	☼	✓	✓	✓	☼	☼		☼	✓	✓	✓
	9–10 Gy x 3 fx 12 Gy x 2 fx 16–24 Gy x 1 fx											✓
	6.2–6.4 Gy x 5 fx								☼			
EBRT Boost Techniques												
EBRT with simultaneous integrated boost	See footnote b.		☼	✓	✓	☼	☼		☼	☼	☼	
EBRT with sequential SBRT boost	*Prostate:* 1.8 Gy x 23–28 fx *Boost:* 6 Gy x 3 fx 9.5 Gy x 2 fx			☼	☼	☼						

(✓ Preferred; ☼ Acceptable based on clinical and medical need; Regimens shaded gray are not recommended)

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PRINCIPLES OF RADIATION THERAPY
FOOTNOTES

[a] Based on the HYDRA individual patient meta-analysis of seven phase III randomized trials, isodose moderately hypofractionated radiotherapy (MHFRT) and dose-escalated MHFRT both have similar efficacy compared with conventionally fractionated radiotherapy (CFRT), but dose-escalated MHFRT was associated with higher toxicity. Isodose regimens (eg, 60 Gy in 20 fractions) are preferred for MHFRT for localized prostate cancer. Alternative regimens (eg, 70 Gy in 28 fractions) may be used, but patients should be counseled on potentially increased risk of toxicity.[1]

[b] EBRT to whole prostate 2.2 Gy x 35 fx plus micro-boost to MRI-dominant lesion to ≤95 Gy (fractions ≤2.7 Gy).[9] The micro-boost technique with level 1 data was established for a modestly hypofractionated regimen but has been extrapolated reasonably to other regimens in ongoing clinical trials. Care must be taken in doing so outside of clinical trials in order to respect normal tissue toxicity risk and above all in prioritizing normal organ tolerances over micro-boost coverage.

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PRINCIPLES OF RADIATION THERAPY

Table 1: Below are examples of regimens that have shown acceptable efficacy and toxicity. The optimal regimen for an individual patient warrants evaluation of comorbid conditions, voiding symptoms, and toxicity of therapy. Additional fractionation schemes may be used as long as sound oncologic principles and appropriate estimate of BED are considered.

See treatment pages and Principles of ADT (PROS-G) for other recommendations, including recommendations for neoadjuvant/concomitant/adjuvant ADT.

Brachytherapy	Preferred Dose/Fraction	Definitive RT						Post-Treatment RT			Advanced Disease	
								Post-RP		Post-RT	Primary	Metastases
		Low	FIR	UIR	High	Very-High	Regional	aRT	sRT	sRT	mCSPC M0 CRPC mCRPC	MDT
Monotherapy												
Iodine 125	145 Gy	☼	✓	✓	☼					✓		
Palladium 103	125 Gy											
Cesium 131	115 Gy											
Iridium 192	13.5 Gy x 2 implants 9.5 Gy BID x 2 implants											
EBRT + Brachytherapy Boost (combined with 45-50.4 Gy in 25-28 fx or 37.5 Gy in 15 fx)												
Iodine 125	110–115 Gy			☼	✓	☼						
Palladium 103	90–100 Gy											
Cesium 131	85 Gy											
Iridium 192	15 Gy x 1 fx 10.75 Gy x 2 fx											

(✓ Preferred; ☼ Acceptable based on clinical and medical need; Regimens shaded gray are not recommended)

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PRINCIPLES OF SURGERY

Pelvic Lymph Node Dissection

- For patients undergoing RP:
  - Pelvic lymph node dissection (PLND) can be considered in patients with favorable intermediate-risk prostate cancer.
  - PLND is recommended in patients with unfavorable intermediate, high, very-high-risk, and regional prostate cancer.
  - A PLND can be excluded in patients with low predicated probability of nodal metastases by nomograms, although some patients with lymph node metastases will be missed. There is no single evidence-based threshold for performing PLND. Based on the risk of complications with PLND and extra time to perform the procedure, the published thresholds range from 2% to 7%.[1-4]
  - A patient who is above the threshold for performing a PLND, but has a negative PSMA PET scan should still undergo PLND. In two studies, the sensitivity of PSMA PET for pelvic lymph node involvement among patients undergoing RP and PLND was low (about 40%), and the negative predictive value was about 81%.[5,6] Thus, basing the decision to perform PLND on a negative PSMA PET scan could result in missing 19% of patients with positive lymph nodes.
  - PLND can be performed using an open, laparoscopic, or robotic technique and can provide staging and prognostic information.
- Extended PLND provides more complete staging and may cure some patients with microscopic metastases; therefore, an extended PLND is recommended when PLND is performed.
- An extended PLND includes removal of all node-bearing tissue from an area bound by the external iliac vein anteriorly, the pelvic sidewall laterally, the bladder wall medially, the floor of the pelvis posteriorly, Cooper's ligament distally, and the internal iliac artery proximally.

Radical Prostatectomy

- RP is an appropriate therapy for any patient not on an active surveillance program with clinically localized prostate cancer that can be completely excised surgically, who has a life expectancy of ≥10 years, and who has no serious comorbid conditions that would contraindicate an elective operation.
- High-volume surgeons in high-volume centers generally provide better outcomes.
- Blood loss can be substantial with RP, but can be reduced by using laparoscopic or robotic assistance or by careful control of the dorsal vein complex and periprostatic vessels when performed as open surgery.
- Urinary incontinence can be reduced by preservation of urethral length beyond the apex of the prostate and avoiding damage to the distal sphincter mechanism. Bladder neck preservation may decrease the risk of incontinence. Anastomotic strictures increase the risk of long-term incontinence.
- Recovery of erectile function is directly related to age at RP, preoperative erectile function, and the degree of preservation of the cavernous nerves. Replacement of resected nerves with nerve grafts has not been shown to be beneficial. Early restoration of erections may improve late recovery.

Secondary Radical Prostatectomy

- Secondary RP is an option for highly selected patients with local recurrence after external beam RT (EBRT), brachytherapy, or cryotherapy in the absence of metastases, but the morbidity (ie, incontinence, loss of erection, anastomotic stricture) is high and the operation should be performed by surgeons who are experienced with secondary RP.

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PRINCIPLES OF LOCAL SECONDARY THERAPY POST-RADIATION

Local Secondary Therapy for Recurrent Prostate Cancer After Definitive Radiotherapy:

- **Patients with biopsy-proven recurrence in the prostate after prior RT and without distant metastatic disease can be considered for local therapy. Monitoring is also an option (see PROS-10). ADT may also be included, but it should not be reflexively ordered.**
- **The Panel recommends that patients receive multidisciplinary counseling about the risks and benefits of each of these options in the context of the available comparative literature on this topic.[1,2]**
- **Local therapy options for patients with recurrence in the prostate include[a,b,c]:**
  - **Cryotherapy**
  - **High-intensity focused ultrasound (HIFU)**
  - **Irreversible electroporation (IRE) (category 2B)**
  - **Reirradiation**
  - **RP + PLND**
- **Reirradiation options include LDR brachytherapy, HDR brachytherapy, and SBRT.[1-7]**
- **There is no consensus as to the most appropriate reirradiation volume, and there are published experiences for both focal/partial and whole gland reirradiation. The Panel recommends that patients receiving local therapy for RT recurrence are treated within the context of clinical trials when available and/or at experienced centers.**

[a] Reirradiation with LDR brachytherapy, HDR brachytherapy, and SBRT is supported by phase II trials with >36 months of median follow-up and should be strongly considered as an option in patients with >2 years interval from prior radiation who do not have ongoing moderate or higher grade radiation-associated toxicities. Phase II data with >36 months of median follow-up are available, albeit to a more limited scope, for secondary RP, cryotherapy, and HIFU.
[b] Currently, all phase II and/or prospectively accrued registries reporting oncologic outcomes with >36 months of follow-up have pursued whole gland treatments. Focal therapy for prostate-only recurrences is the subject of active investigation. See Principles of Focal/Subtotal Therapy or Whole Gland Ablative Therapy (PROS-I).
[c] There are no prospective data to guide the use and duration of ADT with reirradiation for prostate-only recurrences. The Panel recommends extrapolation from the definitive treatment setting to suggest that adding ADT or prolonging it in this setting likely offers an improvement in biochemical control. See Principles of Radiation Therapy (PROS-J).

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PRINCIPLES OF METASTASIS-DIRECTED THERAPY (MDT)

General

- **MDT is the treatment of metastatic sites with a local therapy to improve oncologic outcomes, not simply to provide symptom palliation. MDT has been studied primarily with radiotherapy and with highly selected use of surgical resection in the form of lymph node dissections. MDT is not recommended to be performed with other local therapies.**
- **MDT is mostly commonly delivered in the form of MDRT, which specifically refers to the use of higher than palliative dose radiotherapy to provide durable local control of the areas targeted. This may be for the intended purpose of delaying the initiation of systemic therapy, improving PFS, radiographic PFS, or overall survival. Similar to systemic therapies in the polymetastatic setting, MDRT is uncommonly curative and should be viewed as a form of cytotoxic therapy, and it may need to be repeated.**

Disease State

- **The settings for which MDT may have utility with variable evidence quality are shown in Table 1:**

Table 1.

Treatment Regimen	mCSPC		mCRPC	
	Synchronous (de novo) Oligometastatic	Metachronous Oligorecurrent[a]	Oligometastatic	Oligoprogressive
MDT	☼	✓	✓	☼

(✓ Preferred; ☼ Acceptable based on clinical and medical need)

[a] Limited data suggest that lymphadenectomy may be beneficial for select patients with pelvic nodal recurrence after radical prostatectomy if they do not have evidence of distant metastases.

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PRINCIPLES OF METASTASIS-DIRECTED THERAPY (MDT)

Number of Metastatic Sites

- **The number of metastatic sites to define oligometastatic disease remains an evolving space and is impacted by sensitivity of imaging used. Early studies limited metastatic sites to 1–3 or 1–5 by CT, MRI, or bone scan.[1-4] More recent studies have included up to 10 metastatic sites by PSMA-PET imaging (NCT04787744, NCT06150417, NCT03721341). The upper limit is not clearly established and is both a function of oncologic limitations and technical limitations of treating numerous metastatic sites.**
  - **If CT, MRI, or bone scan is used, >5 metastasis is generally an exclusion for MDT.**
  - **If PSMA-PET is used, >10 metastatic sites is generally an exclusion for MDT.**
- **Lymph nodes in the same drainage chain (eg, two adjacent para-aortic lymph nodes) are to be considered 1 metastatic site and would be treated together with one radiation therapy isocenter or plan.**
- **Discrete, non-contiguous bony metastases should be counted separately even if treated with one isocenter (eg, two discrete femur metastases).**
- **The goal of MDT is generally to treat all metastatic sites, which may include regional lymph nodes and potentially treatment of the primary tumor if untreated or evidence of local recurrence. The primary tumor in this setting should be counted as a site.**

Location of Metastatic Sites

- **The location of metastatic sites have generally consisted of lymph node and osseous metastatic sites, but some studies include lung and liver metastases. Brain metastases are generally not included in studies of MDRT in prostate cancer. MDT should not be used to delay systemic therapy when there is visceral organ involvement.**
- **Pelvic lymph nodes are also considered metastatic sites in the context of MDRT for metachronous mCSPC or mCRPC.**

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PRINCIPLES OF METASTASIS-DIRECTED THERAPY (MDT)

Radiotherapy Details (Also see Principles of Radiation Therapy)

- **Use of SBRT (also known in this setting as stereotactic ablative radiotherapy [SABR]) is most commonly recommended for MDRT. Established dose/fraction regimens should be used with strict OAR constraints. Daily 3D image guidance is required and motion management is recommended for thoracic and abdominal sites if relevant. Online adaptive radiotherapy, if available, may be preferred if treatment of a metastatic site is near an OAR with frequent day-to-day setup uncertainty. Appropriate clinical target volume (CTV) and PTV margins should be applied to account for microscopic spread and setup uncertainty.**
- **When treating a lymph node metastasis, one should generally include an elective nodal volume to the involved chain with a simultaneous integrated boost to the radiographic or biopsy confirmed lymph node. If lymph node(s) are in the pelvis, elective nodal irradiation fields should be considered, either ipsilateral or bilateral.**
- **Dose composites with special medical physics consults are required to account for prior radiotherapy or overlap of multiple treatment plans.**
- **If treatment involves more than two treatment plans simultaneously, consideration of splitting the treatment course into multiple sessions of SBRT may be required due to patient tolerance and safety (eg, 4 treatment plans for 4 metastatic sites may need to be delivered as 2 concurrent treatment plans as course 1, and then the other 2 treatment plans are delivered concurrently at a later date as a separate course of SBRT). Repeat simulation may be required based on potential for anatomy changes or use of alternative immobilization for other treatments sites.**

Concurrent Systemic Therapy with MDRT

- **Synchronous (de novo) oligometastatic mCSPC:**
  - **Systemic therapy, most commonly ADT+ARPI, and treatment of the primary tumor with RT is recommended for most patients. The concurrent addition of MDRT may be considered in select patients. For patient tolerance, it is reasonable to asynchronously treat the primary and metastatic sites in separate treatment courses.**
- **Metachronous oligorecurrent mCSPC:**
  - **Oligorecurrent disease detected primarily by PET imaging may be treated by MDRT alone or concurrently with short-term ADT (6 months). The addition of ADT improved clinical and biochemical PFS, but may simply delay the detection of recurrence rather than improve durable control of the disease.[1]**
  - **Trials have not been reported showing a benefit of the addition of triplet chemotherapy or an ARPI in the setting of MDRT+ADT.**
- **Oligometastatic mCRPC:**
  - **First-line treatment with ADT and an ARPI ± MDRT have demonstrated improvements in PFS and radiographic PFS from the addition of MDRT.[4] MDRT should not be used without recommended mCRPC systemic therapies.**
- **Oligoprogressive or oligopersistent mCRPC**
  - **For oligoprogressive or oligopersistent mCRPC, MDRT may be delivered concurrently with mCRPC systemic therapy.**
  - **MDRT should not be used concurrently with PARP inhibition (alone or in combination with abiraterone or enzalutamide) or chemotherapy outside of a clinical trial.**
  - **Use of MDRT may be considered for oligoprogression sequentially after approved systemic mCRPC treatments.**

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PRINCIPLES OF NON-HORMONAL SYSTEMIC THERAPY

- **An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.**

Non-Hormonal Systemic Therapy for M1 Castration-Sensitive Prostate Cancer

- **Triplet therapy with ADT, certain ARPIs, and docetaxel is an option for certain patients with mCSPC who are fit for chemotherapy based on phase 3 studies:**
  - **ADT with docetaxel and abiraterone was compared to ADT alone or with docetaxel in an open-label, randomized, phase 3 study. Radiographic PFS was longer in patients who received abiraterone than in those who did not. The populations receiving the triplet and doublet therapies experienced similar rates of neutropenia, febrile neutropenia, fatigue, and neuropathy, although grade ≥3 adverse events occurred in 63% of patients who received the triplet combination compared with 52% of those receiving ADT and docetaxel.**
  - **ADT with docetaxel and darolutamide was compared with ADT with docetaxel and placebo in a randomized phase 3 trial. OS, time to CRPC, skeletal event-free survival, and time to initiation of subsequent systemic antineoplastic therapy were improved in the patients who received darolutamide. Adverse events of any grade, grade 3–5 adverse events, and serious adverse events occurred at similar incidence levels between the two arms. Many of these were known effects of docetaxel. Exceptions were rash (16.6% vs. 13.5%) and hypertension (13.7% vs. 9.2%), which are known effects of androgen receptor pathway inhibitors and were more frequent in the darolutamide group.**
  - **An open-label, randomized, phase 3 trial compared ADT with enzalutamide to ADT with a first-generation antiandrogen in this setting. Concurrent docetaxel was allowed and used for stratification. OS was improved with the use of enzalutamide over first-generation anti-androgen regardless of the addition of docetaxel. The most common grade ≥3 adverse events were febrile neutropenia associated with docetaxel use (6% in both groups), fatigue (1% in the control group vs. 6% in the enzalutamide group), and hypertension (6% vs. 10%). The incidence of grade 1–3 memory impairment was 4% versus 13%.**
- **The use of myeloid growth factors should follow the NCCN Guidelines for Hematopoietic Growth Factors, based on risk of neutropenic fever.**
- **An additional option for patients with BRCA2-mutated mCSPC is the use of ADT with niraparib and abiraterone.**
  - **ADT with niraparib and abiraterone was compared to ADT plus abiraterone doublet therapy in a randomimzed phase 3 trial. Most patients had high-volume and/or synchronous disease. rPFS was improved in the BRCAm population; OS data are immature. Grade 3/4 adverse events were more common in the niraparib arm (75% vs. 59%) as were treatment-emergent adverse events leading to death (14 participants vs. 7 participants). The most common grade 3/4 adverse events were anemia (29.1% in the niraparib group versus 4.6% in the control group) and hypertension (26.5% versus 18.4%).**
  - **Niraparib/abiraterone can be used following the completion of docetaxel in patients receiving triplet therapy or following EBRT to the primary tumor.**
  - **The use of this combination in patients with low-volume disease is controversial because 78% of the patients in the trial had high-volume disease and there were very few patients with low-volume metachronous disease in particular. This regimen adds considerable toxicity, and patients with low-volume disease should be informed that the benefit in this setting is not clear.**
  - **The fine-particle (category 2B; other recommended option) or standard formulation of abiraterone can be given with single-agent niraparib as a substitute for the combination niraparib/abiraterone tablet.**

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PRINCIPLES OF NON-HORMONAL SYSTEMIC THERAPY

Non-Hormonal Systemic Therapy for M1 CRPC

- ***Chemotherapy***
  - **Docetaxel with concurrent steroid**
    - ◊ **Concurrent steroid includes daily prednisone, which may be omitted on the day of chemotherapy administration when dexamethasone is given.**
    - ◊ **Every–3-week docetaxel with concurrent steroid is the preferred first-line chemotherapy treatment based on phase 3 clinical trial data for patients with symptomatic mCRPC. Adverse events associated with docetaxel include neutropenia, leukopenia, febrile neutropenia, neutropenic infections, fluid retention, hypersensitivity reaction, hepatic function impairment, neuropathy, and other low-grade adverse events (eg, fatigue, nausea, vomiting, alopecia, diarrhea).**
    - ◊ **Only regimens utilizing docetaxel on an every–3-week schedule demonstrated beneficial impact on survival. The duration of therapy should be based on the assessment of benefit and toxicities. In the pivotal trials establishing survival advantage of docetaxel-based chemotherapy, patients received up to 10 cycles of treatment if no progression and no prohibitive toxicities were noted.**
    - ◊ **Docetaxel retreatment can be attempted after progression on an ARPI and docetaxel in patients with mCRPC whose cancer has not demonstrated definitive evidence of progression on prior docetaxel therapy in either the castration-sensitive setting or the mCRPC setting.**
    - ◊ **Rare patients may have exposure to docetaxel without prior ARPI (ie, in the setting of EBRT to the primary tumor for low-volume synchronous mCSPC). Docetaxel rechallenge is not recommended for such patients in the pre-ARPI mCRPC setting.**
  - **Cabazitaxel with concurrent steroid**
    - ◊ **Concurrent steroid includes daily prednisone, which may be omitted on the day of chemotherapy administration when dexamethasone is given.**
    - ◊ **Patients who are not candidates for docetaxel or who are intolerant of docetaxel should be considered for cabazitaxel with concurrent steroid, based on results that suggest clinical activity of cabazitaxel in mCRPC. Cabazitaxel was associated with lower rates of peripheral neuropathy than docetaxel, particularly at 20 mg/m² (12% vs. 25%) and may be appropriate in patients with pre-existing mild peripheral neuropathy. Current data do not support greater efficacy of cabazitaxel over docetaxel.**
    - ◊ **Cabazitaxel at 25 mg/m² with concurrent steroid has been shown in a randomized phase 3 study (TROPIC) to prolong OS, PFS, PSA response, and radiologic response when compared with mitoxantrone and prednisone and is FDA approved in the post-docetaxel second-line setting. Toxicity at this dose was significant and included febrile neutropenia, severe diarrhea, fatigue, nausea/vomiting, anemia, thrombocytopenia, sepsis, and renal failure. A recent trial, PROSELICA, compared cabazitaxel 25 mg/m² every 3 weeks to 20 mg/m² every 3 weeks. Cabazitaxel 20 mg/m² had less toxicity; febrile neutropenia, diarrhea, and fatigue were less frequent. Cabazitaxel at 20 mg/m² had a significantly lower PSA response rate but non-significantly lower radiographic response rate and non-significantly shorter PFS and OS (13.4 vs. 14.5 months) compared to 25 mg/m².**
    - ◊ **Cabazitaxel at 25 mg/m² with concurrent steroid improved radiographic PFS and reduced the risk of death compared with abiraterone or enzalutamide in patients with prior docetaxel treatment for mCRPC in the CARD study.**
    - ◊ **Biweekly cabazitaxel at 16 mg/m² with prophylactic G-CSF significantly reduced the risk of neutropenia/neutropenic complications compared to the 25 mg/m² dose with G-CSF in patients ≥65 years in a phase 3 trial. Clinical outcomes were comparable between the 2 groups.**
    - ◊ **No chemotherapy regimen to date has demonstrated improved survival or quality of life after cabazitaxel, and trial participation should be encouraged.**

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PRINCIPLES OF NON-HORMONAL SYSTEMIC THERAPY

Non-Hormonal Systemic Therapy for M1 CRPC (continued)

- ***Chemotherapy***
  - **Cabazitaxel/carboplatin with concurrent steroid**
    - ◊ **Concurrent steroid includes daily prednisone, which may be omitted on the day of chemotherapy administration when dexamethasone is given.**
    - ◊ **Cabazitaxel 20 or 25 mg/m² plus carboplatin AUC 4 mg/mL per minute with growth factor support can be considered for fit patients with aggressive variant mCRPC (ie, visceral metastases, low PSA and bulky disease, high lactate dehydrogenase (LDH), high CEA, lytic bone metastases, NEPC histology) or unfavorable genomics (defects in at least 2 of *PTEN*, *TP53*, and *RB1*). The most common grade 3–5 adverse events were fatigue, anemia, neutropenia, and thrombocytopenia. Corn PG, et al. Lancet Oncol 2019;20:1432-1443.**
      - – **Cabazitaxel starting dose can be either 20 mg/m² or 25 mg/m² for patients with mCRPC whose cancer has progressed despite prior docetaxel chemotherapy. Cabazitaxel 25 mg/m² with concurrent steroid may be considered for healthy patients who wish to be more aggressive. Growth factor support may be needed with either dose.**
  - **Mitoxantrone with prednisone**
    - ◊ **Mitoxantrone with prednisone may provide palliation but has not been shown to extend survival in two randomized trials. Adverse events associated with mitoxantrone are similar to docetaxel, but with lower rates of grade 3 or 4 neutropenic fevers, cardiovascular events, nausea and vomiting, metabolic disturbances, and neurologic events.**
  - **Increasing PSA should not be used as the sole criteria for progression. Assessment of response should incorporate clinical and radiographic criteria.**
  - **See NCCN Guidelines for Hematopoietic Growth Factors for recommendations on growth factor support.**

- ***PARP Inhibitors with or without ARPIs***
  - **Inactivation of HRR genes has been associated with response to PARP inhibitor (PARPi) therapy, and a number of FDA-approved agents and combinations are now available (Table 1 and Discussion). Although there are multiple specific gene alterations included in the FDA labels for specific PARPi and combinations (see Table 1), the response rates have been most pronounced for patients with *BRCA2* mutation-associated prostate cancers. Treatment responses have been variable for other genes alterations, and this information should be taken into account in decisions about treatment sequencing and patient counseling. For example, responses have been less robust for patients whose cancers carry ATM mutations, and other recommended regimen options may be favored until more data are available on a gene-by-gene level.**
  - **Loss of *BRCA2* may be especially associated with response to PARPi therapy compared to other HRR gene alterations.**
  - **There has been heterogeneity of response to olaparib for non-*BRCA* mutations based on the specific gene mutation (Discussion).**
  - **The fine-particle (category 2B; other recommended option) or standard formulation of abiraterone can be given with single-agent niraparib as a substitute for the combination niraparib/abiraterone tablet.**
  - **The fine-particle abiraterone (category 2B; other recommended option) may be given with olaparib as a substitute for standard abiraterone.**

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PRINCIPLES OF NON-HORMONAL SYSTEMIC THERAPY

Non-Hormonal Systemic Therapy for M1 CRPC (continued)

- ***Immunotherapy***
  - **Sipuleucel-T**
    - ◊ **Sipuleucel-T is only for asymptomatic or minimally symptomatic patients with no liver metastases, life expectancy >6 months, and ECOG performance status 0–1.**
    - ◊ **Sipuleucel-T is not recommended for patients with small cell prostate cancer/NEPC.**
    - ◊ **Sipuleucel-T has been shown in a phase 3 clinical trial to extend mean survival from 21.7 months in the control arm to 25.8 months in the treatment arm, which constitutes a 22% reduction in mortality risk.**
    - ◊ **Sipuleucel-T is well-tolerated; common complications include chills, pyrexia, and headache.**
  - **Pembrolizumab is an option for certain patients with MSI-H/dMMR mCRPC (PROS-18).**
    - ◊ **Pembrolizumab may cause severe, life-threatening immune-mediated adverse reactions, which may include but are not limited to: pneumonitis, colitis, hepatitis, myocarditis, endocrinopathies, exfoliative dermatologic conditions, renal failure and nephritis, and ocular toxicities. See NCCN Guidelines for Management of Immunotherapy-Related Toxicities.**
    - ◊ **Limited data suggest that pembrolizumab may be associated with some benefit in patients with mCRPC and TMB ≥10 mut/mB. Lenis AT, et al. Clin Canc Res 2024;30:3894-3903.**
    - ◊ **Pembrolizumab and berahyaluronidase alfa-pmph subcutaneous injection may be substituted for IV pembrolizumab. Pembrolizumab and berahyaluronidase alfa-pmph has different dosing and administration instructions compared to IV pembrolizumab.**

- ***Other Targeted Agents***
  - **Pan-cancer, tumor-agnostic treatments can be considered for patients with actionable mutations.**

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PRINCIPLES OF NON-HORMONAL SYSTEMIC THERAPY

Table 1. PARP Inhibitors with or without ARPIs

Treatment	FDA-Approved Disease State Indication	Biomarker Pathogenic variant or mutation in gene (germline and/or somatic)	Clinical Trial
Olaparib	mCRPC that has progressed following prior treatment with enzalutamide or abiraterone.	*BRCA2, BRCA1, ATM, BARD1, BRIP1, CDK12, CHEK1, CHEK2, FANCL, PALB2, RAD51B, RAD51C, RAD51D, or RAD54L*	PROfound[1]
Rucaparib	mCRPC that has been treated with androgen receptor-directed therapy and taxane-based chemotherapy	*BRCA2, BRCA1*	TRITON2[2,3]
Olaparib/ Abiraterone	mCRPC	*BRCA2, BRCA1*	PROPel[4]
Talazoparib/ Enzalutamide	mCRPC	*BRCA2, BRCA1, ATM, ATR, CDK12, CHEK2, FANCA, MLH1, MRE11A, NBN, PALB2, or RAD51C*	TALAPRO-2[5,6]
Niraparib/ Abiraterone	mCRPC	*BRCA2, BRCA1*	MAGNITUDE[7,8]

[1] de Bono JD, Mateo J, Fizazi K, et al. Olaparib for metastatic castration-resistant prostate cancer. NEJM 2020;382:2091-2102.
[2] Abida W, Patnaik A, Campbell D, et al. Rucaparib in men with metastatic castration resistant prostate cancer harboring a BRCA1 or BRCA2 gene alteration. J Clin Oncol 2020;38:3763-3772.
[3] Abida W, Campbell D, Patnaik A, et al. Rucaparib for the treatment of metastatic castration-resistant prostate cancer associated with a DNA damage repair gene alteration: final results from the Phase 2 TRITON2 study. Eur Urol 2023;84:321-330.
[4] Saad F, Clarke NW, Oya M, et al. Olaparib plus abiraterone versus placebo plus abiraterone in metastatic castration-resistant prostate cancer (PROpel): final prespecified overall survival results of a randomised, double-blind, phase 3 trial. Lancet Oncol 2023;24:1094-1108.
[5] Agarwal N, Azad AA, Carles J, et al. Talazoparib plus enzalutamide in men with first-line metastatic castration-resistant prostate cancer (TALAPRO-2): a randomised, placebo-controlled, phase 3 trial. Lancet 2023;402:291-303.
[6] Fizazi K, Azad AA, Matsubara N, et al. First-line talazoparib with enzalutamide in HRR-deficient metastatic castration-resistant prostate cancer: the phase 3 TALAPRO-2 trial. 2023;30:257-264.
[7] Chi KN, Rathkopf D, Smith MR, et al. Niraparib and abiraterone acetate for metastatic castration-resistant prostate cancer. J Clin Oncol 2023;41:339-3351.
[8] Chi KN, Sandhu S, Smith MR, et al. Niraparib plus abiraterone acetate with prednisone in patients with metastatic castration-resistant prostate cancer and homologous recombination repair gene alterations: second interim analysis of the randomized phase III MAGNITUDE trial. 2023;34:772-782.

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American Joint Committee on Cancer (AJCC)
TNM Staging System For Prostate Cancer (8th ed., 2017)
***Table 1.* Definitions for T, N, M**
Clinical T (cT)

T	Primary Tumor
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
T1	Clinically inapparent tumor that is not palpable
T1a	Tumor incidental histologic finding in 5% or less of tissue resected
T1b	Tumor incidental histologic finding in more than 5% of tissue resected
T1c	Tumor identified by needle biopsy found in one or both sides, but not palpable
T2	Tumor is palpable and confined within prostate
T2a	Tumor involves one-half of one side or less
T2b	Tumor involves more than one-half of one side but not both sides
T2c	Tumor involves both sides
T3	Extraprostatic tumor that is not fixed or does not invade adjacent structures
T3a	Extraprostatic extension (unilateral or bilateral)
T3b	Tumor invades seminal vesicle(s)
T4	Tumor is fixed or invades adjacent structures other than seminal vesicles such as external sphincter, rectum, bladder, levator muscles, and/or pelvic wall.

Pathological T (pT)

T	Primary Tumor
T2	Organ confined
T3	Extraprostatic extension
T3a	Extraprostatic extension (unilateral or bilateral) or microscopic invasion of bladder neck
T3b	Tumor invades seminal vesicle(s)
T4	Tumor is fixed or invades adjacent structures other than seminal vesicles such as external sphincter, rectum, bladder, levator muscles, and/or pelvic wall

Note: There is no pathological T1 classification.
Note: Positive surgical margin should be indicated by an R1 descriptor, indicating residual microscopic disease.

N	Regional Lymph Nodes
NX	Regional lymph nodes cannot be assessed
N0	No positive regional nodes
N1	Metastases in regional node(s)

M	Distant Metastasis
M0	No distant metastasis
M1	Distant metastasis
M1a	Nonregional lymph node(s)
M1b	Bone(s)
M1c	Other site(s) with or without bone disease

Note: When more than one site of metastasis is present, the most advanced category is used. M1c is most advanced.

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Table 2. **AJCC Prognostic Groups**

Group	T	N	M	PSA (ng/mL)	Grade Group
Stage I	cT1a-c	N0	M0	PSA <10	1
	cT2a	N0	M0	PSA <10	1
	pT2	N0	M0	PSA <10	1
Stage IIA	cT1a-c	N0	M0	PSA ≥10 <20	1
	cT2a	N0	M0	PSA ≥10 <20	1
	pT2	N0	M0	PSA ≥10 <20	1
	cT2b	N0	M0	PSA <20	1
	cT2c	N0	M0	PSA <20	1
Stage IIB	T1-2	N0	M0	PSA <20	2
Stage IIC	T1-2	N0	M0	PSA <20	3
	T1-2	N0	M0	PSA <20	4
Stage IIIA	T1-2	N0	M0	PSA ≥20	1-4
Stage IIIB	T3-4	N0	M0	Any PSA	1-4
Stage IIIC	Any T	N0	M0	Any PSA	5
Stage IVA	Any T	N1	M0	Any PSA	Any
Stage IVB	Any T	Any N	M1	Any PSA	Any

Note: When either PSA or Grade Group is not available, grouping should be determined by T category and/or either PSA or Grade Group as available.

Histopathologic Type
This classification applies to adenocarcinomas and squamous carcinomas, but not to sarcoma or transitional cell (urothelial) carcinoma of the prostate. Adjectives used to describe histologic variants of adenocarcinomas of prostate include mucinous, signet ring cell, ductal, and neuroendocrine, including small cell carcinoma. There should be histologic confirmation of the disease.

Definition of Histologic Grade Group (G)
Recently, the Gleason system has been compressed into so-called Grade Groups.

Grade Group	Gleason Score	Gleason Pattern
1	≤6	≤3+3
2	7	3+4
3	7	4+3
4	8	4+4, 3+5, 5+3
5	9 or 10	4+5, 5+4, 5+5

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ABBREVIATIONS

3D	three-dimensional
ACE-27	Adult Comorbidity Evaluation-27 Index
ADT	androgen deprivation therapy
AI	artificial intelligence
ARPI	androgen receptor pathway inhibitor
aRT	adjuvant RT
ASTRO	American Society for Radiation Oncology
AUC	area under the curve
BCR	biochemical recurrence
BED	biologically effective dose
BMI	body mass index
CBRT	cone beam CT
CEA	carcinoembryonic antigen
CHIP	clonal hematopoiesis of indeterminate potential
CLR	correct localization rate
CRPC	castration-resistant prostate cancer
CSPC	castration-sensitive prostate cancer
ctDNA	circulating tumor DNA
CVD	cardiovascular disease
DCE	dynamic contrast-enhanced
DEXA	dual-energy x-ray absorptiometry
DIL	dominant intraprostatic lesion
DM	distant metastases
dMMR	mismatch repair deficient
DRE	digital rectal examination
DWI	diffusion-weighted imaging
EBRT	external beam radiation therapy
ECG	electrocardiogram
ECHO	echocardiogram
ENI	elective nodal irradiation
FDG	fluorodeoxyglucose
FFS	failure-free survival
FIR	favorable intermediate-risk
FLA	focal laser ablation
FRAX	Fracture Risk Assessment Tool
FT	focal therapy
GC	genomic classifier
HDR	high dose rate
HIFU	high-intensity focused ultrasound
HRD	homologous recombination deficiency
HRR	homologous recombination repair
IGRT	image-guided radiation therapy
IMRT	intensity modulated RT
IRE	irreversible elecroporation
IRF	intermediate risk factor
ITT	intention to treat
LDH	lactate dehydrogenase
LDR	low dose rate
LHRH	luteinizing hormone-releasing hormone
LT-ADT	long-term androgen deprivation therapy
mCRPC	metastatic castration-resistant prostate cancer
mCSPC	metastatic castration-sensitive prostate cancer
MDT	metastasis-directed therapy
MDRT	metastasis-directed radiotherapy
MFS	metastasis-free survival
MMAI	multimodal artificial intelligence
MMR	mismatch repair
mpMRI	multiparametric MRI
MSI	microsatellite instability
MSI-H	microsatellite instability-high

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ABBREVIATIONS

NEPC	neuroendocrine prostate cancer
NNT	number needed to treat
OAR	organ at risk
OS	overall survival
PARPi	PARP inhibitor
PCSM	prostate cancer-specific mortality
PCSS	prostate cancer-specific survival
PDT	photodynamic therapy
PFS	progression-free survival
PLND	pelvic lymph node dissection
PNRT	prophylactic nodal radiotherapy
PPV	positive predictive value
PSA	prostate-specific antigen
PSADT	prostate-specific antigen doubling time
PSMA	prostate-specific membrane antigen
PTV	planning target value
QOL	quality of life
rh	radiohybrid
RFA	radiofrequency ablation
ROI	region of interest
RP	radical prostatectomy
rPFS	radiographic progression-free survival
RTOG	Radiation Therapy Oncology Group
SABR	stereotactic ablative radiotherapy
SBRT	stereotactic body radiation therapy
sHR	subdistribution hazard ratio
SPECT	single-photon emission computed tomography
SRE	skeletal-related event
sRT	secondary RT
ST-ADT	short-term androgen deprivation therapy
TMB	tumor mutational burden
TULSA	transurethral ultrasound ablation
VUS	variant of uncertain significance

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NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN Categories of Preference	
Preferred	Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended	Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances	Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

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Discussion

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Overview

An estimated 313,780 new cases of prostate cancer will be diagnosed in the United States in 2025, accounting for 30% of new cancer cases in men.[1] It is the most common cancer in men in the United States, who currently have a 1 in 8 lifetime risk of developing prostate cancer.[1] The incidence of prostate cancer declined by approximately 40% from 2007 to 2014, but since that time has increased at a rate of 3% annually. These trends largely reflect the changes in prostate-specific antigen (PSA) screening recommendations. The decrease in PSA screening that followed the 2012 USPSTF recommendations against routine testing was associated with a rise in the diagnosis of regional and metastatic disease.[2-10]

Researchers further estimate that prostate cancer will account for 11% of male cancer deaths in the United States in 2025, with an estimated 35,770 deaths.[1] The age-adjusted death rate from prostate cancer declined by 52% from 1993 to 2017, but the death rate has become more stable in recent years, with a 0.5% annual decrease from 2012 through 2022.[1] For all stages combined, the 5-year relative survival rate for prostate cancer is 97%.[1] The comparatively low death rate suggests that increased public awareness with earlier detection and treatment has affected mortality from this prevalent cancer, but is also complicated by screening-related lead-time bias and detection of indolent cancers. Maintenance of this low death rate is threatened by the rising prostate cancer incidence and diagnosis of advanced disease.

Unfortunately, large inequities exist in incidence of and mortality from prostate cancer across racial and ethnic groups. The incidence rate in Black individuals is 67% higher than in white individuals, with prostate cancer accounting for 44% of cancer diagnoses in Black men and a 1 in 6 lifetime risk of a prostate cancer diagnosis.[11] Black individuals are also more likely to be diagnosed with more aggressive disease and are less likely to have had a PSA test within the past year.[11] The mortality rate from prostate cancer in this population is two to four times higher than all other racial and ethnic groups; prostate cancer accounts for 17% of male cancer deaths in the United States.[1,11] However, the overall prognosis by race appears similar when patients are treated with the same guideline-concordant care.[12]

Use of PSA for early detection of potentially fatal prostate cancer coupled with the use of imaging and the consideration of risk calculators and/or biomarkers to improve the specificity of screening should decrease the risk of overdetection (see the NCCN Guidelines for Prostate Cancer Early Detection, available at www.NCCN.org). This reduced overdetection along with the use of active surveillance in appropriate patients should reduce overtreatment AND preserve the relatively low rates of prostate cancer mortality.

Guidelines Update Methodology

The complete details of the Development and Update of the NCCN Guidelines are available at www.NCCN.org.

Literature Search Criteria

Prior to the update of the NCCN Guidelines for Prostate Cancer, an electronic search of the PubMed database was performed to obtain key literature in prostate cancer published since the previous Guidelines update, using the search term “prostate cancer.” The PubMed database was chosen because it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.[13]

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Clinical Trial, Phase III; Clinical Trial, Phase IV; Guideline; Practice Guideline; Randomized Controlled Trial; Meta-Analysis; Systematic

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Reviews; and Validation Studies. The data from key PubMed articles as well as articles from additional sources deemed as relevant to these Guidelines as discussed by the Panel during the Guidelines update have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

Sensitive/Inclusive Language Usage

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation.[14] NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight-biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, instead focusing on organ-specific recommendations. This language is both more accurate and more inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms men, women, female, and male when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to predominantly represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

Initial Prostate Cancer Diagnosis

Initial suspicion of prostate cancer is based on an abnormal digital rectal exam (DRE) or an elevated PSA level. A separate NCCN Guidelines Panel has written guidelines for prostate cancer early detection (see the NCCN Guidelines for Prostate Early Detection, available at www.NCCN.org). Definitive diagnosis requires biopsies of the prostate, usually performed by a urologist using a needle under transrectal ultrasound (TRUS) guidance. A pathologist assigns a Gleason primary and secondary grade to the biopsy specimen. Clinical staging is based on the TNM (tumor, node, metastasis) classification from the AJCC Staging Manual, Eighth Edition.[15] NCCN treatment recommendations are based on risk stratification that includes TNM staging rather than on AJCC prognostic grouping.

Pathology synoptic reports (protocols) are useful for reporting results from examinations of surgical specimens; these reports assist pathologists in providing clinically useful and relevant information. The NCCN Guidelines Panel favors pathology synoptic reports from the College of American Pathologists (CAP) that comply with the Commission on Cancer (CoC) requirements.[16]

Estimates of Life Expectancy

Estimates of life expectancy have emerged as a key determinant of primary treatment, particularly when considering observation. Life expectancy can be estimated for groups of individuals, but it is difficult to extrapolate these estimates to an individual patient. Life expectancy can be estimated using the Social Security Administration Life Insurance Tables,[17] the Memorial Sloan Kettering Male Life Expectancy tool,[18] or the University of California San Francisco (UCSF) Lee Schonberg Index[19] and adjusted for individual patients by adding or subtracting 50% based on whether one believes the patient is in the healthiest quartile or the unhealthiest quartile, respectively.[20] As an example, the Social Security Administration Life Expectancy for a 65-year-old American male is 17.5 years. If judged to be in the upper quartile of health, a life expectancy of 26.3 years is assigned. If judged to be in the lower quartile of health, a life expectancy of 8.8 years is assigned. Thus, treatment recommendations could change dramatically using the NCCN Guidelines if a 65-year-old

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patient was assessed to be in excellent health or if their health is in decline.

Quality-of-Life and Shared Decision-Making

Patient quality of life (QOL, also referred to as health-related quality of life [HRQOL]) should be taken into consideration when discussing and deciding upon management strategies for all types of cancer. In regards to prostate cancer, addressing QOL is especially important due to early detection and increased survival rates.[21] The management of localized prostate cancer can have risks and side effects related to radiation, surgery, and hormone therapy—including functional effects on urinary, bowel, and sex organs.[22-27] Some side effects have been linked to lasting psychological effects; for example an association has been shown between erectile dysfunction and depressive symptoms.[28,29] Additionally, the use of androgen deprivation therapy (ADT) is associated with 23% higher rates of depression compared to patients who did not receive ADT.[29] Use of a standardized patient-reported outcomes instrument, such as the abbreviated Expanded Prostate Cancer Index Composite (EPIC-26), can aid in the assessment of baseline urinary, bowel, and sexual function and serve as a basis for shared-decision making.[21]

Shared decision-making should be collaborative and informative. This approach is especially applicable when multiple treatment options are considered clinically acceptable and when some treatment options have greater potential for adverse effects on QOL than others. When patients have actively participated in making decisions about their medical care, they generally report that they feel empowered, have better rates of adherence to the medical plan, and are less likely to display regrets about the decisions that were made.[30] When shared-decision making is utilized, patients can make informed decisions with more realistic expectations and greater satisfaction over the long term.

Prostate Cancer Genetics

Family history of prostate cancer raises the risk of prostate cancer.[31-34] In addition, prostate cancer has been associated with hereditary breast and ovarian cancer (HBOC) syndrome (due to germline mutations in homologous DNA repair genes) and Lynch syndrome (resulting from germline mutations in DNA mismatch repair [MMR] genes).[34-39] In fact, approximately 11% of patients with prostate cancer and at least 1 additional primary cancer carry germline mutations associated with increased cancer risk.[40] Therefore, the Panel recommends a thorough review of personal and family history for all patients with prostate cancer at the time of initial diagnosis.[41,42]

The presence of germline mutations has implications for family genetic counseling, cancer risk syndromes, and assessment of personal risk for subsequent cancers. Some patients with prostate cancer and their families may be at increased risk for breast and ovarian cancer, melanoma, and pancreatic cancer (HBOC); colorectal cancers (Lynch syndrome); and other cancer types. While DNA repair defects carry a worse prognosis for those with metastatic prostate cancer, it does not appear to be prognostic in localized disease.[43] Finally, there are possible treatment implications for patients with DNA repair defects with advanced disease (see *Treatment Options for Patients with DNA Repair Gene Mutations*, below).

Prostate cancer is often associated with somatic mutations that occur in the tumor but not in the germline. Both germline and tumor mutations are discussed herein.

Homologous DNA Repair Genes

Somatic mutations in DNA repair pathway genes occur in up to 19% of localized prostate tumors and 23% of metastatic CRPC (mCRPC) tumors, with most mutations found in *BRCA2* and *ATM*.[44,45] These tumor mutations are often associated with germline mutations. For example, 42% of

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patients with mCRPC and somatic mutations in *BRCA2* were found to carry the mutation in their germlines.[44] In localized prostate cancer, that number was 60%.[45]

Overall, germline DNA repair mutations have been reported with the lowest frequencies seen in patients with lower-risk localized prostate cancer (1.6%–3.8%), higher frequencies in those with higher-risk localized disease (6%–8.9%), and the highest frequencies in those with metastatic disease (7.3%–16.2%).[44,46-52] One study found that 11.8% of patients with metastatic prostate cancer have germline mutations in 1 of 16 DNA repair genes: *BRCA2* (5.3%), *ATM* (1.6%), *CHEK2* (1.9%), *BRCA1* (0.9%), *RAD51D* (0.4%), *PALB2* (0.4%), *ATR* (0.3%), and *NBN*, *PMS2*, *GEN1*, *MSH2*, *MSH6*, *RAD51C*, *MRE11A*, *BRIP1*, or *FAM175A*.[51]

An additional study showed that 9 of 125 patients with high-risk, very-high-risk, or metastatic prostate cancer (7.2%) had pathogenic germline mutations in *MUTYH* (4), *ATM* (2), *BRCA1* (1), *BRCA2* (1), and *BRIP1* (1).[48] In this study, the rate of metastatic disease among those with a mutation identified was high (28.6%, 2 of 7 patients). Although having a relative with breast cancer was associated with germline mutation identification ($P = .035$), only 45.5% of the mutation carriers in the study had mutations that were concordant with their personal and family history. Another study also found that a family history of breast cancer increased the chances of identifying a germline DNA repair gene mutation in patients with prostate cancer (OR, 1.89; 95% CI, 1.33–2.68; $P = .003$).[53] In a study of an unselected cohort of 3607 patients with a personal history of prostate cancer who had germline genetic testing based on clinician referral, 11.5% had germline mutations in *BRCA2*, *CHEK2*, *ATM*, *BRCA1*, or *PALB2*.[54]

More than 2% of Ashkenazi Jews carry germline mutations in *BRCA1* or *BRCA2*, and these carriers have a 16% chance (95% CI, 4%–30%) of developing prostate cancer by the age of 70.[55] In a study of 251 unselected Ashkenazi Jewish patients with prostate cancer, 5.2% had germline mutations in *BRCA1* and *BRCA2*, compared with 1.9% of control Ashkenazi Jewish males.[56]

Germline *BRCA1* or *BRCA2* mutations have been associated with an increased risk for prostate cancer in numerous reports.[38,39,56-66] In particular, *BRCA2* mutations have been associated with a 2- to 6-fold increase in the risk for prostate cancer, whereas the association of *BRCA1* mutations and increased risks for prostate cancer are less consistent.[38,39,56,58,60,65,67,68] In addition, limited data suggest that germline mutations in *ATM, PALB2,* and *CHEK2* increase the risk of prostate cancer.[69-72] Furthermore, prostate cancer in individuals with germline *BRCA* mutations (*BRCA*m) appears to occur earlier, and in patients with metastatic prostate cancer, has a more aggressive phenotype, and is associated with significantly reduced survival times than in patients without these mutations.[43,67,73-79]

DNA Mismatch Repair Genes

Tumor mutations in *MLH1*, *MSH2*, *MSH6*, and *PMS2* may result in tumor microsatellite instability (MSI) and deficient MMR (dMMR; detected by immunohistochemistry) and are sometimes associated with germline mutations and Lynch syndrome. Patients with Lynch syndrome may have an increased risk for prostate cancer. In particular, studies show an increased risk for prostate cancer in patients who are older and have germline *MSH2* mutations.[80,81]

In a study of >15,000 patients with cancer treated at Memorial Sloan Kettering Cancer Center who had their tumor and matched normal DNA sequenced and tumor MSI status assessed, approximately 5% of 1048 patients with prostate cancer had MSI-high (MSI-H) or MSI-indeterminate tumors, 5.6% of whom were found to have Lynch syndrome (0.29% of patients with prostate cancer).[35] In another prospective case series, the tumors of 3.1% of 1033 patients with prostate cancer demonstrated MSI-

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H/dMMR status, and 21.9% of these patients had Lynch syndrome (0.68% of the total population).[82] In a study of an unselected cohort of 3607 patients with a personal history of prostate cancer who had germline genetic testing based on clinician referral, 1.7% had germline mutations in *PMS2*, *MLH1, MSH2,* or *MSH6*.[54]

Effect of Intraductal/Cribriform or Ductal Histology

Ductal prostate carcinomas are rare, accounting for approximately 1.3% of prostate carcinomas.[83] Intraductal prostate cancer may be more common, especially in higher risk groups, and may be associated with a poor prognosis.[84] It is important to note that there is significant overlap in diagnostic criteria and that intraductal, ductal, and invasive cribriform features may coexist in the same biopsy. By definition, intraductal carcinoma includes cribriform proliferation of malignant cells as long as they remain confined to a preexisting gland that is surrounded by basal cells. These features are seen frequently with an adjacent invasive cribriform component and would be missed without the use of basal cell markers.

Limited data suggest that acinar prostate adenocarcinoma with invasive cribriform pattern, intraductal carcinoma of prostate (IDC-P), or ductal adenocarcinoma component have increased genomic instability.[85-88] In particular, tumors with these histologies may be more likely to harbor somatic MMR gene alterations than those with adenocarcinoma histology.[88-90] In addition, limited data suggest that germline homologous DNA repair gene mutations may be more common in prostate tumors of ductal or intraductal origin[91,92] and that intraductal histology is common in germline *BRCA2* mutation carriers with prostate cancer.[93] Overall, the Panel believes that the data connecting histology and the presence of genomic alterations are stronger for intraductal than ductal histology at this time.

Genetic Testing Recommendations

Germline Testing Based on Family History, Histology, and Risk Groups

The Panel recommends inquiring about family and personal history of cancer and known germline variants at time of initial diagnosis. Germline testing should be considered in appropriate individuals where it is likely to impact the prostate cancer treatment and clinical trial options, management of risk of other cancers, and/or potential risk of cancer in family members. Criteria for germline testing from the following NCCN Guidelines (available at www.NCCN.org) should be reviewed at time of initial diagnosis and, if relevant, at recurrence:

- The NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate (page CRIT-6)
- The NCCN Guidelines for Genetic/Familial High-Risk Assessment: Colorectal, Endometrial, and Gastric (page HRS-3).

If criteria are met, multigene testing is recommended (see GENE-1 in the NCCN Guidelines for Genetic/Familial High-Risk Assessment: Breast, Ovarian, Pancreatic, and Prostate).

Genetic counseling resources and support are critical, and post-test genetic counseling is recommended if a germline mutation (pathogenic/likely pathogenic variant) is identified. Cascade testing for relatives is critical to inform the risk for familial cancers in all relatives. Post-test genetic counseling is recommended if there is a positive family history but no pathogenic variant OR if only germline variants of uncertain significance (VUS) are identified. This is to ensure accurate understanding of family implications and review indications for additional testing and/or follow up (including clinical trials of reclassification). Resources are available to check the known pathologic effects of genomic variants (eg, https://brcaexchange.org/about/app; https://www.ncbi.nlm.nih.gov/clinvar/). Information regarding germline mutations in patients with metastatic

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disease can be used to inform future treatments or to determine eligibility for clinical trials.

Somatic Tumor Testing Based on Risk Groups

Tumor testing recommendations are as follows:

1. Tumor testing for somatic homologous recombination gene mutations (eg, *BRCA1, BRCA2, ATM, PALB2, FANCA, RAD51D*, *CHEK2, CDK12*) can be considered in patients with regional (N1) prostate cancer and is recommended for those with metastatic disease.
2. Tumor testing for MSI-H or dMMR can be considered in patients with regional or metastatic castration-sensitive prostate cancer (CSPC) and is recommended in the mCRPC setting.
3. Tumor mutational burden (TMB) testing is recommended in patients with mCRPC.

The Panel strongly recommends a metastatic biopsy for histologic and molecular evaluation. When unsafe or unfeasible, plasma ctDNA assay is an option, preferably collected during biochemical (PSA) and/or radiographic progression in order to maximize diagnostic yield. Caution is needed when interpreting ctDNA-only evaluation due to potential interference from clonal hematopoiesis of indeterminate potential (CHIP), which can result in a false-positive biomarker signal.[94]

If MSI testing is performed, testing using an NGS assay validated for prostate cancer is preferred.[95-97] If MSI-H or dMMR is found, the patient should be referred for genetic counseling to assess for the possibility of Lynch syndrome. MSI-H or dMMR indicate eligibility for pembrolizumab for patients with mCRPC (see *Pembrolizumab*, below).

Post-test genetic counseling is also recommended if pathogenic/likely pathogenic somatic mutations in any gene that has clinical implications are also identified in germline (eg, *BRCA1, BRCA2, ATM, PALB2, CHEK2, MLH1, MSH2, MSH6, PMS2).* Virtually none of the NGS tests is designed or validated for germline assessment. Therefore, over-interpretation of germline findings should be avoided. If a germline mutation is suspected, the patient should be recommended for genetic counseling and follow-up dedicated germline testing.

Inequities Based on Race and Ethnicity

A CDC analysis of population-based cancer registries found that the incidence of prostate cancer was 67% higher in Black individuals than in white individuals in the United States.[11] Data also show that Black Americans are more likely to be diagnosed with more advanced disease and less likely to receive PSA screening.[11,98-101] Furthermore, the mortality rate from prostate cancer in the Black American population is two to four times higher than all other racial and ethnic groups.[1,11] Moreover, Black American patients with low-risk or favorable intermediate-risk prostate cancer have an increase in all-cause mortality after treatment, potentially due to cardiovascular complications after ADT.[102]

Significant inequities also exist for other racial and ethnic groups. For example, Hispanic patients are more likely than non-Hispanic white patients to present with higher risk localized or metastatic prostate cancer.[103,104] Furthermore, Hispanic individuals are less likely than non-Hispanic white individuals to receive treatment and more likely to experience treatment delays.[103,104] Similarly, Asian American, Native Hawaiian, and Pacific Islander individuals are more likely to be diagnosed at a higher risk group than white individuals.[105]

The main drivers for these racial and ethnic inequities are lower rates of prostate cancer screening, diagnostic procedures, and overall access to health care; treatment disparities; inadequate health insurance coverage; and other factors related to social determinants of health (eg, environmental exposures, patient and physician behaviors, delays in

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diagnosis).[98,100,106-112] For example, an analysis of SEER data from 2004 to 2011 determined that Black and Hispanic individuals had a decreased likelihood of receiving treatment than white individuals even when diagnosed with high-risk disease.[113] Importantly, data show that Black patients who receive equitable prostate cancer treatment have similar outcomes to white patients.[12,114-116]

Risk Stratification for Clinically Localized Disease

Unlike in some cancers where treatments for early-stage disease are guided by predictive biomarkers, such as ER/PR or HER2 status in breast cancer, in localized prostate cancer treatment is based largely on risk stratification of prognosis.

NCCN and other risk classification schemas, such as CARPA[117] or STAR-CAP,[118] are prognostic and have not been shown to be predictive of benefit to a specific treatment. Thus, common decision points, such as conservative management versus radical therapy, radiotherapy alone versus the addition of short-term ADT, the use of radiotherapy with short-term versus long-term ADT, or the addition of abiraterone acetate/prednisone to radiotherapy plus long-term ADT are based on clinical trial inclusion criteria, expert opinion, and estimates of absolute benefit and harm from a given therapy in the context of NCCN risk groups.

There are newer risk classification schemas that have been shown to outperform NCCN risk groups,[118,119] as well as tools (ie, imaging, gene expression biomarkers) that together improve risk stratification. These tools should not be ordered reflexively. They are recommended only when they will have the ability to change management. Improved risk stratification can better identify patients who may derive greater or lesser absolute benefit from a given treatment.

NCCN Risk Groups

The NCCN Guidelines have, for many years, incorporated a risk stratification scheme that uses a minimum of stage, Gleason grade, and PSA to assign patients to risk groups. Risk group stratification has been published widely and validated, and provides a better basis for treatment recommendations than clinical stage alone.[120-122]

Prostate cancer grading continues to evolve. The grading system was updated during the 2014 International Society of Urological Pathology (ISUP) Consensus Conference.[123] Several changes were made to the assignment of Gleason pattern based on pathology. The new system assigns Grade Groups from 1 to 5, derived from the Gleason score.

- Grade Group 1: Gleason score ≤6; only individual discrete well-formed glands
- Grade Group 2: Gleason score 3+4=7; predominantly well-formed glands with lesser component of poorly formed/fused/cribriform glands
- Grade Group 3: Gleason score 4+3=7; predominantly poorly formed/fused/cribriform glands with lesser component of well-formed glands
    - For cases with >95% poorly formed/fused/cribriform glands or lack of glands on a core or at radical prostatectomy, the component of <5% well-formed glands is not factored into the grade.
- Grade Group 4: Gleason score 4+4=8; 3+5=8; 5+3=8
    - Only poorly formed/fused/cribriform glands; or
    - Predominantly well-formed glands and lesser component lacking glands (poorly formed/fused/cribriform glands can be a more minor component); or
    - Predominantly lacking glands and lesser component of well-formed glands (poorly formed/fused/cribriform glands can be a more minor component)

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- Grade Group 5: Gleason score 9–10; lack gland formation (or with necrosis) with or without poorly formed/fused/cribriform glands
    - For cases with >95% poorly formed/fused/cribriform glands or lack of glands on a core or at radical prostatectomy, the component of <5% well-formed glands is not factored into the grade.

The new Grade Group system was initially validated in two separate cohorts of patients treated for prostate cancer with either radical prostatectomy or radiation: one with >26,000 patients and one with 5880 patients.[124,125] Both studies found that Grade Groups predicted the risk of recurrence after primary treatment with modest discrimination. For instance, in the larger study, the 5-year biochemical recurrence-free progression probabilities after radical prostatectomy for Grade Groups 1 through 5 were 96% (95% CI, 95–96), 88% (95% CI, 85–89), 63% (95% CI, 61–65), 48% (95% CI, 44–52), and 26% (95% CI, 23–30), respectively. The separation between Grade Groups was less pronounced in the radiation therapy (RT) cohort, likely because of increased use of ADT in the higher risk groups. In another study of the new ISUP Grade Group system, all-cause mortality and prostate cancer-specific mortality were higher in patients in Grade Group 5 than in those in Grade Group 4.[126] Additional studies have supported the validity of this new system.[127-132]

The NCCN Panel has accepted the Grade Group system to inform better treatment discussions compared to those using Gleason score. Patients remain divided into low-, intermediate-, high-, and very-high-risk groups. If a patient meets the criteria for intermediate risk, they are then further risk stratified into favorable and unfavorable subsets based on the number of intermediate risk factors, grade group, and the percentage of positive biopsy cores, as delineated in the algorithm above.

The NCCN Guidelines Panel recognized that heterogeneity exists within each risk group.[118] For example, an analysis of 12,821 patients showed that those assigned to the intermediate-risk group by clinical stage (T2b–T2c) had a lower risk of recurrence than those categorized according to Gleason score (7) or PSA level (10–20 ng/mL).[133] A similar trend of superior recurrence-free survival was observed in patients placed in the high-risk group by clinical stage (T3a) compared to those assigned by Gleason score (8–10) or PSA level (>20 ng/mL), although it did not reach statistical significance. Other studies have reported differences in outcomes in the high-risk group depending on risk factors or primary Gleason pattern.[134,135] Evidence also shows heterogeneity in the low-risk group, with PSA levels and percent positive cores affecting pathologic findings after radical prostatectomy.[136,137]

In a retrospective study, 1024 patients with intermediate-risk prostate cancer were treated with radiation with or without neoadjuvant and concurrent ADT.[138] Multivariate analysis revealed that primary Gleason pattern 4, number of positive biopsy cores ≥50%, and presence of >1 intermediate-risk factors (IRFs; ie, T2b-c, PSA 10–20 ng/mL, Gleason score 7) were significant predictors of increased incidence of distant metastasis. The authors used these factors to separate the patients into unfavorable and favorable intermediate-risk groups and determined that the unfavorable intermediate-risk group had worse PSA recurrence-free survival and higher rates of distant metastasis and prostate cancer-specific mortality than the favorable intermediate-risk group. The use of active surveillance in patients with favorable intermediate-risk prostate cancer is discussed below (see *Active Surveillance in Favorable Intermediate Risk*). The NCCN Panel has included the separation of the intermediate risk group into favorable and unfavorable subsets in their risk stratification scheme.

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Nomograms

A nomogram is a predictive instrument that takes a set of input data (variables) and makes predictions about an outcome. Nomograms may potentially predict more accurately for the individual patient than risk groups, because they utilize granular prognostic variables. The Partin tables are an early example of a nomogram for patients with clinically localized prostate cancer.[139-142] The tables give the probability (95% CI) that a patient with a certain clinical stage, Gleason score, and PSA will have a cancer of each pathologic stage. Nomograms can be used to inform treatment decision-making for patients contemplating active surveillance,[143-145] radical prostatectomy,[146-149] neurovascular bundle preservation[150-152] or omission of pelvic lymph node dissection (PLND) during radical prostatectomy,[153-156] brachytherapy,[146,157-159] or external beam RT (EBRT).[146,160] Biochemical progression-free survival (PFS) can be reassessed postoperatively using age, diagnostic serum PSA, and pathologic grade and stage.[146,161-163] Potential success of adjuvant or post-recurrence RT after unsuccessful radical prostatectomy can be assessed using a nomogram.[146,164]

None of the current models predicts with perfect accuracy, and only some of these models predict metastasis[145,146,161,165,166] and cancer-specific death.[147,149,167-169] Given the competing causes of mortality, many patients who sustain PSA recurrence will not live long enough to develop clinical evidence of distant metastases or to die from prostate cancer. Not all PSA recurrences are clinically relevant; thus, PSADT may be a more useful measure of risk of death.[170] The NCCN Guidelines Panel recommends that NCCN risk groups be used to begin the discussion of options for the treatment of clinically localized prostate cancer and that nomograms be used to provide additional and more individualized information.

Advanced Risk Stratification Tools

The Panel strongly advocates the maximum utilization of all routine clinical, pathologic, and patient information to risk stratify patients. This includes the information that forms the basis of the NCCN risk groups, as well as other routine information that can be derived from standard tests, such as PSA density, PSA velocity, percentage or absolute pattern 4, and intraductal and cribriform architecture. Other tools have been developed that, variably, have been shown to be superior to standard clinical tools at estimating prognosis, and these are referred to as advanced risk stratification tools. These currently include multigene expression tests and digital pathology-based artificial intelligence tests, as discussed in more detail below.

These advanced risk stratification tools should only be considered when they have the potential ability to change disease management; they should not be ordered reflexively. Importantly, the Panel notes that there are risks to using tools that lack robust validation to change management decisions, as they may drive patients or providers to choose inappropriate options. For example, in one trial, 200 patients with very low to favorable-intermediate risk prostate cancer (excluding those with grade group 2 and >3 positive cores and limiting inclusion of those with PSA 10–20 ng/mL to only those with PSA density <0.15) were randomized to standard counseling with or without the 17-gene genomic prostate score (GPS) assay.[171] Patients with lower health literacy who received the GPS test were 7-fold less likely to choose active surveillance compared to controls. Therefore, if advanced tools are used, the Panel recommends only tools that have high-quality, long-term clinical trial data, ideally across multiple clinical trials.

Currently, the Panel recommends using tests with high-quality, prospective validation. For predictive tools, ideally this validation is from dedicated prospective biomarker-focused trials (eg, NRG GU006).

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Alternatively, this can be from post-hoc assessment of the biomarker in completed prospective trials, ideally multiple such trials. For prognostic tools, the Panel generally recommends the use of prognostic tests that are validated in well-designed prospective studies with clinically meaningful endpoints based on disease settings that guide a specific treatment indication based on a specific score or result. These studies can be either prospective integral or integrated clinical trial(s) (eg, NRG GU009, NRG GU010) or post-hoc correlative analyses of prospective trials.

Tumor Multigene Expression Testing

Gene testing of a tumor offers the potential of added insight into the biologic behavior of a cancer that could thereby aid in clinical decision-making.

Several tissue-based molecular assays have been developed to improve risk stratification. The 22-gene genomic classifier (GC; Decipher) is discussed in more detail in the *Principles of Risk Stratification* in the algorithm above; others are discussed below. The training and development of each assay is distinct, the genes analyzed are unique, and there is heterogeneity in the robustness of validation. It is important to understand that companies may make invalid or inaccurate claims about these tests, which can create confusion for patients and providers. Furthermore, the resulting test score, such as low or high risk, may be discordant between different tests due to multiple factors.

It is clear that use of tissue-based molecular assays will often change decisions about disease management.[172-175] However, what is unclear from such studies is if the change in management improved patient outcomes or was appropriate. Furthermore, it is increasingly clear that each assay should be evaluated independently because they do not capture the same biology. For example, a study of over 50,000 patients was performed to compare three commonly used gene expression signatures via creation of derived signatures for each test, and the correlation between the signatures was poor (R2, 0.32–0.36).[176]

While multiple gene expression tests have made claims to improve the safety and/or efficacy of active surveillance, such an improvement has not been born out in prospective studies. For example, evaluation of diagnostic biopsy tissue from patients enrolled in the Canary PASS multicenter active surveillance cohort suggested that results of a molecular assay were not associated with adverse pathology in combination with clinical variables nor was there an association with upgrading in surveillance biopsies.[177] While this study utilized GPS, no other gene expression or digital pathology-based tool has positive prospective validation in this setting.

Currently, the above-mentioned validation criteria for prognostic biomarker tests has been reached by the 22-gene GC prognostic assay, which has reported outcomes from multiple post-hoc analyses of randomized trials.[178-181] The relevant disease settings and more details for this tool are available in the *Principles of Risk Stratification* in the algorithm above. There are multiple biomarker-directed randomized trials utilizing the 22-gene GC assay that have completed enrollment or are ongoing (eg, NRG GU006, 009, 010).

Multiple retrospective studies suggest that the 17-gene GPS tool (previously called Oncotype Dx) may be prognostic for patients with localized prostate cancer.[182,183] GPS was also studied in the prospective Canary PASS active surveillance cohort with post-hoc biomarker analysis.[177] GPS results were obtained from 432 patients, 101 of whom underwent radical prostatectomy after an initial period of active surveillance. The authors concluded that adding GPS to a model containing PSA density and diagnostic grade group did not significantly improve adverse pathology stratification over the clinical variables alone (HR, 1.17; 95% CI, 1.00–1.43; P = .066). Additionally, there was no

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association observed between GPS and subsequent biopsy upgrade (*P* = 0.48). Event rates and sample size may have impacted the results. Additionally, as previously referenced, the GPS test was evaluated in a randomized trial and demonstrated that it decreased the relative odds of choosing active surveillance by approximately 50% with variable statistical significance depending on analysis method (*P* = .029 when excluding participants with inadequate biopsy specimens who did not receive a planned GPS result; *P* = .067 for all patients in an intention-to-treat analysis).[171]

Similarly, multiple retrospective institutional and tumor registry studies suggest that the 31-gene cell cycle progression test (CCP; Prolaris) may have prognostic value for patients with localized prostate cancer.[184-186] No prospective trials have been published to date.

Numerous other tests have been developed with variable evidence quality that are currently outside of the Discussion presented here.

Digital Pathology-Based Tools

More recently a new class of advanced risk stratification tool has been developed that utilizes artificial intelligence (AI) to analyze digital pathology slides from a patient's prostate biopsy or prostatectomy specimen. Analogous to the multigene prognostic assays, these tools are primarily focused to improve risk stratification. The multimodal AI (MMAI) test (ArteraAI Prostate) is one such test, which gained FDA approval in July 2025. The MMAI test was validated in multiple post-hoc analyses of completed randomized trials.[187-189]

The MMAI test also has developed a potential predictive model to guide the use of short-term ADT. The predictive model was validated in RTOG 9408, a randomized trial of low-dose radiotherapy with or without 4 months of ADT, and was shown to identify patients more likely to benefit from ADT.[190] A similar, but unique model also was developed that may predict the benefit of short-term versus long-term ADT and validated in RTOG 9202.[191] Given the considerable changes in staging, grading, and treatment since the start of RTOG 9202 and 9408 (>30 years ago), it is unclear if the predictive models still work in contemporarily treated patients, and further validation is recommended by the Panel.

The relevant disease settings and more details for this tool are available in the *Principles of Risk Stratification* in the algorithm above.

Initial Clinical Assessment and Staging Evaluation

For patients with a life expectancy of ≤5 years without clinical symptoms who have low- and intermediate-risk prostate cancer, further imaging and treatment should be delayed until symptoms develop, at which time imaging can be performed and ADT should be given. Those with a life expectancy ≤5 years who fall into the high- or very-high-risk groups should undergo bone imaging and, if indicated by nomogram prediction of lymph node involvement, pelvic +/- abdominal imaging.

For symptomatic patients and those with unfavorable intermediate-risk, high-risk, and very-high-risk prostate cancer and a life expectancy of >5 years, bone and soft tissue imaging is appropriate:

- PSMA-PET/CT or PSMA-PET/MRI can be considered for bone and soft tissue (full body) imaging.
    - Because of the increased sensitivity and specificity of PSMA-PET tracers for detecting micrometastatic disease compared to CT, MRI, and bone scan at both initial staging and biochemical recurrence, PSMA-PET/CT or PSMA-PET/MRI can serve as a more effective front-line imaging tool for these patients.
- Bone imaging can be achieved by conventional technetium-99m-MDP bone scan.

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- Plain films, CT, MRI, PSMA-PET/CT or PSMA PET/MRI, or PET/CT or PET/MRI with F-18 sodium fluoride, C-11 choline, F-18 fluciclovine can be considered for equivocal results on initial bone imaging.
- Soft tissue imaging of the pelvis, abdomen, and chest can include chest CT and abdominal/pelvic CT or abdominal/pelvic MRI. Multiparametric MRI (mpMRI) is preferred over CT for pelvic staging.

mpMRI may detect larger and clinically significant prostate cancer (Grade Group ≥2), extracapsular extension, and seminal vesicle invasion (T staging) and is preferred over CT for abdominal/pelvic staging. mpMRI has been shown to be equivalent to CT scan for pelvic lymph node evaluation.

See *Imaging Techniques* below for a more detailed discussion.

Imaging Techniques

Imaging techniques are useful for staging and for detecting metastases and tumor recurrence. Current clinical imaging techniques for prostate cancer include conventional radiography (ie, x-rays), ultrasound, CT, MRI, single photon emission computed tomography (SPECT, scintigraphy), and PET. Some of these modalities have the ability to assess both anatomy and tumor function/biology. For example, functional MR sequences can be added to conventional anatomic MR sequences in a clinical examination such as diffusion-weighted imaging (DWI) to assess tumor cellularity or MR spectroscopy (MRS) to assess tumor metabolism.

Different modalities can also be merged to maximize prostate cancer assessment. For example, the functional information obtained with PET can be combined with the spatial and anatomic information from either CT (ie, PET/CT) or MRI (ie, PET/MRI) to inform about the locations of tumor foci for diagnosis or therapy response. Another example of the advantage of combining modalities is MR-ultrasound fusion guided biopsy (eg, MR-TRUS), where MRI datasets containing information on suspicious lesions identified by the radiologist are used by the urologist to navigate ultrasound-guided biopsies of the prostate for more accurate diagnosis.[192] More details on each technique are outlined in the algorithm under *Principles of Imaging*, above.

Multiparametric MRI (mpMRI)

The use of mpMRI in the staging and characterization of prostate cancer has increased in the last few years. mpMRI examinations typically include three sequences: T2-weighted imaging, DWI, and dynamic contrast enhancement (DCE) imaging. There has been increased interest in biparametric imaging that excludes the use of gadolinium contrast in prostate MRI examinations; however, more data are needed to identify the risk groups who would benefit most from this approach.[193] In general, it is recommended that mpMRI be performed on a 3 Tesla (3T) magnetic strength MRI scanner. This is the highest strength scanner in routine clinical use and provides the best possible evaluation of prostate cancer.

Additional instrumentation can be used, such as an endorectal coil (ERC) to improve image quality. If a lower strength (eg, 1.5T) MRI is required for a patient because of indwelling medical device incompatibility with 3T MRI, an ERC is recommended. Use of ERC in routine prostate imaging is controversial. Current data suggest that a 3T exam with ERC may not be significantly better than a 3T exam without ERC. Moreover, there may not be a significant difference in image interpretation between a 1.5T with ERC and 3T without ERC.[194] The use of ERC in prostate MRI also introduces new problems into the clinical workflow including patient discomfort, prostate distortion, increased scanner time and expense, and requirement of someone experienced to place the ERC.

Evidence supports the implementation of mpMRI in several aspects of prostate cancer management.[192] *First*, mpMRI helps detect larger and/or

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more clinically significant cancers (ie, Grade Group ≥2).[195] mpMRI has been incorporated into MRI-TRUS fusion-targeted biopsy protocols, which has led to an increase in the diagnosis of high-grade cancers with fewer biopsy cores, while reducing detection of low-grade and insignificant cancers.[196-198] In fact, a recently published clinical study identified that MRI-targeted biopsy synergized with conventional systematic biopsy to identify more clinically significant cancers.[199] *Second*, mpMRI aids in better assessment of extracapsular extension (T staging), with high negative predictive values (NPVs) in patients with low-risk disease.[200] mpMRI results may inform decision-making regarding nerve-sparing operation.[201] *Third*, mpMRI is equivalent to CT scan for staging of pelvic lymph nodes.[202,203] *Finally*, mpMRI outperforms bone scan and targeted x-rays for detection of bone metastases, with a sensitivity of 98% to 100% and specificity of 98% to 100% (vs. sensitivity of 86% and specificity of 98%–100% for bone scan plus targeted x-rays).[204]

PET Imaging

The use of PET/CT or PET/MRI imaging using tracers other than F-18 fluorodeoxyglucose (FDG) for staging of small-volume recurrent or metastatic prostate cancer has rapidly expanded in recent years.[192] Currently, there are several PET tracers that are FDA approved for use in patients with prostate cancer: Ga-68 PSMA-11 (PSMA-HBED-CC), F-18 piflufolastat (DCFPyL), F-18 flotufolastat PSMA (rh-PSMA-7.3), C-11 choline, F-18 fluciclovine, and F-18 sodium fluoride. Although these tracers are approved for the evaluation of patients with biochemical recurrence, the PSMA tracers are also approved for patients at initial staging with suspected metastatic disease. Tracer distribution in patients with prostate cancer can be imaged with either PET/CT or PET/MRI modalities. Although CT and MRI are equivalent in the assessment of lymphadenopathy, PET/MRI has the added advantage over PET/CT with enhanced tissue contrast that is especially important in evaluation of pelvic anatomy and prostate cancer assessment. Table 1 in the *Principles of Imaging* above summarizes the FDA-cleared PET imaging tracers studied in prostate cancer.

PSMA-PET refers to a growing body of radiopharmaceuticals that target prostate specific membrane antigen (PSMA) on the surface of prostate cells. Because of the high density of PSMA receptors on the surface of cancer cells relative to adjacent prostate, PSMA-PET has the advantage of high signal-to-noise relative to adjacent tissues. The mechanistic role of androgen receptor signaling in PSMA regulation is still being investigated, as multiple reports in animals and humans suggest that androgen modulation can affect PSMA expression and may even be dichotomous in patients with castration-sensitive versus castrate-resistant disease.[205-207] There are multiple PSMA radiopharmaceuticals at various stages of investigation. At this time, the NCCN Guidelines only recommend the currently FDA-approved PSMA tracers: F-18 piflufolastat, F-18 flotufolastat, and Ga-68 PSMA-11. PSMA-PET/CT or PSMA-PET/MRI can be considered as an alternative to standard imaging of bone and soft tissue for initial staging, the detection of biochemically recurrent disease, and as workup for progression with bone scan plus CT or MRI for the evaluation of bone, pelvis, and abdomen.

Studies suggest that PSMA-PET imaging has a higher sensitivity than C-11 choline or F-18 fluciclovine PET imaging, especially at very low PSA levels.[208-215] The reported sensitivity and specificity for PSMA-11 PET/CT in the detection of nodal involvement in primary staging of patients with intermediate-, high-, and very-high-risk disease is 40% and 95%, respectively.[216] The patient-level positive predictive value (PPV) in detection of lesions in patients with biochemical recurrence (BCR) is 92%.[217] Similarly, the reported sensitivity and specificity for piflufolastat PET/CT in the detection of nodal involvement in primary staging of patients with unfavorable intermediate-, high-, and very-high-risk disease are 31% to 42% and 96% to 99%, respectively.[218,219] The patient-level

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correct localization rate (CLR; patient-level PPV validated by anatomic lesion co-localization) for piflufolastat PET/CT is 85% to 87%.[220] Thus, PSMA-11 and piflufolastat are considered equivalent. F-18 flotufolastat PSMA is part of a novel class of tracers referred to as radiohybrid (rh) ligands. These rh ligands have the unique advantage of offering two binding sites for radionuclides (ie, F-18 or Ga-68), which increases its flexibility in imaging. In addition, the presence of a chelator in these rh ligands also allows for chelation of Lu-177 for its use as a theranostic as well as imaging agent.

Because of the increased sensitivity and specificity of PSMA-PET tracers for detecting micrometastatic disease compared to CT, MRI, and bone scan at both initial staging and biochemical recurrence, PSMA PET/CT or PSMA-PET/MRI can serve as a more effective front-line imaging tool for these patients.

PET/CT or PET/MRI detect small-volume disease in bone and soft tissues.[221,222] The reported sensitivity and specificity of C-11 choline PET/CT in restaging patients with biochemical recurrence ranges from 32% to 93% and from 40% to 93%, respectively.[223-232] The reported sensitivity and specificity of F-18 fluciclovine PET/CT ranges from 37% to 90% and from 40% to 100%, respectively.[229,233,234] A prospective study compared F-18 fluciclovine and C-11 choline PET/CT scans in 89 patients, and agreement was 85%.[229] Thus, choline and fluciclovine are considered equivalent in the evaluation of patients with biochemical recurrence. The Panel believes that F-18 fluciclovine PET/CT or PET/MRI or C-11 choline PET/CT or PET/MRI may be used in patients with biochemical recurrence after primary treatment for further soft tissue and/or bone evaluation after bone scan, chest CT, and abdominal/pelvic CT or abdominal/pelvic MRI.

The use of these PET tracers can lead to changes in clinical management. The FALCON trial showed that results of F-18 fluciclovine PET/CT in 104 patients with biochemical recurrence after definitive therapy resulted in a change in disease management for 64% of patients.[235] In addition, the LOCATE trial demonstrated that fluciclovine frequently changed disease management plans for patients with biochemical recurrence.[236] In a similar fashion, data also show that PSMA-PET has the ability to change radiation treatment planning in 53% (N = 45) of patients with high- and very-high-risk prostate cancer using PSMA-11 as well as change disease management in over half of a prospective cohort of 635 patients with BCR.[237,238] However, whether changes to treatment planning because of PET tracers have an impact on long-term survival remains to be studied.

The increasing use of PSMA-PET has identified the potential for considerable biological diversity among disease foci within a given individual with prostate cancer, especially in the mCRPC setting. This heterogeneity can be detected with a combination of PSMA-PET and fluorodeoxyglucose (FDG)-PET. Initial data suggest that metastases with PSMA-negative/FDG-positive mismatches may exist in patients with mCRPC undergoing Lu-177-PSMA-617 radioligand therapy and that patients with these mismatches may have worse outcomes. To overcome the limitations of PSMA-PET, the Panel currently recommends the use of MRI or a dedicated high resolution diagnostic contrast-enhanced CT in those with a negative PSMA-PET scan, because the non-contrast CT component of PSMA-PET/CT is insufficient to detect visceral metastatic disease, especially in the liver, due to its high physiological uptake.

F-18 sodium fluoride targets osteoblast activity where the fluoride is deposited into new bone formation, thus limiting use of this agent to the detection of osseous metastases. Fluoride PET/CT has greater sensitivity than standard bone scintigraphy in the detection of bone metastases, with 77% to 94% sensitivity, 92% to 99% specificity, and 82% to 97% PPV.[239] However, emerging evidence indicates that other tracers such as PSMA are at least equivalent to fluoride in the detection of osseous metastases with the added advantage of soft tissue metastasis detection.[240]

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Histologic or radiographic confirmation of involvement detected by PET imaging is recommended whenever feasible due to the presence of false positives. Although false positives exist, literature suggests that these are outweighed by the increase in true positives detected by PET relative to bone scintigraphy. To reduce the false-positive rate, physicians should consider the intensity of PSMA-PET uptake and correlative CT findings in the interpretation of scans. Several reporting systems have been proposed but have not been validated or widely used.[241,242] Moreover, although PET imaging may change treatment,[236] it may not change oncologic outcome. Earlier detection of bone metastatic disease, for instance, may result in earlier use of newer and more expensive therapies, which may not improve oncologic outcomes or OS.

Risks of Imaging

Medical imaging is a critical tool in the evaluation and comprehensive care of patients with malignancy. However, as with any medical procedure, imaging is not without risk. Some of these risks are concrete and tangible, while others are less clear. Risks associated with imaging include exposure to ionizing radiation, adverse reaction to contrast media, false-positive scans, and overdetection. Inappropriate use of imaging also has been identified as a significant contributor to health care costs in the United States and worldwide. Therefore, imaging should be performed only when medically appropriate, and in a manner that reduces risk (eg, minimizing radiation dose). An algorithmic approach to the use of imaging, such as by NCCN and the Appropriateness Criteria developed by the American College of Radiology,[243] can assist in medical decision-making.

Exposure to Ionizing Radiation

Deterministic and stochastic are two types of effects from exposure to ionizing radiation by x-ray, CT, or PET/CT. Deterministic effects are those that occur at a certain dose level and include events such as cataracts and radiation burns. No effect is seen below the dose threshold. Medical imaging is always performed almost below the threshold for deterministic effects. Stochastic effects tend to occur late, increase in likelihood as dose increases, and have no known lower “safe” limit. The major stochastic effect of concern in medical imaging is radiation-induced malignancy. Unfortunately, no direct measurements are available to determine risk of cancer arising from one or more medical imaging events, so risks are calculated using other models (such as from survivors of radiation exposure). The literature is conflicting about the precise risk of secondary malignancies in patients undergoing medical imaging procedures. There is a small but finite risk of developing secondary malignancies because of medical imaging procedures, and the risk is greatest in young patients. However, the absolute risk of fatal malignancy arising from a medical imaging procedure is very low and is difficult to detect given the prevalence of cancer in the population and the multiple factors that contribute to oncogenesis.[244] Efforts should be made to minimize dose from these procedures, which begin with judicious use of imaging only when justified by the clinical situation. Harm may arise from not imaging a patient, from disease non-detection or erroneous staging.

Adverse Reaction to Contrast Media

Many imaging studies use contrast material delivered by oral, intravenous, or rectal routes. The use of contrast material may improve study performance, but reactions to contrast material may occur and they should therefore be used only when warranted. Some patients develop adverse reactions to iodinated intravenous contrast material. Most reactions are mild cutaneous reactions (eg, urticaria, pruritus) but occasionally severe reactions can be life-threatening (bronchospasm or anaphylaxis). The risk of severe reaction is low with non-ionic contrast materials.[245] Both iodinated CT contrast material and gadolinium-based MR contrast materials can be problematic in patients with reduced renal function. Gadolinium MR contrast media, in particular, is contraindicated in patients with acute renal failure or stage V chronic kidney disease (glomerular

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filtration rate [GFR] <15).[245] Patients in this category are significantly more likely to develop nephrogenic systemic fibrosis (NSF). Centers performing imaging studies with contrast materials should have policies in place to address the use of contrast in these patients.

False-Positive Scans and Overdetection

Every imaging test has limitations for sensitivity, specificity, and accuracy that involve both the nature of the imaging modality as well as the interpretation by the physician. Harm can arise when a tumor or tumor recurrence is not detected (ie, false negative), but harm to the patient and added expense to the medical system also can result from false-positive scans. Extensive workup of imaging findings that may be benign or indolent (ie, overdetection) can lead to significant patient anxiety, additional and unnecessary imaging, and invasive procedures that carry their own risks for adverse outcomes.

Accurate and medically relevant interpretation of imaging studies requires familiarity and expertise in the imaging modality, attention to detail in image review, knowledge of tumor biology, and familiarity with treatment options and algorithms. Challenging cases are best addressed through direct communication, either physician-to-physician or in a multidisciplinary tumor board setting.

The increased false positive detection of benign rib lesions with PSMA PET/CT is well established. If these are called metastatic foci then patients may be inappropriately recategorized from localized prostate cancer where they may receive curative-intent local therapy, to now receive doublet systemic therapy. Thus, expert interpretation of PSMA PET/CT scans is essential to avoid misclassification.

Observation

Observation involves monitoring the course of prostate cancer with a history and physical exam no more often than every 12 months (without surveillance biopsies) until symptoms develop or are thought to be imminent. If patients under observation become symptomatic, an assessment of disease burden can be performed, and treatment or palliation can be considered. Observation thus differs from active surveillance. The goal of observation is to maintain quality of life (QOL) by avoiding noncurative treatment when prostate cancer is unlikely to cause mortality or significant morbidity. The main advantage of observation is avoidance of possible side effects of routine biopsies and unnecessary definitive therapy or ADT. However, patients may develop urinary retention or pathologic fracture without prior symptoms or increasing PSA level.

Observation is applicable to patients who are older or in declining health with comorbidity that are more likely than prostate cancer to be the cause of death. Johansson and colleagues[246] observed that only 13% of patients developed metastases 15 years after diagnosis of T0–T2 disease and only 11% had died from prostate cancer. Because prostate cancer will not be treated for cure for patients with shorter life expectancies, observation for as long as possible is a reasonable option based on physician discretion. When symptoms develop or are imminent, patients can begin palliative treatment.

The SPCG-4 and PIVOT trials compared the use of observation versus radical prostatectomy (RP) in patients with localized prostate cancer.[247-249] Results from SPCG-4 did not demonstrate a significant reduction in prostate cancer-specific mortality with RP compared to observation in patients with low-risk disease.[247] In PIVOT, there was no difference in all-cause mortality between these two management strategies in those with low-risk disease, strongly implying that treatment of low-risk prostate cancer should generally be avoided. RP was superior to observation for patients with intermediate-risk disease. However, RP did not significantly improve death from any cause in those with high-risk disease in either trial.[247,249]

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Active Surveillance

The NCCN Guidelines Panel remains concerned about the problems of overtreatment related to the increased frequency of diagnosis of prostate cancer from widespread use of PSA for early detection or screening (see the NCCN Guidelines for Prostate Cancer Early Detection, available at www.NCCN.org). The debate about whether it is necessary to diagnose and treat every individual who has prostate cancer is fueled by the high prevalence of prostate cancer upon autopsy of the prostate[250]; the high frequency of positive prostate biopsies in individuals with normal DREs and serum PSA values[251]; the contrast between the incidence and mortality rates of prostate cancer; and the need to treat an estimated 37 patients with screen-detected prostate cancer[252,253] or 100 patients with low-risk prostate cancer[254] to prevent one death from the disease.

Several large, population-based prostate cancer screening trials have shown that properly applied early detection protocols can reduce prostate cancer mortality (see the NCCN Guidelines for Prostate Cancer Early Detection, available at www.NCCN.org). However, wide-spread screening has led to overdetection of clinically insignificant prostate cancer (grade group 1). A growing body of high-level evidence supports the use of mpMRI to reduce unnecessary biopsies in individuals with an elevated PSA who are not likely to have clinically significant prostate cancer. Its use should decrease the risk of overdetection while maintaining the detection of potentially fatal prostate cancer. These improvements coupled with the use of active surveillance in appropriate patients should reduce overtreatment while preserving the relatively low rates of prostate cancer mortality.

Models of prostate cancer detection and progression from the early 2000's estimated that 23% to 42% of all U.S. screen-detected cancers were overtreated[255] and that PSA detection was responsible for up to 12.3 years of lead-time bias.[256] Encouragingly, the proportion of patients with low-risk prostate cancer choosing active surveillance has increased over time, and was at approximately 60% in 2021.[257-259] This trend has also been observed in the Veterans Affairs Integrated Health Care System, in which active surveillance increased from 2005 to 2015: from 4% to 39% of those <65 years and from 3% to 41% of those ≥65 years.[260]

Active surveillance involves actively monitoring the course of the disease with the expectation to deliver curative therapy if the cancer progresses. Unlike observation, active surveillance is mainly applicable to patients with sufficient life expectancy who may benefit from early detection of cancer progression and radical treatment. Because these patients have a longer life expectancy, they should be followed closely, and treatment should start promptly should the cancer progress so as not to miss the chance for cure.

Early versions of active surveillance were termed active monitoring. This approach differs from active surveillance as it primarily only relies on PSA monitoring, whereas active surveillance typically involved serial prostate biopsies and potential for serial imaging.

The ProtecT study, which randomized 1643 patients with localized prostate cancer to active monitoring, radical prostatectomy, or RT, found no significant difference in the primary outcome of prostate cancer mortality at a median of 10 years follow-up.[261] Of 17 prostate cancer deaths (1% of study participants), 8 were in the active surveillance group, 5 were in the surgery group, and 4 were in the radiation group (P = .48 for the overall comparison). However, a 12.2% absolute increase in the rate of disease progression and a 3.4% absolute increase in the rate of metastases or prostate cancer death were seen in the active surveillance group.[261,262] Approximately 23% of participants had Gleason scores 7–10, and 5 of 8 deaths in the active surveillance group were in this subset. Longer follow-up has been reported with a median follow-up of 15-years, and overall survival remains similar across the 3 groups.[263] However,

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metastases were more common in the active monitoring group (9.4% vs. ~5% in the surgery and RT groups), primarily driven by the patients with intermediate-risk disease who were included in the study. By 15-years, approximately 70% of patients in the active monitoring arm had received treatment.

An updated analysis from the ProtecT trial of 712 patients that had pathology available for re-review was reported.[264] The authors demonstrated that 13.1% of patients had cribriform-positive disease, which significantly increased the risk of developing metastases (HR, 3.61; 95% CI, 1.60–8.11). RT with neoadjuvant ADT significantly reduced metastasis risk (HR, 0.35; 95% CI, 0.16–0.78; $P = .04$), whereas surgery did not significantly improve long-term outcomes compared with active monitoring (15-year cumulative incidence in patients with cribriform-positive disease: 26% for surgery, 25% for active monitoring, and 8% for RT plus ADT).

Patient-reported outcomes were compared among the 3 groups of ProtecT.[25,265] The surgery group experienced the greatest negative effect on sexual function and urinary continence, whereas the RT group had a slight decrease in bowel function. It has been noted that the RT utilized in the trial was 3D radiotherapy without image guidance or use of hydrogel spacers.[266] In addition, studies have shown that active surveillance does not adversely impact psychological well-being or QOL.[25,267-271]

Several large active surveillance cohort studies have shown that >50% of those eligible for active surveillance may safely avoid treatment, and thus the possible associated side effects of treatment, for at least 10 years (see *Table 1* below).] For example, in one study, 55% of the population remained untreated at 15 years.[272] Although a proportion of patients on active surveillance will eventually undergo treatment, the delay does not appear to impact cure rates, and numerous studies have shown that active surveillance can be a safe option for many patients.[272-282] In fact, a 2015 meta-analysis of 26 active surveillance cohort studies that included 7627 patients identified only 8 prostate cancer deaths and 5 cases of metastasis.[283]

For low-risk prostate cancer, active surveillance is preferred.

The favorable intermediate risk grouping was not developed to identify optimal active surveillance, but rather to identify patients with intermediate-risk disease who can safely receive RT alone without ADT. The favorable-intermediate-risk group includes patients that are well suited for active surveillance, such as those with Grade group 1 and PSA <10 ng/mL, with a low PSA density. However, it also includes patients with multiple cores of Grade group 2 disease with cribriform features and a high PSA density. Patients should be counseled that unselected Grade group 2 intermediate-risk prostate cancer has been shown to have unacceptably high risks of metastasis when managed by observation, active monitoring, or active surveillance.[284] Patients with low-volume disease (ie, 1–2 positive cores), low percent or absolute Gleason pattern 4, lack of cribriform features, and low PSA density are likely better candidates for active surveillance among patients with Grade group 2 disease.[264]

Therefore, for select patients with favorable intermediate-risk disease, a recommendation for active surveillance must be based on careful individualized weighing of several factors: life expectancy, general health condition, disease characteristics, potential side effects of treatment, and patient preference. Shared decision-making, after appropriate counseling on the risks and benefits of the various options, is critical. Data in these settings are discussed under *Patient Selection*, below.

The Panel believes there is an urgent need for further clinical research regarding the criteria for recommending active surveillance, the criteria for reclassification on active surveillance, and the schedule for active surveillance especially as it pertains to prostate biopsies, which pose an increasing burden.

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Patient Selection

NCCN risk groups are the main determinant used to select patients who can safely defer treatment.

Active Surveillance in Low-Risk Disease

Panel consensus is that active surveillance is preferred for most patients with low-risk prostate cancer and a life expectancy ≥10 years based on the data discussed above. However, the Panel recognizes that there is heterogeneity across the low-risk group, and that some factors may be associated with an increased probability of near-term grade reclassification including high PSA density, a high number of positive cores (eg, ≥3), high gene expression risk (from tissue-based molecular tumor analysis).[285-287] Of note, core involvement in the major active surveillance cohort studies was generally low (see *Table 1* below) and very few patients with grade group 2 disease were included. Therefore, in very select patients with low-risk prostate cancer, upfront treatment with radical prostatectomy or RT may be appropriate based on shared decision-making with the patient. Patient anxiety should be addressed so that it is not a deterrent to the appropriate use of active surveillance.

Active Surveillance in Favorable Intermediate-Risk Disease

The literature on outcomes of active surveillance in patients with intermediate-risk prostate cancer is limited because few prospective studies of active surveillance included patients with intermediate-risk prostate cancer (see *Table 1*, below).

In prospective studies that included patients with intermediate-risk disease, favorable prostate cancer-specific survival rates of 94% to 100% were seen for the full cohorts.[272,274,276] However, with extended follow-up, the Toronto group has demonstrated inferior metastasis-free survival for patients with intermediate-risk prostate cancer (15-year metastasis-free survival for cases of Gleason 6 or less with PSA <10 ng/mL, 94%; Gleason 6 or less with PSA 10–20 ng/mL, 94%; Gleason 3+4 with PSA 20 ng/mL or less, 84%; and Gleason 4+3 with PSA 20 ng/mL or less, 63%).[288] Furthermore, there was an unacceptably high 15-year distant metastasis rate (16.4%) that largely occurred between 10 to 15 years after diagnosis.[288] Likewise, Canary PASS included a small subset of patients with grade group 2 disease, but only 2 patients had 15 years of follow-up. Two patients with grade group 2 prostate cancer developed metastatic disease between 9 and 12 years,[289] with an estimated 15-year risk of distant metastasis of approximately 10%.[284] Thus, 5 to 10 years of follow-up may be insufficient to assess active surveillance safety.

Overall, the Panel interpreted these data to show that a subset of patients with favorable intermediate-risk prostate cancer and life expectancy >10 years may be considered for active surveillance, especially with grade group 1 disease. However, the precise inclusion criteria and follow-up protocols need continued refinement. Patients must understand that a significant proportion of those clinically staged as having favorable intermediate-risk prostate cancer may have higher risk disease.[290-293] Particular consideration to active surveillance may be appropriate for those patients with a low percentage of Gleason pattern 4 cancer, low tumor volume, low PSA density, and/or low genomic risk (from tissue-based molecular tumor analysis), but should be approached with caution, include informed decision-making, and use close monitoring for progression. How to optimize active surveillance for patients with grade group 2 prostate cancer remains an active area of investigation.

Confirmatory Testing

Confirmatory testing is critical to the process of active surveillance because a subset of patients will be found to no longer be candidates for active surveillance after confirmatory testing. Active surveillance starts after confirmatory testing. Since an initial prostate biopsy may underestimate tumor grade or volume, confirmatory testing is strongly

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recommended within the first 6 to 12 months of diagnosis for patients who are considering active surveillance.

Before starting on an active surveillance program, mpMRI with calculation of PSA density should be considered to confirm candidacy for active surveillance if not performed during initial workup.[294] Patients with PI-RADS 4 or 5 on mpMRI have an increased risk of biopsy progression during active surveillance.[295]

In patients with low- and favorable-intermediate-risk disease, molecular tumor analysis has not been shown in prospective studies to clearly improve long-term outcomes for patients receiving active surveillance. One study examined the role of molecular tumor analysis for predicting upgrading on surveillance biopsy or the presence of adverse pathology on eventual radical prostatectomy in patients in an active surveillance cohort.[177] In this study, results of the molecular testing did not significantly improve risk stratification over the use of clinical variables alone.

If results of mpMRI and/or molecular testing are concerning, a repeat biopsy may be appropriate.

Early confirmatory testing may not be necessary in patients who have had a complete workup including mpMRI prior to diagnostic biopsy. However, all patients should undergo a confirmatory prostate biopsy within 1 to 2 years of their diagnostic biopsy.

Active Surveillance Program

The current NCCN recommendations for the active surveillance program include PSA no more often than every 6 months unless clinically indicated; DRE no more often than every 12 months unless clinically indicated; repeat prostate biopsy no more often than every 12 months unless clinically indicated; and consideration of repeat mpMRI no more often than every 12 months unless clinically indicated. mpMRI may be particularly useful to exclude the presence of anterior cancer when PSA levels increase and systematic prostate biopsies remain negative.[296]

Results of a study of 211 patients with Grade Group 1 prostate cancer who had initial and repeat mpMRIs and PSA monitoring suggest that a negative initial mpMRI predicts a low risk of Gleason upgrading by systematic biopsy.[297] In addition, PSA velocity was significantly associated with subsequent progression in those with an initial negative mpMRI. In contrast, those with high-risk visible lesions on mpMRI before initiation of active surveillance had an increased risk of progression. A meta-analysis of 43 studies found the sensitivity and NPV for mpMRI to be 0.81 and 0.78, respectively.[298] An analysis of patients in Canary PASS found that mpMRI had an NPV and PPV for detecting Grade Group ≥2 cancer of 83% and 31%, respectively.[299] Another study found the NPV of mpMRI to be 80%.[300]

In patients with a suspicious lesion on mpMRI, MRI-US fusion biopsy improves the detection of higher grade (Grade Group ≥2) cancers. Early experience supports the utilization of mpMRI in biopsy protocols to better risk stratify patients under active surveillance.[301-303] However, more recent studies have shown that a significant proportion of high-grade cancers are detected with systematic biopsy and not targeted biopsy in patients on active surveillance.[304-306]

Whereas the intensity of surveillance may be tailored on an individual basis (eg, based on life expectancy and risk of reclassification), most patients should have prostate biopsies incorporated as part of their monitoring, but no more often than every 12 months, because PSA kinetics may not be reliable for predicting progression. Repeat biopsy is useful to determine whether higher Gleason grade exists, which may influence prognosis and hence the decision to continue active surveillance or proceed to definitive local therapy.[307] A repeat prostate biopsy should also be considered if the prostate exam changes, if mpMRI (if done)

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suggests more aggressive disease, or if PSA increases. However, literature suggests that as many as 5% of patients undergoing prostate biopsy will suffer an adverse event, including serious ones such as infections requiring hospitalization.[308-310] Therefore, many clinicians choose to wait 2 years for a biopsy if there are no signs of progression.

Patients should be transitioned to observation (see *Observation*, above) when life expectancy is <10 years and they have low risk disease.

Considerations for Treatment of Patients on Active Surveillance

Reliable parameters of prostate cancer progression await the results of ongoing clinical trials. PSADT is not considered reliable enough to be used alone to detect disease progression.[311] If repeat biopsy shows Grade Group ≥3 disease, or if tumor is found in a greater number of biopsy cores or in a higher percentage of a given biopsy core, cancer progression may have occurred. Grade reclassification on repeat biopsy is the most common factor influencing a change in management approach from active surveillance to treatment. Other factors affecting decisions to actively treat patients on active surveillance include increase in tumor volume and a rise in PSA density. Considerations for a change in management strategy should be made in the context of the patient's life expectancy. Patient anxiety over continued active surveillance should be addressed if treatment does not appear warranted.

Each of the major active surveillance series has used different criteria for reclassification.[272,275,278-281,312-315] Reclassification criteria were met by 23% of patients with a median follow-up of 7 years in the Toronto experience,[313] 36% of patients with a median follow-up of 5 years in the Johns Hopkins experience,[275] and 16% of patients with a median follow-up of 3.5 years in the University of California, San Francisco (UCSF) experience[281] (also see *Table 1*, below). Uncertainty regarding reclassification criteria and the desire to avoid missing an opportunity for cure drove several reports that dealt with the validity of commonly used reclassification criteria. The Toronto group demonstrated that a PSA trigger point of PSADT <3 years could not be improved upon by using a PSA threshold of 10 or 20, PSADT calculated in various ways, or PSA velocity >2 ng/mL/y.[316] The Johns Hopkins group used biopsy-demonstrated reclassification to Gleason pattern 4 or 5 or increased tumor volume on biopsy as their criteria for reclassification. Of 290 patients on an annual prostate biopsy program, 35% demonstrated reclassification at a median follow-up of 2.9 years.[317] Neither PSADT (area under the curve [AUC], 0.59) nor PSA velocity (AUC, 0.61) was associated with prostate biopsy reclassification. Both groups have concluded that PSA kinetics cannot replace regular prostate biopsy, although treatment of most patients who demonstrate reclassification on prostate biopsy prevents evaluation of biopsy reclassification as a criterion for treatment or reduction of survival. Treatment of all patients who developed Gleason pattern 4 on annual prostate biopsies has thus far resulted in only two prostate cancer deaths among 1298 patients (0.15%) in the Johns Hopkins study.[275] However, it remains uncertain whether treatment of all who progressed to Gleason pattern 4 was necessary. Studies remain in progress to identify the best trigger points when interventions with curative intent may still be successful.

The patients in the Toronto cohort were followed for a median of 6.8 years.[313] Of the 30% (n = 145) of patients who progressed, 8% had an increase in Gleason grade, 14% had a PSADT <3 years, 1% developed a prostate nodule, and 3% were treated because of anxiety. One hundred thirty-five of these 145 patients were treated: 35 by radical prostatectomy, 90 by external beam RT (EBRT) with or without ADT, and 10 with ADT alone. The 10- and 15-year actuarial cause-specific survival rates for the entire cohort were 98.1% and 94.3%, respectively.[272] Only 15 of 993 (1.5%) patients had died of prostate cancer, an additional 13 patients (1.3%) had developed metastatic disease, and only 36.5% of the cohort

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had received treatment by 10 years. In an analysis of 592 patients enrolled in this cohort who had ≥1 repeat prostate biopsies, 31.3% of cases were upgraded. Fifteen percent of upgraded cases were upgraded to Gleason ≥8, and 62% of total upgraded cases proceeded to active treatment.[318] Another analysis of this cohort revealed that metastatic disease developed in 13 of 133 patients with Gleason 7 disease (9.8%) and 17 of 847 patients with Gleason ≤6 disease (2.0%).[319] PSADT and the number of positive scores were also predictors of increased risk for the development of metastatic disease.

In comparison, among 192 patients on active surveillance who underwent delayed treatment at a median of 2 years after diagnosis in the Johns Hopkins experience, 5-year biochemical PFS was 96% for those who underwent radical prostatectomy and 75% for those who underwent radiation.[315] The two groups were similar by pathologic Gleason grade, pathologic stage, and margin positivity. All patients treated by radical prostatectomy after progression on active surveillance had freedom from biochemical progression at a median follow-up of 37.5 months, compared to 97% of those in the primary radical prostatectomy group at a median follow-up of 35.5 months. A later publication from this group showed that 23 of 287 patients who were treated after active surveillance (8%) experienced biochemical recurrence, and the rate was independent of the type of treatment.[275] Several studies have shown that delayed radical prostatectomy does not increase the rates of adverse pathology.[279,320-322]

Radical Prostatectomy

Radical prostatectomy is an appropriate option for patients whose cancer appears clinically localized to the prostate and can safely undergo general anesthesia and an operation. Because of potential perioperative morbidity, radical prostatectomy should generally be reserved for patients whose life expectancy is ≥10 years.

Radical prostatectomy was compared to watchful waiting in a randomized trial of 695 patients with early-stage prostate cancer (mostly T2) in the SPCG-4 trial.[247,323] With a median follow-up of 12.8 years, those assigned to the radical prostatectomy group had significant improvements in disease-specific survival, OS, and risk of metastasis and local progression.[323] The reduction in mortality was confirmed at 18-year follow-up, with an absolute difference of 11%.[247] Overall, 8 patients needed to be treated to avert one death; that number fell to 4 for patients <65 years of age. Longer follow-up results were also reported, in which the cumulative incidence of death from prostate cancer was 19.6% and 31.3% in the radical prostatectomy and watchful waiting groups, respectively, at 23 years, with a mean increase of 2.9 years of life in the radical prostatectomy group.[248] However, in the small subset of patients with high-risk disease, there was no improvement in survival between treatment groups. A report of 30-year follow-up of this trial showed that the absolute difference in death due to prostate cancer between groups increased over time.[324] Radical prostatectomy resulted in a 48% lower risk of death from prostate cancer and a mean of 2.2 life-years gained. The authors concluded that treating six patients would result in one prostate cancer death averted. The results of this trial offer high-quality evidence to support radical prostatectomy alone as a treatment option for patients with intermediate-risk clinically localized prostate cancer.

While the SPCG-4 and PIVOT trials comparing observation to surgery did not clearly demonstrate an improvement in outcomes among those with high-risk disease, these trials enrolled few patients and are not adequately powered to rigorously assess the role of surgery in high-risk disease.[248,249] Trials that have utilized surgery in high-risk disease generally demonstrate a 70% to 80% risk of recurrence and thus surgery should be viewed as one component of a multi-modality approach.[325] For example, in the CALGB 90203 trial of patients who received surgery in the control arm,

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69% received adjuvant or secondary treatment, most commonly radiotherapy with or without ADT.

There remains no high-level evidence for the use of radical prostatectomy for patients with NCCN very-high-risk, regional/node-positive, or metastatic disease. Ideally surgery is used in the context of a clinical trial or when radiotherapy plus ADT is contraindicated. Surgery should be discussed as one component of a planned multimodality approach in these patients. There is an ongoing trial, SPCG-15, which aims to compare RP versus RT plus ADT in patients with high-risk prostate cancer.[326]

Radical prostatectomy is also a treatment option for select patients experiencing a local recurrence after primary EBRT, but morbidity (incontinence, erectile dysfunction, bladder neck contracture, and rectal injury) remain significantly higher than when radical prostatectomy is used as initial therapy.[327,328] Patient selection is important, and post-RT recurrence radical prostatectomy should only be performed by highly experienced surgeons.

Operative Techniques and Adverse Effects

The retropubic and the perineal approaches are reasonable to perform a radical prostatectomy. Surgeons who perform a high-volume of operations in high-volume centers generally achieve superior outcomes.[329,330] Laparoscopic and robot-assisted radical prostatectomy are commonly used and are considered comparable to conventional approaches in experienced hands.[331-333] In a cohort study using SEER Medicare-linked data on 8837 patients, minimally invasive compared to open radical prostatectomy was associated with shorter length of hospital stay, less need for blood transfusions, and fewer surgical complications, but rates of incontinence and erectile dysfunction were higher.[334] A second large study reported no difference in overall complications, readmission, and additional cancer therapies between open and robot-assisted radical prostatectomy, although the robotic approach was associated with higher rates of genitourinary complications and lower rates of blood transfusion.[335] Oncologic outcomes of a robotic versus open approach were similar when assessed by use of additional therapies[334] or rate of positive surgical margins,[336] although longer follow-up is necessary. A meta-analysis of 19 observational studies (n = 3893) reported less blood loss and lower transfusion rates with minimally invasive techniques than with open operation.[336] Risk of positive surgical margins was the same. Two more recent meta-analyses showed a statistically significant advantage in favor of a robotic approach compared to an open approach in 12-month urinary continence[337] and potency recovery.[338] Early results from a randomized controlled phase 3 study comparing robot-assisted laparoscopic radical prostatectomy and open radical retropubic prostatectomy in 326 patients were published in 2016.[339,340] Urinary function and sexual function scores and rates of postoperative complications did not differ significantly between the groups at 6, 12, and 24 months after surgery. Rates of positive surgical margins were similar, based on a superiority test (10% in the open group vs. 15% in the robotic group). Assessment of oncologic outcomes from this trial will be limited because postoperative management and additional cancer therapies were not standardized between the groups.[339]

Return of urinary continence after radical prostatectomy may be improved by preserving the urethra beyond the prostatic apex and by avoiding damage to the distal sphincter mechanism. Bladder neck preservation may allow more rapid recovery of urinary control.[341] Anastomotic strictures that increase the risk of long-term incontinence are less frequent with modern surgical techniques. Recovery of erectile function is related directly to the degree of preservation of the cavernous nerves, age at surgery, and preoperative erectile function. Improvement in urinary and sexual function has been reported with nerve-sparing techniques.[342,343]

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Replacement of resected nerves with nerve grafts does not appear to be effective for patients undergoing wide resection of the neurovascular bundles.[344] The ability of mpMRI to detect extracapsular extension can aid in decision-making in nerve-sparing surgery.[201]

Common adverse events after RP include urinary incontinence, erectile dysfunction, penile shortening, and urethral or bladder neck contracture or stricture. The rates of occurrence vary by patient population (eg, by age and risk group), surgeon skill, and surgical technique (eg, unilateral vs. bilateral nerve sparring). In the ProtecT trial and PACE-A, which both randomized patients to RP versus RT, RP was associated with 40% to 60% risk of erectile dysfunction and 20% to 50% risk of urinary incontinence requiring at least one pad per day.[25,345]

Pelvic Lymph Node Dissection

PLND can be considered with a radical prostatectomy for patients with favorable intermediate, unfavorable intermediate, high, or very-high-risk prostate cancer. If performed, it is recommended to be an extended PLND.[346,347] An extended PLND includes removal of all node-bearing tissue from an area bounded by the external iliac vein anteriorly, the pelvic side wall laterally, the bladder wall medially, the floor of the pelvis posteriorly, Cooper's ligament distally, and the internal iliac artery proximally. Removal of more lymph nodes using the extended technique has been associated with increased likelihood of finding lymph node metastases, thereby providing more complete staging.[348-350] A survival advantage with more extensive lymphadenectomy has been suggested by several studies, possibly due to elimination of microscopic metastases,[349,351-353] although definitive proof of oncologic benefit is lacking.[354] PLND can be performed safely laparoscopically, robotically, or as an open procedure, and complication rates should be similar among the three approaches.

Radiation Therapy

RT techniques used in prostate cancer can be described generally by the following: delivery method (ie, internal vs. EBRT), source of radiation (ie, photon vs. proton), planning technique (ie, 3D vs. intensity modulated RT [IMRT]), image guidance (none vs. cone beam CT [CBCT] or MRI), adaptation (ie, online adaptive treatment), and fractionation (ie, conventional vs. moderate hypofractionation vs. stereotactic body radiotherapy).

External Beam Radiation Therapy

Over the past several decades, EBRT techniques have evolved to allow higher doses of radiation to be administered safely. EBRT can be delivered either with photons, also called x-rays, or proton beam. Photons are used in >95% of cases in the United States.

Proton Therapy

Proton therapy and x-ray–based therapies like IMRT can deliver highly conformal doses to the prostate. Proton-based therapies will deliver less low-dose radiation to some of the surrounding normal tissues not immediately adjacent to the prostate, such as muscle, bone, vessels, and fat. These tissues do not routinely contribute to the morbidity of prostate radiation and are relatively resilient to radiation injury; therefore, the benefit of decreased dose to these types of normal, non-critical tissues has not been clearly proven.

Only one randomized trial has been performed, the PartiQOL trial, and results are reported in abstract only.[355] This trial randomized 450 patients with low- and intermediate-risk prostate cancer to either proton or photons. After a median follow-up of 5-years, there were no significant differences in patient-reported quality of life, physician reported toxicity, or biochemical recurrence. Thus, both treatments appear to offer excellent tumor control and good safety profile.

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The costs associated with proton beam facility construction and proton beam treatment are high compared to the expense of building and using the more common photon linear accelerator-based practice.[356] These higher costs may translate into higher charges for protons, a point that should be considered when tumor control and side effects appear similar between photon and proton beam therapy for prostate cancer, especially if these charges are incurred by the patient.

The NCCN Panel believes no clear evidence supports a benefit or decrement to proton therapy over IMRT for either treatment efficacy or long-term toxicity. Conventionally fractionated or moderately hypofractionated prostate proton therapy can be considered a reasonable alternative to x-ray–based regimens at clinics with appropriate technology, physics, and clinical expertise. Insufficient data is present to recommend their delivery with ultra-hypofractionation.

Photon (X-ray) Beam

Treatment planning in the 1980s used two-dimensional planning techniques. By the 1990s this was largely replaced by three-dimensional planning. In the late 1990s IMRT planning techniques were increasingly utilized, and as of 2010 nearly all centers utilize IMRT planning. This technique has been shown to improve dose conformality.

In parallel, improvements in image guidance, also referred to as image guided RT (IGRT), have occurred. The purpose of IGRT is to ensure the planned dose is delivered as intended. Given the potential changes in anatomy day to day due to bladder and rectal filling, IGRT is essential to ensure accurate alignment of the prostate and target volumes and avoidance of organs at risk. IGRT techniques often include use of implanted fiducials, electromagnetic targeting and tracking, or endorectal balloon, which have variably been shown to decrease complications and allow margin reduction.[357] More recently, the use of real-time tracking and online adaptive radiotherapy have also demonstrated the ability to improve dosimetry and enable margin reduction, which may result in reduced toxicity.[358]

The advancements in treatment planning and IGRT have enabled greater comfort and confidence to test increasingly hypofractionated regimens. Historically, conventionally fractionated RT was utilized with 1.8 to 2.0 Gy per fraction over 8 to 9 weeks of daily RT. This schema has been compared in numerous randomized trials to moderately hypofractionated RT in 2.5 to 4 Gy/fraction, which has been shown to be noninferior when using iso-effective doses.[359-368] Recently, the HYDRA individual patient meta-analyses of these trials demonstrated that 60 Gy in 20 fractions, or regimens with a similar iso-effective dose, should be preferred.[369] Reduction from >40 treatments to 20 treatments reduces both financial toxicity and increases patient convenience.

The relatively slow proliferation rate of prostate cancer is reflected in a low α/β ratio, most commonly reported between 1 and 4.[370] Because the α/β ratio for prostate cancer is similar to or lower than the surrounding tissues responsible for most of the toxicity reported with radiation, appropriately designed radiation treatment fields and schedules using ultra hypofractionated regimens were hypothesized to result in similar cancer control rates without increased risk of late toxicity.

SBRT is a technique that delivers highly conformal, high-dose radiation in five or fewer treatment fractions, which are safe to administer only with precise, image-guided delivery.[371] Numerous prospective studies demonstrated safety and efficacy of SBRT in the setting of localized prostate cancer.[372] Two large randomized phase 3 trials have reported results. HYPO-RT-PC trial enrolled 1200 patients with intermediate- and high-risk prostate cancer and demonstrated noninferiority of 42.7 Gy in seven fractions to 78.0 Gy in 39 fractions with respect to FFS.[373] Likewise, PACE-B compared 5-fraction SBRT with a dose of 40 Gy to the prostate and 36.25 Gy to the planning target volume versus moderate

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hypofractionated RT with 62 Gy in 20 fractions.[374] Results showed noninferiority in tumor control and generally similar safety profiles.

SBRT regimens should utilize IMRT planning and daily IGRT. SBRT is an appropriate standard RT method at clinics with appropriate technology, physics, and clinical expertise.

The Panel lists fractionation schemes that have shown acceptable efficacy and toxicity within the *Principles of Radiation Therapy*.

Relative contraindications or considerations to EBRT include prior pelvic irradiation, active inflammatory disease of the rectum, permanent indwelling Foley catheter, very low bladder capacity, chronic moderate or severe diarrhea, and bladder outlet obstruction requiring a suprapubic catheter.

Hydrogel Per-Rectal Spacers

Biomaterials have been developed, tested, and FDA approved to serve as spacer materials when inserted between the rectum and prostate.[375,376] In a randomized phase 3 multicenter clinical trial of patients undergoing image-guided IMRT (IG-IMRT), the risk of late (3-year) grade 2 or higher, physician-recorded rectal complications declined from 5.7% to 0% in the control versus hydrogel spacer group.[377] The control group had a significant reduction in bowel QOL decline. No significant differences in urinary adverse events were noted in those receiving hydrogel placement versus controls. Results of a secondary analysis of this trial suggest that use of a perirectal spacer may decrease the sexual side effects of radiation.[378] Spacer implantation may be associated with rare complications such as rectum perforation and urethral damage.[379,380] Retrospective data also support its use in similar patients undergoing brachytherapy. Overall, the Panel believes that biocompatible and biodegradable perirectal spacer materials may be implanted between the prostate and rectum in patients undergoing RT in order to displace the rectum from high radiation dose regions. Patients with obvious rectal invasion or visible posterior extension should not undergo perirectal spacer implantation.

EBRT for Low-Risk Disease

Active surveillance is preferred for low-risk disease. If EBRT is to be used, it should be given as monotherapy without a brachytherapy boost, without nodal irradiation, and without ADT. Ideally it is given with SBRT or moderately hypofractionated RT.

EBRT for Intermediate-Risk Disease

The favorable/unfavorable distinction within the intermediate-risk group should be used to help guide treatment for these patients. Generally, patients with favorable intermediate-risk disease should be treated with EBRT alone without a brachytherapy boost, without nodal irradiation, and without ADT.[381] Use of SBRT or moderately hypofractionated RT are preferred. Patients with unfavorable intermediate-risk disease generally should be considered for the addition of short-term ADT.[382,383] Studies have shown that the addition of a focal EBRT boost may further improve biochemical control in these patients.[384] Similarly, the addition of a brachytherapy boost may improve biochemical control, at the expense of increased toxicity.[385] Trials to date that have enrolled patients with intermediate-risk disease have not shown improvements in disease control from the use of elective nodal irradiation (ENI), and have generally seen an increase in toxicity.[386]

EBRT for Patients with High-Risk or Very-High-Risk Disease

EBRT represents the only local therapy for patients with high- or very-high-risk disease that has been shown to improve overall survival when compared to ADT alone.[387-390] Furthermore, the addition of ADT, specifically long-term ADT of >18 months, has been shown to improve overall survival as compared to RT alone or RT with short-term ADT.[382]

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There are mixed results regarding the benefit of ENI in this patient population. Three multicenter cooperative group trials have not demonstrated improvement in outcomes from ENI; GETUG-01, RTOG 9413, and recently in abstract form RTOG 0924.[386,391,392] However, one small single center trial from India, POP-RT, showed that pelvic radiation can improve biochemical FFS and DFS compared with prostate-only radiation in patients with high- and very-high-risk prostate cancer.[393] With the recent results of NRG/RTOG 0924 (ClinicalTrials.gov ID NCT01368588) demonstrating no improvement in distant metastasis, prostate cancer-specific mortality, or overall survival from ENI in an approximately 2500 patient trial of unfavorable-intermediate and high-risk prostate cancer, the role of ENI remains unclear.

More recently, the STAMPEDE trial has reported a post-hoc analysis of the benefit of adding abiraterone acetate/prednisone to EBRT plus long-term ADT in patients with NCCN very-high-risk prostate cancer.[394] Results demonstrated improvement in metastasis-free survival from the addition of abiraterone at the expense of increased grade ≥3 side effects. Notably, the patients enrolled in this trial had markedly higher rates of metastasis and death from prostate cancer than those observed in other trials of high-risk prostate cancer conducted by NRG, EORTC, or TROG. Given some patients had PSAs >1000 ng/mL on the STAMPEDE trial but were called non-metastatic, it remains unclear the true value of the addition of abiraterone in contemporary very-high-risk disease. ENZARAD, a randomized trial of patients with high- and very-high-risk and clinical node positive disease, recently reported results in abstract form and demonstrated that the addition of the APRI enzalutamide to RT+ADT did not improve metastasis-free survival or overall survival.[395] Ongoing trials utilizing other ARPIs have not yet reported their results, such as ATLAS (NCT02531516).

EBRT for Node-Positive Disease

EBRT with neoadjuvant, concurrent, and/or adjuvant ADT and abiraterone is the preferred option for patients with clinical N1 disease based upon the previously discussed STAMPEDE trial.[394] Other options in this setting include EBRT with ADT, ADT alone or with abiraterone, and, for select patients, RP plus PLND (see *Radical Prostatectomy*, above).

Post-Recurrence EBRT

The use of EBRT in patients with biochemical recurrence is discussed under *Post-Radical Prostatectomy Approaches*, below.

Brachytherapy

Brachytherapy involves placing radioactive sources into the prostate tissue. Brachytherapy has been used traditionally for low-risk disease because earlier studies found it less effective than EBRT for high-risk disease.[121,396] However, increasing evidence suggests that technical advancements in brachytherapy may provide a role for contemporary brachytherapy in higher-risk localized and locally advanced prostate cancer.[397]

There are currently two methods for prostate brachytherapy: low dose-rate (LDR) and high dose-rate (HDR). LDR brachytherapy consists of placement of permanent seed implants in the prostate. The short range of the radiation emitted from these low-energy sources allows delivery of adequate dose levels to the cancer within the prostate, with excessive irradiation of the bladder and rectum avoided. Post-implant dosimetry should be performed to document the quality of an LDR implant.[398] HDR brachytherapy, which involves temporary insertion of a radiation source, is a newer approach.

Two groups have observed a lower risk of urinary frequency, urgency, and rectal pain with HDR brachytherapy compared with LDR brachytherapy (permanent seed implant).[399,400] Vargas and colleagues[401] reported that

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HDR brachytherapy results in a lower risk of erectile dysfunction than LDR brachytherapy. Commonly prescribed doses for LDR and HDR brachytherapy are listed in the *Principles of Radiation Therapy* within the algorithm above.

For patients with very large or very small prostates, symptoms of bladder outlet obstruction (high International Prostate Symptom Score), or a previous transurethral resection of the prostate (TURP), seed implantation may be more difficult. These patients also have an increased risk of side effects. Neoadjuvant ADT may be used to shrink the prostate to an acceptable size; however, increased toxicity is expected from ADT, and prostate size may not decline in some patients. The potential toxicity of ADT must be weighed against the possible benefit of target reduction.

Ideally, the accuracy of brachytherapy treatment should be verified by daily prostate localization with techniques of IGRT: CT, ultrasound, implanted fiducials, or electromagnetic targeting/tracking. Endorectal balloons may be used to improve prostate immobilization. Perirectal spacer materials (discussed under *External Beam Radiation Therapy*, above) may be used when the previously mentioned techniques are insufficient to reduce side effects due to anatomic geometry or other patient-related factors (eg, medication usage, comorbid conditions). Patients with obvious rectal invasion or visible cT3 and posterior extension should not undergo perirectal spacer implantation.

Brachytherapy Alone for Localized Disease

Active surveillance is preferred for low-risk disease, but brachytherapy alone is an option for select patients with low-risk disease. Brachytherapy alone is an option for patients with favorable intermediate-risk and for carefully selected patients with unfavorable intermediate-risk prostate cancer, depending on life expectancy. Patients with high-risk cancers are not generally considered candidates for brachytherapy alone. Either LDR or HDR brachytherapy can be used in this setting; however, there are concerns regarding the efficacy of both single fraction HDR and, more recently, two fraction HDR as monotherapy.[402]

There are no reported trials comparing SBRT or other contemporary EBRT methods to brachytherapy. ASCENDE-SBRT is comparing a brachytherapy boost versus SBRT.[403]

Brachytherapy Boost

LDR or HDR brachytherapy can be added as a boost to EBRT plus ADT in patients with unfavorable intermediate- or high-risk prostate cancer being treated with curative intent and for carefully selected patients with very-high-risk disease. Combining EBRT and brachytherapy allows dose escalation. This combination has demonstrated improved biochemical control over EBRT plus ADT alone in randomized trials, but with higher toxicity.[404-406]

The randomized ASCENDE-RT trial compared two methods of dose escalation in 398 patients with intermediate- or high-risk prostate cancer: dose-escalated EBRT boost to 78 Gy or LDR brachytherapy boost.[385] All patients were initially treated with 12 months of ADT and pelvic EBRT to 46 Gy. An ITT analysis found that the primary endpoint of biochemical PFS was 86% versus 75% at 7 years for the LDR boost versus EBRT boost arms (log-rank $P < .001$). Toxicity was higher in the brachytherapy arm, with the cumulative incidence of grade 3 genitourinary events at 5 years of 18.4% for brachytherapy boost and 5.2% for EBRT boost ($P < .001$).[407] A trend for increased gastrointestinal toxicity with brachytherapy boost was also seen (cumulative incidence of grade 3 events at 5 years, 8.1% vs. 3.2%; $P = .12$). However, at 6-year follow-up, health-related QOL was similar between the groups in most domains, except that physical and urinary function scales were significantly worse in the LDR arm.[408] Whereas the toxicity is increased with the use of brachytherapy boost, this and other randomized controlled trials have not shown an improvement in OS or cancer-specific survival.[409]

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The optimal duration of ADT with EBRT plus brachytherapy is unclear. The TROG RADAR trial randomized patients to receive 6 versus 18 months of ADT, and a subset of participants received EBRT with an HDR boost.[410] In the subset who received EBRT plus HDR, the use of 18 months of ADT improved overall survival compared to 6 months. More recently, the TRIP/TRIGU0907 randomized trial of EBRT plus LDR compared 6 versus 30 months of ADT.[411] The study found no significant difference in biochemical progression between arms, both were <10% at 7-year follow-up.

To address concerns for increased toxicity incidence associated with brachytherapy boost from historical trial data, careful patient selection and contemporary planning associated with lesser toxicity, such as use of recognized organ at risk dose constraints, use of high-quality ultrasound and other imaging, and prescription of dose as close as possible to the target without excessive margins, should be implemented.

Post-Recurrence Brachytherapy

Brachytherapy can be considered in patients with biochemical recurrence after EBRT. Results of a phase 2 study of post-recurrence HDR brachytherapy after EBRT included relapse-free survival, distant metastases-free survival, and cause-specific survival rates of 68.5%, 81.5%, and 90.3%, respectively, at 5 years.[412] Toxicities were mostly grade 1 and 2 and included gastrointestinal toxicity and urethral strictures, and one case of Grade 3 urinary incontinence. In another prospective phase 2 trial, the primary endpoint of grade ≥3 late treatment-related gastrointestinal and genitourinary adverse events at 9 to 24 months after post-recurrence brachytherapy was below the unacceptable threshold, at 14%.[413]

Data on the use of brachytherapy after permanent brachytherapy are limited, but the Panel agrees that it can be considered for carefully selected patients. Decisions regarding the use of brachytherapy in the recurrent-disease setting should consider comorbidities, extent of disease, and potential complications. Brachytherapy in this setting is best performed at high-volume centers.

Radiation for Distant Metastases

EBRT can be used for multiple purposes in the setting of metastatic disease. Historically, it was only used as a means of palliation, often for bone pain, obstruction, or bleeding. A short course of 8 Gy x 1 is as effective for palliation as, and less costly than, 30 Gy in 10 fractions.[414] In a randomized trial of 898 patients with bone metastases, grade 2–4 acute toxicity was observed less often in the 8-Gy arm (10%) than in the 30-Gy arm (17%) ($P = .002$); however, the retreatment rate was higher in the 8-Gy group (18%) than in the 30-Gy group (9%) ($P < .001$).[415] In another study of 425 patients with painful bone metastases, a single dose of 8 Gy was noninferior to 20 Gy in multiple fractions in terms of overall pain response to treatment.[416] The SCORAD randomized trial did not show noninferiority for ambulatory status of single-fraction 8-Gy EBRT to 20 Gy in 5 fractions.[417]

Eight Gy as a single dose is as effective for short-term (ie, 12 week) pain palliation at any bony site as longer courses of radiation, but re-treatment rates are higher. Other regimens (eg, 30 Gy in 10 fractions) may be used as alternative palliative dosing depending on clinical scenario. Neither regimen has demonstrated clear long-term pain or cancer control and are intended for transient pain control palliation.

EBRT may also be used as an oncologic treatment and not solely for the palliation of symptoms in the metastatic setting. The topic of metastasis-directed radiation therapy is discussed in detail below (see *MDT for Oligometastatic CSPC* and *MDT for mCRPC*).

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Comparison of Treatment Options for Localized Disease

Multiple randomized trials have compared RT to RP. The ProtecT trial demonstrated that, with a median follow-up of 15 years, patients with low and intermediate risk prostate cancer had similar rates of developing metastasis or death.[263,265] A post-hoc subset analysis of the ProtecT trial demonstrated that patients with cribriform features on active monitoring had significantly reduced rates of distant metastasis when treated with RT plus ADT, but not with RP.[264]

The ProtecT trial demonstrated that urinary incontinence rates and impotence rates were significantly higher with RP than RT. There were small but significant increases in bowel side effects with RT as compared to surgery. The RT utilized on this trial was 3D conformal radiotherapy without contemporary image guidance or rectal spacer use.

More recently, results from PACE-A, which included patients with intermediate-risk prostate cancer receiving either SBRT versus RP, were reported.[345] No spacer gels were utilized. The study demonstrated that, with more modern RT, there were smaller differences in patient-reported bowel quality of life as compared to surgery. Furthermore, those receiving surgery were >3-fold more likely to develop impotence, and >8-fold more likely to develop urinary incontinence requiring ≥1 pads per day.

Recently, a cross-trial comparison using two phase 3 randomized trials was performed.[418] Trials were matched by era, country, and randomization. NRG/RTOG 0521 randomized patients with high-risk prostate cancer to RT plus 24 months of ADT with or without 6 cycles of docetaxel, and CALGB randomized patients with high-risk prostate cancer to surgery with or without 6 cycles of docetaxel with concurrent ADT in the neoadjuvant setting. The patients in the RT trial were generally older and had more aggressive prognostic features. This study demonstrated that patients receiving RT with ADT had significantly reduced risk of developing distant metastasis than patients treated with a surgical approach, but there were no significant differences in death from prostate cancer. As this was not a true randomized trial of the two treatment approaches, confounding may impact the results.

Numerous non-randomized studies have been reported that should be interpreted with caution given the high risk for selection bias and residual confounding. Barocas et al compared radical prostatectomy, EBRT, and active surveillance in 2550 patients and found that, after 3 years, radical prostatectomy was associated with a greater worsening in urinary and sexual function than either EBRT or active surveillance.[419] However, radical prostatectomy was also associated with an improvement in urinary irritative symptoms that was not observed with radiation or active surveillance. Health-related QOL measures including bowel and hormonal function were similar among the groups, as was disease-specific survival.

Chen et al compared radical prostatectomy, EBRT, and brachytherapy against active surveillance in 1141 patients.[23] As in the Barocas study, radical prostatectomy was associated with greater declines in sexual and urinary function than other treatments at 3 months. In this study, EBRT was associated with worse short-term bowel function, and both EBRT and brachytherapy were associated with worsened urinary obstructive and irritative symptoms. By 2 years, however, differences among the groups compared with active surveillance were insignificant. Results of a systematic review showed similar findings to these studies.[420]

Another study examined patient-reported outcomes in >2000 patients with localized prostate cancer managed by radical prostatectomy, brachytherapy, EBRT with or without ADT, or active surveillance.[24] By 5 years, most functional differences were minimal between management approaches. However, radical prostatectomy was associated with worse incontinence in the full cohort and with worse sexual function in those with

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unfavorable intermediate-, high-, or very-high-risk disease than in those treated with EBRT and ADT.

Other Local Therapies

Many therapies have been investigated for the treatment of localized prostate. Cryotherapy was investigated as a potential whole gland treatment option. A study by Donnelly and colleagues included 244 patients with T2 or T3 disease who were randomly assigned to either cryotherapy or EBRT.[421] In this small trial, all patients received neoadjuvant ADT. Patients who received cryotherapy reported poorer sexual function.[422] For patients with locally advanced cancer, cryoablation was associated with a lower 8-year biochemical progression-free rate compared to EBRT in a small trial of 62 patients.[423]

Other ablative therapies (eg, high-intensity focused ultrasound [HIFU], transurethral ultrasound ablation [TULSA], irreversible electroporation [IRE], photodynamic therapy [PDT]) are not recommended as primary therapy for localized prostate cancer due to lack of long-term randomized data comparing these treatments to radiation or radical prostatectomy.[424-430] Single arm phase I or II trials have shown high rates of residual or recurrent cancer (20%–70%) and clinically meaningful toxicity (ie, incontinence, obstruction, erectile dysfunction). Furthermore, monitoring strategies appear to be compromised (ie, MRI accuracy) and secondary treatment may carry higher risk of complications. These ablative treatments can be utilized in the context of a clinical trial, ideally a randomized trial compared them to a current standard treatment option. Companies and institutions may make non–evidenced-based claims to the evidence, efficacy, and safety of focal therapy.

At this time, the only local therapy options that can be considered are cryotherapy, HIFU, and IRE (category 2B) in the setting of non-metastatic RT recurrence due to the overall lack of high-quality evidence in this setting for any modality. Also see *Post-Radiation Recurrence*, below.

Disease Monitoring

Please refer to the NCCN Guidelines for Survivorship (available at www.NCCN.org) for recommendations regarding common consequences of cancer and cancer treatment (eg, anxiety, depression, trauma, and distress; hormone-related symptoms; sexual dysfunction) and on the promotion of physical activity, weight management, and proper immunizations in cancer survivors. Prostate-cancer specific survivorship recommendations pertaining to bone health and cardiovascular disease are included in the algorithm above and are discussed in more detail in the section on *Adverse Effects of Traditional ADT*, below.

Patients After Initial Definitive Therapy

For patients initially treated with curative intent, serum PSA levels should be measured every 6 to 12 months for the first 5 years and then annually. PSA testing every 3 months may be better for patients at high risk of recurrence. When prostate cancer recurred after radical prostatectomy, Pound and colleagues found that 45% of patients experienced recurrence within the first 2 years, 77% within the first 5 years, and 96% by 10 years.[431]

Patients with N1 CSPC on ADT

The intensity of clinical monitoring for patients on ADT for N1 CSPC is determined by the response to initial ADT, EBRT, or both. Follow-up evaluation of these patients should include physical examination and PSA measurement every 3 to 6 months. Imaging for symptoms or increasing PSA. The relative risk for bone metastasis or death increases as PSADT falls. Bone imaging should be performed more frequently in patients with shorter PSADTs.[432]

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Patients with Localized Disease Under Observation

Patients with localized disease on observation follow the same monitoring recommendations as patients with N1 CSPC who are on ADT.

Workup for Progression

Castrate levels of testosterone should be documented if clinically indicated in patients with signs of progression, with adjustment of ADT as necessary. If serum testosterone levels are <50 ng/dL, the patient should undergo disease workup with bone and soft tissue imaging (see *Imaging Techniques* above for more details).

Post-Radical Prostatectomy Approaches

Patients who have undergone radical prostatectomy should be monitored as described above. However, some patients will have adverse pathologic features, positive lymph nodes, or biochemical persistence or recurrence. Options for these patients include adjuvant radiation (the use of EBRT immediately following radical prostatectomy in the setting of an undetectable postoperative PSA) with or without ADT, secondary radiation (the use of EBRT after radical prostatectomy when PSA is detectable) with or without ADT, or monitoring. Decisions about when to initiate post-radical prostatectomy radiation and whether to include ADT for such patients are complex. The Panel recommends use of nomograms and consideration of age and comorbidities, clinical and pathologic information, PSA levels, and PSADT to individualize treatment discussion. The various scenarios are discussed in more detail below.

Bone and soft tissue imaging and prostate bed biopsy should be considered in patients with PSA persistence/recurrence after RP if their life expectancy is >5 years. The utility of imaging for patients with an early biochemical recurrence after radical prostatectomy depends on disease risk before operation and pathologic stage, Gleason grade, PSA, and PSADT after recurrence. Patients with low- and intermediate-risk disease and low postoperative serum PSA levels have a very low risk of positive bone scans or CT scans.[433,434] In a series of 414 bone scans performed in 230 patients with biochemical recurrence after radical prostatectomy, the rate of a positive bone scan for patients with PSA >10 ng/mL was only 4%.[435] In another study, the probability of a positive bone scan for a patient not on ADT after radical prostatectomy was <5% unless the PSA increased to 40 to 45 ng/mL.[436] Thus bone imaging is appropriate when patients develop symptoms or when PSA levels are increasing rapidly. A prostate bed biopsy can also be considered. However, PSMA PET/CT imaging has increased sensitivity at low PSA levels post-RP, and approximately 30% to 40% of patients will have detectable disease on PSMA PET/CT at PSA levels of 0.2 to 0.5 ng/mL.

Early Secondary or Adjuvant Treatment for Adverse Features After Radical Prostatectomy

Patients with undetectable PSA levels and without lymph node metastases following radical prostatectomy can be considered for immediate adjuvant treatment if adverse pathologic features are present (ie, positive margins, seminal vesicle invasion, extracapsular extension). However, monitoring these patients is the preferred option (category 1). Waiting for a detectable and rising PSA or PSA ≥0.1 ng/mL before initiating secondary radiation is preferred in this setting, since this approach avoids over-treatment and subsequent potential side effects and does not appear to compromise oncologic outcomes.[437,438]

In the RADICALS-RT trial, 1396 patients with adverse features after radical prostatectomy were randomized to receive immediate adjuvant treatment or to be monitored with a policy to initiate secondary treatment if PSA levels reached 0.1 ng/mL or rose three consecutive times.[439] Participants were followed for a median 4.9 years and no differences were seen in 5-year biochemical PFS and freedom from non-protocol hormone therapy. There was no statistically significant difference in the primary

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endpoint of freedom from distant metastasis between the arms at 10 years (93% in the adjuvant arm vs. 90% in the early secondary therapy arm; 95% CI, 0.43–1.07; *P* = .095) and there was no difference in OS.[438] Importantly, urinary incontinence and grade 3–4 urethral strictures were more frequent in the adjuvant therapy group.

The GETUG-AFU 17 trial and the TROG 08.03/ANZUP RAVES trial were both terminated early for unexpectedly low event rates, but similarly found no evidence of oncologic benefit with increased risk of genitourinary toxicity and erectile dysfunction when an adjuvant therapy approach was used.[440,441] One small randomized trial in Finland (n = 126) saw a significant improvement in 10-year survival for biochemical recurrence with the use of adjuvant therapy in patients with positive margins or extracapsular extension (HR, 0.26; 95% CI, 0.14–0.48; *P* < .001).[442] However, there was no difference in OS, and grade 3 adverse events were significantly higher in the adjuvant arm.

Several meta-analyses have also shown that adjuvant radiotherapy improved biochemical control, but not overall survival, in patients with adverse features and that the approach leads to a higher incidence of adverse events, though some noted that careful patient selection is warranted.[437,443-445] Real world evidence supports the safety of this approach for most patients.[446,447]

The Panel notes immediate adjuvant radiation may be reasonable for patients with multiple adverse features, a population that was under-represented in the above-mentioned clinical trials. True noninferiority of the delayed, early secondary therapy approach has not been demonstrated in patients with a high risk of recurrence after surgery.

Adjuvant or Early Secondary Therapy for pN1

Patients with positive lymph nodes found during or after radical prostatectomy have several disease management options to consider. Immediate adjuvant treatment with EBRT with or without ADT is one approach for these patients based on historic data.[448] However, retrospective data show that initial observation may be safe in some patients with pN1 disease at radical prostatectomy, because 28% of a cohort of 369 patients remained free from biochemical recurrence at 10 years.[449] Therefore, another option is monitoring with consideration of early secondary treatment for a detectable and rising PSA or PSA >0.1 ng/mL. However, RADICALS-RT, GETUG-AFU 17, and TROG 08.03/ANZUP RAVES have very few pN1 patients that were enrolled.[439-441]

An analysis of the National Cancer database identified 8074 patients with lymph node metastases after radical prostatectomy. [450] The results indicate that immediate adjuvant EBRT and ADT improved OS compared to monitoring for those who also had adverse pathologic features but not for those without these features. Therefore, use of early secondary radiotherapy can be considered for patients with pN1 disease, with careful patient selection and monitoring.

The addition of abiraterone to EBRT with ADT can also be considered for patients with positive lymph after surgery (category 2B). However, these patients comprised <5% of patients on the STAMPEDE trial, and caution is warranted with extrapolation of data from cN1 disease.[394]

Secondary Treatment for PSA Persistence or Biochemical Recurrence After Radical Prostatectomy

The timing of secondary treatment for patients whose only evidence of cancer after radical prostatectomy is a persistent or increasing PSA is influenced by PSA levels, PSADT, and the underlying comorbidities of the patient. PSADT should be calculated to inform nomogram use and counseling for patients with PSA persistence or recurrence after radical prostatectomy. Early secondary treatment (at a PSA of 0.1–0.2 ng/mL) is recommended for most patients who experience a biochemical recurrence after RP. However, patients who have a delayed recurrence (eg, many

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years post-RP) with very slow PSA doubling times (ie, >24 months) can consider delaying secondary treatment, especially if they have limited life expectancy.

For those with PSA persistence or recurrence after RP without evidence of M1 disease, secondary treatment with EBRT is the preferred approach. Large retrospective cohort studies support the use of EBRT in the setting of biochemical recurrence, because it is associated with decreased all-cause mortality and increased prostate cancer-specific survival.[451,452]

Numerous phase 3 randomized trials have evaluated the role of hormone therapy with secondary EBRT in this setting, as well as a harmonized aggregate meta-analysis.[453-457] In general, these studies have demonstrated that 6 to 24 months of hormone therapy improved outcomes primarily for patients receiving late secondary RT (PSA levels >0.5 ng/mL), but a smaller or no benefit is clear at lower PSA levels. For example, RTOG 9601 demonstrated an OS benefit with 2 years of bicalutamide with secondary EBRT, but enrolled patients primarily receiving late secondary EBRT.[458] In contrast, the RADICALS-HD trial randomized patients to post-operative EBRT with or without 6 months of ADT and enrolled mostly adjuvant or early secondary EBRT patients. They showed that ADT did not improve MFS or OS. The DADSPORT meta-analysis did not show an OS benefit for the addition of ADT in patients receiving early secondary RT, although an MFS and PCSS benefit were seen.[459]

Based on these data, short-term ADT is generally recommended to balance toxicity, QOL, and tumor control. However, long-term ADT may be preferred for patients receiving late secondary RT and/or those with multiple adverse pathologic features or lymph node involvement.

There remains no randomized data to support the addition of an ARPI to EBRT and ADT. However, abiraterone is an option for patients with lymph node metastases (category 2B), based on extrapolation from STAMPEDE.[394]

Monitoring patients without evidence of M1 disease in the setting of PSA persistence or a rising PSA post-RP is another option, especially if the risk of rapid progression appears to be low based on factors such as PSA levels, PSADT, and results of nomograms and with consideration of patient age and comorbidities. It is important to note that some patients have detectable PSA after radical prostatectomy due to benign prostate tissue in the prostate fossa. They have low stable PSAs and a very low risk of prostate cancer progression.[460,461] Serial PSA measurements can be helpful for stratifying patients at highest risk of progression and metastases.

Post-Radiation Recurrence

The 2006 Phoenix definition was revised by ASTRO and the RTOG in Phoenix: 1) PSA rise by 2 ng/mL or more above the nadir PSA is the standard definition for biochemical recurrence after EBRT with or without hormonal therapy; and 2) A recurrence evaluation should be considered when PSA has been confirmed to be increasing after radiation even if the rise above nadir is not yet 2 ng/mL, especially in candidates for additional local therapy who are young and healthy.[462] Retaining a strict version of the ASTRO definition allows comparison with a large existing body of literature. Rapid increase of PSA may warrant evaluation (prostate biopsy) prior to meeting the Phoenix definition, especially in younger or healthier patients.

Workup for PSA recurrence or a positive DRE after RT for a patients with a life expectancy >5 years typically includes PSADT calculation, bone and soft tissue imaging, and consideration of a prostate/seminal vesicle biopsy. Biopsy is particularly useful if the staging workup does not reveal

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metastatic disease and local recurrence is suspected. The junction of the seminal vesicle and prostate is a common site of recurrence.

Local radiation recurrences are most responsive to additional therapy when PSA levels at the time of treatment are low (<5 ng/mL). Options for therapy for those without evidence of distant metastases include monitoring with physical exam, PSA every 3 to 6 months, and imaging for symptoms or increasing PSA. For those deemed to be at higher risk of rapid progression, treatment options are ADT and local secondary therapy with or without ADT. For those with positive lymph nodes, abiraterone can be added. Treatment should be individualized based on the patient's risk of progression and the potential benefits and risks involved with each option.

Local secondary therapy options in the setting of RT recurrence include cryotherapy,[463] HIFU,[464-469] IRE[470,471] (category 2B), reirradiation, and RP with PLND. The Panel recommends that patients receive multidisciplinary counseling about the risks and benefits of each of these options in the context of the available comparative literature on this topic.[472,473] Reirradiation options include LDR brachytherapy, HDR brachytherapy, and SBRT based on phase II and other prospective and retrospective studies.[412,474-476] Radical prostatectomy after RT recurrence can result in long-term disease control, but is often associated with impotence and urinary incontinence.[477] Overall, there is limited evidence for these approaches, and the Panel recommends that patients receiving local therapy for RT recurrence are treated within the context of clinical trials when available and/or at experienced centers.

Second Biochemical Recurrence

Patients who have been treated for recurrence after definitive therapy for localized prostate cancer may continue to experience a rise in PSA levels without developing metastases. The Panel refers to this disease setting as *second biochemical recurrence*. A staging workup with imaging is recommended. For patients without evidence of lymph node or distant metastases, risk stratification is essential to determine treatment options and is based mainly on PSA levels and PSADT. The definition of high-risk second biochemical recurrence differs between clinical trials but is generally considered to include patients with a PSADT ≤9 months and other adverse prognostic features.

For individuals deemed to be a low risk of rapid disease progression, monitoring until diagnosis of metastatic disease is preferred. ADT monotherapy is also an option, and PSA levels and PSADT should be considered when deciding whether to begin ADT.

For patients deemed to be at high risk of rapid disease progression, treatment intensification can be considered based on results of two trials as discussed below. Monitoring and ADT are equally reasonable options for these patients, and shared decision-making is key.

In the phase 3 EMBARK trial, 1068 patients with prostate cancer and high-risk biochemical recurrence were randomly assigned into one of three groups using a 1:1:1 ratio.[478] The “combination group” received enzalutamide (160 mg/day) plus leuprolide, the “leuprolide-alone group” received placebo plus leuprolide, and the “monotherapy group” received enzalutamide alone. Eligibility for the trial included the following high-risk criteria: M0 by CT, MRI, or bone scan; PSADT ≤9 months; PSA ≥2 ng/mL above nadir after RT or ≥1 ng/mL after radical prostatectomy with or without postoperative RT; and patients were not considered a candidate for pelvic-directed therapy. Patients were monitored for a median follow-up time of 60.7 months. The trial demonstrated that the primary endpoint, metastasis-free survival, was significantly improved for enzalutamide plus leuprolide (HR for metastasis or death, 0.42; 95% CI, 0.30–0.61; $P < .001$) and for enzalutamide alone (HR, 0.63; 95% CI, 0.46–0.87; $P = .005$) when

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compared to leuprolide alone. The most common adverse events associated with combination therapy and enzalutamide monotherapy were hot flashes and fatigue. Enzalutamide monotherapy was also significantly associated with gynecomastia (45% compared with 8% to 9% in the combination and leuprolide alone groups), nipple pain (15% compared with 1%–3%), and breast tenderness (14% compared with 1%). Based on these results, the Panel includes enzalutamide with or without leuprolide as options for patients with high-risk biochemical recurrence after maximal pelvic therapy in the absence of metastatic disease.

In the phase 3 PRESTO trial, 503 patients with biochemically recurrent prostate cancer were randomly assigned into one of three groups using a 1:1:1 ratio.[479] The three arms of the study included: ADT alone (control arm A), ADT plus apalutamide (experimental arm B), or ADT plus apalutamide plus abiraterone acetate/prednisone (AAP; experimental arm C). The dose of apalutamide was 240mg/day. To be eligible for the trial, patients had prostate cancer with biochemical recurrence after radical prostatectomy and the following high-risk criteria: PSADT ≤9 months; PSA ≥0.5 ng/mL; and prior adjuvant or secondary RT or not considered a candidate for RT. The planned duration of treatment was 52 weeks. The primary endpoint of PSA progression-free survival (PSA-PFS) was met in patients in both experimental arms. PSA progression was defined as serum PSA >0.2 ng/mL after completion of treatment. At the first interim analysis (median follow-up of 21.3 and 21.5 months in arms B and C, respectively), both experimental arms had significant prolongation of PSA-PFS compared to the control arm (ADT plus apalutamide, 24.9 months; ADT plus apalutamide plus AAP, 26.0 months; ADT alone, 20.3 months). For the comparison of ADT with and without apalutamide, the HR was 0.52 (95% CI, 0.35–0.77; P = .00047), and for the comparison of ADT plus apalutamide/AAP the HR was 0.48 (95% CI, 0.32–0.71; P = .00008). There was no difference in median time to testosterone recovery across treatment arms. The most prevalent treatment-associated adverse events across treatment arms were hot flashes (78%), fatigue (55%), injection site reaction (33%), and hypertension (31%). Hypertension was the most common grade ≥3 adverse event, experienced in 7%, 8%, and 19% of patients in arms A, B, and C, respectively. Even though the study was not designed to directly compare the two experimental arms, there was not shown to be further clinical benefit with the addition of AAP to apalutamide with ADT. Based on the results of this study, the Panel includes apalutamide plus ADT as a category 2B option for patients with non-metastatic disease and high-risk of biochemical recurrence after radical prostatectomy who have received maximal pelvic therapy.

Androgen Deprivation Therapy

ADT is administered as primary systemic therapy for regional or advanced disease and as neoadjuvant/concomitant/adjuvant therapy in combination with radiation in patients with unfavorable-intermediate or higher risk localized prostate cancers. It is also used as palliative therapy for symptomatic patients with a life expectancy ≤5 years.

Placing patients with early prostate cancer on ADT should not be routine practice.[480,481] Bicalutamide monotherapy is not recommended to be used and is recommended when used to be in combination with ADT as per the FDA label.

Medical castration (ie, LHRH agonist or antagonist) and surgical castration (ie, bilateral orchiectomy) are equally effective options for ADT. Castrate levels of serum testosterone (<50 ng/dL; <1.7 nmol/L) should be achieved, because low nadir serum testosterone levels were shown to be associated with improved cause-specific survival in the PR-7 study.[482] Patients who do not achieve adequate suppression of serum testosterone (<50 ng/dL) with medical or surgical castration can be considered for additional hormonal manipulations (with antiandrogens, LHRH antagonists, or steroids), although the clinical benefit remains uncertain. Monitoring

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testosterone levels 12 weeks after first dose of LHRH therapy and upon increase in PSA should be considered.

The LHRH antagonist, relugolix, has been studied as ADT in patients with localized and advanced prostate cancer.[483,484] In the HERO study, 622 patients received relugolix (120 mg orally once daily) and 308 received leuprolide (injections every 3 months) for 48 weeks. Sustained castrate levels of testosterone (<50 ng per deciliter) through 48 weeks was superior with relugolix over leuprolide (96.7%; 95% CI, 94.9–97.9 vs. 88.8% [95% CI, 84.6–91.8]; $P < .001$ for superiority). Time to castration and testosterone recovery were both faster in the relugolix arm. The incidence of major adverse cardiovascular events was 2.9% in the relugolix arm and 6.2% in the leuprolide arm (HR, 0.46; 95% CI, 0.24–0.88).

ADT for Initial Treatment of Clinically Localized (N0,M0) Disease

ADT should not be used as monotherapy in clinically localized prostate cancer unless there is a contraindication to definitive local therapy, such as life expectancy less than 5 years and comorbidities (see *Palliative ADT*, below).

In the clinically localized setting, ADT using an LHRH agonist—alone or with a first-generation antiandrogen—or an LHRH antagonist can be used as a neoadjuvant, concurrent, and/or adjuvant to EBRT in patients with unfavorable intermediate-, high-, or very-high-risk prostate cancer, as described in more detail below.

ADT used as neoadjuvant treatment before radical prostatectomy is strongly discouraged outside of a clinical trial.

Neoadjuvant, Concurrent, and/or Adjuvant ADT with EBRT for Unfavorable Intermediate-Risk Disease

The MARCAP consortium performed an individual patient meta-analysis of nearly every randomized trial assessing the use and prolongation of ADT with EBRT in localized prostate cancer.[382] They demonstrated that the addition of 4 to 6 months of ADT improved OS for patients with intermediate- and high-risk disease. Additionally, the prolongation of adjuvant ADT for a total duration of 18 to 36 months further improved OS for patients primarily with high-risk prostate cancer. However, the prolongation of neoadjuvant ADT for multiple months did not significantly improve biochemical control, MFS, or OS. Furthermore, post-hoc analyses of RTOG 9408 and subgroup analyses of RTOG 0815 demonstrate that patients with multiple intermediate-risk features have a greater absolute benefit from ADT than those with a single intermediate-risk factor.[381,383]

Thus, the Panel recommends 4 to 6 months of ADT when EBRT is given to patients as initial treatment of unfavorable intermediate-risk prostate cancer.

Two randomized phase 3 trials assessed the optimal sequencing of ADT with EBRT: RTOG 9413 and the Ottawa 0101 trials.[485] These trials were pooled in an individual patient meta-analysis and demonstrated that concurrent/adjuvant ADT significantly improved biochemical control, distant metastasis, and MFS over a neoadjuvant/concurrent approach. Thus, concurrent/adjuvant ADT is preferred over neoadjuvant ADT in this setting.

Neoadjuvant, Concurrent, and/or Adjuvant ADT with EBRT for High-Risk or Very-High-Risk Disease

ADT combined with EBRT is an effective treatment for patients with high-risk or very-high-risk disease, as discussed in the *Radiation Therapy* section above. Combination therapy was consistently associated with

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improved disease-specific survival and OS compared to single-modality treatment in randomized phase 3 studies.[382,388,389,486-488]

Evidence favors long-term over short-term ADT for patients with high- and very-high-risk disease. MARCAP performed an individual patient level meta-analysis of every randomized trial that compared short-term versus long-term ADT with radiotherapy.[382] This study demonstrated that long-term ADT significantly improved MFS and OS over short-term ADT, with a greater absolute benefit in patients with high-/very high-risk disease as compared with intermediate-risk disease.

While the TROG RADAR trial demonstrated that in patients who received EBRT+HDR brachytherapy boost had an OS benefit from 18 months of ADT versus 6 months, the ASCENDE-RT trial of EBRT+LDR brachytherapy boost utilized only 12 months as their fixed duration of ADT.[385,410] Thus, it may be reasonable for select patients with high-risk disease (not very high-risk) to use 12 months of ADT if receiving EBRT and LDR brachytherapy per the ASCENDE-RT trial regimen.

ADT for Initial Treatment of Regional Disease

One option for patients with regional disease is EBRT with 2 to 3 years of neoadjuvant/concurrent/adjuvant ADT with or without abiraterone. Neoadjuvant/concurrent/adjuvant ADT options are an LHRH agonist, an LHRH agonist with a first-generation antiandrogen, or an LHRH antagonist. Abiraterone should not be co-administered with an antiandrogen.

Data regarding the treatment of cN1 disease is limited. Exploratory analysis of STAMPEDE suggested that there may be benefit to the use of RT in this setting.[489] Failure-free survival was better in those for whom RT was planned than for those receiving ADT alone (HR, 0.48; 95% CI, 0.29–0.79).

Another option for the treatment of patients initially diagnosed with node-positive disease is primary ADT. Primary ADT options are orchiectomy, an LHRH agonist, an LHRH agonist with a first-generation antiandrogen, or an LHRH antagonist; or orchiectomy, LHRH agonist, or LHRH antagonist with abiraterone.

Palliative ADT

Palliative ADT can be given to patients with a life expectancy of ≤5 years who have regional or metastatic prostate cancer. Palliative ADT also can be given to patients with disease progression during observation, usually when symptoms develop or when changes in PSA levels suggest that symptoms are imminent. The options in this setting are orchiectomy, LHRH agonist, or LHRH antagonist.

Primary ADT for mCSPC

ADT is strongly recommended in combination therapy for mCSPC (see Management of mCSPC, below). The use of ADT monotherapy in this setting is discouraged unless there are clear contraindications to combination therapy. If ADT monotherapy is given, intermittent ADT can be considered to reduce toxicity. Close monitoring of PSA and testosterone levels and possibly imaging is required when using intermittent ADT, especially during off-treatment periods, and patients may need to switch to continuous ADT upon signs of disease progression. ADT options for mCSPC include orchiectomy, LHRH agonist, or LHRH antagonist with various ARPIs and/or docetaxel. Metastasis-directed therapy or EBRT to the primary tumor are also options for certain patients. Specific recommendations depend on prior therapy and volume of disease as described in more detail below (see *Management of mCSPC*).

In patients with overt metastases in weight-bearing bone who are at risk of developing symptoms associated with the flare in testosterone with initial LHRH agonist alone, antiandrogen therapy should precede or be

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coadministered with LHRH agonist for at least 7 days to diminish ligand binding to the androgen receptor.[490,491] LHRH antagonists rapidly and directly inhibit the release of androgens, unlike LHRH agonists that initially stimulate LHRH receptors prior to hypogonadism. Therefore, no initial flare is associated with these agents and coadministration of antiandrogen is unnecessary.

Intermittent Versus Continuous ADT

ADT is associated with substantial side effects, which generally increase with the duration of treatment. Intermittent ADT is an approach based on the premise that cycles of androgen deprivation followed by re-exposure may delay “androgen independence,” reduce treatment morbidity, and improve QOL.[492,493] Some patients who have no ADT-related morbidity may find the uncertainty of intermittent ADT not worthwhile. Intermittent ADT requires close monitoring of PSA and testosterone levels, especially during off-treatment periods, and patients may need to switch to continuous therapy upon signs of disease progression.

Intermittent ADT in Non-Metastatic Disease

The Canadian-led PR.7 trial was a phase 3 trial of intermittent versus continuous ADT in patients with non-metastatic prostate cancer who experienced biochemical recurrence after primary or post-recurrence EBRT.[494] One thousand three hundred eighty-six patients with PSA >3 ng/mL were randomly assigned to intermittent ADT or continuous ADT. At a median follow-up of 6.9 years, the intermittent approach was noninferior to continuous ADT with respect to OS (8.8 vs. 9.1 years, respectively; HR, 1.02; 95% CI, 0.86–1.21). More patients died from prostate cancer in the intermittent ADT arm (120 of 690 patients) than in the continuous ADT arm (94 of 696 patients), but this was balanced by more non-prostate cancer deaths in the continuous ADT arm. Physical function, fatigue, urinary problems, hot flashes, libido, and erectile dysfunction showed modest improvement in the intermittent ADT group. The population of participants was heterogenous, so it remains unclear which of these asymptomatic patients benefited from treatment. It is possible that many of these patients could have delayed ADT without harm. The population had a low disease burden, and 59% of deaths in the trial were not related to prostate cancer. Follow-up longer than 6.9 years may be required for disease-specific deaths to out-balance deaths by other causes.

The multinational European ICELAND trial randomized 702 participants with locally advanced or biochemically recurrent prostate cancer to continuous or intermittent ADT.[495] Clinical outcomes, which included time to PSA progression, PSA PFS, OS, mean PSA levels over time, QOL, and adverse events, were similar between the arms.

A 2015 meta-analysis identified 6 randomized controlled trials comparing continuous with intermittent ADT in patients with locally advanced prostate cancer and found no difference in mortality and progression and an advantage of the intermittent approach in terms of QOL and adverse effects.[496]

Therefore, the Panel believes that patients receiving ADT for M0 recurrent disease can consider intermittent ADT based on shared decision making.

Intermittent ADT in Metastatic Disease

Hussain and colleagues[497] conducted the SWOG (Southwest Oncology Group) 9346 trial to compare intermittent and continuous ADT in patients with metastatic disease. After 7 months of induction ADT, 1535 patients whose PSA dropped to 4 ng/mL or below (thereby demonstrating androgen sensitivity) were randomized to intermittent or continuous ADT. At a median follow-up of 9.8 years, median survival was 5.1 years for the intermittent ADT arm and 5.8 years for the continuous ADT arm. The HR for death with intermittent ADT was 1.10 with a 90% CI between 0.99 and 1.23, which exceeded the prespecified upper boundary of 1.20 for noninferiority. The authors stated that the survival results were

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inconclusive, and that a 20% greater mortality risk with the intermittent approach cannot be ruled out. The study demonstrated better erectile function and mental health in patients receiving intermittent ADT at 3 months, but the difference became insignificant thereafter, most likely due to contamination on the intermittent arm of patients who may have returned to continuous ADT at the prespecified time points. A secondary analysis of SWOG 9346 showed that intermittent ADT did not reduce endocrine, bone, or cognitive events, whereas it led to an unexpected increase in the incidence of ischemic and thrombotic events.[498]

In a post-hoc stratification analysis of the trial, patients with minimal disease had a median survival of 5.4 years when receiving intermittent ADT versus 6.9 years when receiving continuous ADT (HR, 1.19; 95% CI, 0.98–1.43).[497] The median survival was 4.9 years in the intermittent ADT arm compared to 4.4 years in the continuous ADT arm for patients with extensive disease (HR, 1.02; 95% CI, 0.85–1.22). These subgroup analyses are hypothesis-generating.

A population-based analysis that included 9772 patients ≥66 years with advanced prostate cancer showed that intermittent ADT reduced the risks of total serious cardiovascular events by 36%, heart failure by 38%, and pathologic fracture by 48%, compared with continuous ADT.[499] Furthermore, several meta-analyses of randomized controlled trials reported no difference in survival between intermittent ADT and continuous ADT,[500-502] although another analysis concluded that the non-inferiority of intermittent to continuous ADT in terms of survival has not been clearly demonstrated.[503] Still, the intermittent approach leads to marked improvement in QOL compared to the continuous approach in most studies, and the Panel believes that intermittent ADT should be strongly considered in appropriate patients with metastatic disease.

Adverse Effects of Traditional ADT

ADT has a variety of adverse effects including hot flashes, vasomotor instability, loss of libido, erectile dysfunction, shrinkage of penis and testicles, loss of muscle mass and strength, fatigue, anemia, breast enlargement and tenderness/soreness, depression and mood swings, hair loss, osteoporosis, greater incidence of clinical fractures, obesity, insulin resistance, alterations in lipids, and greater risk for diabetes, acute kidney injury, and cardiovascular disease.[504-506] The intensity and spectrum of these side effects vary greatly. In general, the side effects of continuous ADT increase with the duration of treatment. In addition, some forms of ADT may result in lower risk than others. For example, post-hoc analysis of a randomized trial of LHRH antagonist versus LHRH agonist found lower risk of cardiac events in patients with existing cardiac disease treated with LHRH antagonist.[507] In addition, relugolix was associated with a lower risk of major adverse cardiovascular events than leuprolide in the phase 3 HERO study, although the FDA considered these results in HERO to be exploratory and therefore did not allow for these data to be included in the prescribing information for relugolix.[483] Results from the randomized controlled PRONOUCE trial suggest that there is no difference in cardiovascular safety between LHRH agonists and antagonists.[508] However, all patients had cardiovascular risk factors optimized or addressed in both arms, which does not reflect real world practice.

Some evidence suggests that orchiectomy may be safer than an LHRH agonist. The heart and T lymphocytes have receptors for LHRH. Therefore, LHRH agonists may affect cardiac contractility, vascular plaque stability, and inflammation.[509] Four hundred twenty-nine patients with metastatic prostate cancer who underwent orchiectomy were compared to 2866 patients who received LHRH agonist between 1995 and 2009.[510] Orchiectomy was associated with lower risk of fracture, peripheral arterial disease, and cardiac-related complications, although risk was similar for

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diabetes, deep vein thrombosis, pulmonary embolism, and cognitive disorders.

Evidence suggests that a link between ADT and cognitive decline, dementia, or future Alzheimer's disease may exist, although data are inconsistent, the risk is low, and the link remains to be proven.[511-518] Some data suggests that the risk of cognitive effects may be increased with the addition of a second-generation ARPI.[519]

Patients and their medical providers should be advised about these risks prior to treatment. Many side effects of ADT are reversible or can be avoided or mitigated. For example, physical activity can counter many of these symptoms and should be recommended (see NCCN Guidelines for Survivorship, available at www.NCCN.org). Specific recommendations for bone and cardiovascular health are included in the *Principles of Survivorship in Prostate Cancer* in the algorithm above and in the text that follows.

Bone Health During ADT

Medical or surgical ADT is associated with greater risk for osteoporosis and clinical fractures. In large population-based studies, for example, ADT was associated with a 21% to 54% relative increase in fracture risk.[520-522] Longer treatment duration conferred greater fracture risk. Age and comorbidity also were associated with higher fracture incidence. In a population-based cohort of 3295 patients, surgical castration was associated with a significantly lower risk of fractures than medical castration using an LHRH agonist (HR, 0.77; 95% CI, 0.62–0.94; $P = .01$).[509] ADT increases bone turnover and decreases bone mineral density,[523-526] a surrogate for fracture risk in patients with non-metastatic disease. Bone mineral density of the hip and spine decreases by approximately 2% to 3% per year during initial therapy. Most studies have reported that bone mineral density continues to decline steadily during long-term therapy. ADT significantly decreases muscle mass,[527] and treatment-related sarcopenia appears to contribute to frailty and increased risk of falls in patients who are older.

The NCCN Guidelines Panel recommends screening and treatment for osteoporosis according to guidelines for the general population from the Bone Health and Osteoporosis Foundation, as outlined in the algorithm above.[528]

Earlier randomized controlled trials demonstrated that bisphosphonates increase bone mineral density, a surrogate for fracture risk, during ADT.[529-531] In 2011, the FDA approved denosumab as a treatment to prevent bone loss and fractures during ADT. Denosumab binds to and inhibits the receptor activator of NF-κB ligand (RANKL) to blunt osteoclast function and delay generalized bone resorption and local bone destruction. Approval was based on a phase 3 study that randomized 1468 patients with non-metastatic prostate cancer undergoing ADT to either biannual denosumab or placebo. At 24 months, denosumab increased bone mineral density by 6.7% and reduced fractures (1.5% vs. 3.9%) compared to placebo.[532] Denosumab also was approved for prevention of SREs in patients with bone metastasis (see *Chemotherapy, Immunotherapy, and Targeted Therapy*).

Currently, treatment with denosumab, zoledronic acid, or alendronate is recommended when the absolute fracture risk warrants drug therapy, as outlined in the algorithm above. A baseline DEXA scan before start of therapy and a follow-up DEXA scan after 1 year of therapy is recommended by the International Society for Clinical Densitometry to monitor response. Use of biochemical markers of bone turnover is not recommended. There are no existing guidelines on the optimal frequency of vitamin D testing.

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Diabetes and Cardiovascular Disease

Cardiovascular disease and diabetes are leading causes of morbidity and mortality in the general population. Cardiovascular disease is also a leading cause of death in patients with prostate cancer.[533] In a landmark population-based study, ADT was associated with higher incidence of diabetes and cardiovascular disease.[534] After controlling for other variables, which included age and comorbidity, ADT with an LHRH agonist was associated with increased risk for new diabetes (HR, 1.44; $P < .001$), coronary artery disease (HR, 1.16; $P < .001$), and myocardial infarction (HR, 1.11; $P = .03$). More recent data show that the main cause for the increased risk in cardiovascular disease in this population is the use of ADT and ARPIs.[535,536]

Importantly, increased physical activity may decrease the symptoms and cardiovascular side effects of patients treated with ADT.[537] Management of modifiable cardiovascular disease risk factors can also decrease the risk of subsequent cardiovascular events.

ADT may also prolong the QT/QTc interval. Providers should consider whether the benefits of ADT outweigh the potential risks in patients with congenital long QT syndrome, congestive heart failure, and frequent electrolyte abnormalities, and in patients taking drugs known to prolong the QT interval. Electrolyte abnormalities should be corrected, and periodic monitoring of electrocardiograms and electrolytes should be considered.

Based on the observed adverse metabolic effects of ADT and the association between ADT and higher incidence of diabetes and cardiovascular disease, screening for and intervention to prevent/treat diabetes and cardiovascular disease are recommended for patients receiving ADT, as described in the *Principles of Survivorship in Prostate Cancer*, in the algorithm above.

Management of mCSPC

ADT with treatment intensification is strongly recommended for patients with metastatic castration-sensitive prostate cancer (mCSPC). The use of ADT monotherapy in this setting is discouraged unless there are clear contraindications to combination therapy. Treatment intensification options include doublet therapy of ADT with an androgen receptor pathway inhibitor (ARPI; abiraterone, apalutamide, darolutamide, or enzalutamide); triplet therapy of ADT with docetaxel and one of the same ARPIs; metastasis-directed therapy (MDT) for oligometastases (see *MDT for Oligometastatic CSPC*, below); ADT with niraparib and abiraterone for *BRCA2*-mutated tumors; or ADT with EBRT to the primary tumor with or without docetaxel, abiraterone, apalutamide, or enzalutamide for low-metastatic burden (see *EBRT to the Primary Tumor in Low-Metastatic-Burden M1 Disease*, above). The specific recommended therapy options vary depending on whether the metastases were diagnosed in the synchronous or metachronous setting and on whether disease is oligometastatic, low-volume metastatic, or high-volume metastatic.

The data supporting doublet or triplet therapy in this setting are discussed below. For some of the combinations recommended by the Panel, supporting data are limited. They are included based on extrapolation from the studies of other agents, since the Panel considers the four ARPIs with approval in prostate cancer to be generally interchangeable.

Doublet Therapies for mCSPC

Abiraterone Acetate in mCSPC

In February 2018, the FDA approved abiraterone in combination with prednisone for mCSPC. This approval was based on two randomized phase 3 clinical trials of abiraterone and low-dose prednisone plus ADT in patients with newly diagnosed metastatic prostate cancer or high-risk or node-positive disease (STAMPEDE and LATITUDE) that demonstrated improved OS over ADT alone.[538]

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In LATITUDE, 1199 patients with high-risk mCSPC were randomized to abiraterone with prednisone 5 mg once daily or matching placebos. High-risk disease was defined as at least two of the following: Gleason score 8–10, ≥3 bone metastases, and visceral metastases.[538] Efficacy was demonstrated at the first interim analysis, and the trial was unblinded. The primary endpoint of OS was met and favored abiraterone (HR, 0.62; 95% CI, 0.51–0.76; *P* < .0001). Estimated 3-year OS rates improved from 49% to 66% at 30-month follow-up. Secondary endpoints were improved and included delayed castration-resistant radiographic progression (from median 14.8–33.2 months), PSA progression (7.4–33.2 months), time to pain progression, and initiation of chemotherapy. After the first interim analysis, 72 patients crossed over from placebo to abiraterone. Final OS analysis of LATITUDE after a median follow-up of 51.8 months showed median OS was significantly longer in the abiraterone group than in the placebo group (53.3 vs. 36.5 months; HR, 0.66; 95% CI, 0.56–0.78; *P* < .0001).[539]

Adverse events were higher with abiraterone and prednisone but were generally mild in nature and largely related to mineralocorticoid excess (ie, hypertension, hypokalemia, edema), hormonal effects (ie, fatigue, hot flushes), and liver toxicity.[538] Cardiac events, such as atrial fibrillation, were rare but slightly increased with abiraterone. The overall discontinuation rate due to side effects was 12%. Patient-reported outcomes were improved with the addition of abiraterone, with improvements in pain intensity progression, fatigue, functional decline, prostate cancer-related symptoms, and overall health-related QOL.[540] A limitation of this trial is that only 27% of placebo-treated patients received abiraterone or enzalutamide at progression, and only 52% of these patients received any life-prolonging therapy.[538]

The second randomized trial (STAMPEDE) of 1917 patients with CSPC demonstrated similar OS benefits.[541] However, unlike LATITUDE, STAMPEDE eligibility permitted patients with high-risk N0,M0 disease (2 of 3 high-risk factors: stage T3/4, PSA >40, or Gleason score 8–10; n = 509), or N1,M0 disease (pelvic nodal metastases; n = 369) in addition to patients with metastatic disease, who made up the majority of patients (n = 941). Most patients were newly diagnosed, while a minority had recurrent, high-risk, or metastatic disease after local therapy (n = 98). Thus, STAMPEDE was a heterogeneous mix of patients with high-risk, non-metastatic, node-positive, or metastatic disease. In patients with M1 disease, treatment with abiraterone plus prednisone was continued until progression. In patients with N1 or M0 disease, 2 years of abiraterone plus prednisolone was used if curative-intent EBRT was utilized. OS was improved in the overall population (HR, 0.63; 95% CI, 0.5–0.76; *P* < .0001) and in the M1 and N1 subsets, without any heterogeneity of treatment effect by metastatic status. The survival benefit of abiraterone was larger in patients <70 years of age than those ≥70 years (HR, 0.94 vs. HR, 0.51). Patients ≥70 years also suffered increased toxicities, which suggests heterogeneity in clinical benefits by age and comorbidity. The secondary endpoint of FFS, which included PSA recurrence, was improved overall (HR, 0.29; *P* < .0001) and in all subgroups regardless of M1 (HR, 0.31), N1 (HR, 0.29), or M0 (HR, 0.21) status. No heterogeneity for FFS was observed based on subgroups or by age. In this trial, subsequent life-prolonging therapy was received by 58% of those in the control group, which included 22% who received abiraterone and 26% who received enzalutamide. Thus, these data reflect a survival advantage of initial abiraterone in newly diagnosed patients compared with deferring therapy to the CRPC setting.

Adverse events in STAMPEDE were similar to that reported in LATITUDE, but were increased in patients ≥70 years, with higher incidences of grade 3–5 adverse events with abiraterone (47% vs. 33%) and 9 versus 3 treatment-related deaths. Severe hypertension or cardiac disorders were noted in 10% of patients and grade 3–5 liver toxicity in 7%, which

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illustrates the need for blood pressure and renal and hepatic function monitoring.

Taken together, these data led the NCCN Panel to recommend abiraterone with 5-mg once-daily prednisone as a treatment option with ADT for patients with newly diagnosed mCSPC (category 1). Alternatively, the fine-particle formulation of abiraterone can be used with 4 mg methylprednisolone PO twice daily (category 2B; see *Abiraterone Acetate in mCRPC*, below).

The standard formulation of abiraterone can be given at 250 mg/day and administered following a low-fat breakfast as an alternative to the dose of 1000 mg/day after an overnight fast (see *Abiraterone Acetate in mCRPC*, below).[542] The cost savings may reduce financial toxicity and improve adherence in those who will not take or cannot afford the standard dose.

Apalutamide in mCSPC

The double-blind phase 3 TITAN clinical trial randomized 1052 patients with mCSPC to ADT with apalutamide (240 mg/day) or placebo.[543] Participants were stratified by Gleason score at diagnosis, geographic region, and previous docetaxel treatment. The median follow-up was 22.7 months. Both primary endpoints were met: radiographic PFS (68.2% vs. 47.5% at 24 months; HR for radiographic progression or death, 0.48; 95% CI, 0.39–0.60; $P < .001$) and OS (82.4% vs. 73.5% at 24 months; HR for death, 0.67; 95% CI, 0.51–0.89; $P = .005$). Adverse events that were more common with apalutamide than with placebo included rash, hypothyroidism, and ischemic heart disease. Health-related QOL was maintained during treatment.[544] At final analysis of TITAN, median OS was improved with apalutamide plus ADT compared with ADT alone after a median follow-up of 44 months (not reached vs. 52.2 months; HR, 0.65; 95% CI, 0.53–0.79; $P < .001$).[545]

Apalutamide is a category 1 option for patients with mCSPC. The FDA approved this indication in September 2019.

Enzalutamide in mCSPC

The open-label randomized phase 3 ENZAMET clinical trial compared enzalutamide (160 mg/day) plus ADT (LHRH analog or surgical castration) with a first-generation antiandrogen (bicalutamide, nilutamide, or flutamide) plus ADT in 1125 patients with mCSPC.[546] Stratification was by volume of disease, planned use of early docetaxel, planned use of bone antiresorptive therapy, comorbidity score, and trial site. The primary endpoint of OS was met at the first interim analysis with median follow-up of 34 months (HR for death, 0.67; 95% CI, 0.52–0.86; $P = .002$). Enzalutamide also improved secondary endpoints, such as PFS using PSA levels and clinical PFS. An additional analysis was triggered at 470 deaths.[547] After a median follow-up of 68 months, the 5-year OS rate was again lower in the first-generation antiandrogen group than in the enzalutamide group (HR, 0.70; 95% CI, 0.58–0.84; $P < .0001$). The median OS was not reached.

In the double-blind randomized phase 3 ARCHES clinical, 1150 patients with mCSPC were randomized to receive ADT with either enzalutamide (160 mg/day) or placebo. Participants were stratified by disease volume and prior docetaxel use. The primary endpoint was radiographic PFS, which was improved in the enzalutamide group after a median follow-up of 14.4 months (19.0 months vs. not reached; HR, 0.39; 95% CI, 0.30–0.50; $P < .001$).[548] At the final, prespecified OS analysis, median OS was not met in either group, but a 34% reduction in the risk of death was observed in those receiving enzalutamide versus placebo (HR, 0.66; 95% CI, 0.53–0.81; $P < .001$).[549] This result could be an underestimate of the effect of enzalutamide, since approximately 32% of the patients assigned placebo crossed over to enzalutamide after unblinding.

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The safety of enzalutamide in these trials was similar to that seen in previous trials in the castration-resistant setting. Adverse events associated with enzalutamide in these trials included fatigue, seizures, and hypertension.[546,548]

Enzalutamide is a category 1 option for patients with mCSPC. The FDA approved this indication in December 2019.

Darolutamide in mCSPC

The phase 3 ARANOTE trial assessed darolutamide with ADT compared with placebo and ADT in 669 patients with mCSPC.[550] The primary endpoint of radiological PFS was improved in the darolutamide arm compared with the placebo arm (HR, 0.54; 95% CI, 0.41–0.71; $P < .0001$). The benefit was consistent across the low- and high-volume subgroups. Some of the secondary endpoints were also met, including delayed time to mCRPC and time to pain progression. However, a significant improvement in OS was not evident in the current follow-up (HR, 0.81; 95% CI, 0.59–1.12).

The FDA approved this indication for darolutamide in June 2025. The Panel include darolutamide with ADT as an option for patients with low- and high-volume mCSPC. Because an OS benefit has not been demonstrated for darolutamide doublet therapy, it is not a category 1 recommendation at this time.

Docetaxel in mCSPC

Docetaxel with ADT has been studied as an upfront option for patients with mCSPC in two phase 3 trials (ECOG 3805/CHAARTED and STAMPEDE).[551,552] CHAARTED randomized 790 patients with mCSPC to docetaxel (75 mg/m^2 IV every 3 weeks x 6 doses) plus ADT or ADT alone.[552] After a median follow-up of 53.7 months, the patients in the combination arm experienced a longer OS than those in the ADT arm (57.6 vs. 47.2 months; HR, 0.72; 95% CI, 0.59–0.89; $P = .002$).[553]

Subgroup analysis showed that the survival benefit was more pronounced in the 65% of participants with high-volume disease (HR, 0.63; 95% CI, 0.50–0.79; $P < .001$). Patients with low metastatic burden in CHAARTED did not derive a survival benefit from the inclusion of docetaxel (HR, 1.04; 95% CI, 0.70–1.55; $P = .86$).

The STAMPEDE trial, a multi-arm, multi-stage phase 3 trial, included patients with both M0 and M1 CSPC.[551] The results in the M1 population confirmed the survival advantage of adding docetaxel (75 mg/m^2 IV every 3 weeks x 6 doses) to ADT seen in the CHAARTED trial. In STAMPEDE, extent of disease was not evaluated in the 1087 patients with metastatic disease, but the median OS for all patients with M1 disease was 5.4 years in the ADT-plus-docetaxel arm versus 3.6 years in the ADT-only arm (a difference of 1.8 years between groups compared with a 1.1-year difference in CHAARTED).

Patients with low metastatic burden did not have definitively improved survival outcomes in the ECOG CHAARTED study or a similar European trial (GETUG-AFU 15).[552,554,555] Furthermore, the triplet options of ADT with docetaxel and either abiraterone or darolutamide showed improved OS over ADT with docetaxel (see below). The Panel therefore does not include docetaxel with ADT as an option for patients with mCSPC. Rather, patients with high-volume mCSPC who are fit for chemotherapy should be considered for triplet therapy.

Triplet Therapies for mCSPC

Data from the PEACE-1 and ARASENS trials indicate that triplet therapies of ADT with docetaxel and an ARPI —either abiraterone or darolutamide— improve OS over ADT with docetaxel in patients with high-volume metastatic CSPC.[556,557] These trials are discussed below. Both of these combinations are included as category 1, preferred options for patients with high-volume mCSPC, and their use is encouraged for patients with

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high-volume disease who are fit for chemotherapy. However, the Panel notes that no studies have compared doublet therapies (ADT plus an ARPI; discussed above) to triplet therapies. Therefore, doublet therapies are also suitable options for patients with high-volume metastatic disease.

These triplet combinations are also included as options in the low-volume, synchronous CSPC setting based on results of a meta-analysis, although their use in this setting is controversial and should be reserved for patients who desire aggressive treatment.[558] No data support the use of triplet therapy in low-volume metachronous mCSPC, and they are not recommended in this setting.

Docetaxel Plus Abiraterone in CSPC

PEACE-1 was an international, open-label, randomized, phase 3 study conducted in seven European countries.[556] Using a 2 × 2 factorial design, 1173 patients with de novo metastatic prostate cancer were randomized at a 1:1:1:1 ratio to standard of care (ADT alone or with docetaxel), standard of care with RT, standard of care with abiraterone, or standard of care with radiation and abiraterone. The two primary endpoints of the trial were radiographic PFS and OS. Adjusted Cox regression modelling showed no interaction between abiraterone and RT, so data were pooled for the analysis of abiraterone efficacy. Consistent with results of older studies, at a median follow-up of 3.5 years, radiographic PFS was longer in patients who received abiraterone than in those that did not (HR, 0.54; 99.9% CI, 0.41–0.71; $P < .0001$) as was OS (HR, 0.82; 95.1% CI, 0.69–0.98; $P = .030$). An OS benefit with abiraterone was also seen in the subset of patients with high metastatic burden as defined by CHAARTED criteria (HR, 0.77 ;95% CI, 0.62–0.96), but was not seen in those with low metastatic burden (HR, 0.93 ;95% CI, 0.69–1.28).

As part of the analysis, the efficacy of abiraterone was assessed in the population that received docetaxel. As in the overall population, radiographic PFS (HR, 0.50; 99.9% CI, 0.34–0.71; $P < .0001$) and OS (HR, 0.75; 95.1% CI, 0.59–0.95; $P = .017$) were longer in those receiving all three therapies compared with those only receiving ADT and docetaxel. The populations receiving the triplet and doublet therapies experienced similar rates neutropenia, febrile neutropenia, fatigue, and neuropathy, although grade ≥3 adverse events occurred in 63% of patients who received the triplet combination compared with 52% of those receiving ADT and docetaxel.

Based on these data, the Panel includes this regimen as a category 1 option for patients with high-volume mCSPC.

Docetaxel Plus Darolutamide in CSPC

The international, phase 3 trial ARASENS trial, the second phase 3 trial evaluating a triplet therapy, randomized 1306 patients with mCSPC to receive ADT and docetaxel with either darolutamide or matching placebo.[557] The primary endpoint, OS, was improved in the darolutamide group at 4 years (62.7%; 95% CI, 58.7–66.7) compared with the placebo group (50.4%; 95% CI, 46.3–54.6). The risk of death was lower in the darolutamide group by about 32% (HR, 0.68; 95% CI, 0.57–0.80; $P < .001$). The addition of darolutamide also showed significant benefits over placebo for secondary efficacy endpoints, including time to CRPC (HR, 0.36; 95% CI, 0.30–0.42; $P < .001$), skeletal event–free survival (HR, 0.61; 95% CI, 0.52–0.72; $P < .001$), and time to initiation of subsequent systemic antineoplastic therapy (HR, 0.39; 95% CI, 0.33–0.46; $P < .001$). Subgroup analysis showed a similar improvement in OS in those with high-volume disease as defined by CHAARTED criteria as in the overall population.[559] An OS benefit was not observed in those with low metastatic burden (HR, 0.68; 95% CI, 0.41–1.13), but median survival was not reached in either arm.

Adverse events of any grade, grade 3 to 5 adverse events, and serious adverse events occurred at similar incidence levels between the two arms. Many of these were known effects of docetaxel. The most frequent

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adverse events were alopecia (40.5% of patients in the darolutamide arm vs. 40.6% with placebo), neutropenia (39.3% vs. 38.8%), fatigue (33.1% vs. 32.9%), and anemia (27.8% vs. 25.1%). Exceptions were rash (16.6% vs. 13.5%) and hypertension (13.7% vs. 9.2%), which are known effects of androgen receptor pathway inhibitors and were more frequent in the darolutamide group.

The FDA approved this indication in August 2022, and the Panel includes this regimen as a category 1 option for patients with high-volume mCSPC.

PARP Inhibitor Combination for mCSPC

Niraparib Plus Abiraterone in BRCA2-Mutated mCSPC

ADT with PARP inhibitors alone or with ARPIs have been options for patients with HRR-mutated mCRPC for several years (see *Treatment Options for Patients with DNA Repair Gene Mutations*, below). The randomized, double-blind, phase 3 AMPLITUDE trial assessed ADT with niraparib and abiraterone compared with ADT plus abiraterone doublet therapy in 696 patients with homologous recombination repair (HRR)-deficient mCSPC.[560] Most patients had high-volume disease (78%) and/or synchronous disease (87%) and most had *BRCA2* (42%) or *BRCA1* (7.2%) gene alterations. Sixteen percent of patients had received docetaxel. After a median follow-up of 30.8 months, the primary endpoint of median radiographic PFS (rPFS) was met in the *BRCA* subset (not reached vs. 26.0 months; HR, 0.52; 95% CI, 0.37–0.72; $P < .0001$) as well as in the full intention-to-treat population (not reached vs. 29.5 months; HR, 0.63; 95% CI, 0.49–0.80; $P = 0.0001$). This effect was largely driven by the *BRCA2*-mutated subset. OS data are immature, and the trial did not include formal cross-over to PARP inhibition for patients in the abiraterone group. Roughly one-third of participants in the control group received a PARP inhibitor following progression to mCRPC at the time of analysis. Grade 3/4 adverse events were more common in the niraparib arm (75% vs. 59%) as were treatment-emergent adverse events leading to death (14 participants vs. 7 participants). The most common grade 3/4 adverse events were anemia (29.1% in the niraparib group versus 4.6% in the control group) and hypertension (26.5% versus 18.4%).

The FDA approved the combination of niraparib plus abiraterone with ADT for patients with *BRCA2*-mutated mCSPC in December 2025. It is included in these Guidelines as an option for patients with high-volume, synchronous or metachronous, mCSPC. It is also included for patients with synchronous or metachronous low-volume mCSPC (category 2B), but the Panel cautions that this regimen adds considerable toxicity and the benefit in this population is not yet clear due to the low numbers of such patients included in the AMPLITUDE trial. The panel notes that the niraparib plus abiraterone combination can be used following the completion of docetaxel in patients receiving triplet therapy or following EBRT to the primary tumor.

EBRT to the Primary Tumor in Synchronous Low-Volume M1 Disease

Patients with newly diagnosed, low-volume metastatic prostate cancer can be considered for ADT with EBRT to the primary tumor based on results from the randomized controlled phase 3 STAMPEDE trial.[561] In this multicenter, international study, 2061 patients were randomized to lifelong ADT with or without EBRT to the primary tumor (either 55 Gy in 20 daily fractions over 4 weeks or 36 Gy in 6 weekly fractions over 6 weeks). The primary outcome of OS by intention-to-treat (ITT) analysis was not met (HR, 0.92; 95% CI, 0.80–1.06; $P = .266$), but EBRT improved the secondary outcome of failure-free survival (FFS; HR, 0.76; 95% CI, 0.68–0.84; $P < .0001$). In a pre-planned subset analysis, outcomes of patients with high-metastatic burden (defined as visceral metastases; ≥4 bone metastases with ≥1 outside the vertebral bodies or pelvis; or both) and

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those with low-metastatic burden (all others) were determined. EBRT improved OS (adjusted HR, 0.68; 95% CI, 0.52–0.90), prostate cancer-specific survival (adjusted HR, 0.65; 95% CI, 0.47–0.90), FFS (adjusted HR, 0.59; 95% CI, 0.49–0.72), and PFS (adjusted HR, 0.78; 95% CI, 0.63–0.98) in patients with low-metastatic burden, but not in patients with high metastatic burden. Long-term results have been reported, confirming the benefit of RT to the primary tumor in the setting of ADT with or without docetaxel.[562]

Abiraterone may be added to ADT with EBRT to the primary tumor in patients with low-volume synchronous metastases based on results of the PEACE-1, open-label, randomized trial.[563] In this trial, participants were randomized 1:1:1:1 to ADT alone or with docetaxel (standard of care, SOC), SOC with abiraterone, SOC with radiation to the prostate, or SOC with abiraterone and radiation to the prostate. Results demonstrated that the addition of RT to the primary tumor in patients with low-volume disease treated with abiraterone led to improvements in median radiographic PFS (4.4 vs. 7.5 years). RT to the primary tumor also reduced rates of serious genitourinary toxicity regardless of disease volume, and time to castration resistance was delayed in the full population. Thus, some patients with high-volume disease may also benefit from RT to the primary tumor.

In PEACE-1, the benefits of RT to the primary tumor were only seen in patients receiving abiraterone, not in those receiving ADT alone or with docetaxel.[563] However, in a secondary analysis of the STAMPEDE trial, the benefits of RT on OS and FFS in patients with low-volume disease were seen regardless of planned docetaxel use.[564] The Panel therefore includes the addition of docetaxel to ADT and EBRT to the primary tumor as a category 2B recommendation for patients with low-volume synchronous mCSPC.

MDT for Oligometastatic CSPC

Treatment of metastatic sites with local therapy with the intent to improve oncologic outcomes (eg, delaying the initiation of systemic therapy or ADT; improving PFS, radiographic PFS, or OS) is known as metastasis-directed therapy, or MDT. MDT has been primarily studied as metastasis-directed RT (MDRT) and with the highly selected use of surgical lymph node dissection. MDRT is delivered at a higher-than-palliative dose to provide durable local control of the areas targeted and is the most used form of MDT.

MDT is used in patients with oligometastatic disease. The number of metastatic sites to define oligometastatic disease remains an evolving space and is impacted by the sensitivity of the imaging modality used. Early studies included patients with 1 to 3 or 1 to 5 metastatic sites by CT, MRI, or bone scan.[565-568] More recent studies allow for up to 10 metastatic sites by PSMA-PET imaging (NCT04787744, NCT06150417, NCT03721341). The upper limit is not clearly established and is both a function of oncologic limitations and technical limitations of treating numerous metastatic sites. It should be noted that the goal of MDT is generally to treat all metastatic sites, which may include regional lymph nodes and, potentially, the primary tumor if untreated or if there is evidence of local recurrence. The primary tumor in this setting should be counted as a site. The Panel notes that general exclusion limits of >5 and >10 metastases by CT, MRI, or bone scan or by PSMA-PET imaging, respectively, are appropriate.

The best evidence supporting the use of MDT in the CSPC setting comes from randomized phase 2 studies in patients with metachronous oligorecurrent disease.[565,568-570] These trials mostly used MDRT and showed that the approach improved ADT-free survival or PFS over monitoring or ADT. For example, the ORIOLE trial included 54 previously treated patients with 1 to 3 metastases by conventional imaging who

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were randomized to receive MDRT or observation.[570] Median PFS was better in the MDRT group than in the observation group (not reached vs. 5.8 months; HR, 0.30; 95% CI, 0.11–0.81; *P* = .002), and the treatment was well tolerated. STOMP randomized 62 patients with biochemically recurrent CSPC and ≤3 metastases to surveillance or to MDT with surgery or RT.[569] The median ADT-free survival was improved in the MDT arm after a median 3-year follow-up (13 vs. 21 months; HR, 0.60; 80% CI, 0.40–0.90; log-rank *P* = .11). In a combined, longer-term analysis of ORIOLE and STOMP, median PFS was longer with MDT compared with observation (pooled HR, 0.44; 95% CI, 0.29–0.66; *P* < .001) after a median follow-up of 52.5 months.[565]

The EXTEND trial included a more heterogeneous group of patients, as 24 of the 87 enrolled patients had no prior definitive therapy to the prostate and 7 patients had mCRPC.[568] Participants had ≤5 metastases amenable to MDRT and were randomized to MDRT with intermittent ADT or to intermittent ADT alone. After a median follow-up of 22 months, median PFS was improved in the MDRT/ADT group compared with the ADT-only group (not reached vs. 15.8 months; HR, 0.25; 95% CI, 0.12–0.55; *P* < .001). Analysis of a separate basket of participants in EXTEND who received continuous ADT have also been reported.[571] Results showed that the inclusion of MDT improved the primary endpoint of PFS improved in the participants who received continuous ADT (47 vs. 22 months; HR, 0.50; one-sided *P* = .036) and in the combined group of intermittent or continuous ADT (36 months vs. 17 months; HR, 0.45; *P* <.001).

The SABR-COMET phase 2 study, which enrolled patients with breast, lung, colorectal, and prostate cancers who had a controlled primary tumor and 1 to 5 metastases amenable to MDRT, showed an improvement on OS with an MDT approach.[572] SABR-COMET included 16 patients with prostate cancer; 14 were randomized to the MDRT arm, and 2 were randomized to receive palliative RT. Patients in both arms received palliative systemic therapy as appropriate. After a median follow-up of 51 months, improvements were seen in 5-year OS rate (17.7% vs. 42.3%; 95% CI, 0.28-0.56; stratified log-rank *P* = .006) in the total population of 99 patients. A post-hoc sensitivity analysis was used to address the imbalance in the distribution of patients with prostate cancer between the two arms of the study. When patients with prostate cancer were excluded from the analysis, the 5-year OS rate continued to trend in favor of the MDRT group (16.2% vs. 33.1%; stratified log-rank test *P* = .085).

The benefit of adding ADT to MDRT in patients with oligorecurrent CSPC was assessed in the randomized, phase 2 RADIOSA trial.[567] The 105 enrolled patients were randomized to 6 months of ADT with MDRT or MDRT alone. After a median follow-up of 31 months, the median clinical PFS was improved in the group receiving ADT (32.2 vs. 15.1 months; HR, 0.43; 95% CI, 0.26–0.72; *P* = .001).

Based on these data, the Panel recommends MDT with or without ADT as an option for patients with metachronous oligometastatic CSPC. These patients may alternatively be treated with ADT plus systemic therapy for low-volume mCSPC with or without concurrent MDT.

There is currently no randomized evidence in the synchronous oligometastatic setting, just single-arm prospective trials and retrospective cohorts. However, the Panel believes that concurrent MDT with recommended systemic therapy can be considered in select patients with synchronous oligometastatic disease.

Progression to and Management of CRPC

Most advanced disease eventually stops responding to traditional ADT and is categorized as castration-resistant (also known as castration-recurrent). CRPC is defined as prostate cancer that progresses clinically, radiographically, or biochemically despite castrate levels of serum

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testosterone (<50 ng/dL).[573] Patients whose disease progresses to CRPC during primary ADT should receive a laboratory assessment to assure a castrate level of testosterone (<50 ng/dL; <1.7 nmol/L). Imaging tests may be indicated to monitor for signs of distant metastases. Factors affecting the frequency of imaging include individual risk, age, overall patient health, PSA velocity, and Gleason grade.

For patients who develop CRPC, ADT with orchiectomy or an LHRH agonist or antagonist should be continued to maintain castrate serum levels of testosterone (<50 ng/dL).

Patients with CRPC and no signs of distant metastasis on conventional imaging studies (M0) can consider monitoring with continued ADT if PSADT is >10 months (preferred), because these patients will have a relatively indolent disease history.[574] Secondary hormone therapy with continued ADT is an option mainly for patients with shorter PSADT (≤10 months) as described below.

For patients who develop mCRPC, metastatic lesion biopsy is recommended, as is MSI/MMR testing, if not previously performed. If MSI-H or dMMR is found, referral to genetic counseling should be made to assess for the possibility of Lynch syndrome. These patients should also have germline and tumor testing to check for mutations in homologous recombination repair (HRR) genes (eg, *BRCA1*, *BRCA2*, *ATM*, *PALB2*, *FANCA, RAD51D, CHEK2, CDK12*) if not done previously.[575] This information may be used for genetic counseling, cascade germline testing for family members, early use of platinum chemotherapy, and understanding eligibility for biomarker-directed treatments or clinical trials. TMB testing is also recommended for patients with mCRPC.

ADT is continued in patients with mCRPC while additional therapies, including secondary hormone therapies, chemotherapies, immunotherapies, radiopharmaceuticals, and/or targeted therapies, are applied sequentially or concurrently, as discussed in the sections that follow; all patients should receive best supportive care. The Panel defined treatment options for patients with mCRPC based on previous exposure to ARPIs (abiraterone, enzalutamide, darolutamide, or apalutamide) and docetaxel. Abiraterone given as part of neoadjuvant/concomitant/adjuvant ADT with EBRT is not considered prior ARPI therapy.

The decision to initiate therapy in the CRPC setting after disease progression on one or more treatments should be based on the available high-level evidence of safety, efficacy, and tolerability of these agents and the application of this evidence to an individual patient. Prior exposures to therapeutic agents should be considered. Data to inform the optimal sequence for delivery of these agents in patients with mCRPC is evolving (see *Sequencing of Therapy in CRPC*, below). Choice of therapy is based largely on clinical considerations, which include patient preferences, prior treatment, presence or absence of visceral disease, symptoms, and potential side effects.

NCCN recommends that patients being treated for CRPC be closely monitored with radiologic imaging (ie, CT, bone imaging), PSA tests, and clinical exams for evidence of progression. Therapy should be continued until clinical progression or intolerability, with consideration of the fact that even in cases where PSA remains undetectable, bone imaging may reveal progression.[576,577] The sequential use of these agents is recommended in patients who remain candidates for further systemic therapy. The Panel also notes that Pan-cancer, tumor-agnostic treatments can be considered for patients with mCRPC who have actionable mutations. Clinical trial and best supportive care are additional options.

Secondary Hormone Therapy for CRPC

Research has shown enhancement of autocrine and/or paracrine androgen synthesis in the tumor microenvironment of patients receiving

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ADT.[578,579] Androgen signaling consequent to non-gonadal sources of androgen in CRPC refutes earlier beliefs that CRPC was resistant to further hormone therapies. The development of novel ARPIs demonstrating efficacy in the non-metastatic CRPC and mCRPC settings dramatically changed the paradigm of CRPC treatment over the past two decades.

Abiraterone Acetate in mCRPC

In April 2011, the FDA approved the androgen synthesis inhibitor, abiraterone, in combination with low-dose prednisone, for the treatment of patients with mCRPC who have received prior chemotherapy containing docetaxel. This FDA approval in the post-docetaxel, mCRPC setting was based on the results of a phase 3, randomized, placebo-controlled trial (COU-AA-301) in patients with mCRPC previously treated with docetaxel-containing regimens.[580,581] Patients were randomized to receive either abiraterone 1000 mg orally once daily (n = 797) or placebo once daily (n = 398), and both arms received daily prednisone. In the final analysis, median survival was 15.8 versus 11.2 months in the abiraterone and placebo arm, respectively (HR, 0.74; 95% CI, 0.64–0.86; $P < .0001$).[581] Time to radiographic progression, PSA decline, and pain palliation also were improved by abiraterone.[581,582]

FDA approval in the pre-docetaxel setting occurred in December 2012, and was based on the randomized phase 3 COU-AA-302 trial of abiraterone and prednisone (n = 546) versus prednisone alone (n = 542) in patients with asymptomatic or minimally symptomatic, mCRPC.[583] Most participants in this trial were not taking narcotics for cancer pain and none had visceral metastatic disease or prior ketoconazole exposure. The coprimary endpoint of radiographic PFS was improved from 8.3 to 16.5 months with abiraterone (HR, 0.53; $P < .001$). OS was improved at final analysis with a median follow-up of 49.2 months (34.7 vs. 30.3 months; HR, 0.81; 95% CI, 0.70–0.93; $P = .003$).[584] Key secondary endpoints of time to symptomatic deterioration, time to chemotherapy initiation, time to pain progression, and PSA PFS improved significantly with abiraterone treatment; PSA declines (62% vs. 24% with >50% decline) and radiographic responses (36% vs. 16% RECIST responses) were more common.

The most common adverse reactions with abiraterone/prednisone (>5%) were fatigue (39%); back or joint discomfort (28%–32%); peripheral edema (28%); diarrhea, nausea, or constipation (22%); hypokalemia (17%); hypophosphatemia (24%); atrial fibrillation (4%); muscle discomfort (14%); hot flushes (22%); urinary tract infection; cough; hypertension (22%, severe hypertension in 4%); urinary frequency and nocturia; dyspepsia; or upper respiratory tract infection. The most common adverse drug reactions that resulted in drug discontinuation were increased aspartate aminotransferase and/or alanine aminotransferase (11%–12%), or cardiac disorders (19%, serious in 6%).

Based on the studies described here, abiraterone is a category 1, preferred option for mCRPC without prior ARPI therapy. It can also be considered in patients with mCRPC following progression on another ARPI, although other therapies are preferred in this setting.

In May 2018, the FDA approved a novel, fine-particle formulation of abiraterone, in combination with methylprednisolone, for the treatment of patients with mCRPC. In studies of healthy males, this formulation at 500 mg was shown to be bioequivalent to 1000 mg of the originator formulation.[585,586] In a phase 2 therapeutic equivalence study, 53 patients with mCRPC who were not treated previously with abiraterone, enzalutamide, radium-223, or chemotherapy (docetaxel for mCRPC completed ≥1 year prior to enrollment was allowed) were randomized to 500 mg daily of the new, fine-particle formulation plus 4 mg methylprednisolone orally twice daily or to 1000 mg of the originator formulation daily plus 5 mg prednisone orally twice daily.[587]

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Bioequivalence of these doses was confirmed based on serum testosterone levels, PSA response, and abiraterone pharmacokinetics. The rates of total and grade 3/4 adverse events were similar between the arms, with musculoskeletal and connective tissue disorders occurring more frequently in the originator-treated patients (37.9% vs. 12.5%). The Panel believes that the fine-particle formulation of abiraterone can be used instead of the original formulation of abiraterone in the treatment of patients with mCRPC (category 2A).

Abiraterone should be given with concurrent steroid (either oral prednisone 5 mg twice daily or oral methylprednisolone 4 mg twice daily, depending on which formulation is given) to abrogate signs of mineralocorticoid excess that can result from treatment. These signs include hypertension, hypokalemia, and peripheral edema. Thus, monitoring of liver function, potassium and phosphate levels, and blood pressure readings on a monthly basis is warranted during abiraterone therapy. Symptom-directed assessment for cardiac disease also is warranted, particularly in patients with pre-existing cardiovascular disease.

A randomized phase 2 noninferiority study of 75 patients with mCRPC compared 1000 mg/day abiraterone after an overnight fast with 250 mg/day after a low-fat breakfast.[542] The primary endpoint was log change in PSA, with secondary endpoints of PSA response (≥50%) and PFS. The primary endpoint favored the low-dose arm (log change in PSA, -1.59 vs. -1.19), as did the PSA response rate (58% vs. 50%), with an equal PFS of 9 months in both arms. Noninferiority of the low dose was established according to the predefined criteria. Therefore, abiraterone can be given at 250 mg/day administered following a low-fat breakfast, as an alternative to the dose of 1000 mg/day after an overnight fast in patients who will not take or cannot afford the standard dose. The cost savings may reduce financial toxicity and improve adherence. Food impacts absorption unpredictably; therefore, side effects should be monitored and standard dosing (1000 mg on empty stomach) utilized if excess toxicity is observed on modified dosing (250 mg with food).

Abiraterone with Dexamethasone in mCRPC

Switching from prednisone to dexamethasone 0.5 mg/day can be considered for patients with mCRPC with disease progression on either formulation of abiraterone. Trials show improved PSA responses and PFS and acceptable safety using this strategy.[588,589]

The SWITCH study was a single-arm, open-label, phase 2 study of this approach with 26 enrolled patients.[588] The primary endpoint, the proportion of patients with a PSA decline ≥30% in 6 weeks, was 46.2%. No significant toxicities were observed, and two radiologic responses were seen. In another study, 48 consecutive patients with mCRPC, with disease progression on abiraterone with prednisone, were switched to abiraterone with 0.5 mg/day dexamethasone.[589] The primary endpoint of median PFS was 10.35 months, and PSA levels decreased or stabilized in 56% of patients after switching to dexamethasone.

Enzalutamide in M0 and M1 CRPC

In August 2012, the FDA approved enzalutamide, a next-generation antiandrogen, for treatment of patients with mCRPC who had received prior docetaxel chemotherapy. Approval was based on the results of the randomized, phase 3, placebo-controlled AFFIRM trial.[590,591] AFFIRM randomized 1199 patients to enzalutamide (160 mg daily) or placebo in a 2:1 ratio and the primary endpoint was OS. Median survival was improved with enzalutamide from 13.6 to 18.4 months (HR, 0.63; $P < .001$). Survival was improved in all subgroups analyzed. Secondary endpoints were also improved significantly, which included the proportion of patients with >50% PSA decline (54% vs. 2%), radiographic response (29% vs. 4%), radiographic PFS (8.3 vs. 2.9 months), and time to first skeletal-related event (SRE) (16.7 vs. 13.3 months). QOL measured using validated

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surveys was improved with enzalutamide compared to placebo. Adverse events were mild, and included fatigue (34% vs. 29%), diarrhea (21% vs. 18%), hot flushes (20% vs. 10%), headache (12% vs. 6%), and seizures (0.6% vs. 0%). The incidence of cardiac disorders did not differ between the arms. Patients in the AFFIRM study were maintained on LHRH agonist/antagonist therapy and could receive bone supportive care medications. The seizure risk in the enzalutamide FDA label was 0.9% versus 0.6% in the manuscript.[590,592]

Another phase 3 trial studied enzalutamide in the pre-chemotherapy setting. The PREVAIL study randomly assigned 1717 patients with chemotherapy-naïve metastatic prostate cancer to daily enzalutamide or placebo.[593,594] The study was stopped early due to benefits shown in the treatment arm. Compared to the placebo group, the enzalutamide group showed improved median PFS (20.0 vs. 5.4 months) and median OS (35.3 vs. 31.3 months). Improvements in all secondary endpoints were also observed (eg, the time until chemotherapy initiation or first SRE).

Thus, enzalutamide represents a category 1, preferred treatment option for patients with mCRPC without prior ARPI therapy. It can also be considered in patients with mCRPC with prior exposure to another ARPI, although other therapies are preferred in this setting.

The randomized, double-blind, placebo-controlled phase 3 PROSPER trial assessed the use of enzalutamide in 1401 patients with non-metastatic CRPC.[595] Patients with PSADT ≤10 months were stratified according to PSADT (<6 vs. ≥6 months) and use of bone-sparing agents and randomized 2:1 to enzalutamide (160 mg/day) plus ADT or placebo plus ADT. Enzalutamide improved the primary endpoint of metastasis-free survival over placebo (36.6 vs. 14.7 months; HR for metastasis or death, 0.29; 95% CI, 0.24–0.35; $P < .0001$). Median OS was longer in the enzalutamide group than in the placebo group (67.0 vs. 56.3 months; HR for death, 0.73; 95% CI, 0.61–0.89; $P = 0.001$).[596] Adverse events included fatigue (33% vs. 14%), hypertension (12% vs. 5%), major adverse cardiovascular events (5% vs. 3%), and mental impairment disorders (5% vs. 2%). Patient-reported outcomes from PROSPER indicate that enzalutamide delayed pain progression, symptom worsening, and decrease in functional status, compared with placebo.[597]

The FDA expanded its approval for enzalutamide to include patients with non-metastatic CRPC in July 2018, and the Panel believes that patients with M0 CRPC can be offered enzalutamide, if PSADT is ≤10 months (category 1, preferred option).

Patients receiving enzalutamide have no restrictions for food intake and concurrent prednisone is permitted but not required.[590]

Apalutamide in M0 CRPC

The FDA approved apalutamide for treatment of patients with non-metastatic CRPC in February 2018. This approval was based on the phase 3 SPARTAN trial of 1207 patients with M0 CRPC and PSADT ≤10 months.[598] Participants were stratified according to PSADT (>6 vs. ≤6 months), use of bone-sparing agents, and the presence of metastatic pelvic lymph nodes (N0 vs. N1). After a median follow-up of 20.3 months, apalutamide at 240 mg/day with ADT improved the primary endpoint of metastasis-free survival over placebo with ADT (40.5 vs. 16.2 months; HR for metastasis or death, 0.28; 95% CI, 0.23–0.35; $P < .001$). Adverse events included rash (24% vs. 5.5%), fracture (11% vs. 6.5%), and hypothyroidism (8% vs. 2%). In a prespecified exploratory analysis of SPARTAN, health-related QOL was maintained in both the apalutamide and placebo groups.[599]

After a median follow-up of 52 months, final OS analysis showed that participants in SPARTAN experienced an improved median OS with apalutamide versus placebo (73.9 vs. 59.9 months; HR, 0.78; 95% CI, 0.64–0.96; $P = .016$).[600] This longer OS reached prespecified statistical

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significance, even though 19% of participants crossed over from placebo to apalutamide.

Apalutamide is a category 1, preferred option for patients with M0 CRPC if PSADT is ≤10 months.

Darolutamide in M0 CRPC

The FDA approved darolutamide for treatment of patients with non-metastatic CRPC in July 2019. The phase 3 ARAMIS study randomized 1509 patients with M0 CRPC and PSADT ≤10 months 2:1 to darolutamide (600 mg twice daily) or placebo.[601] Participants were stratified according to PSADT (>6 vs. ≤6 months) and the use of osteoclast-targeted agents. The median follow-up time was 17.9 months. Darolutamide improved the primary endpoint of metastasis-free survival compared to placebo (40.4 vs. 18.4 months; HR for metastasis or death, 0.41; 95% CI, 0.34–0.50; $P < .001$).

Patients in the placebo group of ARAMIS crossed over to darolutamide (n = 170) or received other life-prolonging therapy (n = 137). Final analysis occurred after a median follow-up time of 29.0 months. The risk of death was 31% lower in the darolutamide group than in the placebo group (HR for death, 0.69; 95% CI, 0.53–0.88; $P = .003$).[602] OS at 3 years was 83% (95% CI, 80–86) in the darolutamide group compared with 77% (95% CI, 72–81) in the placebo group. Adverse events that occurred more frequently in the treatment arm included fatigue (12.1% vs. 8.7%), pain in an extremity (5.8% vs. 3.2%), and rash (2.9% vs. 0.9%). The incidence of fractures was similar between darolutamide and placebo (4.2% vs. 3.6%).[601]

Darolutamide is a category 1, preferred option for patients with M0 CRPC if PSADT is ≤10 months.

Other Secondary Hormone Therapies

Other options for secondary hormone therapy include a first-generation antiandrogen, antiandrogen withdrawal, corticosteroid, or ketoconazole (adrenal enzyme inhibitor) with hydrocortisone.[603-605] However, none of these strategies has been shown to prolong survival in randomized clinical trials. In the mCRPC setting, these options should only be used for select patients who are not candidates for other recommended mCRPC therapies.

A randomized phase 2 trial, TRANSFORMER, compared the effect of bipolar androgen therapy (BAT) with that of enzalutamide on PFS in 195 patients with asymptomatic, mCRPC with prior progression on abiraterone.[606] BAT involves rapid cycling between high and low serum testosterone to disrupt the adaptive upregulation of the androgen receptor that occurs with low testosterone levels. Patients in the BAT arm received testosterone cypionate 400 mg intramuscularly once every 28 days. The PFS was 5.7 months in both arms (HR, 1.14; 95% CI, 0.83–1.55; $P = .42$). Crossover was allowed after disease progression, and OS was similar between the groups. BAT resulted in more favorable patient-reported QOL. The Panel awaits more data on this approach.

Chemotherapy, Immunotherapy, and Targeted Therapy for mCRPC

Research has expanded the therapeutic options for patients with mCRPC. In addition to the hormonal and radiopharmaceutical therapies described in other sections, options include chemotherapy, immunotherapy, and targeted therapy. As noted above, selection of therapy depends on patient preferences, prior treatment exposures, the presence or absence of symptoms, the location of metastases, the presence of certain biomarkers, and consideration of potential side effects.

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Docetaxel

Docetaxel was FDA-approved for mCRPC in May 2004. Two randomized phase 3 studies evaluated docetaxel-based regimens in symptomatic or rapidly progressive CRPC (TAX 327 and SWOG 9916).[607-609] TAX 327 compared docetaxel (every 3 weeks or weekly) plus prednisone to mitoxantrone plus prednisone in 1006 patients.[608] Every-3-week docetaxel resulted in higher median OS than mitoxantrone (18.9 vs. 16.5 months; $P = .009$). This survival benefit was maintained at extended follow-up.[609] The SWOG 9916 study showed improved survival with docetaxel when combined with estramustine compared to mitoxantrone plus prednisone.[607]

Thus, docetaxel is a category 1 option for treatment of docetaxel-naïve mCRPC. It is the preferred option post-ARPI in patients without prior docetaxel exposure.

The standard regimen is 75 mg/m^2 every 3 weeks. An alternative to every-3-week docetaxel is a biweekly regimen of 50 mg/m^2. This regimen is based on a large randomized phase 2 trial of 346 patients with mCRPC randomized to either every-2-week docetaxel or every-3-week docetaxel, each with maintenance of ADT and prednisone.[610] Patients treated with the every-2-week regimen survived an average of 19.5 months compared to 17.0 months with the every-3-week regimen ($P = .015$). Time to progression and PSA decline rate favored every-2-week therapy. Tolerability was improved with every-2-week docetaxel; febrile neutropenia rate was 4% versus 14% and other toxicities and overall QOL were similar.

The duration of docetaxel therapy should be based on the assessment of benefit and toxicities. Treatment with ≥8 cycles of docetaxel may be associated with better OS than fewer cycles in the mCRPC setting.[611]

Retrospective analysis from the GETUG-AFU 15 trial suggests that docetaxel may benefit some patients with CRPC who received docetaxel in the CSPC setting.[612] Thus, the Panel believes that docetaxel can be given as a rechallenge after progression on an ARPI in the mCRPC setting if the patient's cancer did not demonstrate definitive evidence of progression on prior docetaxel therapy in either the castration-sensitive setting or the mCRPC setting.

Adverse events associated with docetaxel include neutropenia, leukopenia, febrile neutropenia, neutropenic infections, fluid retention, hypersensitivity reaction, hepatic function impairment, neuropathy, and other low-grade adverse events (eg, fatigue, nausea, vomiting, alopecia, diarrhea).

Cabazitaxel

In June 2010, the FDA approved cabazitaxel, a semi-synthetic taxane derivative, for patients with mCRPC previously treated with a docetaxel-containing regimen. An international randomized phase 3 trial (TROPIC) randomized 755 patients with progressive mCRPC to receive cabazitaxel 25 mg/m^2 or mitoxantrone 12 mg/m^2, each with daily prednisone.[613] A 2.4-month improvement in OS was demonstrated with cabazitaxel compared to mitoxantrone (HR, 0.72; $P < .0001$). The improvement in survival was balanced against a higher toxic death rate with cabazitaxel (4.9% vs. 1.9%), which was due, in large part, to differences in rates of sepsis and renal failure. Febrile neutropenia was observed in 7.5% of patients treated with cabazitaxel versus 1.3% of patients treated with mitoxantrone. The incidences of severe diarrhea (6%), fatigue (5%), nausea/vomiting (2%), anemia (11%), and thrombocytopenia (4%) also were higher in patients treated with cabazitaxel, which indicated the need for vigilance and treatment or prophylaxis in this setting to prevent febrile neutropenia. The survival benefit was sustained at an updated analysis with a median follow-up of 25.5 months.[614] Furthermore, results of a post-hoc analysis of this trial suggested that the occurrence of grade ≥3 neutropenia after

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cabazitaxel treatment was associated with improvements in both PFS and OS.[615]

The multicenter CARD study was a randomized, open-label clinical trial that compared cabazitaxel with either abiraterone or enzalutamide in 255 patients with mCRPC who had previously received docetaxel and either abiraterone or enzalutamide.[616] Cabazitaxel at 25 mg/m^2 with concurrent steroid improved the primary endpoint of radiographic PFS (8.0 vs. 3.7 months; HR, 0.54; $P < .0001$) and reduced the risk of death (13.6 vs. 11.0 months; HR, 0.64; $P = .008$) compared with abiraterone or enzalutamide in these patients. Cabazitaxel was also associated with an increased rate of pain response and delayed time to pain progression and SREs.[617]

The phase 3 open-label, multinational, noninferiority PROSELICA study compared 20 mg/m^2 cabazitaxel with 25 mg/m^2 cabazitaxel in 1200 patients with mCRPC who progressed on docetaxel.[618] The lower dose was found to be noninferior to the higher dose for median OS (13.4 months [95% CI, 12.19–14.88] vs. 14.5 months [95% CI, 13.47–15.28]), and grade 3/4 adverse events were decreased (39.7% vs. 54.5%). In particular, grade ≥3 neutropenia rates were 41.8% and 73.3% for the lower and higher dose groups, respectively.

Results from the phase 3 FIRSTANA study suggest that cabazitaxel has clinical activity in patients with chemotherapy-naïve mCRPC.[619] Median OS, the primary endpoint, was similar between 20 mg/m^2 cabazitaxel, 25 mg/m^2 cabazitaxel, and 75 mg/m^2 docetaxel (24.5 months, 25.2 months, and 24.3 months, respectively). Cabazitaxel was associated with lower rates of peripheral sensory neuropathy than docetaxel, particularly at 20 mg/m^2 (12% vs. 25%). However, the Panel does not currently recommend cabazitaxel in docetaxel-naïve patients.

Based on these data, cabazitaxel is included in these Guidelines as a category 1, preferred option after exposure to docetaxel and an ARPI in patients with mCRPC. Cabazitaxel at 20 mg/m^2 every 3 weeks, with or without growth factor support, is the recommended dose for fit patients. Cabazitaxel at 25 mg/m^2 may be considered for healthy patients who opt for more aggressive treatment. Biweekly cabazitaxel at 16 mg/m^2 with prophylactic G-CSF is an option for patients ≥65 years based on results from the phase 3 CABASTY trial.[620] Using the lower dose significantly reduced the risk of neutropenia/neutropenic complications compared to the 25 mg/m^2 dose with G-CSF. Clinical outcomes were comparable between the 2 groups.

Cabazitaxel should be given with concurrent steroids (daily prednisone or dexamethasone on the day of chemotherapy). Physicians should follow current guidelines for prophylactic white blood cell growth factor use, particularly in this heavily pretreated population. In addition, supportive care should include antiemetics (prophylactic antihistamines, H2 antagonists, and corticosteroids prophylaxis) and symptom-directed antidiarrheal agents. Cabazitaxel was tested in patients with hepatic dysfunction in a small, phase I, dose-escalation study.[621] Cabazitaxel was tolerated in patients with mild to moderate hepatic impairment. However, cabazitaxel should not be used in patients with severe hepatic dysfunction. Cabazitaxel should be stopped upon clinical disease progression or intolerance.

Cabazitaxel/Carboplatin

Cabazitaxel 20 mg/m² plus carboplatin AUC 4 mg/mL per minute with growth factor support can be considered post-ARPI for fit patients with aggressive variant mCRPC (visceral metastases, low PSA and bulky disease, high lactate dehydrogenase [LDH], high carcinoembryonic antigen [CEA], lytic bone metastases, and neuroendocrine prostate cancer [NEPC] histology) or unfavorable genomics (defects in at least 2 of *PTEN*, *TP53*, and *RB1*). This recommendation is based on a phase 1–2, open label, randomized study.[622] In the phase 2 portion, 160 patients were

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randomized to receive cabazitaxel alone or with carboplatin, and the primary endpoint was investigator-assessed PFS. In the ITT population, median PFS was 4.5 months in the cabazitaxel arm versus 7.3 months in the cabazitaxel/carboplatin arm (HR, 0.69; 95% CI, 0.50–0.95; *P* = .018). The most common grade 3–5 adverse events (fatigue, anemia, neutropenia, and thrombocytopenia) were all more common in the combination arm. Post-hoc analyses showed that patients with aggressive variant disease had a longer median PFS in the combination arm than the cabazitaxel arm (7.5 vs. 1.7 months; *P* = .017). Patients without aggressive variant tumors, on the other hand, had similar median PFS regardless of treatment (6.5 vs. 6.3 months; *P* = .38).

Sipuleucel-T

In April 2010, sipuleucel-T became the first in a new class of cancer immunotherapeutic agents to be approved by the FDA. This autologous cancer "vaccine" involves collection of the white blood cell fraction-containing, antigen-presenting cells from each patient; exposure of the cells to the prostatic acid phosphatase-granulocyte macrophage colony-stimulating factor (PAP-GM-CSF recombinant fusion protein); and subsequent reinfusion of the cells. The pivotal study was a phase 3, multicenter, randomized, double-blind trial (D9902B).[623] Five hundred twelve patients with minimally symptomatic or asymptomatic mCRPC were randomized 2:1 to receive sipuleucel-T or placebo. Eighteen-point two percent of patients had received prior chemotherapy, which included docetaxel; eligibility requirements included no chemotherapy for 3 months and no steroids for 1 month prior to enrollment. Median survival in the vaccine arm was 25.8 months compared to 21.7 months in the control arm. In a subset analysis, both those who did and those who did not receive prior chemotherapy benefited from sipuleucel-T treatment. Sipuleucel-T treatment resulted in a 22% reduction in mortality risk (HR, 0.78; 95% CI, 0.61–0.98; *P* = .03). Common complications included mild to moderate chills (54.1%), pyrexia (29.3%), and headache (16.0%), which usually were transient.

A prospective registry of patients with mCRPC, PROCEED, enrolled 1976 patients from 2011 to 2017, who were followed for a median of 46.6 months.[624] The safety and tolerability of sipuleucel-T were consistent with previous findings, and the median OS was 30.7 months (95% CI, 28.6–32.2 months).

Sipuleucel-T is included in these Guidelines as an option for asymptomatic patients with mCRPC regardless of prior ARPI or docetaxel, although the Panel notes that the data supporting its use post-docetaxel is limited. Benefit of sipuleucel-T has not been reported in patients with visceral metastases and is not recommended if visceral metastases are present. Sipuleucel-T is also not recommended for patients with small cell prostate cancer/NEPC. The Panel prefers that sipuleucel-T be used as a therapy for asymptomatic or minimally symptomatic patients with mCRPC, so that disease burden is lower and immune function is potentially more intact. Patients should have good performance level (ECOG 0–1), estimated life expectancy >6 months, and no liver metastases. Clinicians and patients should be aware that the usual markers of benefit (decline in PSA and improvement in bone or CT scans) are not seen. Therefore, benefit to the individual patient cannot be ascertained using currently available testing.

Pembrolizumab

The FDA approved the tumor-agnostic use of pembrolizumab, an anti-PD-1 antibody, for treatment of patients with unresectable or metastatic MSI-H or dMMR solid tumors who have progressed on prior treatment and who have no satisfactory alternative treatment options in May 2017. This approval was based on the treatment of 149 patients across five clinical studies involving MSI-H or dMMR colorectal (n = 90) or non-colorectal (n = 59) cancer for an objective response rate of 40% (59/149).[592] All patients

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received ≥1 prior regimen. Among the non-colorectal cohorts, two patients had mCRPC: one achieved a partial objective response, and the other achieved stable disease for >9 months.

Outcomes of additional patients with mCRPC treated with pembrolizumab have been reported.[82,625-629] In an early study, 10 patients with CRPC and non-visceral metastases (bone = 7; lymph nodes = 2; bone and liver = 1) who had disease progression on enzalutamide were treated with pembrolizumab and enzalutamide.[625] Some of the patients also had experienced disease progression on additional therapies (docetaxel for CSPC, abiraterone, and/or sipuleucel-T). Three of the 10 patients showed a near complete PSA response. Two of these three patients had radiographically measurable disease and achieved a partial radiographic response (including a response in liver metastases). Of the remaining patients, three showed stable disease, and four displayed no evidence of clinical benefit. Genetic analysis of biopsy tissue revealed that one patient whose disease showed PSA response had an MSI-H tumor, whereas the other patient with responsive disease and two with non-responsive disease did not. The nonrandomized phase Ib KEYNOTE-028 trial included 23 patients with advanced, progressive prostate cancer, of whom 74% had received ≥2 previous therapies for metastatic disease.[627] The objective response rate by investigator review was 17.4% (95% CI, 5.0%–38.8%), with four confirmed partial responses. Eight patients (34.8%) had stable disease. Treatment-related adverse events occurred in 61% of patients after a median follow-up of 7.9 months; 17% of the cohort experienced grade 3/4 events (ie, grade 4 lipase increase, grade 3 peripheral neuropathy, grade 3 asthenia, grade 3 fatigue).

KEYNOTE-199 was a multi-cohort, open-label phase II study in 258 patients with mCRPC and prior treatment with docetaxel and at least one ARPI that assessed pembrolizumab in patients regardless of MSI status.[630] Cohorts 1 and 2 included patients with programmed cell death ligand 1 (PD-L1)–positive (n = 133) and PD-L1–negative (n = 66) prostate cancer, respectively. Cohort 3 included those with bone-predominant disease with positive or negative PD-L1 expression (n = 59). The primary endpoint of overall response rate (ORR) was 5% (95% CI, 2%–11%) in cohort 1 and 3% (95% CI, <1% to 11%) in cohort 2. Responses were durable (range, 1.9 to ≥21.8 months).

The most common adverse events from pembrolizumab are fatigue, pruritus, diarrhea, anorexia, constipation, nausea, rash, fever, cough, dyspnea, and musculoskeletal pain. Pembrolizumab also may be associated with immune-mediated side effects, which include colitis, hepatitis, endocrinopathies, pneumonitis, or nephritis.

Based on the available data, the Panel includes pembrolizumab as an option for patients with MSI-H or dMMR mCRPC (category 2B). The prevalence of MMR deficiency in metastatic CPRC is estimated at 2% to 5%,[44,626,631] and testing for MSI-H or dMMR can be performed using DNA testing or immunohistochemistry. If tumor MSI-H or dMMR is identified, the Panel recommends referral to genetic counseling for consideration of germline testing for Lynch syndrome.

In June 2020, the FDA granted accelerated approval for pembrolizumab's tumor-agnostic use in patients with unresectable or metastatic TMB-high (TMB-H) [≥10 mutations/megabase (mut/Mb)] solid tumors that have progressed following prior treatment and who have no satisfactory alternative treatment options. Results from prospective biomarker analysis of the multicohort, non-randomized, open-label, phase 2 KEYNOTE-158 trial support this approval, but this trial did not include any patients with prostate cancer.[632] One retrospective study found that 1.5% of patients with prostate cancer had TMB-H tumors.[631] Of those patients, 8 had received an immune checkpoint inhibitor; 4 patients (50%) experienced a reduction in PSA of ≥50% that lasted at least 1 week. The Panel therefore

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notes that pembrolizumab may be associated with some benefit in patients with mCRPC and TMB ≥10 mut/Mb.

Mitoxantrone

Two randomized trials assessed the role of mitoxantrone in patients with mCRPC.[633,634] Although there was no improvement in OS, palliative responses and improvements in QOL were seen with mitoxantrone.

Mitoxantrone can be used for palliation in symptomatic patients with mCRPC who cannot tolerate other therapies after disease progression on prior docetaxel and an ARPI.

Treatment Options for Patients with DNA Repair Gene Mutations

Early studies suggest germline and somatic mutations in HRR genes (eg, *BRCA1, BRCA2, ATM, PALB2, FANCA, RAD51D, CHEK2*) may be predictive of the clinical benefit of poly-ADP ribose polymerase (PARP) inhibitors.[635-637] PARP inhibitors are oral agents that exert their activity through the concept of synthetic lethality.[638] PARP inhibitor therapy options are discussed below.

DNA repair defects have also been reported to be predictive for sensitivity to platinum agents in CRPC and other cancers.[639-643] Platinum agents have shown some activity in patients with CRPC without molecular selection.[644] Studies of platinum agents in patients with CRPC that have DNA repair gene mutations are needed.

Results of one study suggested that patients with mCRPC and germline mutations in DNA repair genes may have better outcomes if treated with abiraterone or enzalutamide than with taxanes.[52] However, it should be noted that the response in patients with mCRPC and HRR gene mutations to standard therapies is similar to the response in patients without mutations.[645,646]

Patients with *CDK12* mutations tend to have aggressive disease, with high rates of metastases and short OS. Their disease also does not respond well to hormonal therapy, PARP inhibitors, or taxanes. Two large, multi-institutional, retrospective studies have shown that 11% to 33% of patients with mCRPC and *CDK12* mutations experienced disease response to PD-1 inhibitors (ie, nivolumab, pembrolizumab), some with durable responses.[647,648] There are also limited data from phase 2 trials indicating that ipilimumab plus nivolumab may have some activity against *CDK12*-mutated mCRPC.[649,650] The Panel awaits more data on the use of PD-1 inhibition in patients with *CDK12* mutations.

Olaparib

Preliminary clinical data using olaparib suggested favorable activity of this agent in patients with HRR gene mutations, but not in those without HRR mutations.[636,637,651] The phase 3 PROfound study was a randomized trial evaluating olaparib 300 mg twice daily versus physician's choice of abiraterone or enzalutamide in patients with mCRPC and progression on at least one novel hormonal agent (abiraterone or enzalutamide) and up to one prior taxane agent (permitted but not required).[652] Patients were required to have a somatic or germline HRR gene mutation, and were allocated to one of two cohorts: cohort A comprised patients with *BRCA1/2* or *ATM* mutations, and cohort B comprised patients with a mutation in at least one of 12 other HRR genes (*BARD1, BRIP1, CDK12, CHEK1, CHEK2, FANCL, PALB2, PPP2R2A, RAD51B, RAD51C, RAD51D*, or *RAD54L*). The primary endpoint of improving radiographic PFS with olaparib versus abiraterone/enzalutamide was met in cohort A (HR, 0.34; 95% CI, 0.25–0.47; $P < .001$), and radiographic PFS was also superior in the entire study population encompassing cohorts A+B (HR, 0.49; 95% CI, 0.38–0.63; $P < .001$).

In addition, final analysis of PROfound showed that OS was improved with olaparib versus abiraterone/enzalutamide in cohort A (HR, 0.69; 95% CI,

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0.50–0.97; *P* = .02), despite the fact that 86 of 131 patients (66%) crossed over to olaparib after disease progression in the control arm.[653]

The Panel notes that there may be heterogeneity of response to olaparib based on which gene has a mutation. Efficacy in PROfound appears to be driven by the cohort of patients with at least one alteration in *BRCA2*, *BRCA1*, or *ATM*, and in particular by patients with *BRCA2* or *BRCA1* mutations based on exploratory gene-by-gene analysis.[653] Patients with *BRCA2* mutations in PROfound experienced an OS benefit with olaparib (HR, 0.59; 95% CI, 0.37–0.95), whereas the HR for OS in patients with *ATM* mutations was 0.93 (95% CI, 0.53–1.75).[653] Furthermore, there were few patients in PROfound with mutations in some of the genes. For example, only 4 patients had *BRIP1* mutations (2 in olaparib arm and 2 in control arm), 2 patients had *RAD51D* mutations (both in olaparib arm), and no patients had *RAD51C* mutations.[652] Patients with *PPP2R2A* mutations in PROfound experienced an unfavorable risk-benefit profile.

As a result of the favorable efficacy data from the PROfound trial, the FDA approved olaparib (300 mg twice daily) in May 2020 for use in patients with mCRPC and deleterious or suspected deleterious germline or somatic HRR gene mutations in at least one of 14 genes (*BRCA1, BRCA2, ATM, BARD1, BRIP1, CDK12, CHEK1, CHEK2, FANCL, PALB2, RAD51B, RAD51C, RAD51D*, or *RAD54L*) and who had previously received treatment with enzalutamide or abiraterone.

Adverse events that may occur with olaparib treatment include anemia (including that requiring transfusion), fatigue, nausea or vomiting, anorexia, weight loss, diarrhea, thrombocytopenia, creatinine elevation, cough, and dyspnea. Rare but serious side effects may include thromboembolic events (including pulmonary emboli), drug-induced pneumonitis, and a theoretical risk of myelodysplasia or acute myeloid leukemia.[652]

Since some patients in PROfound had prior taxane therapy, olaparib use might be reasonable in patients with mCRPC before or after docetaxel treatment. The Panel therefore recommends olaparib as an option for patients with mCRPC, previous ARPI, and an HRRm regardless of prior docetaxel therapy. The HRR genes to be considered for use of olaparib are *BRCA1, BRCA2, ATM, BARD1, BRIP1, CDK12, CHEK1, CHEK2, FANCL, PALB2, RAD51B, RAD51C, RAD51D* and *RAD54L*. Olaparib is included in the Guidelines as a category 1 recommendation for BRCAm mCRPC and is a preferred option for these patients if they have not yet received docetaxel.

Any commercially available analytically and clinically validated somatic tumor and germline assays can be used to identify patients for treatment. The Panel strongly recommends a metastatic biopsy for histologic and molecular evaluation; a plasma circulating tumor DNA (ctDNA) assay can be used if a metastatic biopsy is unsafe or not feasible.

Careful monitoring of complete blood counts and hepatic and renal function, along with type and screens and potential transfusion support and/or dose reductions as needed for severe anemia or intolerance are recommended during olaparib therapy.

Rucaparib

Rucaparib is another PARP inhibitor approved for use in patients with mCRPC. This agent received accelerated FDA approval in May 2020 based on the preliminary favorable data from the TRITON2 clinical trial. In that open-label, single-arm, phase 2 trial, patients with mCRPC harboring a deleterious or suspected deleterious germline or somatic *BRCA1* or *BRCA2* mutation, who had previously received therapy with an ARPI plus one taxane chemotherapy, were treated with rucaparib 600 mg twice daily.[654] The primary endpoint of TRITON2 was the objective response rate in patients with measurable disease, and was 43.5% (95% CI, 31.0%–56.7%) in this *BRCA1/2*-mutated population. Median radiographic

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PFS, a key secondary endpoint, was 9.0 months (95% CI, 8.3–13.5 months). The most common adverse events were asthenia/fatigue, nausea, and anemia/decreased hemoglobin, with grade ≥3 anemia/decreased hemoglobin in 25.2% of participants. Final analysis of TRITON2 confirmed results of the earlier analysis.[655]

In the randomized phase 3 TRITON3 study, patients with mCRPC and a germline or somatic *BRCA1/2* or *ATM* mutation who have previously received an ARPI but no chemotherapy for mCRPC were randomized 2:1 to rucaparib versus physician's choice of therapy (abiraterone, enzalutamide, or docetaxel).[656] The primary endpoint of TRITON3, the median duration of imaging-based PFS, was significantly longer at 62 months in the group of 270 participants assigned to receive rucaparib than in the 135 participants who received a control medication (10.2 vs. 6.4 months; HR, 0.61; 95% CI, 0.47–0.80; $P < .001$). This effect was also seen in the 201 patients and 101 patients in each group with a *BRCA*m (11.2 vs. 6.4 months; HR, 0.50; 95% CI, 0.36–0.69). For those with *ATM* mutations, an exploratory analysis suggested a possible improvement as well (8.1 vs. 6.8 months; HR, 0.95; 95% CI, 0.59–1.52). As in TRITON2, the most frequent adverse events with rucaparib were fatigue and nausea.

The Panel recommends rucaparib as an option for patients with mCRPC, a *BRCA1* or *BRCA2* mutation, and prior treatment with an ARPI. It is a category 1, preferred option pre-docetaxel. Rucaparib should not be used in patients with HRR gene mutations other than *BRCA1/2*.[657] Adverse events that may occur with rucaparib include anemia (including that requiring transfusion), fatigue, asthenia, nausea or vomiting, anorexia, weight loss, diarrhea or constipation, thrombocytopenia, increased creatinine, increased liver transaminases, and rash. Rare but serious side effects of rucaparib include a theoretical risk of myelodysplasia or acute myeloid leukemia, as well as fetal teratogenicity.[654,657]

The preferred method of selecting patients for rucaparib treatment is somatic and germline analysis of *BRCA1* and *BRCA2* from a metastatic biopsy. A ctDNA sample can be used if biopsy is unsafe or not feasible.

As with olaparib, careful monitoring of complete blood counts and hepatic and renal function, along with type and screens and potential transfusion support and/or dose reductions as needed for severe anemia or intolerance are recommended during treatment with rucaparib.

Olaparib Plus Abiraterone

Pre-clinical data suggest that PARP-1 promotes androgen receptor activity.[658] Additional pre-clinical data show that androgen receptor inhibitors can down-regulate DNA repair genes, creating a situation similar to that of HRR mutation.[659,660] These results suggest that the combination of PARP inhibition with androgen receptor inhibition may have an enhanced antitumor effect and that this effect may not be limited to patients with HRR mutations. In fact, a randomized phase 2 trial showed that the combination of abiraterone with olaparib increased radiographic PFS over abiraterone and placebo in patients with mCRPC regardless of HRR status (ITT population: HR, 0.65; 95% CI, 0.44–0.97; $P = .034$).[637]

The PROpel trial was an international, double-blind, phase 3 trial comparing abiraterone and olaparib with abiraterone and placebo in 796 patients with mCRPC regardless of HRR mutation status.[661] Prior docetaxel in the localized or mCSPC setting was allowed, but patients were untreated for CRPC. The primary endpoint, imaging-based PFS by investigator assessment in the ITT population, was significantly longer in the abiraterone/olaparib group than in the abiraterone/placebo group (24.8 vs. 16.6 months; HR, 0.66; 95% CI, 0.54–0.81; $P < .001$). HRR mutations were identified in tumors of 226 patients; 552 patients did not have HRR tumor mutations. The HR for the primary endpoint in those with HRR mutations was 0.50 (95% CI, 0.34–0.73). The safety profile of the olaparib/abiraterone combination was as expected based on the known

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safety profiles of the individual drugs, with the most common adverse events being anemia, fatigue/asthenia, and nausea.

Final OS data from PROpel showed that OS was not significantly improved with the abiraterone/olaparib combination therapy in the full cohort after a median follow up of approximately 36.5 months (42.1 vs 34.7 months; HR, 0.81; 95% CI, 0.67–1.00; *P* = .054).[662] In a post-hoc exploratory analysis, the BRCAm population saw an OS benefit, with a median OS of 23.0 months in the abiraterone arm and not reached in the combination arm (HR, 0.29; 95% CI, 0.14–0.56). A smaller OS benefit was seen in the HRRm group overall, and no OS benefit was evident in the non-HRRm and the non-BRCAm/other HRRm subgroups.

In May 2023, the FDA approved the combination of olaparib with abiraterone for the treatment of adult patients with BRCAm mCRPC. Based on the results of PROpel, olaparib/abiraterone is included in the NCCN Guidelines as an option for patients with mCRPC and a pathogenic *BRCA1* or *BRCA2* mutation (germline and/or somatic) who have not yet received an ARPI (category 1).

Talazoparib Plus Enzalutamide

Talazoparib is another PARP inhibitor; it has had an FDA indication in breast cancer. The open-label, international phase 2 TALAPRO-1 trial included 127 patients with an HRR mutation and progressive, mCRPC, all of whom received at least one dose of talazoparib.[663] The objective response rate after a median follow-up of 16.4 months was 29.8% (95% CI, 21.2–39.6). The most common grade 3–4 treatment-emergent adverse events were anemia (31%), thrombocytopenia (9%), and neutropenia (8%).

As noted above (see *Olaparib Plus Abiraterone*), pre-clinical data suggest that the PARP inhibition combined with androgen receptor inhibition may have an enhanced antitumor effect that may not be limited to those with HRR mutations. The randomized, double-blind, phase 3 TALAPRO-2 study compared enzalutamide plus talazoparib with enzalutamide plus placebo in 805 patients with untreated mCRPC.[664] HRR gene alteration status and treatment with docetaxel and/or abiraterone in the castration-sensitive setting were used to stratify the randomization. The primary endpoint was radiographic PFS in the ITT population. At the planned primary analysis, median radiographic PFS was not reached (95% CI, 27.5 months–not reached) for the talazoparib group and was 21.9 months (95% CI, 16.6–25.1) for the control group (HR, 0.63; 95% CI, 0.51–0.78; *P* < .0001).

HRR mutations were present in 21% of TALAPRO-2 participants, with *BRCA* alterations being the most common.[664] The HR for radiographic PFS in the HRR-deficient subgroup was more strongly in favor of the talazoparib combination than in the HRR-proficient/unknown population (0.46 [95% CI, 0.30–0.70; *P* = .0003] vs. 0.70 [95% CI, 0.54–0.89; *P* = .0039]). Among HRR mutations, talazoparib conferred a 77% lower risk of radiographic progression or death in those with tumor mutations in *BRCA1* or *BRCA2* (HR, 0.23; 95% CI, 0.10–0.53; *P* = .0002), whereas the corresponding reduction was 34% (HR, 0.66; 95% CI, 0.39–1.12; *P* = .12) in those with non-*BRCA* HRR alterations.

Prior therapy also affected the radiographic PFS outcomes in this trial.[664] In the 179 participants in TALAPRO-2 who had received docetaxel in earlier disease settings, the HR for radiographic PFS was 0.51 (95% CI, 0.32–0.81; *P* = .0034). In the small population of 50 participants in the ITT population who had received prior novel hormonal therapy, the corresponding HR was non-significant at 0.57 (95% CI, 0.28–1.16; *P* = .12).

Final results from an HRRm-only cohort of TALAPRO-2 at a median follow-up of 44.2 months showed that median OS was improved with the combination compared with enzalutamide alone (45.1 vs. 31.1 months;

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HR, 0.62; 95% CI, 0.48–0.81; two-sided *P* = .0005).[665] Median radiographic PFS was also improved with the talazoparib/enzalutamide group compared with the enzalutamide group (30.7 vs. 12.3 months; HR, 0.47; 95% CI, 0.36–0.61; *P* < .0001).

The safety profile of enzalutamide plus talazoparib was consistent with the known safety profiles of the individual drugs, with the most common adverse events in those who received talazoparib being anemia, neutropenia, and fatigue. However, hematologic adverse events were of higher grades and occurred more frequently than would be expected with talazoparib alone. Overall, the combination had significant toxicity, with dose interruption due to adverse events in 75% of participants in the talazoparib group compared with 23% in the placebo group. Dose reductions due to adverse events occurred in 56% and 7% of the talazoparib and placebo groups, respectively.

Based on the results from the first TALAPRO-2 cohort, the FDA approved talazoparib plus enzalutamide for HRRm mCRPC in June 2023. The Panel includes talazoparib plus enzalutamide as a category 1 treatment option for patients with mCRPC and a pathogenic mutation (germline and/or somatic) in one of certain HRR and other DNA repair genes (*BRCA1, BRCA2, ATM, ATR, CDK12, CHEK2, FANCA, MLH1, MRE11A, NBN, PALB2*, or *RAD51C*) who have not yet had treatment with an ARPI. Use of talazoparib/enzalutamide for those who have received prior ARPI therapy without prior docetaxel is controversial (category 2B) because a benefit of this combination over use of a PARP inhibitor alone has not been shown in this setting, but responses are likely.

Niraparib Plus Abiraterone

Another PARP inhibitor, niraparib, has also been studied in combination with androgen inhibition in the setting of mCRPC. The randomized, double-blinded phase 3 MAGNITUDE trial compared niraparib plus abiraterone to placebo plus abiraterone in 423 patients with mCRPC and HRR mutations and an additional 247 patients without HRR mutations.[666] Prior chemotherapy and novel hormonal therapy were allowed in the mCSPC or M0 CRPC settings, and were received by 3.1% and 20.1% of the total HRRm cohort, respectively.

The primary endpoint of MAGNITUDE was radiographic PFS. After a median follow-up of 18.6 months, radiographic PFS was improved for those receiving niraparib in the HRRm group overall (16.5 vs. 13.7 months; HR, 0.73; 95% CI, 0.56–0.96; *P* = .022) as well as in the BRCAm subgroup (16.6 vs. 10.9 months; HR, 0.53; 95% CI, 0.36–0.79; *P* = .001). However, radiographic PFS was not improved in the subgroup of patients with non-*BRCA* HRR mutations (HR, 0.99; 95% CI, 0.68–1.44). For the cohort without HRR mutations, futility was declared based on prespecified criteria. The secondary endpoints of time to symptomatic progression and time to initiation of cytotoxic chemotherapy were improved with the combination therapy in the HRRm and BRCAm cohorts.

A second interim analysis of MAGNITUDE included a prespecified, inverse probability censoring weighting analysis of OS, which was designed to account for the receipt of subsequent therapies, including PARP inhibitors.[667] Results of this analysis suggest that there may be an OS benefit for the combination therapy (HR, 0.54; 95% CI, 0.33–0.90; nominal *P* = .0181).

The incidence of grade 3/4 adverse events was higher with the combination of niraparib plus abiraterone compared with placebo and abiraterone (67.0% vs. 46.4%).[666] Anemia (28.3% vs. 7.6%) and hypertension (14.6% vs. 12.3%) were the most reported grade ≥3 adverse events. Overall, the combination was tolerable and QOL was maintained.

Based on these results, the FDA approved niraparib plus abiraterone for the treatment of patients with BRCAm mCRPC in August 2023. The Panel includes niraparib plus abiraterone as a treatment option for patients with

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mCRPC and a pathogenic *BRCA1* or *BRCA2* mutation (germline and/or somatic) who have not yet had treatment in the setting of mCRPC. This is a category 1 recommendation for those without prior ARPI. Use of niraparib/abiraterone for those who have received a prior ARPI without prior docetaxel is controversial (category 2B) because a benefit of this combination over use of a PARP inhibitor alone has not been shown in this setting.

Radiopharmaceuticals for mCRPC

Lutetium Lu 177 vipivotide tetraxetan

Lu-177-PSMA-617 is a radiopharmaceutical that is administered intravenously and is indicated for PSMA-positive mCRPC that has been treated with androgen receptor pathway inhibition and taxane-based chemotherapy. The active moiety is a radionuclide that delivers radiation to PSMA-expressing and surrounding cells, which induces DNA damage and leads to cell death. The approval of Lu-177-PSMA-617 was based on the international, open-label phase III VISION trial of 831 patients with mCRPC and PSMA-positive metastatic lesions. Patients in VISION were previously treated with at least one androgen receptor-directed therapy and one or two taxane-based chemotherapy regimens.[668] Patients had at least one PSMA-positive metastatic lesion and no PSMA-negative lesions determined by Ga-68 labeled PSMA-11 PET/CT imaging. Patients were randomized in a 2:1 ratio to receive standard of care (abiraterone, enzalutamide, bisphosphonates, RT, denosumab, and/or glucocorticoids) and Lu-177-PSMA-617 (7.4 GBq or 200 mCi every 6 weeks for 4–6 cycles) or standard of care alone.

The median OS was improved in the Lu-177-PSMA-617 group compared to the control group (15.3 vs. 11.3 months; HR, 0.62; 95% CI, 0.52–0.74; $P < .001$). Similarly, the median PFS was improved in the Lu-177-PSMA-617 group compared to the control group (8.7 vs. 3.4 months; HR, 0.40; 99.2% CI, 0.29–0.57; $P < .001$). The incidence of grade ≥3 adverse events (particularly anemia, thrombocytopenia, lymphopenia, and fatigue) was significantly higher in the Lu-177-PSMA-617 group compared to the control group. [668]

The FDA approved Lu-177-PSMA-617 in the post-ARPI setting in March 2022.

Another randomized controlled phase 3 trial, PSMAfore, assessed the efficacy of Lu-177-PSMA-617 pre-docetaxel in 468 patients with mCRPC who experienced disease progression on an ARPI.[669] The primary endpoint of rPFS in the ITT population was improved with Lu-177-PSMA-617 compared with a change in ARPI. At a median of 24.1 months after randomization, median radiographic PFS was 11.6 months in the Lu-177-PSMA-617 group versus 5.6 months in the control group (HR 0.49; 95% CI, 0.39–0.61). There was no difference in OS, but OS data are difficult to interpret because patients were allowed to crossover from an ARPI change to Lu-177-PSMA-617 upon radiographic progression, and 57% of patients in the control arm did so. Importantly, there were fewer grade 3–5 toxicities in Lu-177-PSMA-617 arm.

Based on the results of the PSMAfore trial, the FDA expanded the indication for Lu-177-PSMA-617 to include adult patients with PSMA-positive mCRPC who have progressed after ARPI therapy and are considered appropriate candidates for delaying taxane-based chemotherapy.

The NCCN Panel recommends Lu-177-PSMA-617 as a treatment option for patients with one or more PSMA-positive lesion and/or metastatic disease that is predominately PSMA-positive and with no dominant PSMA-negative metastatic lesions who have been treated previously with an ARPI and a taxane-based chemotherapy (category 1) or who have prior ARPI therapy and are considered appropriate to delay taxane-based chemotherapy.

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PSMA-negative lesions are defined as metastatic disease that lacks PSMA uptake including bone with soft tissue components ≥1.0 cm, lymph nodes ≥2.5 cm in short axis, and solid organ metastases ≥1.0 cm in size. The Panel notes that Ga-68 PSMA-11, F-18 piflufolastat PSMA, or F-18 flotufolastat PSMA can be used to identify patients eligible for treatment with Lu-177–PSMA-617.

Radium-223

In May 2013, the FDA approved radium-223 dichloride, an alpha particle-emitting radioactive agent. This first-in-class radiopharmaceutical was approved for treatment of mCRPC in patients with symptomatic bone metastases and no known visceral metastatic disease. Approval was based on clinical data from a multicenter, phase 3, randomized trial (ALSYMPCA) that included 921 patients with symptomatic CRPC, two or more bone metastases, and no known visceral disease.[670] Fifty-seven percent of the patients received prior docetaxel and all patients received best supportive care. Patients were randomized in a 2:1 ratio to 6 monthly radium-223 intravenous injections or placebo. Compared to placebo, radium-223 significantly improved OS (median 14.9 vs. 11.3 months; HR, 0.70; 95% CI, 0.058–0.83; $P < .001$) and prolonged time to first SRE (median 15.6 vs. 9.8 months). Preplanned subset analyses showed that the survival benefit of radium-223 was maintained regardless of prior docetaxel use.[671] ITT analyses from ALSYMPCA showed that radium-223 also may reduce the risk of symptomatic SREs.[672] Grade 3/4 hematologic toxicity was low (3% neutropenia, 6% thrombocytopenia, and 13% anemia), likely due to the short range of radioactivity.[670] Fecal elimination of the agent led to generally mild non-hematologic side effects, which included nausea, diarrhea, and vomiting. Radium-223 was associated with improved or slower decline of QOL in ALSYMPCA.[673]

The multicenter, international, double-blind, placebo-controlled, phase 3 ERA 223 trial randomized patients with bone-metastatic chemotherapy-naïve CRPC to abiraterone with or without radium-223.[674] The patients were asymptomatic or mildly symptomatic. The primary endpoint of symptomatic skeletal event-free survival in the ITT population was not met. In fact, the addition of radium-223 to abiraterone was associated with an increased frequency of bone fractures compared with placebo.

The randomized PEACE-3 trial also compared radium-223 with or without an ARPI in patients who were ARPI-naive.[675] Radium-223 with enzalutamide was compared with enzalutamide therapy alone in 446 patients with mildly symptomatic mCRPC. The use of bone-protecting agents (denosumab or zoledronic acid) was made mandatory following results from ERA 223. The primary endpoint of radiological PFS was improved in the combination arm compared with enzalutamide alone (16.4 vs. 19.4 months; HR, 0.69; 95% CI, 0.54–0.87; $P = .0009$). At a preplanned interim OS analysis, median OS was also improved with the addition of radium-223 (35.0 vs. 42.3 months; HR, 0.69; 95% CI 0.52–0.90; $P = .0031$). Grade ≥3 adverse events occurred more commonly in the combination arm (65.6% vs. 55.8%) and included hypertension (in 34% of the combination arm), fatigue (6%), fracture (5%), anemia (5%), and neutropenia (5%). Fractures were also more common in the combination arm (24.3% vs. 13.4%).

In an earlier safety analysis of PEACE-3, the cumulative incidence of fractures at 1.5 years in patients who received a bone-protecting agent was 2.8% in participants receiving radium-223 plus enzalutamide and 3.9% in those receiving enzalutamide alone.[676] In the absence of bone agents, these numbers were 45.9% and 22.3%, respectively. This result suggests that radium-223 combined with an ARPI may be safe if preventive administration of a bone agent is used.

Radium-223 is a category 1 option to treat symptomatic bone metastases without visceral metastases in patients with mCRPC regardless of prior therapy. Radium-223 plus enzalutamide is included as an option for

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patients with bone-metastatic CRPC without prior exposure to an ARPI. Hematologic evaluation should be performed according to the FDA label before treatment initiation and before each subsequent dose.[592] Radium-223 given in combination with chemotherapy (such as docetaxel) outside of a clinical trial has the potential for additive myelosuppression.[592] Its use in combination with docetaxel or any other systemic therapy except ADT or enzalutamide should be pursued with caution. It should not be used in patients with visceral metastases. All patients receiving radium-223 should be given concomitant denosumab or zoledronic acid.

MDT for mCRPC

Although most data supporting the use of MDT for oligometastatic prostate cancer is in the oligorecurrent CSPC setting, as discussed in detail above (see *MDT for Oligometastatic CSPC*), MDT has also been studied in the oligoprogressive CRPC. As noted above, the EXTEND trial included 7 patients with oligoprogressive CRPC, although results of MDT in this subset specifically were not reported.[568]

ARTO is a phase 2 study that included 157 patients with CRPC and 1 to 3 metastatic lesions who were randomized to receive abiraterone alone or abiraterone with MDRT.[566] The rate of biochemical response (defined as a ≥50% decrease in PSA levels at 6 months compared to baseline), which was the primary endpoint, was 92% in the MDRT group compared with 68.3% in the control group (OR, 4.22; 95% CI, 2.12–8.38; $P < .001$). PFS, a secondary endpoint was also improved with the use of MDRT (HR, 0.35; 95% CI, 0.21–0.57; $P < .001$). A subgroup analysis of ARTO further suggested that MDRT for oligoprogressive mCRPC may result in similar PFS as second-line systemic therapy.[677]

Initial results from the phase 2 GROUQ-PCS-9 were presented at the 2025 ASCO Genitourinary Cancers Symposium.[678] MDRT added to ADT and enzalutamide in oligometastatic CRPC (with 1–5 metastases) led to improvements in radiological PFS, biochemical PFS, and time to next line of therapy compared to ADT with enzalutamide alone.

The Panel includes MDT with mCRPC systemic therapy as an option for patients with oligometastatic or oligoprogressive CRPC regardless of prior therapy.

Small Cell/Neuroendocrine Prostate Cancer

De novo small cell carcinoma in untreated prostate cancer occurs rarely and is very aggressive.[679] Treatment-associated small cell prostate cancer/NEPC that occurs in patients with mCRPC is more common.[680] In a multi-institution prospective series of 202 consecutive patients with mCRPC, all of whom underwent metastatic biopsies, small cell/neuroendocrine histology was present in 17% of patients.[680] Patients with small cell/neuroendocrine tumors and prior abiraterone and/or enzalutamide had a shorter OS when compared with those with adenocarcinoma and prior abiraterone and/or enzalutamide (HR, 2.02; 95% CI, 1.07–3.82). Genomic analysis showed that DNA repair mutations and small cell/neuroendocrine histology were almost mutually exclusive.

Small cell/neuroendocrine carcinoma of the prostate should be considered in patients with disease that no longer responds to ADT and who test positive for metastases. These relatively rare tumors are associated with low PSA levels despite large metastatic burden and visceral disease.[681] Those with initial Grade Group 5 are especially at risk. Biopsy of accessible metastatic lesions to identify patients with small cell/neuroendocrine histomorphologic features is recommended in patients with mCRPC.

These patients may be treated with cytotoxic chemotherapy (ie, cisplatin/etoposide, carboplatin/etoposide, docetaxel/carboplatin, cabazitaxel/carboplatin).[622,682,683] Physicians should consult the NCCN Guidelines for Small Cell Lung Cancer for additional options in the first and

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subsequent lines of therapy (available at www.NCCN.org), because the behavior of small cell/neuroendocrine carcinoma of the prostate is similar to that of small cell carcinoma of the lung.

Additional Treatment Options for Bone Metastases

In a multicenter study, 643 patients with CRPC and asymptomatic or minimally symptomatic bone metastases were randomized to intravenous zoledronic acid every 3 weeks or placebo.[684] At 15 months, fewer patients in the zoledronic acid 4-mg group than patients in the placebo group had SREs (33% vs. 44%; $P = .02$). An update at 24 months also revealed an increase in the median time to first SRE (488 vs. 321 days; $P = .01$).[685] No significant differences were found in OS. Other bisphosphonates have not been shown to be effective for prevention of disease-related skeletal complications.

The randomized TRAPEZE trial used a 2 X 2 factorial design to compare clinical PFS (pain progression, SREs, or death) as the primary outcome in 757 patients with bone-metastatic CRPC treated with docetaxel alone or with zoledronic acid, 89Sr, or both.[686] The bone-directed therapies had no statistically significant effect on the primary outcome or on OS in unadjusted analysis. However, adjusted analysis revealed a small effect for 89Sr on clinical PFS (HR, 0.85; 95% CI, 0.73–0.99; $P = .03$). For secondary outcomes, zoledronic acid improved the SRE-free interval (HR, 0.78; 95% CI, 0.65–0.95; $P = .01$) and decreased the total SREs (424 vs. 605) compared with docetaxel alone.

Denosumab was compared to zoledronic acid in a randomized, double-blind, placebo-controlled study in patients with CRPC.[687] The absolute incidence of SREs was similar in the two groups; however, the median time to first SRE was delayed by 3.6 months by denosumab compared to zoledronic acid (20.7 vs. 17.1 months; $P = .0002$ for noninferiority; $P = .008$ for superiority). The rates of important SREs with denosumab were similar to zoledronic acid and included spinal cord compression (3% vs. 4%), need for radiation (19% vs. 21%), and pathologic fracture (14% vs. 15%). Treatment-related toxicities reported for zoledronic acid and denosumab were similar and included hypocalcemia (more common with denosumab 13% vs. 6%), arthralgias, and osteonecrosis of the jaw (ONJ, 1%–2% incidence).

Therefore, denosumab every 4 weeks (category 1, preferred) or zoledronic acid every 3 to 4 weeks is recommended for patients with CRPC and bone metastases to prevent or delay disease-associated SREs. SREs include pathologic fractures, spinal cord compression, operation, or EBRT to bone. The optimal duration of zoledronic acid or denosumab in patients with CRPC and bone metastases remains unclear. A multi-institutional, open-label, randomized trial in 1822 patients with bone-metastatic prostate cancer, breast cancer, or multiple myeloma found that zoledronic acid every 12 weeks was noninferior to zoledronic acid every 4 weeks.[688] In the every-12-week and every-4-week arms, 28.6% and 29.5% experienced at least 1 SRE within 2 years of randomization, respectively.

Use of zoledronic acid in patients with CSPC and bone metastases is not associated with lower risk for SREs.[689] Therefore, the routine use of these agents in bone-metastatic CSPC is not recommended. Bone antiresorptive agents should, however, be used for SRE prevention in patients with CSPC if they have treatment-related bone loss (see *Principles of Survivorship, Bone Health in Prostate Cancer*, in the algorithm above).

Oral hygiene, baseline dental evaluation for high-risk individuals, and avoidance of invasive dental surgery during therapy are recommended to reduce the risk of ONJ.[690] Most, but not all, patients who develop ONJ have preexisting dental problems.[691] If invasive dental surgery is necessary, therapy should be deferred until the dentist confirms that the patient has healed completely from the dental procedure. Supplemental

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calcium and vitamin D are recommended to prevent hypocalcemia in patients receiving either denosumab or zoledronic acid.

Monitoring of creatinine clearance is required to guide dosing of zoledronic acid. Zoledronic acid should be dose reduced in patients with impaired renal function (estimated creatinine clearance 30–60 mL/min) and held for creatinine clearance <30 mL/min. Denosumab may be administered to patients with impaired renal function or even patients on hemodialysis; however, the risk for severe hypocalcemia and hypophosphatemia is greater, and the dose, schedule, and safety of denosumab have not yet been defined. A single study of 55 patients with creatinine clearance <30 mL/min or on hemodialysis evaluated the use of 60-mg-dose denosumab.[592] Hypocalcemia should be corrected before starting denosumab, and serum calcium monitoring is required for denosumab and recommended for zoledronic acid, with repletion as needed.

Radium-223 is a category 1 option to treat symptomatic bone metastases in patients with mCRPC without visceral metastases (see *Radium-223*, above). The use of palliative RT is also an option.

Clinical research on the prevention or delay of disease spread to bone continues. A phase 3 randomized trial of 1432 patients with non-metastatic CRPC at high risk of bone involvement showed that denosumab delayed bone metastasis by 4 months compared to placebo.[692] OS was not improved, and the FDA did not approve denosumab for the prevention of bone metastases.

Considerations for Visceral Metastases

The Panel defines visceral metastases as those occurring in the liver, lung, adrenal gland, peritoneum, or brain. Soft tissue/lymph node sites are not considered visceral metastases. In general, there are fewer data on treatment of patients with CRPC and visceral metastases than for those without visceral metastases.

Sequencing of Therapy in CRPC

The number of treatment options for patients with CRPC has expanded rapidly over the past several years. Although the optimal sequence of therapies remains undefined, some data that can help with treatment selection in some cases continues to emerge.

After abiraterone or enzalutamide, data suggest that giving the alternate ARPI may not be the optimal strategy considering the availability of other treatment options, including chemotherapy. The CARD trial, for instance, showed that treatment with cabazitaxel significantly improved clinical outcomes over enzalutamide or abiraterone in patients with mCRPC who had been previously treated with docetaxel and the alternate hormonal therapy (abiraterone or enzalutamide).[616] Furthermore, data suggest cross-resistance between abiraterone and enzalutamide.[693-696] Results of a randomized, open-label, phase 2, crossover trial suggest that the sequence of abiraterone followed by enzalutamide may be more efficacious than the reverse.[697]

Some data inform the sequencing of therapies in patients with PSMA-positive mCRPC. The multicenter, unblinded, randomized phase 2 TheraP trial compared PSA response after Lu-177-PSMA-617 versus cabazitaxel in 200 patients with PSMA-positive mCRPC who previously received docetaxel.[698] Prior androgen receptor-directed therapy was permitted. Among the ITT population, the PSA response rate was 66% in the Lu-177-PSMA-617 arm compared with 37% in the cabazitaxel arm (difference 29%; 95% CI, 16–42; $P < .0001$). These numbers were 66% and 44%, respectively, in those who received treatment (difference 23%; 95% CI, 9–37; $P = .0016$). Furthermore, grade 3–4 adverse events were less frequent in the Lu-177-PSMA-617 arm than in the cabazitaxel arm (33% vs. 53%). Results from the phase 3 PSMAfore trial as discussed above (see *Lutetium Lu 177 vipivotide tetraxetan*) showed that Lu-177-PSMA-617

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improved rPFS compared with switching to a different ARPI in docetaxel-naïve patients.[669]

Data for patients with HRRm mCRPC are more limited, but comparative effectiveness research suggests that olaparib may result in superior radiographic PFS than cabazitaxel in patients with *BRCA1* or *BRCA2* mutations and prior treatment with docetaxel.[699] Furthermore, data from PROfound and TRITON3 suggest that a PARP inhibitor is preferred over a different ARPI in patients with BRCAm mCRPC and prior ARPI exposure.[652,656]

No chemotherapy regimen has demonstrated improved survival or QOL after cabazitaxel or cabazitaxel/carboplatin in patients with mCRPC of adenocarcinoma histology, although several systemic agents other than mitoxantrone have shown palliative and radiographic response benefits in clinical trials (ie, carboplatin, cyclophosphamide, doxorubicin, vinorelbine, carboplatin/etoposide, docetaxel/carboplatin, gemcitabine/oxaliplatin, paclitaxel/carboplatin).[700-709] No survival benefit for any these combination regimens over sequential single-agent regimens has been demonstrated, and toxicity is higher. Treatment with these regimens could be considered after an informed discussion between the physician and an individual patient about treatment goals and risks/side effects and alternatives, which must include best supportive care. In patients not able to receive life prolonging therapy, prednisone and dexamethasone at low doses may provide palliative benefits.[710] Participation in a clinical trial is encouraged.

Summary

The intention of these guidelines is to provide a framework on which to base treatment decisions. Prostate cancer is a complex disease, with many controversial aspects of management and with limited data to support some of the treatment recommendations. Several variables (including adjusted life expectancy, disease characteristics, predicted outcomes, and patient preferences) must be considered by the patient and physician to tailor prostate cancer therapy for the individual patient.

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Table 1: Selected Active Surveillance Experiences with Large Patient Cohorts

Cohort	Toronto[272,313,319]	Johns Hopkins[275,312,314,711]	UCSF[281]		Canary PASS[289]	Cooley/Catalona Meta-Dataset[277]	PRIAS[278]
			Initial Cohort	Newer Cohort			
Number of patients	993	1298	321	810	2155	6775	5302
Median age (y)	68	66	63	62	63	64	66
Core involvement and risk groups	% of cohort with ≤2 positive cores, 69 25% IR, D'Amico criteria	Median # positive cores, 1	Mean % positive cores, 20.3%	Not available	Median % positive cores, 12.5% 17% NCCN IR/HR	% of cohort with ≤2 positive cores, 77.6	% of cohort with ≤2 positive cores, 99
Median follow-up (months)	77	60	43	60	86	80	120
Conversion to treatment*	36.5% (10-y) 45% (15-y)	50% (10-y)	24% (3-y)	40% (5-y)	49% (10-y)	33% (6.7-y)	52% (5-y) 73% (10-y)
Systemic progression Lymph node involvement and/or metastasis	6.6% systemic progression in IR group 3.1% (1.8% distant metastases; 1.3% pos. lymph nodes)	0.15% distant metastases 0.08% pos. lymph nodes	0% distant metastases 0.2% pos. lymph nodes	0.1%	0.5% distant metastases 0.5% pos. lymph nodes	0.4%	0.2%
Cancer-specific survival	98% (10-y)	99.9% (10-y)	100% (5-y)	100% (5-y)	99.9% (10-y)	99.8% (6.7-y)	>99% (10-y)
Overall survival	80% (10-y)	93% (10-y)	98% (10-y)	98% (5-y)	94.3% (10-y)	—	—
***Reason for conversion to treatment (% of entire cohort)**							
Gleason grade change	9.5%	15.1%	38%	—	43% (10-y)	49%	34% (5-y) 41% (20-y)[a]
PSA increase	11.7%	—	26%	—	—	8.5%	—
Tumor volume increase	—	—	—	—	—	7.2%	—
Personal choice	1.6%	8%	8%	—	—	5% (anxiety)	5%

IR = intermediate risk; HR = high risk.
[a]Protocol-based reclassification (included change in Gleason grade, number of positive cores, or cT stage)

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